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For reviewer's eyes only

The Forgotten Half of the Patent Clause

*How Federal Regulations Doubled the Cost of
American Living
and Rewrote the Constitution*

How a Constitutional Bargain Was Cancelled —
Restoring It Frees Buyers from Surveillance,
Stops Robots Replacing America's Drivers & Operators,
and Returns the Joy of Operating Real Machines

A constitutional law litigation strategy
and a citizen's playbook — in one book.

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Foreword by Gord Magill

Author, *End of the Road: Inside the War on Truckers*
Substack, *Autonomous Truck(er)s* • Host, *Voice of GO(r)D*

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A Note to the Executive Who Received This Book

I sent you this advance copy myself. Here is why.

Your business runs on the professional driver. A driver buys a meal, takes a shower, parks for the night, and has the truck serviced. A driverless truck still needs service, but it never walks inside.

This book makes the constitutional case for keeping those drivers on the road. The Constitution lets Congress protect inventors for “limited Times.” That limit predates the first patent law by two and a half years, so it binds everything Congress does. When a patent expires, the public is supposed to be free to build what it covered. Since about 1970, federal rules have blocked Americans from rebuilding proven, patent-expired machines, including the older heavy trucks owner-operators trust and can repair. The book shows how that barrier can be challenged in court, and what it has cost: a wealth transfer of at least \$22 trillion, about \$3,385 per household each year. A win would put rebuilt trucks, and their drivers, back on the road.

The foreword is by Gord Magill, a working trucker and the author of *End of the Road: Inside the War on Truckers*. He has discussed trucking twice on The Tucker Carlson Show, and on C-SPAN's Washington Journal, War Room, and The Ingraham Angle. On September 18, 2026, I recorded an interview about this book with Timothy Dooner for his trucking podcast; the episode airs this fall.

What I am asking. Skyhorse Publishing, which published *The Real Anthony Fauci* and *Melania*, will publish this book if about 3,000 copies are committed in advance. Three partners at 1,000 copies each would do it. A short, non-binding letter saying your company would stock about 1,000 copies once it is published would make the difference. No money comes to me. Any order would go through your normal buying channels, from the publisher or your usual distributor, on standard wholesale terms.

Why the timing matters. Goldman Sachs Research projects the cost of autonomous trucking falling below a conventional new truck around 2028 and reaching about \$2.03 a mile by 2035. A remanufactured pre-2000 truck runs about \$2.11. A case filed now reaches the Supreme Court between 2030 and 2033. On Goldman's own line, the autonomous truck costs about \$2.40 a mile in 2030 and about \$2.18 in 2033. So in 2030 the remanufactured truck wins clearly; by 2033 it is a coin flip. Worse, the falling autonomous cost depends on volume the platforms have not yet won. A ruling in 2030 checks a fleet that has barely begun; a ruling in 2033 checks one with five more years of scale. That difference is why this book needs to be in print now.

If you are short on time, start here:

Chapter 18. The Last Human Driver — the per-mile cost of a driverless truck against a remanufactured one

Picture Section: The Forgotten Half in 23 Images — the whole argument in ten minutes

Foreword, by Gord Magill

Chapter 15. The Trucker's Burden — the constitutional case for the American cab

Two full chapters, a summary of the book, and an interest sign-up are at https://marti124.github.io/Patent_BkSignup/. Colleagues who want their own copy can ask me for the confidential online version, which they can read today, or a printed copy, which takes three to four weeks. Please keep it within your company; the book is not yet public.

I would be glad to talk whenever it suits you.

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End of inserted note — the book resumes on the next page

A note before you begin

This review copy does not
contain the book's endnotes,
glossary, and key terms.

That section runs to
243 pages and lives online,
where every citation is a
live link:

**[marti124.github.io/
Patent_EndNotes/](https://marti124.github.io/Patent_EndNotes/)**

*The last two pages of this
copy explain why.*

TABLE OF CONTENTS

Author's Review Copy

Front Matter

Foreword — By Gord Magill.....	7
Preface.....	17
Picture Section: The Forgotten Half in 23 Images.....	41

PART I — The Bargain and the Betrayal

Chapter 1. Four Problems, One Remedy.....	61
Chapter 2. Jefferson's Bargain.....	81
Chapter 3. How the Bargain Was Cancelled: The Regulatory Ratchet.....	91
Chapter 4. The Anti-Backsliding Provision: How 42 U.S.C. § 6295(o)(1) Locked the Ratchet.....	101
Chapter 5. The Wall Itself: Seventeen Laws That Hold the Patent Bargain Hostage.....	113
Chapter 6. Why Nobody Noticed: How a Constitutional Violation Hides in Plain Sight.....	125
Chapter 7. You No Longer Need to Break a Law to Challenge One.....	145
Chapter 8. The \$22 Trillion Wealth Transfer.....	153
Chapter 9. Why Wealthy Nations Break the Patent Promise That Poor Nations Honor.....	163
Chapter 10. The Environmental Question: Why the Patent Bargain Does Not Unleash 1970s Emissions.....	173

PART II — The Historical Witness: When the Bargain Worked

Chapter 11. When the Bargain Worked: Pre-1900 American Prosperity.....	187
Chapter 12. The Bargain Still Works: Bicycles, Calculators, and the Political Warning of Firearms.....	203

PART III — The Modern Injury: Six American Kitchen Tables

Chapter 13. The \$72,495 Pickup Truck: Cars, Trucks, and the Household Budget.....215

Chapter 14. How Prohibited Expired Patents Double Your Grocery Bill.....225

Chapter 15. The Trucker’s Burden: A Constitutional Brief for the American Cab.....235

Chapter 16. Healthy Lighting and the Anti-Backsliding Trap.....251

Chapter 17. The Surveillance Premium.....261

Chapter 18. The Last Human Driver.....271

PART IV — Where the Patent Bargain Only Partly Reaches

Chapter 19. The Hindered (not Blocked) Sector: Pharmaceuticals and Medical Devices.....287

PART V — Restoring the Patent Bargain: Who Can Sue and How

Chapter 20. Sovereign and Public-Fiduciary Plaintiffs.....297

Chapter 21. Direct and Associational Plaintiff Categories.....305

Chapter 22. Why You Want to Be the Plaintiff: The Personal, Political, and Institutional Case for Filing.....313

Chapter 23. The Filable Case: Why 2026 and Not Before.....325

Chapter 24. The Patent Clause Prong: “Limited Times” as Constitutive Language.....333

Chapter 25. The Takings Clause Prong: Property the Government Cannot Cancel for Free.....355

Chapter 26. How a Remake Is Built and Certified.....377

PART VI — Testing the Argument

Chapter 27. Objections Considered.....389

Chapter 28. Reason the Constitutional Argument
Yourself: The Open-Book Examination.....429

PART VII — After the Ruling

Chapter 29. The Policy Framework After a
Successful Ruling.....475

Chapter 30. If a President Read This Book.....485

Chapter 31. Five Years On: The Plaintiffs’ Recoveries.....509

Chapter 32. Five Years On: The Household Recovery.....521

PART VIII — Keeping the Bargain Whole

Chapter 33. Hybrid Legislative Voting: How to Stop
the Ratchet from Resetting.....533

Chapter 34. What You Can Do: A Citizen’s Playbook.....555

PART IX — Why an Outsider Saw It

Chapter 35. When Outsiders Advance Constitutional
Law and Other Disciplines.....563

Back Matter

Epilogue.....579

Endnotes587

Glossary — electronic edition

Key Terms — electronic edition

Appendix A. SCOTUS Case Master Table — electronic
edition

Appendix B. A Note on AI-Assisted Authorship and the U.S.
Copyright Office’s January 2025 Guidance — electronic edition

Appendix C. LLM Responses to Prompt 7 in the Open-Book
Examination — electronic edition

Foreword

*"My uncle has a country place
That no one knows about
He says it used to be a farm
Before the Motor Law
And on Sundays I elude the eyes
And hop the turbine freight
To far outside the wire
Where my white-haired uncle waits"*

And so begins the 1981 song 'Red Barchetta' by Canadian progressive rock band Rush, a musical prophecy influenced by a dystopian short story once published in the vaunted car guy magazine 'Road and Track', way back in 1973. The song describes a young man escaping the confines of what some dissidents today call the 'Bug Hive' of urbanity, and re-connecting with the spirit of life itself through operation of a well preserved though forbidden motor car, a 'Red Barchetta' of the title. The authorities are not amused, and the song continues with the protagonist speeding along to evade capture.

'Red Barchetta' may have seemed pessimistic to the average person in 1981, where the dictates brought about by the massive expansion of the regulatory state in 1970 were still somewhat nascent, and had yet to metastasize on the body politic to the cancerous and deadly levels they have grown to in 2026. Cars and trucks were still somewhat inexpensive, easy to work on, and modifiable in ways accessible to the average enthusiast of the time; they did not require software updates, nor were drivers emasculated by digital intermediations of their operation. Many of those enthusiasts were teenagers who, unlike today, obtained their drivers licenses as soon as possible,

and were not caught under the spell of the digital realm via ubiquitous Satanic Rectangles, whose implicit communications reek of compliance and conformity. Other versions of those rectangles now dominate all new vehicles, and the programming within them makes the cars into very expensive and difficult to maintain surveillance iPads on wheels.

The profound and notable resonance of 'Red Barchetta' in 2026, along with other works of dystopian forecasting such as *1984* by George Orwell, naturally elicit the question of 'How did we get here?', and, 'What can we do about it?' Though 'Red Barchetta' may speak to a perceived rebellious or contrarian inclination, it resonates because the 'Motor Law' described in the song is not just about motors, but about nearly every facet of modern life now hobbled by the friction of technological interruption which produces nothing of value and interfere with nearly every human activity and interaction while also capturing highly detailed bits of information about your life to be fed into a voracious surveillance panopticon. The song also resonates because working class people in the west are being kneecapped by massive cost of living increases that are a consequence of so many of these 'Motor Laws' brought to us by Safetyism and Climate Alarmism, a pair of wholly manufactured ideologies which have now been weaponized for profit by corporate America, and are backed by a regulatory state which this book argues has interfered in a long lost constitutional bargain.

In "The Forgotten Half of The Patent Clause - How Federal Regulations Doubled The Cost of American Living and Rewrote The Constitution" Roleigh Martin brings a number of very specific answers to 'How did we get here?' and likewise proposes a Constitutional Challenge that includes

many legitimate plaintiff categories in ‘What can WE do about it?’

Martin approached me to write this foreword for him not only because of our overlapping projects as writers trying to solve problems, but because I'm a member of one of the plaintiff categories Martin describes in the book, truckers, and the trucking industry has certainly been at the very shitty end of the government regulation stick. Anecdotes from my own book, *“End Of The Road - Inside The War on Truckers”* might serve as a small view of millions of examples of truckers being economically harmed by the ratcheting regulatory state, and also encapsulates how a return to the original understanding of the Patent Clause can help truckers avoid that stick, and all of the financial burdens whipped on to us by its very shitty end.

I tell about how a couple of dozen brand new trucks equipped with the latest in mandated emission control technology were sent to run on Canada's Ice Roads during the 2009 Winter Road season. Those units failed miserably, constantly breaking down, stalling out and becoming immobilized, thus greatly endangering the frozen lakes over which fuel and other supplies are hauled to remote diamond mines. The engineering solution was to make them run at higher RPMs and thus burn far more fuel than normal, losing any gains made by the new emission control systems. The older 1993 Peterbilt I was driving that season had none of these problems, and burned 30-40% less fuel than those newer trucks, a substantial cost differential that can break an owner-operator. I also discuss how 75% of the business at a Freightliner service center near where I live now is dedicated to solving emission control system problems, as well as how a logging truck I operated had sat for six months due to the

complexity of the bugs within the computer system which controlled these wicked and costly emissions systems.

The chapter of *'End of The Road'* these anecdotes are relayed from is called *'Punishment by Castration'*, and a great many of the examples in Roleigh's book follow the same pattern - a new mandate is brought to bear on some piece of equipment, be it a truck, a tractor, a water heater for your house, or anything else that requires energy to operate - which is the Castration. The purchase cost of that piece of equipment goes up, often quite substantially, as does the cost of maintaining the fragile systems imposed upon them to meet the dictates of 'those who worship the sky' - which are the Punishments. These mandates are part of what Roleigh calls a 'regulatory ratchet' which turns in only one direction. This ratchet combines with and enforces a mandated 'wall' that legally blocks anyone from building the older and cheaper versions of any of this equipment, which is where we run into the Patent Clause, and what Roleigh describes as a 'broken bargain' between what the Constitution grants us in Article 1, Section 8, Clause 8, and the forgotten part of that clause, 'limited times'. The patents for a 3406B Caterpillar engine expired in the 90s and the patent on every component to a 1993 Peterbilt expired in 2013. Under the 'limited times' part of the patent clause, on which Roleigh goes into great and granular detail in this book, anyone should be able to remanufacture that Caterpillar engine and the 93 Peterbilt. You can't use those trademarked brand names, but the designs for those technologies passed into the commons for anyone to build on their own a long time ago, but the regulators stand between you and your constitutional right to build and sell that patent expired equipment, and make a few miles without a selective catalytic reduction system and its Pig Piss Diesel Exhaust Fluid ruining your motor. Multiply these regular breakdowns across millions of

pieces of equipment in America, and the costs to the trucking industry (as well as farming, construction, and various others) reach unknown and untold billions in downtime and extra costs that come out of the pockets of hardworking truckers, and the wider economy. That 93 Peterbilt would also be exempt from the Fourth Amendment breaking 'Eye of Sauron' forced on truckers via the Electronic Logging Device mandate, an anti-human and emasculating system of surveillance which essentially tells a trucker when he may shit or when he may eat. An anti-American, anti-worker, and totalitarian abomination if there ever was one.

Not only truckers, but any other consumer in America who is sick and tired of the cost of new vehicles and their maintenance, along with these vehicles being turned into data harvesting machines for the surveillance state, would benefit from the 'limited times' aspect of the patent bargain being returned to them. And if you don't believe me or Roleigh Martin, none other than Jim Farley, the CEO of Ford, made the case for me in an interview given to the Detroit Free Press in June of 2026.

"But you don't want people repairing their own vehicles?"

Farley's reply: "No, that's, that's fine, not for warranty work, though. These are very complicated cars, and we don't think that's safe, for many of the repairs on our vehicles, someone at home like myself could never do it. I have no problem working on a '73 Bronco, but to work on a brand-new Bronco? I need all sorts of specialty tools. That's something that, um, you know, we would put people's lives at risk."

Farley may not have meant to give away the game here, but he did, and it is worth noting that even CEOs understand the joy, utility, and economic advantage in

working on your own equipment. I wonder if the irony is lost on Farley, as Henry Ford very explicitly made his cars accessible to the working man through their reasonable price and ease of maintenance. What a shame Farley has besmirched the legacy of Ford as he works hand in hand with the government to take these price and ease of maintenance advantages away from you, protected as Ford is by the 'regulatory wall' described by Martin in this book.

Regulatory walls are not the only culprit Martin takes to task as he walks us through the breaking of this guarantee between the Constitution and the American people. There are numerous court cases cited throughout in painstaking detail showing how the 'limited times' provision was purposely buried or ignored, and various groups of 'Bootleggers and Baptists' formed coalitions of convenience to create regulatory capture that screwed everyone out of their rights to patent expired equipment. This was partially done in order to satisfy the nebulous quasi-religious moral yearnings of various advocacy groups - most of whom today practice the new secular orthodoxy of progressivism, which is often at odds with the material needs of most people. These advocacy groups, often leftists in political orientation, have found themselves in very cozy relationships with the pimps of technology who are busy hemming us all into an inescapable surveillance panopticon. Martin also shows us how remanufacturing patent expired equipment works in the rest of the world, and how the savings benefits accrue to those countries' citizens. Why not revitalize American manufacturing with a new Golden Age of building patent expired equipment that people actually want and can afford? Many critics of NeoLuddites like me say that no one would purchase such older model equipment, but it's not even legal to make it, much less legal for me or any

other consumer to purchase brand new. How can these critics make such a claim when the consumer is prevented from making such a choice? Why not let the market make that call instead of the regulatory state?

Martin is not just here to accurately and precisely lay out how American society is being hosed with cost of living increases intimately related to the results of this breaking of the patent clause; he also shows us a novel approach to launching a Supreme Court challenge. Though I am personally not a fan of AI content anything, Martin has acted as a maestro in directing high end AI engines, specifically built for case law and legal research, to support any potential challenger to the regulatory wall that stands between the citizens and their rights to expired patents. With the Loper Bright vs Raimondo Supreme Court case in 2024 bringing us an end to the Chevron Deference doctrine, Martin believes now is the time to reforge the bargain promised to us, and this book is a very thorough examination of that broken promise and provides us a roadmap to repair it, which includes detailed breakdowns for a number of plaintiff categories, and letter templates for interested and passionate patriots to take action.

Martin talks about the 'ratcheting' effect of the regulatory state, and the anti-backsliding measures that our wicked regulators have built into their dictates. I believe there is a related 'ratchet' placed into our minds by our corporate-government-media ecosystem, and that is the heavily implied inevitability of 'the future.' Yes, there is a future, but who are they to dictate to us what it looks like, or that it may not be imbued with what worked in the past? Do we have the capacity to imagine something better for ourselves? Perhaps a near term future where American truckers aren't going broke because of the multiple pressures they experience across the board, over and above the looming

imposition of autonomous trucks, because they can buy a cheaply remanufactured patent expired rig that worked just fine when it was originally built? The farmers who don't lose their shirts because they have to pay ten times the cost of what a tractor or combine harvester costs in other countries? Lost beauty returned to our roads, as consumers are allowed to purchase re-manufactured cars from the 1960s or 70s, when the car companies primary design calculation was not compliance with the moralizing dictates of climate alarmism, which render today's cars into a beige lozenge shaped sameness, devoid of any human level aesthetic considerations whatsoever? What investments in the future are not being made because we are barred from the financial savings of purchasing cheaper and proven equipment? Do you remember when going for a drive was a respite from the constant desire for your attention, one of the last places an American could have some peace and quiet? Wouldn't you like to not have a screen staring you in the face while taking in your birthright of the American road trip? What of a return to a Golden Age of manufacturing in America when a right guaranteed to us by the Founders is restored? The future looks very bright without a wall blocking out the sunshine.

At a time when the public is growing increasingly skeptical of the excesses of the surveillance state, the loss of their savings and economic standing to ever increasing inflation and growing costs of living, and farmers are having public fights to obtain the 'right to repair' their own equipment, Roleigh Martin's book, which is really a challenge to the status quo and an instruction manual on how to change it, couldn't be more timely. There are futures available to us by honoring the bargains of the past, and from the cost savings realized when the parasites of regulatory capture are burned from the body politic by the rising Phoenix of the Constitution. In the hands of the right group of

dedicated patriots, this book could be the beginning of a consumer rights revolution, and a necessary escape valve out of our growing technocratic imperium.

Are you willing to elude the eyes and go outside the wire?

Gord Magill

August 2026, Ithaca, NY

Preface

Why a Co-inventor of an AI Patent Wrote a Book About What Happens After Patents Expire

Two words in the Constitution's Patent Clause — *limited Times* — were written to make every patent monopoly end, so that once a patent expires the public may freely make the invention. For most of American history that is what happened, and prices fell. But since about 1970 a wall of federal regulation has quietly cancelled the public's half of that bargain for roughly 58 percent of commerce. The patent on a truck, a tractor, a furnace, or a light bulb expires on schedule. No rule forbids remaking it outright. But a faithful remake of a decades-old engineering design must meet standards written for new designs, which it can rarely satisfy without ceasing to be that design. In practice that is a ban: the monopoly ends, and the original maker's protection from inexpensive patent-expired remanufacturing outlives it, with no end in sight. The result is a cost-of-living crisis that is, at its root, a constitutional one.

The reader does not have to take that argument on my word. In one of its chapters, this book puts the argument to an open examination. It hands the relevant Supreme Court holdings, the governing statutes, and a deliberately balanced brief — both sides given equal space — to independent artificial-intelligence models. It asks each, on its own, whether a real constitutional injury exists. I put that examination to eleven of them, built by rival companies and trained on different data. Among them was OpenCase, a legal-research model built for practicing lawyers and trained on the corpus of United States statutes and Supreme Court opinions. Every one agreed: on the authorities provided, the Patent Clause's forgotten

half states a genuine constitutional question that deserves its day in federal court. That is not proof the case will win. It is evidence that the question is real. The examination comes later in the book for a reason: the chapters before it supply everything it draws on, and the reader who follows the path laid out in the reading table below will arrive at it prepared — and able to run it personally.

Before You Begin: What to Read If Time Is Short

Read the Foreword first. Its author, Gord Magill, is a trucker who writes for truckers. No plaintiff category stands closer to this case than the men and women who drive semi-trucks for a living. The book's highest ambition is that this legal argument sees its day in court. The drivers are the likeliest ones to carry it there. Start with their voice.

Magill is not a marginal voice. He is a working trucker who became one of the most widely heard chroniclers of the industry. He has written for *Newsweek*, *The American Conservative*, and *American Affairs*; publishes the *Autonomous Truck(er)s* Substack; and hosts the *Voice of GO(r)D* podcast. His 2026 book, *End of the Road: Inside the War on Truckers*, carries blurbs from Matthew B. Crawford, Oren Cass, and the executive vice president of the Owner-Operator Independent Drivers Association. He has carried the trucking story onto the two largest platforms of the populist right. He appeared twice on *The Tucker Carlson Show* — among the most-watched independent programs in the country, which through mid-2025 averaged on the order of a million views an episode on YouTube alone — and on Steve Bannon's *WarRoom*, which streams several times each weekday and has ranked for years among the most popular political

podcasts in the country. When such a writer agrees to introduce a constitutional argument, he carries it into the rooms where the coalition this book must reach already gathers. The Endnote Commentary gives the citations.

After the Foreword, the table below lists the essential sections, in reading order, for the reader whose time is short. The order is deliberate — the environmental answer comes before the international evidence, so the savings arrive with the objection already retired. Stop when satisfied. Rank 4 exists for lawyers and the legally curious; ordinary readers may skip it without loss.

Finish this Preface first, even if you intend to jump. Two of its sections do work the table cannot. *The Question, the Author, and the Lateral Move* says who is making this argument and why the question was available to him and not to the specialists who had every reason to ask it sooner. *How This Book Is Documented* says how to check any claim in the book against its source, which matters more here than in most books, because a reader who doubts an argument this large should not have to take a single sentence of it on trust. Ten minutes on the two of them will make everything the table sends you to easier to weigh.

How long will it take? At an average adult reading pace, the Rank 1 sections take about 45 minutes. Ranks 1 and 2 together take about two hours and twenty minutes. Ranks 1 through 3 take about three and a half hours. The full table, all four ranks, takes about five hours. Running the examination yourself — seven prompts through eleven models — adds roughly forty-five minutes of hands-on time beyond the reading. The whole book is a longer commitment; the table is the short road in.

Section Title	Why Read	Rank
Chapter 35. When Outsiders Advance Constitutional Law and Other Disciplines	Read if you doubt that an outsider can move constitutional law forward, with or without the aid of AI, expertly and scrupulously used. If you are not in doubt, skip.	1
Picture Section: The Forgotten Half in 23 Images	The whole argument in 23 images. Ten minutes here shows what the regulations hide.	1
Chapter 1. Four Problems, One Remedy	The map of the book. Four problems, one constitutional remedy, stated plainly.	1
Chapter 6. Why Nobody Noticed: How a Constitutional Violation Hides in Plain Sight	Answers the first question every reader asks: if this is true, why has nobody said so?	2
Chapter 10. The Environmental Question: Why the Patent Bargain Does Not Unleash 1970s Emissions	Read this before the savings chapter. The remedy does not bring back 1970s smog.	2
Chapter 9. Why Wealthy Nations Break the Patent Promise That Poor Nations Honor	Not speculation. Ninety countries already allow remakes, with the savings priced across cars, trucks, semis, tractors, and combines.	2
Chapter 30. If a President Read This Book	The whole book in one story: the problem, the constitutional remedy, and the impact on the nation, told as a short fiction of how the case is won and what the country looks like after.	2
Chapter 24. The Patent Clause Prong: “Limited Times” as Constitutive Language	The first prong. Why “limited Times” means what it says.	3
Chapter 25. The Takings Clause Prong: Property the Government Cannot Cancel for Free	The second prong. What the government owes when it cancels a vested public right.	3
Chapter 34. What You Can Do: A Citizen’s Playbook	What one citizen can do, starting this week.	3
Chapter 20. Sovereign and Public-Fiduciary Plaintiffs	For lawyers: which sovereigns and public fiduciaries can sue.	4
Chapter 21. Direct and Associational Plaintiff Categories	For lawyers: the direct and associational plaintiffs with standing.	4

Section Title	Why Read	Rank
Chapter 23. The Filable Case: Why 2026 and Not Before	For lawyers: why the case is filable in 2026 and was not before.	4
Chapter 26. How a Remake Is Built and Certified	For lawyers and engineers: how a remake is built, certified, and kept honest.	4
Chapter 27. Objections Considered	For lawyers: the strongest objections, answered on what the cases actually hold.	4
Chapter 28. Reason the Constitutional Argument Yourself: The Open-Book Examination	For lawyers: hand the authorities to independent AI models and grade the constitutional question yourself.	4

How Different Readers Might Approach the Rest of the Book

No two readers come to this book for the same reason. After the essentials above, follow the path that fits you.

The household reader. You feel the wall at the dealership and the grocery store. Go next to *The \$72,495 Pickup Truck*, then *How Prohibited Expired Patents Double Your Grocery Bill*, then *The Surveillance Premium*. Finish with *Five Years On: The Household Recovery*. Those four chapters price the injury and the recovery at your own kitchen table. If the wall reaches your medicine cabinet as well as your driveway, add *The Hindered (not Blocked) Sector*, which covers the third of the economy that is not walled off outright but is held back — pharmaceuticals and medical devices, where the patent expires on schedule and the pathway from expiration to the patient does not.

The trucker and the operator. Your livelihood sits behind the wall. Read *The Trucker's Burden* first — it is the constitutional brief for the American cab. Then *The Last Human Driver*, *The Surveillance Premium*, and *The*

\$72,495 Pickup Truck. After those, *Five Years On: The Plaintiffs' Recoveries*. It shows what winning looks like for the drivers who file. Close with *What You Can Do: A Citizen's Playbook*, because the plaintiff categories include you.

The historian. You want the bargain before the betrayal. Read *Jefferson's Bargain*, then *When the Bargain Worked*, then *The Bargain Still Works*. Together they show two centuries in which expired patents entered the public domain and prices fell — and the three product categories where that engine still runs today.

The lawyer. Read Part V whole, in order, from *Sovereign and Public-Fiduciary Plaintiffs* through *How a Remake Is Built and Certified*, then *Objections Considered*, and the examination chapter, *Reason the Constitutional Argument Yourself*, where you can put the authorities to independent AI models and grade the question yourself. Give it two passes: one before you form a view, one after. Then test the foundation against Appendix A, which gives a plain-language note on every Supreme Court case the argument rests on.

The policymaker and the staffer. Start with *The Wall Itself*, which names the seventeen statutes. Then read *If a President Read This Book*, *The Policy Framework After a Successful Ruling*, and *Hybrid Legislative Voting*. The last shows how to keep the ratchet from resetting after the ruling.

The skeptic. Good. The book was built for you. Read *The Environmental Question*, then *Why Wealthy Nations Break the Patent Promise That Poor Nations Honor*, then *Objections Considered*. Then run the examination yourself: *Reason the Constitutional Argument Yourself*

hands the authorities to eleven independent AI models and lets you grade the constitutional question without taking anyone's word for it. The cases are cited for what they hold, and the inferences are flagged as inferences. Judge for yourself.

I. The Question, the Author, and the Lateral Move

The clause itself reads: Article I, Section 8, Clause 8 grants Congress the power “to promote the Progress of Science and useful Arts, by securing for *limited Times* to Authors and Inventors the exclusive Right to their respective Writings and Discoveries.” It was written in 1787, more than two years before the first patent law, so its words are a limit on securing exclusivity by any means, not merely a rule for patent statutes. A secured restriction against competition must expire. The book's claim is that in the roughly 58 percent of commerce built around moving vehicles, motorized equipment, and appliances, those two words have been removed from the Constitution in practice — by regulatory agencies, without an amendment and without a vote. The bill for that removal is measurable.

Households now pay roughly double, relative to income, what their parents paid in 1970 for cars, pickup trucks, tractors, furnaces, light bulbs, and household appliances, whether the doubling is measured against wages or against prices in the roughly ninety nations where patent-expired remanufacturing is lawful. The constitutional argument and the kitchen-table arithmetic are the same argument said in two different languages.

The book is the public-facing brief for the litigant who eventually files this case in federal court. No such case has been filed in the fifty-six years the regulatory wall has

stood. The body of the book explains why the question was never asked, why it can now be asked, what the remedy looks like, and what the citizen reading the book can do to push it toward a courtroom. This preface explains a different question: why I — a co-inventor of an AI patent, and neither a constitutional lawyer nor a litigator — came to be the one to write it.

I am a co-inventor of U.S. Patent No. 5,359,509, an early artificial-intelligence system for adjudicating health-insurance claims, once archived in the permanent research collection of the Smithsonian Institution. I spent thirty-four years inside the Information Technology department of a Fortune 25 health-care insurance company where my primary professional skills were troubleshooting complex code and database design. I got my B.S. from what is now Truman State University, with a triple minor in American literature, Russian literature, and physical science, and an M.A. in sociology from the University of Northern Iowa, with my thesis entitled “Towards an Holistic View of Social Policy Problems: An Heuristic Approach.” In 1978 I earned a diploma in data processing from Western Business College in Portland, Oregon, an institution that no longer exists.

Between 1990 and 1995 I founded, edited, and published ShareDebate International, one of the Internet’s first electronic magazines, documented in Tom Easton’s “Off the Main Sequence” (2006). It was a small journal whose contributors included Milton Friedman, Murray Rothbard, George Gilder, Doug Bandow, Eamonn Butler, Jerry Pournelle, Leonard Peikoff, three Prometheus-Award-winning science-fiction novelists, and Douglas Ostrom of the Japan Economic Institute.

Between 1998 and 2000 I was recognized as a Y2K thought leader, with sixty-one published articles, gave the keynote at the IQPC Y2K conference in London, and recognition by the U.S. President's Y2K Council and the World Bank.

I am a lifetime member of the Sierra Club and a former Northern Nevada chapter treasurer, and a hiking trainer and trip leader for the Sierra Club there. For 24 years I backpacked for about a month (on average) each summer in the High Sierra. Plenty of time to think about the world away from civilization. I am seventy-six and live in Northern Kentucky, and write full-time. I founded the Make American Living Affordable (MALA) movement on X, on one of X's largest communities which I created on election day 2024 and which has grown to more than 115,000 members. The more than 111 long-form articles I have written on the patent clause, Congressional history and complex social problems have drawn more than 7.8 million cumulative impressions.

I offer the biography, in compressed form, because the body of the book argues a thesis that disciplinary specialists — patent attorneys, environmental lawyers, regulatory economists, constitutional scholars — have spent fifty-six years not arguing. Why was this the man who saw it?

The honest answer is that an eclectic and cross-disciplinary background, the kind a career counselor would call unfocused, is the specific cognitive preparation — what Thomas Kuhn called the outsider's vantage on a settled paradigm — needed to see across the field boundaries that specialists are trained to respect. The lateral move from patent law to environmental regulation to constitutional jurisprudence to consumer arithmetic is

invisible to anyone whose paycheck depends on remaining inside one of those fields. It is visible to a person who has spent a working life moving across them. That is a claim about standing rather than about conclusions, and the section that follows puts evidence behind it.

My background is in software — first as a programmer, later as a database programmer. A compiler does not interpret intent. A compiler reads what you wrote, applies the language's rules, and returns either an executable result or a compilation error. It does not consult the spirit of the document; it consults the syntax of the document.

That habit of mind, applied for long enough, becomes how a person reads any structured text. It is how I read the Constitution. When I read Article I, Section 8, Clause 8, I do not read it the way constitutional lawyers are trained to read it. In addition to looking at how the framers might have meant beneath what they wrote, or what the practice has settled into in the centuries since, I also look at it as a “line of code.” The phrase that arrests the compiler-reading mind is *limited Times*. It has a type signature. It is a duration. Durations have a start and an end. After the end, the resource — the right to copy and remanufacture the invention — enters the public domain. None of that is interpretation. It is just reading the line of code as written.

Constitutional lawyers are trained to find meaning in penumbras and emanations — in the long shadow a written document casts after centuries of judicial elaboration. The training is necessary. Most constitutional questions are not, in fact, simple syntax. But a few are. The Patent Clause, on the question of what happens at the end of the limited time, is one of them.

Three decades ago I built the first English-language webpage on how to play Janggi, Korean chess — a game whose name means Generals' Game, and whose rules, like any rulebook, are not the spirit of the game but the game itself. A player who treats them as suggestions does not lose to a better strategist; he loses to the rulebook, because the rulebook is the only thing the position is being evaluated against.

What follows documents the habit of mind that made the lateral move feasible for one person working alone: the relationship to artificial intelligence.

II. The AI Paradox: Using the Tool to Disarm It

A book whose central concern is the cost of living for an ordinary American household cannot ignore that, in the year this book goes to print, the loudest single complaint Americans have about the technology industry is that artificial intelligence is coming for their jobs. The complaint is not unfounded. The first wave of large-language-model deployment is, by any honest measurement, displacing white-collar work in customer service, paralegal research, basic radiology, copywriting, junior accounting, and entry-level computer programming. The trucking industry's roughly five million drivers face a more specific version of the same threat: autonomous-vehicle programs at multiple firms now operate fleet pilots whose internal cost-per-mile projections undercut the human-driver baseline, and the regulatory environment that produced that cost asymmetry was the environment that had already made human-operated trucks expensive enough to be undercut in the first place. The animosity Americans feel toward AI in 2026

is the rational response of people watching the economic ground move under them.

Even within that picture, the math has begun to move in unexpected directions. An Axios report from April 2026 documented that information-technology budgets at multiple firms now exceed payroll, with computing costs running, in at least one named case, well beyond the cost of the employees the AI was meant to replace. The proposition that machine labor will always undercut human labor on price is, even today, an empirical claim under pressure. The most reliable lever for restoring the American worker's competitive footing is not to slow the deployment of the tool. It is to reduce the cost of what the worker's wages must cover to live.

This book is, paradoxically, an AI-assisted answer to the AI-assisted problem.

The mechanism is straightforward. The book argues that restoring the right to remanufacture patent-expired durable goods — pickup trucks built on a 1990s Ford F-150 platform, tractors built on a 1970-era John Deere platform, furnaces built on a turn-of-the-century Carrier platform — lets a buyer who chooses the remanufactured good pay half or less of the new price, while competition pulls down the price of new goods themselves by roughly fifteen percent, as later chapters show. A pickup truck driver whose rig costs \$25,000 instead of \$72,495, whose furnace costs \$3,000 installed instead of \$18,000, whose appliances cost half of what they cost today, is operationally cost-competitive against the autonomous vehicle at a baseline the autonomous vehicle's economics cannot match.

The semi-trucker chapter develops this point in detail: the human-driven, pre-2000-platform rig, freshly remanufactured to vintage-era federal standards, is the cost configuration that thwarts robotic replacement. The remedy the book proposes does not stop AI from being deployed in trucking. It changes the math underneath the deployment, so that the AI's labor-displacing advantage becomes the advantage of a tool, not the advantage of a regulatory subsidy. The same logic applies, in their own forms, to long-haul drivers, delivery drivers, rideshare and taxi operators, agricultural-equipment operators, and the trades that depend on operating equipment rather than being replaced by it.

What makes the paradox most acute is that none of the analysis behind that argument could have been done, by me alone at a kitchen table in Northern Kentucky, without the same artificial-intelligence tools whose downstream consequences the book is trying to mitigate.

I am not a constitutional lawyer. I am not a patent attorney. I have never been licensed to practice law, never clerked at a federal court, never taught at a law school. The case the book makes — across Supreme Court doctrine — is the kind of case that, in the institutional configuration of American constitutional advocacy, would normally be assembled by a research team at a law-school clinic or a public-interest legal organization. I assembled it instead, a retired data engineer working with disciplined access to a small set of AI tools: Anthropic's Claude for prose drafting and structural editing; Abacus AI's DeepAgent for multi-step source retrieval; OpenCase, a legal model trained on Cornell's Legal Information Institute datasets of United States statutes and Supreme Court opinions and indexed daily against open federal and state opinion archives, used to surface candidate authorities and to argue against the

thesis; X's SuperGrok for source material the other models cannot read; GPT-5 Thinking as a cross-model auditor of Claude's output; Perplexity Pro for general research; Nano Banana Pro for visual production; and Meta AI, DeepSeek Pro, Microsoft Copilot, and Gab's Arya as additional cross-model auditors and as respondents to the open-book examination. Every citation any tool produced was checked against the primary source before use.

On January 29, 2025, the U.S. Copyright Office published *Copyright and Artificial Intelligence, Part 2: Copyrightability* (copyright.gov/ai), concluding that works created by human authors using AI tools are eligible for copyright protection when the human provides substantial creative direction, selection, and arrangement. The Office confirmed that “the use of AI to assist in the process of creation or the inclusion of AI-generated material in a larger human-generated work does not bar copyrightability.” All constitutional arguments, litigation strategy, savings methodology, editorial structure, and substantive conclusions in this book are my own original intellectual work; AI tools served as research assistants and drafting aids under my direction across multiple working sessions.

The standard the Office applied is not a formality, and it is worth stating in the terms it used. Copyright attaches to what a human contributes — the selection, the arrangement, the creative direction — and not to material a machine generated on its own. A prompt is not authorship. What is protected is the judgment exercised over the output: what was kept, what was cut, what was wrong, what had to be checked again. On that standard the question of whether this book is mine is not close, and the manuscript record answers it.

The practical consequence for a reader is narrower and more useful than the legal one. A book assembled the way this one was assembled can be audited in a way a conventionally researched book cannot. Every authority is cited with a live link. Every figure is either checked against a primary source or marked as unchecked. Jamie Metzl's openly AI-co-authored *The AI Ten Commandments*, published in 2026, took the same approach to disclosure. The appendix on AI-assisted authorship gives the full account of which tool did what, and the raw model replies to the examination are reproduced in Appendix C without editing. A reader who distrusts the method has more to work with here, not less.

Most people use AI the way a child uses Legos, snapping a few blocks together and deciding it can't do anything sophisticated or new. Carpenters made the same misjudgment about power tools. Black & Decker's portable drill appeared in 1916 and the portable circular saw in 1924, but hand labor ruled the job site until after World War II. Mark Erlich's oral history of the Massachusetts trade (1986) records what happened when the Skilsaw finally arrived in force. The older men resisted, and they had cause. Early models carried marginal safety features. One carpenter watched a workmate nearly cut his leg off. Another hauled his Skilsaw to the roof and threw it off the building. "The saw's too fast," he said. "I'm going to cut by hand." He was judging a tool by its worst version.

The unions saw the sharper problem. The tools were expensive, and if a contractor expected his men to supply them, the jobs would go only to the carpenters who could afford the price. In 1932 the Springfield, Massachusetts, District Council proposed fining any member twenty-five dollars for furnishing electric machinery to an employer.

The fight was never against the saw. It was over who paid for it.

Both questions face anyone using AI now. An off-the-shelf chatbot is a fresh college graduate: quick, confident, and wrong often enough to cost you time. Training it is the first job — settings files, project files, a writing guide, response modes, cross-validation against rival models, self-audits, agent deep dives, and the plain fact that a chatbot is not an agent. The advanced tiers run past a hundred dollars a month, and without the subscription those models stay out of reach entirely.

Training the machine is the smaller half. Take a problem like the forgotten half of the Patent Clause. You have to be conversationalist, mentor, critic, debater, validator, and reader, often inside a single session. You have to bring your own library — history, law, engineering, economics — because the machine cannot tell you which discipline the answer is hiding in. You need a critical mind and the habit of thinking past the obvious. Linear logic checks the chain of reasoning. Lateral logic finds the chain worth checking. Without both, you will get fluent answers and never learn which ones were wrong. The Skilsaw did not make a man a carpenter. It made a good carpenter faster and a careless one dangerous.

As of 30 July 2026 this manuscript stood at forty-five files carrying seven hundred seventy-three saved revisions between them — an average of seventeen per file. The endnote apparatus was on its one hundred sixty-fifth version. A manuscript revised at that rate, for that long, is not the output of a prompt. It is the residue of a person deciding, several thousand times, what to keep, what to cut, what was wrong, and what had to be checked again.

That is a claim about standing, and it needs evidence rather than assertion. The evidence exists — not in law, where the record is thin, but in fields where an outsider's result could be checked by something physical.

In 2023 a physicist with no training in papyrology downloaded scans of a carbonized Herculaneum scroll that hundreds of contestants and the whole papyrological profession already had, looked at them longer than anyone else had, and noticed a faint cracked-mud texture that turned out to be ink — the first evidence of writing ever found inside an unopened scroll. A twenty-one-year-old undergraduate read his blog post, trained a model to find more of it, and recovered the Greek word for purple. Papyrologists at Naples and Pisa then read the text and said whether it was Greek. In 2017 a software engineer put a neural network on the weak Kepler signals the professionals had reasonably skipped and surfaced an eighth planet around Kepler-90; an astronomer worked the problem with him and co-authored the paper, and the planet is listed in NASA's archive alongside every other confirmed world.

The pattern is not that expertise proved unnecessary. Papyrologists read the Greek. An astronomer did the astronomy. What the outsider supplied was the question, and what the field supplied was the judgment about whether the answer was true. That is the whole claim, and the chapter *When Outsiders Advance Constitutional Law and Other Disciplines* sets out twenty cases of it — in law, medicine, geology, astronomy, linguistics, and financial regulation — with the four constitutional cases sharing one feature: the fact that moved the doctrine came from outside the law every time.

So this book does not claim to have been independently verified. It claims to be checkable, which is a different property and a more useful one. Every case, statute, and regulation it relies on is cited so that a reader can open it and see for himself. This book hands the reader the primary sources and invites him to reason the argument independently, and reports what happens when independent reasoners do. A former federal energy regulatory attorney has read successive drafts across a long correspondence, and disagreed with parts of them; I rewrote the first chapter in consequence. Practicing attorneys are now engaging with the argument publicly, under their own names and with their reputations attached. That is not a certification that the argument is right. It is evidence that people with something to lose consider it worth arguing with.

One thing belongs here plainly rather than left for a reader to discover on his own. Artificial intelligence made this analysis possible for one person working alone. Artificial intelligence is also, at this moment, putting fabricated legal citations in front of judges — Damien Charlotin’s public database logs the decisions that caught them, and by August 2026 it had passed nineteen hundred. The failure in every one of those cases has the same shape: somebody put his name on a citation he had not read. The remedy is not to put the tool down. It is to read the citation. Every authority in this book was opened.

The tools were not used as ghost-writers. They were used as a small newsroom. Claude drafted prose and edited structure. OpenCase verified citations against the authoritative federal record and twice flagged circuit-court conflicts my synthesis had missed. SuperGrok read X-published source material the other models could not read. DeepAgent ran multi-step source-retrieval workflows.

GPT-5 Thinking was used as a cross-model audit on Claude's drafts. Every claim of constitutional fact in this book was vetted against a primary source I personally examined. Every citation has a URL.

A word about what kind of author that makes me. Working this way, I was not so much a writer as a composer, editor, proof-reader, and conductor. A conductor plays none of the instruments. He decides what is played, in what order, at what tempo, and what gets cut in rehearsal — and the performance is his. That is the right picture of how this book was made, and it changes what the reader is being asked to do. A conventional book of ideas asks you to follow one man's thinking as it developed over a lifetime. This one asks you to ride along on a journey of exploration, through two centuries of American history and a hundred ninety-five years of Supreme Court rulings on what happens when a patent expires — a journey nobody had taken before, and nobody could have taken alone. Not for want of intelligence; for want of years. A language model can read the whole line of cases, every statute in the wall, and the economic and environmental literature around them in the time it takes a person to read one brief, and it can carry a rider down that path without losing the thread. That is a large part of why the book is new. The chapter *Why Nobody Noticed* explains why the question sat unasked for fifty-six years. The rest of the explanation is here: the reading path that answers it was longer than one working life, until now.

The constitutional question this book asks is not a math problem in the formal sense. But it is a kind of problem — cross-disciplinary, hidden in plain sight, defended by overlapping institutional incentives — that the human alone has likewise been unable to solve for fifty-six years.

The closer analogy is not mathematics but security. Three cases from the last two years share one shape: a system defended by experts, a flaw the experts never found, and a machine that found it by reading the rules. The first is Microsoft's Bing. In March 2026 Microsoft published two critical advisories, each rated 9.8 on a scale of 10, for remote-code-execution flaws in Bing Images — one reachable through the public “search by image” upload that anyone can use. Both credit XBOW, an autonomous system that a year earlier had become the first non-human to reach the top of HackerOne's United States leaderboard of vulnerability researchers. XBOW's founder described Bing image search as one of the best secured systems in the world, very well secured by Microsoft's engineers and hammered by thousands of hackers trying to get in.

The second is OpenBSD, the operating system whose developers audit every line and whose reputation for security is the best in the field. The code that handles internet connections — the TCP stack — carried a flaw for twenty-seven years that let a stranger crash the machine by opening a connection to it. Some of the most careful programmers alive read that code for a generation, and automated bug-hunters ran over it millions of times. In 2026 Anthropic's Claude Mythos Preview found it. The third is SQLite, the small database engine embedded in nearly every phone, browser, and car. In July 2025 Google's threat-intelligence group saw signs that an attacker was preparing to use an unknown flaw in it but could not tell which. They handed the fragments to Big Sleep, an AI agent built by DeepMind and Google's Project Zero, which read the code, found the bug the attacker was about to use, and Google patched it first — the first time, Google said, that an artificial intelligence had foiled an exploit before it was used.

None of these systems was carelessly built. They were opened because a machine can hold every rule of a system in view at once and ask where the rules stop agreeing with each other. That is what this book does to a constitutional grant, the statutes written under it, and the regulations written under those: it reads them together and finds the place where they break.

It is not a coincidence that the year in which the book becomes writable is the year in which the tools became powerful enough to make it writable.

The book exists to argue that ordinary Americans can be paid more, in effective real income, by being charged less for what they buy.

AI is one of the reasons American workers in 2026 feel the floor moving. AI is also the reason this book could be written at all. If the argument succeeds, it hands those workers back half their old purchasing power.

III. The Search No One Has Run

If machines can read a system's rules and find where they break, the natural question is whether anyone has pointed one at the system this book examines: the Constitution, the statutes passed under it, and the regulations written under those. I looked, and as far as I can find, no one has. The nearest thing is the government's own effort. In July 2025 the Washington Post reported that the U.S. DOGE Service had built an AI tool to read roughly 200,000 federal regulations and flag the ones no longer required by law; it had already reviewed more than a thousand regulatory sections at the Department of Housing and Urban Development, and by the account of staff who used it, misread the statute more than once. That is a real use of

the tool. But it asks a question one layer below this book's. It asks whether a statute still requires a rule. It does not ask whether the statute and the rule, stacked together, do something the Constitution forbade.

Academic work has gone a step further in kind. A 2025 dataset built from rulings of Italy's Constitutional Court trains models to recognize conflicts between statutes and constitutional articles — but only conflicts a court has already declared. Nobody, so far as I can find, has run the search downward from a constitutional clause, through the statutes and regulations built on it, to a live conflict no court has yet been asked about. That is the search this book runs, by hand and with the tools described above, on two words in Article I. The point of saying so is not priority. It is that the method is available to anyone, the sources are public, and the field is empty. I would be glad to be shown that someone ran this search first; a reader who knows of such a case should say so, and the correction will go in the next edition.

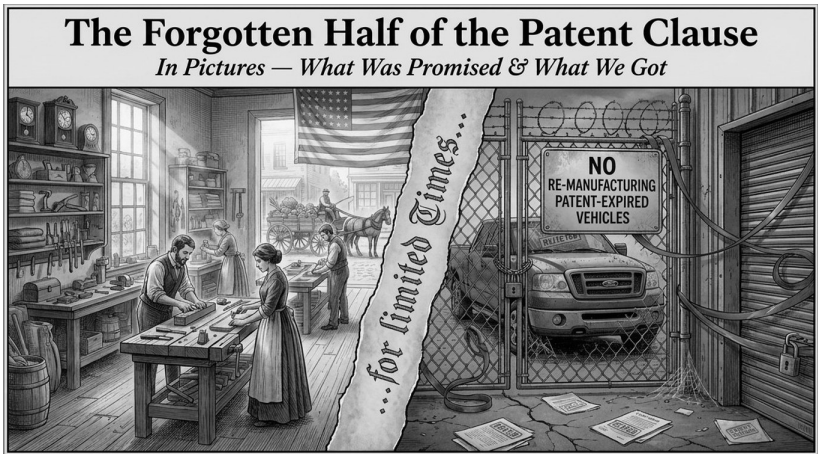
IV. How This Book Is Documented

The running text carries no superscript reference numbers. Sources, court cases, and statutes are gathered in the book's Endnotes — hosted online, in full and continuously clickable, at marti124.github.io/Patent_EndNotes.

They are alphabetized by author surname or by case name, in the Chicago style. Each entry ends with a link — Justia for Supreme Court cases, Cornell's Legal Information Institute for federal statutes, Founders Online for the founding documents, the publisher's own page for everything else — so any reader can go straight to the source with a single tap. Where an in-text citation is

needed, it sits inside the sentence in the form “Yandle (1983) named the pattern,” with the full reference held for the Endnote. An entry that rewards a longer note ends with the words *See Endnote Commentary*; that parallel section, alphabetized the same way, holds the fuller passages and the context the running text leaves out, for the reader who wants depth and the reader who would rather skip it. A page at the back of the book sets out everything the Endnotes contain and how to reach them.

Roleigh Martin
Northern Kentucky



The Forgotten Half of the Patent Clause, in Pictures -What Was Promised & What We Got

Article I, Section 8, Clause 8 of the United States Constitution: “[The Congress shall have Power] To promote the Progress of Science and useful Arts, by securing for limited Times to Authors and Inventors the exclusive Right to their respective Writings and Discoveries.”

Twenty-seven operative words. Two halves. Americans remember only one.

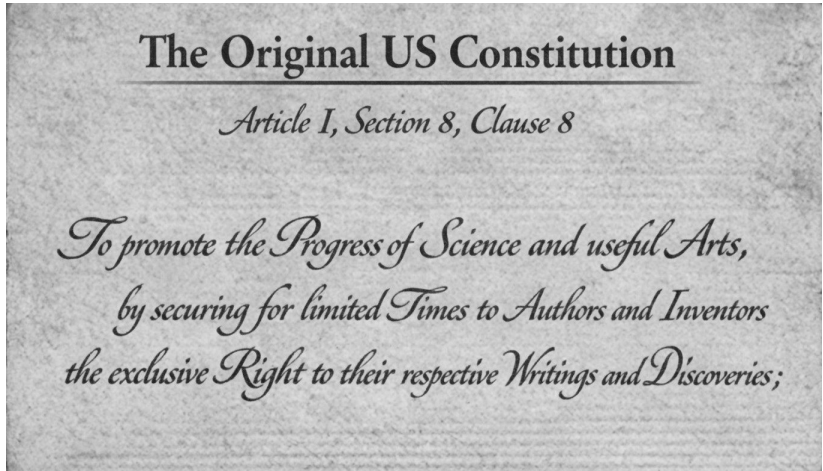
The first half — the “exclusive Right” part every schoolchild learns — grants inventors control over their discoveries for a limited number of years. The second half — the “limited Times” part no one talks about — is what happens when those years run out. The patented invention enters the public domain. It becomes, in the Supreme Court’s words, “free for all to use.” Anyone can build it. Anyone can

improve it. Anyone can sell it. Competition drives the price down, quality up, and the invention finally delivers on the bargain the Founders struck: a temporary monopoly in exchange for a permanent public inheritance. That inheritance is the forgotten half of the Patent Clause, and for most of American history it worked exactly as promised.

This section tells that story in pictures — what the Founders promised, what Americans received for the first 180 years, and what was quietly taken away after 1970 when federal regulations began requiring many newly manufactured products in heavily regulated sectors to meet current-year standards no matter how old their underlying patents. The images show both sides of the ledger: the bicycles, sewing machines, Model Ts, and generic medicines that once flowed freely into the public domain and made American life affordable, alongside the tractors, trucks, appliances, and light bulbs whose expired patents now sit locked behind regulatory walls.

For the full written argument — the Supreme Court cases, the constitutional analysis, and the path to restoring what was lost — this book presents the full case. The pictures will show you what was taken. The rest of the book will show you how to get it back.

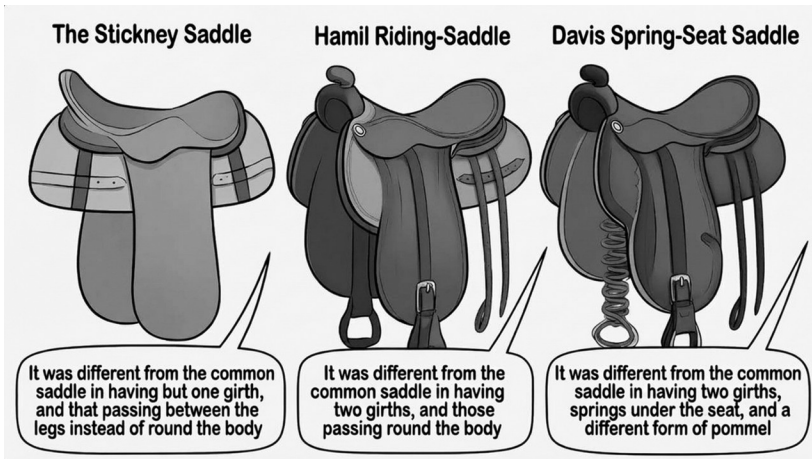
Act I – The Promise



Panel 1- The Patent Clause: Twenty-seven words, written into the Constitution at Article I, Section 8, Clause 8. Everything in this booklet follows from those words.



Panel 2- Washington's deal. The first President signed the Patent Act of 1790. The deal was up to fourteen years of exclusivity. Then the invention belonged to the public. That was the whole point.



Panel 3- The bargain in action. Three 19th-century riding saddle patents, illustrating how later inventors could build on earlier designs once prior patents expired. For much of American history, this is how the patent bargain was understood to work — each expired patent became the raw material for the next one. The bargain was being honored.

See the chapter in this book, “When the Bargain Worked: Pre-1900 American Prosperity.” Cumulative improvements from these expired patents — deeper seats, better padding, shock-absorbing springs, improved girths — pushed everyday range from 20–30 miles a day to 30–40, with endurance riders covering 60, once the designs were free for anyone to build, sell, and improve upon.

Act II – The Break



The opening of congressional committees

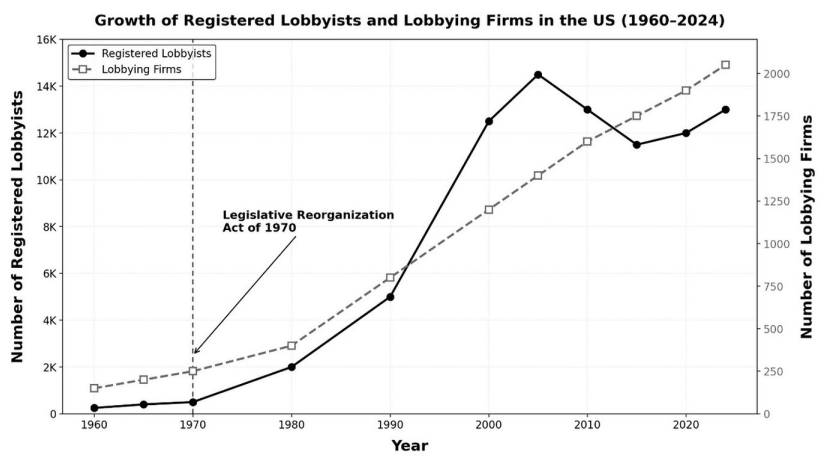
Panel 4 - 1949 vs 2013 committee rooms. In 1949, a congressional committee looked like this. A dozen members, one table, business conducted in the quiet. By 2013, it looked like this. What happened in between was a law passed in 1970. The Legislative Reorganization Act opened the committee doors in the name of transparency. The unintended consequence: the rooms filled up with lobbyists, their staff, and the industries whose legislation they were quietly drafting. Today, when a major bill moves through markup, the language the committee is voting on has often been written — sometimes literally word for word — by the industries the bill is meant to regulate.

In 2013, 70 of 85 lines in a House financial-deregulation bill were taken verbatim from draft language supplied by Citigroup's lobbyists. The committee members voted without knowing who wrote their own bill. Photos from: <https://www.congressionalresearch.org/>

Unless stated otherwise, for all the panels in Act II, see the chapter, “Four Problems, One Remedy,” for the first discussion in the book about this.



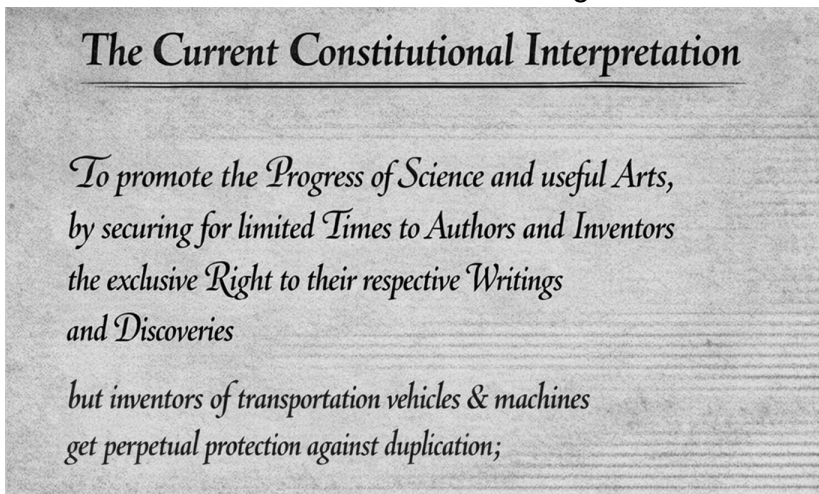
Panel 5 - Why they are called lobbyists. Before 1970, lobbyists were literally kept out of the room — which is why they are called lobbyists. They waited in the lobby. The Legislative Reorganization Act of 1970 opened the committee doors in the name of transparency. The unintended consequence was that the people waiting in the lobby were the ones who walked in first.



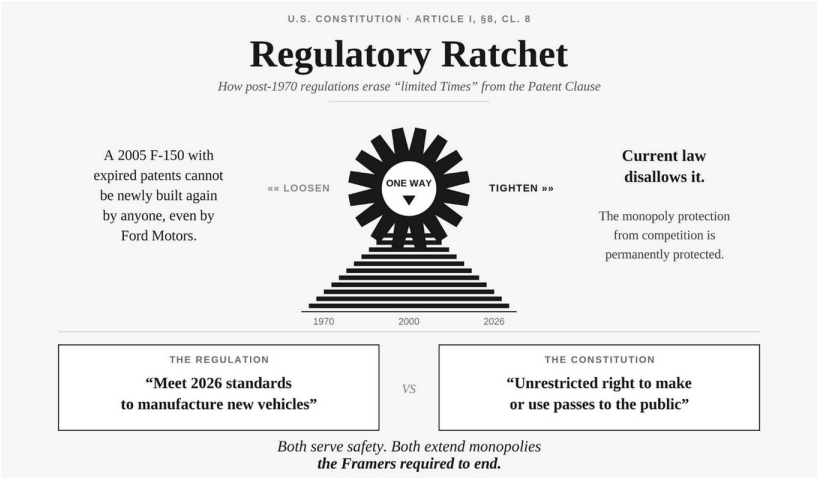
Note: Data for 1960-1990 is estimated based on historical accounts.
Data for 2000-2024 is based on Statista and OpenSecrets reporting.

Panel 6 — The lobbyist explosion. And this is what happened next. In 1971 there were 175 registered lobbyists in Washington. By 2024 there were over 13,000 — a 75-fold increase in the number of people paid to shape federal rules.

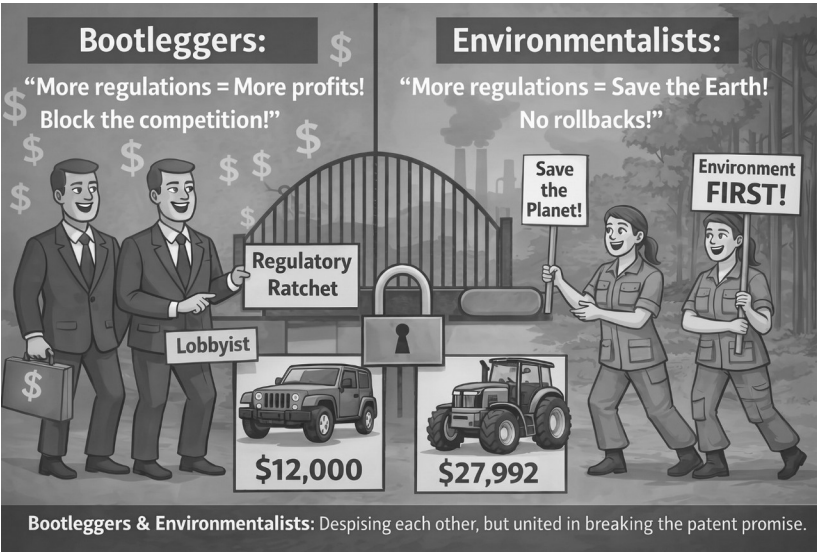
See the chapter, “Why Nobody Noticed How a Constitutional Violation Hides in Plain Sight.”



Panel 7 – The Patent Clause, Rewritten - Beginning in the 1970s, federal regulators did something the Constitution never authorized: they turned “limited Times” into forever for vehicles and machines. No regulation can override the Constitution. But until someone takes the regulators to court, the unconstitutional rules stay in force — and fifty-five years later, they still do.

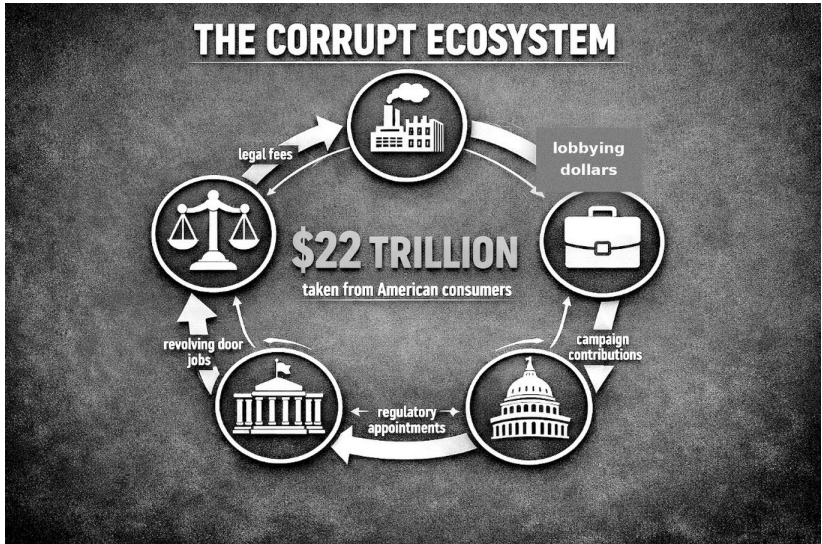


Panel 8 — The ratchet mechanism. The mechanism has a name: the regulatory ratchet. It only tightens, never loosens. Every new standard becomes the permanent floor. By statutory design.



Panel 9 — The unlikely coalition. In 1983, economist Bruce Yandle gave this pattern its name: Bootleggers and Baptists. The metaphor came from Prohibition — bootleggers wanted alcohol illegal because it kept their black-market prices high; Baptists wanted alcohol illegal because they believed it was sinful. The two groups despised each other and never appeared at the same rallies, but they lobbied for the same law and they got it. The pattern repeats whenever a regulation simultaneously enriches incumbents and lets idealists feel righteous. In the post-1970 regulatory ratchet, the bootleggers

are the legacy automakers, appliance manufacturers, and machinery oligopolies who profit from blocking patent-expired competition. The environmentalists are the moral cause that supplies public legitimacy and political cover. Neither group will ever vote for a rollback. That is why fifty-five years of administrations from both parties have left the ratchet untouched — the coalition supporting it includes people who agree on nothing else.



Panel 10 — The corrupt ecosystem. Over 55 years, this machine has transferred an estimated \$22 trillion from American households to the companies the regulations protect. About \$3,385 per household, per year, silently.

See the chapter, “Why Nobody Noticed How a Constitutional Violation Hides in Plain Sight”.

Act III – The Damage

THE 1911 PATENT EXPIRED IN 1928. AMERICANS HAVE BEEN BUILDING IT EVER SINCE.



- Firearms kept what cars and tractors lost — because federal product rules did not target them.

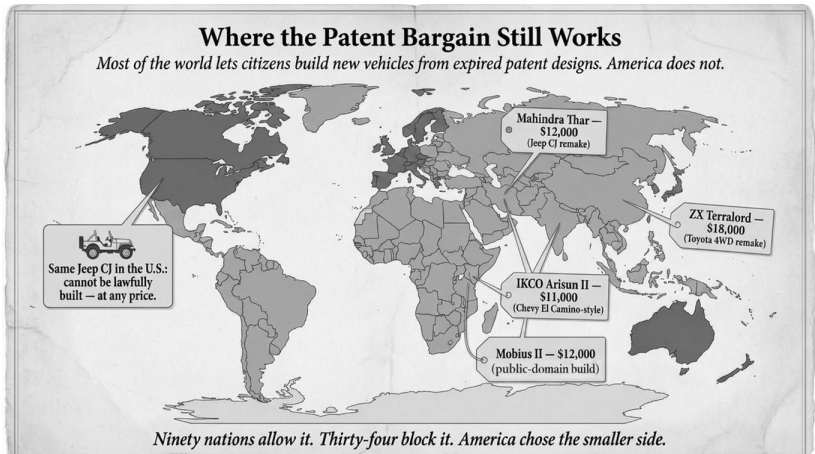
Panel 11 – The success story. Firearms kept what cars and tractors lost — because federal product rules never targeted them. This is what the bargain looks like when the Patent Clause works. The Colt M1911 was John Browning's .45-caliber sidearm, adopted by the U.S. Army in 1911 and carried by American soldiers through both World Wars. Its patent expired in 1928, and dozens of American manufacturers have been building it ever since — competing on price and quality, paying no royalty to anyone.

The price history
of calculators vs
trucks proves
there is a hidden
war on expired
auto-patents



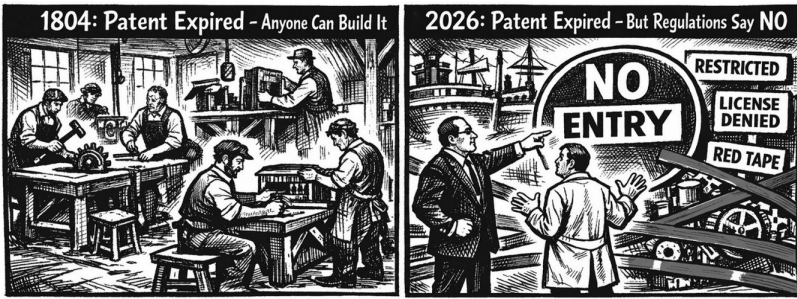
Panel 12 – The Receipt. Same forty years. Same country. Same expired patents. One line goes down because the public domain was honored. The other goes up because it wasn't.

See the chapter, “The \$72,495 Pickup Truck: Cars, Trucks, and the Household Budget”.

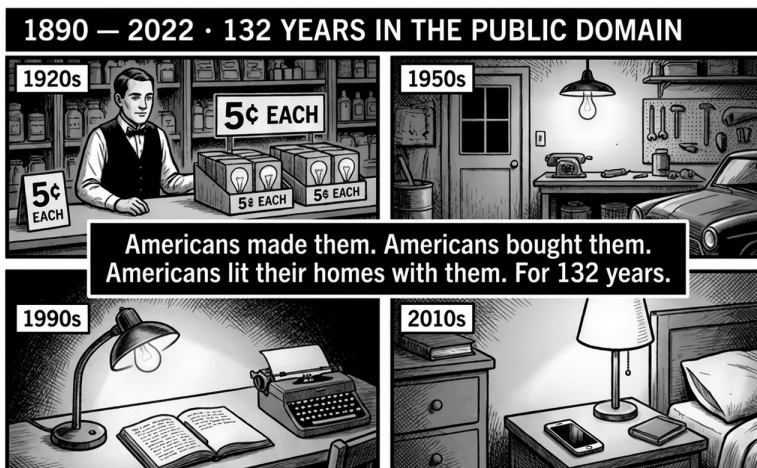
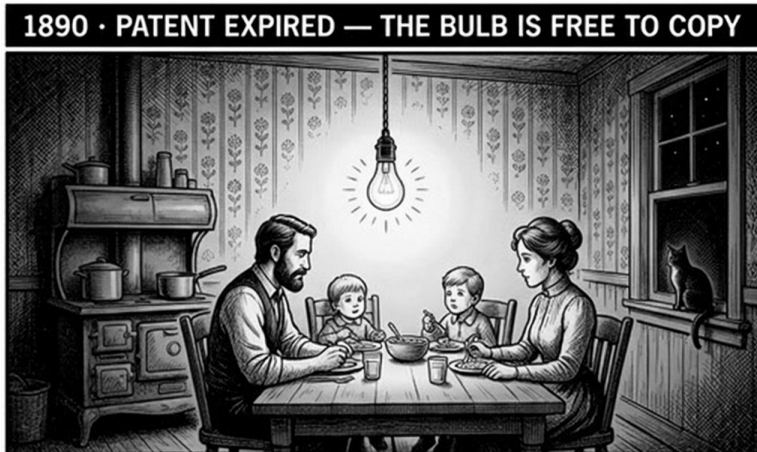


Panel 13- Where the bargain still works. In ninety nations, citizens can buy newly built vehicles based on patent designs that expired more than twenty years ago — Jeep CJs, Toyota Land Cruisers, Mercedes G-Class equivalents — for \$6,000 to \$18,000. Thirty-four wealthy nations block them. America is in the smaller group, by choice and not necessity. The same Mahindra Thar that sells new in India for \$12,000 cannot be lawfully built here at any price.

The breakdown of nations in either category is given in the chapter, “Why Wealthy Nations Break the Patent Promise That Poor Nations Honor.” In-depth linked proof that the vehicles manufactured are either remakes of patent-expired vehicles or made entirely of public domain components are provided in the book. The image shows strong price differences in cars and trucks where the patent bargain works. These nations show even stronger price differences in semi-trucks and farm equipment too. See the Trucking and Farming chapters.



Panel 14 - 1804 vs 2026 workshops. In 1804, the year when the earliest American patents expired. Any American could build a wagon, plow, or machine once the patent expired. In 2026, for many regulated products — especially vehicles and machines — the regulations say no to identical remanufacture for new sale. The constitutional principle is the same. The Constitution is the same. The surrounding regulatory law has changed.





Panel 15 - The withdrawal pattern. The tractor case shows the bargain blocked before it could be exercised. The incandescent bulb case shows something worse: a right Americans had been exercising for a century, withdrawn by regulators in 2023. Two patterns, same constitutional concern.

See the chapter “Healthy Lighting and the Anti-Backsliding Trap”.

Act IV – The Law

The Supreme Court on “limited Times”

Three rulings. Sixty-two years of settled doctrine.

Bonito Boats v. Thunder Craft · 1989

*“After the expiration of a federal patent,
the subject matter of the patent
passes to the free use of the public.”*

Kimble v. Marvel Entertainment · 2015

*“the right to make or use the article, free from all
restriction, passes to the public.”*

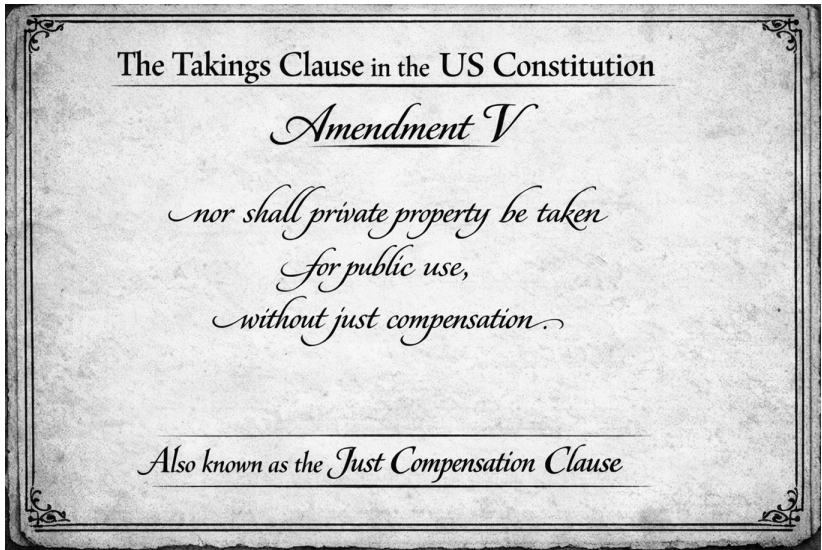
Brulotte v. Thys Co. · 1964

*“a patentee’s use of a royalty agreement that projects
beyond the expiration date of the patent is unlawful per se.”*

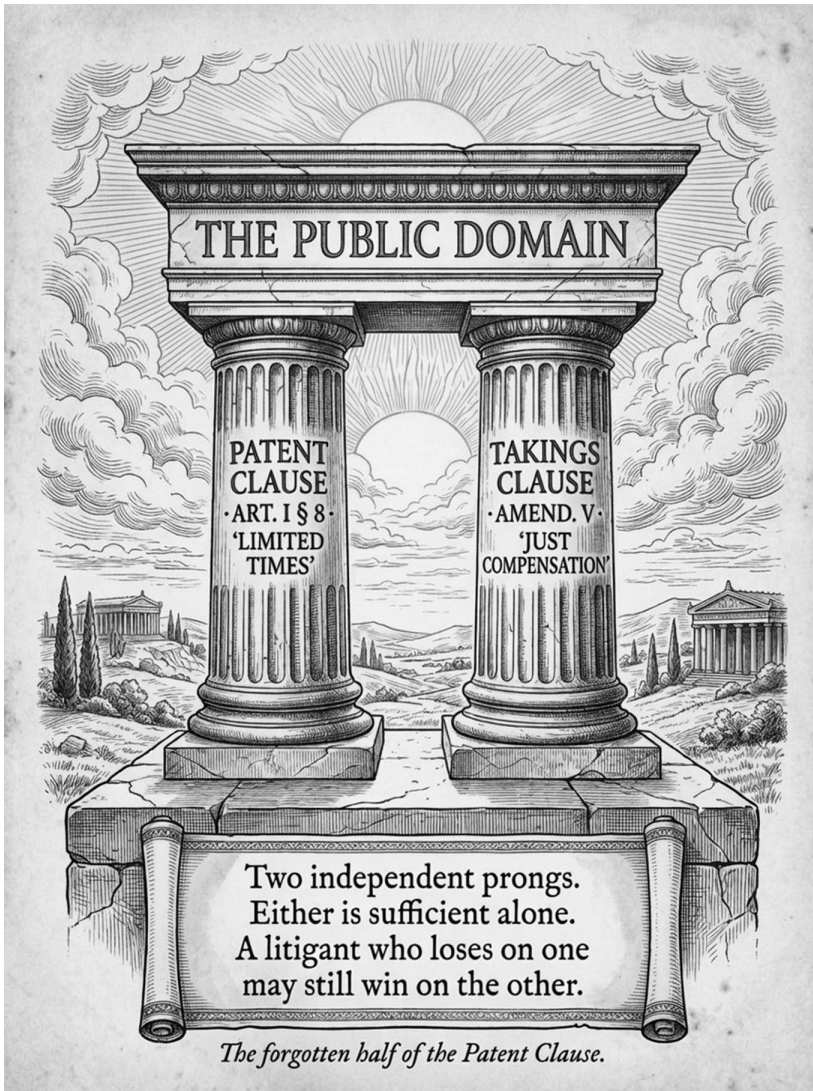
The law is not ambiguous. It has simply not been enforced.

Panel 16 — The Supreme Court on “limited Times.” The Supreme Court has said it plainly, three times over the past sixty-two years. In *Bonito Boats v. Thunder Craft* (1989), the Court held that “after the expiration of a federal patent, the subject matter of the patent passes to the free use of the public.” In *Kimble v. Marvel Entertainment* (2015), the Court confirmed that upon expiration, “the right to make or use the article, free from all restriction, passes to the public.” And in *Brulotte v. Thys Co.* (1964), the Court ruled that “a patentee’s use of a royalty agreement that projects beyond the expiration date of the patent is unlawful per se.” This book argues that the same constitutional principle should apply when regulation bars the public’s post-expiration remanufacture of patented designs. The principle is not ambiguous. This book argues that it has not yet been squarely enforced in this regulatory context.

See the chapter, “The Patent Clause Prong: “Limited Times” as Constitutive Language”.



Panel 17 – The Takings Clause. The Fifth Amendment says private property cannot be taken for public use without just compensation. This book argues that when regulators cancelled the public's right to manufacture expired-patent designs, they took something that belonged to every American. The Supreme Court ruled in *James v. Campbell* (1882) that patents are property protected by the Takings Clause — and confirmed in *Horne v. USDA* (2015) that appropriating personal property, tangible or intangible, is a per se taking requiring compensation. The rule has been settled since 1882. The Constitution already has the remedy. Someone just has to enforce it.



Panel 18 – The two pillars. Two independent constitutional arguments holding up the same case. The Patent Clause says the monopoly cannot last forever. The Takings Clause says what was taken must be returned or paid for. A litigant who loses on one prong can still win on the other.

See the chapters “The Patent Clause Prong: “Limited Times” as Constitutive Language” and “The Takings Clause Prong: Cancellation of Vested Public Rights”.

The Declaratory Judgment Act in US Constitutional Law

A federal court may declare the rights
of any party in a case of
actual controversy, without requiring
anyone to first break the law
they seek to challenge.

28 U.S.C. §2201 • Enacted 1934.

Confirmed: *MedImmune v. Genentech*, 549 U.S. 118 (2007)

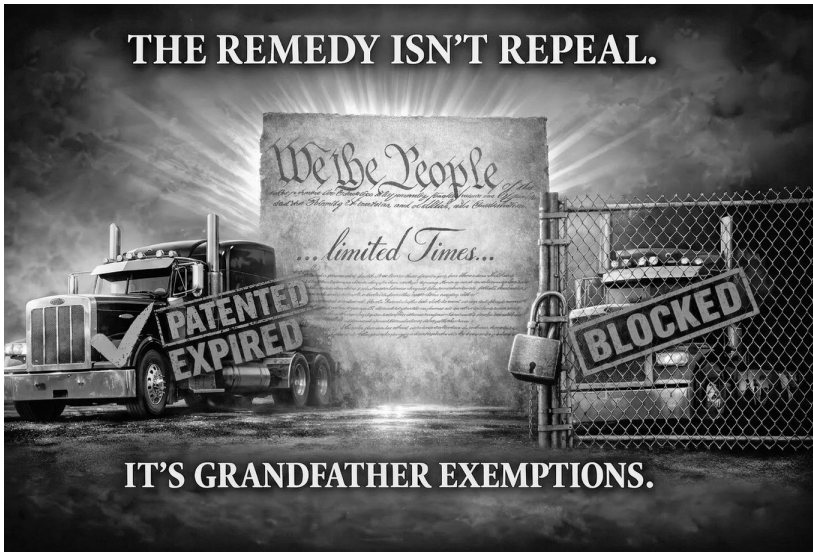
Panel 19 - The Declaratory Judgment Act. You don't necessarily have to violate a law to challenge it. A state attorney general could seek declaratory relief if standing, ripeness, and jurisdiction are satisfied. Under the Declaratory Judgment Act, a federal court may declare the rights of parties in an actual controversy. The legal architecture for that kind of suit has existed since 1934.

See the chapter, “You No Longer Need to Break a Law to Challenge One”.

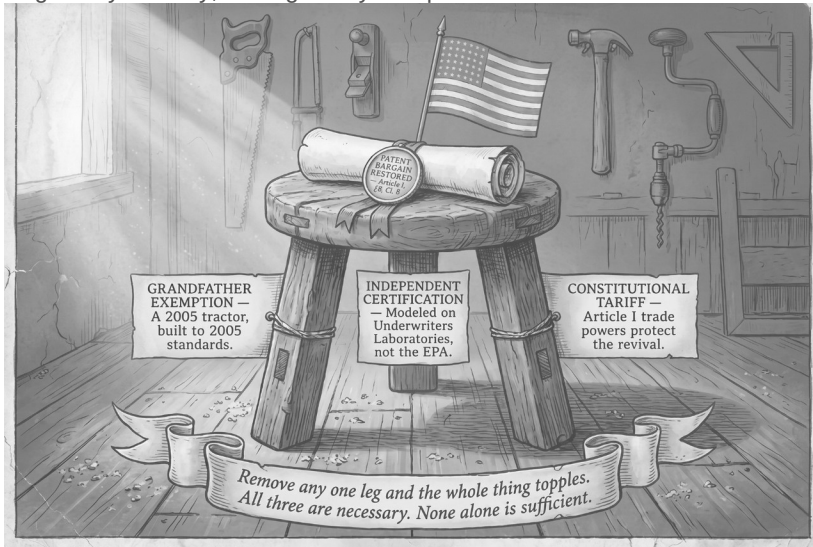
Act V – The Case Filed



Panel 20 - The case filed. We have found no reported decision squarely ruling on this question. It is a question of first impression, supported by thirty-eight Supreme Court citations and the most reform-minded Court in decades. Someone just has to file it.



Panel 21 – The remedy is grandfather exemptions — the right to manufacture a 1999 tractor to 1999 standards, clearly labeled, with the buyer informed. Regulatory honesty, not regulatory collapse.



Panel 22. A judicial solution for the Patent Clause abuse involves only securing grandfather exemptions & independent certifications. A politically optimal compromise is to add Article 1, Section 8 trade protection to remanufacturing being allowed to maximally protect American manufacturing while being faithful to the Constitution.



Panel 23 – That's you! Gregory Watson moved a constitutional amendment to ratification by writing letters for ten years. One person. That's who moved the 27th Amendment from dormant to law. You can be the next Gregory Watson whose letter triggers a historic lawsuit.

Write a letter to a possible plaintiff who may have legal standing such as Governors, State Attorneys General, Sheriffs, Farm Associations, Indian Tribes, Trucking Associations, or Health/Lighting Associations. Nineteen plaintiff categories in all. Click on the menu choices at https://marti124.github.io/Patent_Letters/ to write a letter to those who can take this issue to court. Each has a letter writing app or sample letters to use.

CHAPTER 1

Four Problems, One Remedy

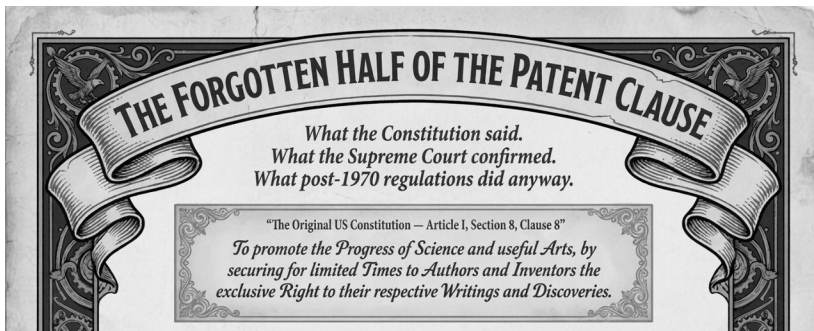
Four problems are eating at American life right now. The cost of living has doubled against the median wage since 1970. Every new car has become a rolling data recorder, and several major automakers have sold drivers' behavior data to brokers who resell it to insurers. Tens of millions of working Americans look at the AI and robotics buildout and see their livelihoods being engineered out of existence. And the people who used to love driving — who knew their machines, who could shift through the gears without using the clutch because their ears knew when it was safe to shift gears, who felt the highway as a form of freedom — have been handed cars that nag them, override them, brake for them, watch them, and update themselves overnight without permission. What is the political remedy for these four problems? What do they have in common that can be rectified without depending on a Congress that has spent fifty years ignoring the American public on every one of them?

Imagine an America where the two words at the heart of the Constitution's Patent Clause were honored.

Those two words are “limited Times.” They are the bargain. The Founders gave Congress the power to grant an inventor a temporary monopoly on what he invented; when the time was up, every American had the right to remanufacture the invention freely. The design entered the public domain the way the Founders promised — and stayed there. The judicial remedy that produces this America is operationally simple: a grandfather exemption under which a remanufactured product built on a public-

domain platform is held to the federal safety and energy-efficiency standards in effect during the platform year the remanufactured product mimics — the same standards the original product was certified to, and the same standards Americans drove, farmed with, and lived with at the time. A 2005-platform pickup truck remanufactured today meets the 2005 National Highway Traffic Safety Administration (NHTSA) safety standards and the 2005 Environmental Protection Agency (EPA) emissions tier; a 1999-platform tractor meets the 1999 standards. Certification runs through independent engineering bodies on the Underwriters Laboratories model rather than through federal-agency control.

In this mirror-universe America, faithful to the Constitution, the dealerships, contractor offices, and hardware-store shelves carry products built on long-expired patent designs.



“The Forgotten Half of the Patent Clause” — title banner with Article I, Section 8, Clause 8 text

A newly remade pickup truck on a proven 2005 platform, a full-size bed, backup camera, the works — \$25,000 instead of \$72,495. For a farmer, a combine sized for the American farm, harvesting around a hundred acres of corn in a ten-hour day, \$108,000 instead of \$646,000. From the HVAC contractor, a residential furnace built on the proven

2005 platform, \$3,000 installed instead of \$18,000. At participating hardware stores, an ordinary incandescent bulb on the shelf next to the LED, ninety cents instead of seven dollars. At a dealership selling semi-trucks, a new American semi built on the 1999 Cummins N14 — the million-mile engine owner-operators trusted, grandfathered by its pre-2000 engine year from the electronic logging device mandate, the diesel particulate filter, and the selective catalytic reduction aftertreatment that came later — \$80,000 instead of \$172,900.

Three of those five sit on a 2005 platform — the year American living can come back to. The bulb and the semi are exceptions: the bulb because its patents expired more than a century ago, the semi because owner-operators have a specific reason to want a 1999 engine, the latest model year grandfathered from the electronic logging device mandate, the diesel particulate filter, and the aftertreatment stacked on every engine built since.

Competition would drive every other price down with them, the generic-drug effect applied to American durable goods. This America is not hypothetical. Roughly ninety other countries already have it. The United States had it once and gave it up. The Constitution still promises it.

The cost of living. A new pickup truck that should cost \$25,000 in real dollars costs \$72,495. A combine that should cost \$108,000 costs \$646,000. A residential HVAC replacement that should cost \$3,000 costs \$18,000. The household that used to run on one paycheck now runs on two and still does not make it.

As of the first quarter of 2026, the average new-car loan term is 70 months and rising; default rates on car loans hit their highest level since 2010; thirty percent of Americans

trading in a vehicle owe more on the trade-in than it is worth, on average \$7,200 underwater. Eight-year car loans on cars that depreciate faster than they amortize are not a personal-finance failing. They are the only way the math works for ordinary buyers in a market whose prices have been engineered upward by federal regulation.

The surveillance subscription. Every new vehicle sold in the United States carries an event data recorder mandated by 49 C.F.R. § 563, a backup camera mandated by Federal Motor Vehicle Safety Standard No. 111, an automatic emergency braking system finalized by NHTSA in 2024, and increasingly a driver-monitoring camera pointed at the driver's face. General Motors, Honda, and Hyundai have been documented selling granular driving-behavior data — location, speed, hard-braking events — to data brokers who resell to insurance companies, without meaningful driver consent. The American buyer is paying for a prosecuting attorney to have live witnesses in his car twenty-four hours a day. The surveillance architecture applies only to newly manufactured vehicles. No legal way exists to buy a new vehicle without it.

The robotic replacement of working Americans. Federal regulation is pricing the human driver out of the market. That is the only reason the autonomous truck and the autonomous taxi can compete. Aurora, Kodiak, and Torc are building toward the displacement of 3.5 million American truck drivers. Waymo is building toward the displacement of personal driving and rideshare work. Both displacements rely on the same structural fact: that human-operated vehicles are now so expensive — because the regulatory wall keeps them so — that even a half-million-dollar autonomous platform looks competitive on capital cost. Remove the wall, which stands only by forgetting the Patent Clause's "limited Times" mandate,

and the math reverses overnight. The patent-expired remanufactured American semi-truck at less than half the price of a new rig leaves no economic room for an autonomous platform to compete. The same holds for autonomous farm equipment, earth-movers, and personal vehicles.

The lost relationship between the driver and the machine. The manual transmission fell from roughly a third of new U.S. vehicle sales in the 1980s to under one percent today. That collapse was not a market death. It was a regulatory death — Corporate Average Fuel Economy standards favor automatics, emissions testing protocols favor automatics, and the safety mandates layered on top assume an automatic as the baseline. The same architecture eliminated the user-serviceable engine, the rebuildable carburetor, and the car that does not auto-brake when the driver knows the obstacle is a paper bag. A generation of Americans grew up knowing what it meant to be in command of a machine, to hear the engine and shift by ear, to feel the road through the steering wheel rather than through a marketing brochure. That experience was engineered out of the new-vehicle market by an architecture that treats the driver as a liability to be managed rather than a person to be trusted with the property he paid for.

The four problems share one cause and one remedy. The cause is a federal regulatory wall, erected piece by piece since 1970, that has quietly nullified the second half of the bargain the Founders wrote into Article I, Section 8, Clause 8 of the U.S. Constitution — the bargain in which inventors receive exclusive rights for limited times in exchange for the public's reciprocal right, when those times expire, to freely make, copy, improve, and remanufacture what was patented. The remedy is the constitutional restoration of

that bargain. It requires no act of Congress. It requires no constitutional amendment. It requires no new legal theory. It requires no one to break a law. The remedy can be obtained by a single plaintiff, with standing, in a single federal courtroom. The doctrine that compels it is not hidden in a footnote or a forgotten precedent. It is in the sentence itself. Article I, Section 8, Clause 8 authorizes the federal government to restrain the public from making what an inventor has invented — and authorizes that restraint for limited times only. The regulatory wall makes the restraint permanent. No clause of the Constitution grants Congress, or any agency Congress creates, the power to bar the making of a patent-expired design forever.

A word on improvement. The expired-patent commons is not a museum; it is a montage. Once a patent's limited times expire, its claimed assembly enters the public domain, and anyone may mix, match, and improve those assemblies freely. Within reason, of course — a reasonableness common across the centuries. Mix the wrong parts and the result can be genuinely dangerous, and no age has condoned that. But every age measured danger against a fixed floor: the manifestly, obviously unsafe. The regulatory ratchet measures against a moving ceiling instead, barring a design not because it is dangerous but because something newer exists.

Improvement does not close the commons behind it. An improvement is itself either unpatentable — the law denies a patent to old elements merely doing what each always did — or patentable only at the inventor's option, which many decline. Either way, the expired matter stays in the commons. And a new patent on a novel combination

reaches only that combination; it cannot withdraw the expired components. The remanufacturer's right survives.

How Challenging Unconstitutional Laws Has Changed

DECLARATORY JUDGMENT ACT & THE PATH TO COURT

1934 **Declaratory Judgment Act · 28 U.S.C. § 2201**

A federal court may declare the rights of any party in an actual controversy — without requiring anyone to break the law first.

2007 **Massachusetts v. EPA**

State attorneys general have quasi-sovereign standing to challenge federal regulations that harm citizens' economic wellbeing.

2007 **MedImmune v. Genentech, 549 U.S. 118**

A challenger need not expose itself to prosecution to establish standing. An imminent, concrete enforcement threat creates actual controversy.

2014 **Susan B. Anthony List v. Driehaus**

Pre-enforcement standing confirmed: injury-in-fact when enforcement is “sufficiently imminent.” No unlawful plant, no prosecution required.

2024 **Loper Bright Enterprises v. Raimondo**

Chevron deference ended. Courts now review agency statutory authority de novo — no more agency thumb on the scale once challengers are in.

A clear path to court — no law broken, no criminal defendant needed.

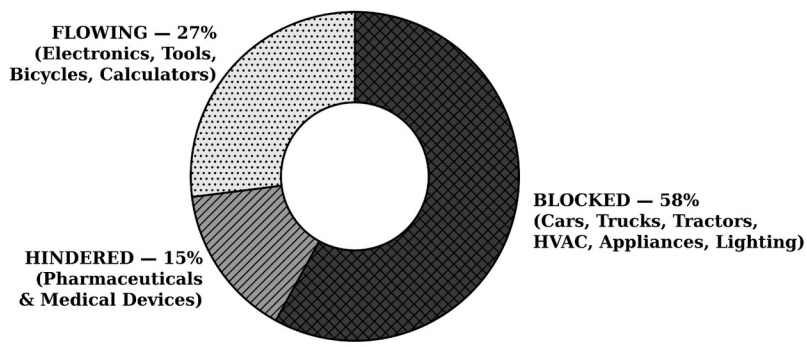
Consider what the Patent Clause actually is. It is the only place in the entire United States Constitution where the federal government is authorized to grant a private party a monopoly. Nowhere else. The Founders distrusted monopolies the way they distrusted standing armies, and for the same reason. They made one exception, and only one, and bounded it with two words: “limited Times.” The bound runs against the whole federal government, not the Patent Office alone.

A monopoly with an expiration date is a bargain; a monopoly whose commercial benefit never ends is the thing the Revolution was fought against. The patent itself still expires on schedule; what does not expire — and what the Constitution was written to prevent — is the original manufacturer's protection from low-cost remanufactured competition. The Supreme Court has remembered those two words for two centuries. Congress, the agencies, and the coalitions that lobby them have not.

Beginning around 1970 the federal regulatory state quietly constructed a permanent shield against remanufacturing competition across an estimated 58 percent of what the American economy produces. That figure is not a rhetorical flourish. It derives from a category-level estimate weighing each class of expired-patent commerce by its share of U.S. manufacturing output and household spending, drawn from Bureau of Labor Statistics consumer-expenditure data, Bureau of Economic Analysis industry accounts, and the Census Bureau's Annual Survey of Manufactures, producing a three-tier breakdown of the expired-patent economy — 58 percent Blocked, 15 percent Hindered, 27 percent Flowing.

The mechanism of the wall was not a repeal of the public's right to remanufacture — that could not have been done openly, and was not tried. It was an erection of a regulatory wall around the products themselves: EPA emissions tiers, NHTSA safety mandates, and Department of Energy efficiency standards layered on platforms whose core mechanical patents expired twenty, fifty, or in some cases over a hundred years ago. The technology is legally free. Nobody can lawfully build it.

Expired-Patent Commerce
The Three-Tier Breakdown



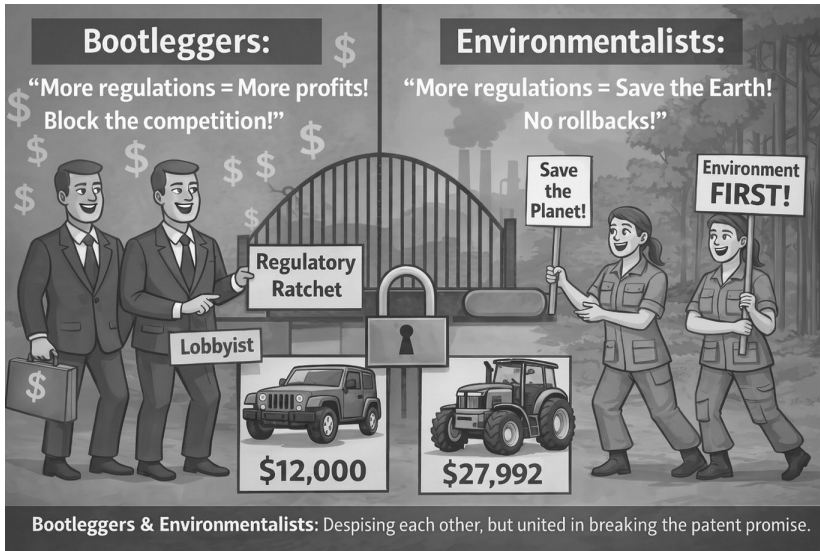
*Three-Tier Breakdown of the Expired-Patent Economy — 58%
Blocked / 15% Hindered / 27% Flowing*

The international evidence runs deeper than the dealer-lot picture. India’s Mahindra Thar, built on the long-expired Jeep CJ design, sells for \$12,000 against a \$45,000 U.S. Wrangler. Patent-expired tractors sell new in Indian and Chinese markets at one-fifth the U.S. price. A Sinotruk HOWO Class 8 truck sells overseas for \$28,000 to \$35,000 against \$161,000 to \$285,000 here. None of this is an argument for importing those vehicles. Their prices are evidence of what Americans should be paying for American-made versions of the same patent-expired platforms.

The cost-of-living relief and the manufacturing-jobs revival arrive together. The household runs on one paycheck and saves the second. The trucker owns his rig outright by his fifth year. The farmer keeps his farm. The American factory hires. The buyer who wants a map on a screen runs the

cable from his phone and pockets the savings. The buyer who wants a manual transmission gets one. The buyer who wants an engine he can service himself gets one. The remedy is not a museum piece. It is the freshly engineered remanufactured vehicle, built today out of any combination of expired patents. A chassis whose patents expired in the early 2000s. A USB port whose Intel patents expired around 2015. A backup camera whose imaging patents expired years ago. A Bluetooth radio whose 1998 patents have expired. A touchscreen whose late-1990s patents have expired. Mixed and matched to deliver the features buyers actually want at a fraction of the new-vehicle price — without the surveillance subscription, without the over-the-air control, and without the patent rent. The public domain is a kit of parts, not a museum.

If the remedy is this large, this constitutional, and this available, why has nobody written this book? Why does no member of Congress, in either party, campaign on it? Why has no federal court been asked the question? The answer has a name. The economist Bruce Yandle in 1983 called it the bootleggers-and-Baptists coalition: a regulatory regime endures when it is supported simultaneously by a moralizing public-interest constituency — the Baptists, who believe in the regulation's stated purpose — and a profit-taking commercial constituency — the bootleggers, who profit from its suppression of competition. The post-1970 American regulatory state is that architecture writ very large.



Bootleggers and Environmentalists cartoon

EPA tiers were written by environmentalists who believed in cleaner emissions and welcomed by manufacturers who understood, as every incumbent does, that stricter standards raise the barrier to cheap competition. Energy-efficiency rules were written by climate advocates and welcomed by appliance incumbents on the same instinct. NHTSA mandates were written by safety advocates; the surveillance-capable hardware they require became an asset the automakers could exploit. Whether or not incumbents ever pictured competition from patent-expired products, the historical pattern is plain: incumbents support barriers to new entrants. Each statute is genuinely supported by both wings of its coalition, which is what makes the coalition durable. The cost falls on the American buyer, in neither wing.

Around that architecture sits the broader corrupt ecosystem the populist coalition has been describing for years: lobbying expenditures and campaign contributions from the regulated industries to the legislators who write

the statutes, regulatory appointments routed through the same industries to the agencies that enforce them, revolving-door employment back into those industries, and the legal-fee and consulting economy that makes a career of the loop.

What no one has yet supplied is the specific constitutional remedy that ends the loop in court rather than in Congress: a Supreme Court ruling that converts the entire post-1970 regulatory ratchet into a violation of an existing constitutional clause that has been on the books since 1789. A successful Patent Clause ruling does not require Congress to act. It does not require lobbyists to lose their jobs voluntarily. It does not require campaign-finance reform. It is a structural counterweight the affected industries cannot lobby away after the fact, because the ruling locates the injury in the Constitution itself. The lobbyists keep their offices. They lose most of what they were lobbying for.

The hidden cause is a constitutional forgetting — the quiet disappearance of two words, “limited Times,” from the operational thinking of Congress and the agencies. The argument has not been made because the patent profession has paid almost no attention to what happens after a patent expires, because the constitutional-law profession has not connected Article I, Section 8, Clause 8 to the modern regulatory state, and because the politicians who campaign on constitutional fidelity have gone quiet on this particular question, every one of them, for reasons the bootleggers-and-Baptists framework explains.

The argument is now ripe because four recent Supreme Court decisions — *Arkansas Game & Fish v. United States* (2012), *Cedar Point Nursery v. Hassid* (2021), *West Virginia v. EPA* (2022), and *Loper Bright Enterprises v.*

Raimondo (2024) — have together broadened what counts as a regulatory taking and stripped agencies of the deference that has shielded the regulatory ratchet for fifty years. The case is a question of first impression. It has simply never been filed. This book is the public-facing brief for the litigant who eventually files it.

The Supreme Court Has Already Spoken

PATENT CLAUSE FOUNDATION

1882 **James v. Campbell**

Patents are property. The federal government cannot appropriate them without just compensation — 144 years of unbroken authority.

1898 **McCormick Harvesting v. Aultman**

Once issued, a patent is the patentee's property — not revocable by the agency.

1945 **Scott Paper v. Marcalus**

Reservation of rights after patent expiration runs counter to the policy and purpose of patent law.

1966 **Graham v. John Deere**

Congress may not remove existing knowledge from the public domain.

1989 **Bonito Boats v. Thunder Craft**

When a patent expires, the design passes into the public domain — free for all to use.

1999 **Florida Prepaid v. College Savings**

Patents are “a species of property” — constitutionally protected by due process.

2015 **Kimble v. Marvel**

The unrestricted right to make or use the article passes to the public.

Three centuries of Supreme Court doctrine.

The Supreme Court Has Already Spoken

TAKINGS CLAUSE & THE MODERN COURT

1984 Ruckelshaus v. Monsanto

EPA's use of submitter's proprietary data was a Fifth Amendment taking requiring just compensation. Intangible IP is protected property.

1992 Lucas v. South Carolina

Regulations eliminating all economically viable use are per se takings.

2015 Horne v. Dept. of Agriculture

Takings Clause covers personal property — physical appropriation is a per se taking.

2021 Cedar Point Nursery v. Hassid

Government-mandated physical occupation, even temporary, is a per se taking.

2022 West Virginia v. EPA

Major Questions: agencies need "clear authorization" for rules of vast economic significance.

2023 Tyler v. Hennepin County

Government may not keep property value beyond what the owner owes.

2024 Loper Bright v. Raimondo

Ended Chevron deference. Courts no longer defer to agency interpretations.

2024 Sheetz v. El Dorado Cty.

Expanded regulatory takings — broadened when government action is compensable.

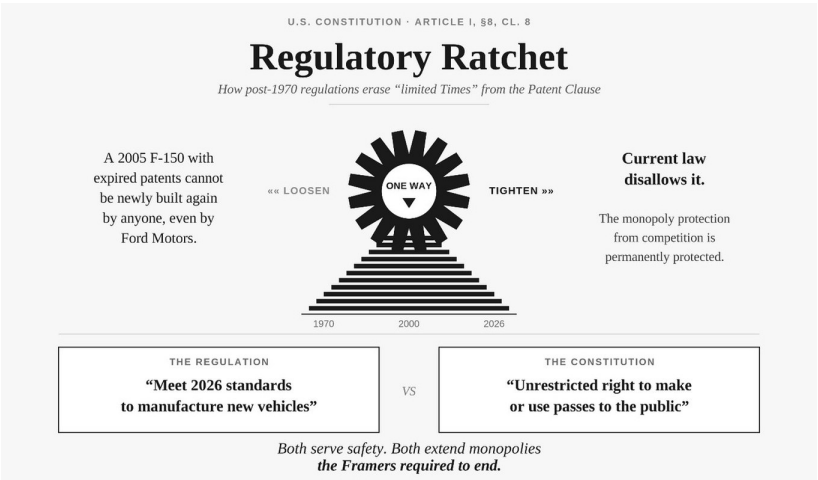
The most favorable Court in a generation.

That Court has not yet been asked. Beginning in 1970, the wall went up beneath it, and it broke the patent bargain on both halves at once. The original manufacturer kept the benefit of a monopoly with no end; the public never received the expired-patent commons the clause had promised it. What the wall holds in place is unequal treatment under the law — the original manufacturer

measured against the standards he met when he built the product, the remanufacturer of his now-expired patent measured against ratcheted standards layered on decades later.

The constitutional injury falls on every class of federally regulated durable good — vehicles, aircraft, industrial and farm machinery, and household appliances and equipment — where the core patents expired decades ago but federal manufacturing mandates imposed after expiration make remanufacturing those platforms legally impossible. Bicycles, hand tools, and product classes without federal manufacturing mandates are not affected; the wall exists only where Washington built it.

The mechanism has a name: the regulatory ratchet. The feature that makes it uniquely hostile to the Patent Clause is that it is designed to be irreversible. When Congress wrote the anti-backsliding provision into 42 U.S.C. § 6295(o)(1) in March of 1987, it told every future administration that an energy-efficiency standard, once set, can never be relaxed. The same one-way motion runs through EPA emissions tiers and NHTSA safety mandates. These are not ordinary rules that tighten under one administration and loosen under the next. They are designed so that each tightening becomes the new permanent floor. Converting the original manufacturer's protection from temporary into permanent is what turns a policy problem into a constitutional one. The anti-backsliding provision itself passed the House without a single recorded individual vote, a story the chapter *Hybrid Legislative Voting* tells in full.



“Regulatory Ratchet” poster — Article I § 8 Cl. 8, one-way regulation diagram

The ratchet is no accident. It was installed. The Legislative Reorganization Act of 1970 fundamentally restructured how Congress works. Before 1970, the number of registered lobbyists in Washington was estimated at between 200 and 400. By 2024 it was 13,070. The 1970 reforms, marketed at the time as sunshine reforms, replaced most committee secrecy and most unrecorded floor votes with open transparent voting and open committee proceedings — and in doing so handed lobbyists the tool they needed to track, pressure, and discipline every legislator on every vote. The political scientists James D’Angelo and Brent Ranalli documented what followed in *Foreign Affairs* (D’Angelo and Ranalli 2019).

The annual volume of legislation more than tripled, from roughly two thousand pages to over seven thousand, and the bills grew too opaque for legislators to read what they were voting on. The fiscal record tells a parallel story. By the Treasury’s own Historical Debt Outstanding data, the federal government has not closed a single fiscal year with

the total national debt lower than the year before since 1957 — a date this book takes, from that record, as the pivot in American budget control. Readers who recall the budget surpluses of 1998 through 2001 as years of debt reduction should consult the chapter Hybrid Legislative Voting, where the Treasury's record is reconciled against the popular memory.

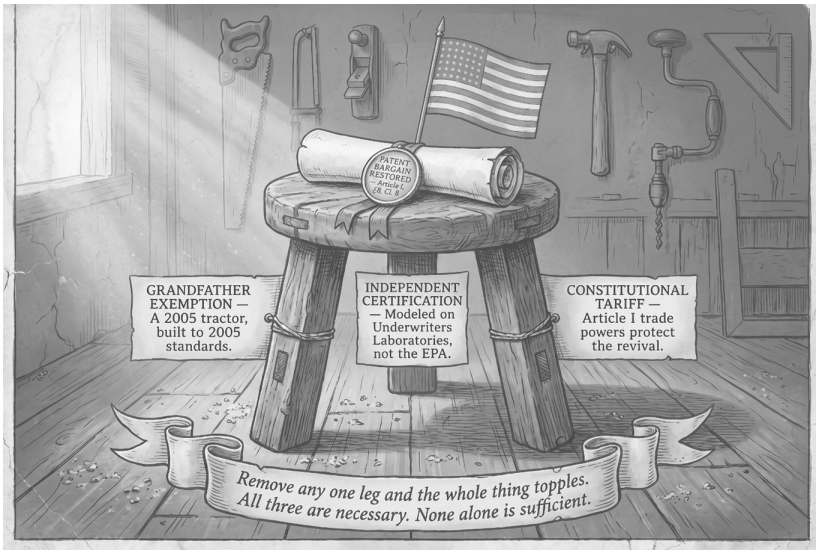
The aggregate harm is large. Between 1970 and the present, the American cost of living roughly doubled relative to household income; the ratio of median household income to median home price fell from 41 percent to 20 percent. Across the categories the regulatory wall encloses, Americans are paying roughly twice what they should be paying. The cumulative transfer, from households to incumbent manufacturers, is on the order of \$22 trillion since 1970, conservatively measured. That works out to approximately \$3,385 per American household per year, every year, silently, until the bargain is restored. That \$22 trillion is the midpoint of two independent analyses, one worked from Bureau of Economic Analysis industry accounts and one from Bureau of Labor Statistics consumer-expenditure data across sixty-plus product categories. A fuller accounting that includes the categories set aside here — housing through construction equipment, food through farm equipment, and most specialized industrial equipment — would bring the true total to \$30 to \$45 trillion over fifty years.

Every figure cited here is a conservative one. Where industry-by-industry evidence suggested the savings would be larger still, the larger figure was set aside rather than carried into the headline. The bolder claim is not made because the constitutional argument does not require it, and because a case of this magnitude is best

built on the floor of the evidence rather than the ceiling. The three-tier percentages are informed estimates, order-of-magnitude accurate rather than precisely calculated, and the Endnotes set out the derivation and its limits.

A successful constitutional remedy restoring remanufacturing rights across the affected sectors would, within roughly a decade, reduce the American cost of living by approximately one-third on average. The operational mechanism — the grandfather exemption and independent engineering certification described above — flows naturally from the constitutional holding itself. No act of Congress is required to put it in place.

The judicial remedy is two legs of a three-legged stool. The third leg protects the domestic manufacturing revival the first two unleash, and it cannot come from a federal court — it has to come from Congress or the President under existing Article I trade authority. A constitutional tariff, a quantitative import restriction, or both keeps foreign producers from undercutting American workers in American plants building patent-expired American platforms. The third leg is therefore not a remanufacturing protection but a domestic-manufacturing protection: every American factory benefits, the patent-expired remanufacturers gaining a price floor for the cost-of-living relief to reach American households, the new-design manufacturers gaining competitive parity to price against domestic competitors rather than foreign undercutters. Remove any one leg and the whole thing topples.



The three-legged stool of patent bargain restoration. The grandfather exemption and independent engineering certification flow directly from a federal court ruling. The tariff or import restriction is a political follow-on under Article I trade powers.

The distribution matters as much as the mean. For households that fully embrace remanufactured alternatives, cost-of-living reductions on major purchases will exceed 50 percent: the trucker, the farmer, the family replacing a pickup truck or an HVAC system. For households that continue to buy newly designed products, the floor reduction from competitive discipline on incumbent prices will be at least 15 percent. The remedy is not a mandate. It is a restoration of choice. It helps everyone, but it helps the most cost-burdened Americans most.

Everything above is the surface. The rest of this book documents the argument beneath it. Part I establishes the constitutional foundation — Jefferson’s bargain, the regulatory ratchet, the anti-backsliding provision that locked it, the seventeen statutes that hold the patent

commons hostage, the \$22 trillion wealth transfer, and the international comparison across roughly ninety countries that already honor the bargain Americans have lost. Part II is the historical witness: pre-1900 American prosperity, and bicycles and calculators as the proof the bargain still works where the wall never went up. Part III is the modern injury at six American kitchen tables — the \$72,495 pickup, the grocery bill, the trucker's burden, residential lighting, the surveillance premium, and the last human driver. Part IV covers the sector the wall only partly reaches: pharmaceuticals and medical devices. Part V is the restoration — who can sue, on what theory, the two constitutional prongs they would carry, and how a remake is built and certified. Part VI tests the argument against the objections a government brief would raise, and hands the reader an open-book examination to run himself. Part VII describes the years after a favorable ruling, for the president, the plaintiffs, and the household. Part VIII is about keeping the bargain whole once it is won. Part IX asks why an outsider saw it, and closes with a citizen's playbook: what every reader can do tomorrow morning.

The case has never been filed. It is a question of first impression. The reader who picks up this book is reading the public-facing brief that the eventual litigant — whoever that turns out to be — will rely on. The book ends where the reader can begin.

CHAPTER 2

Jefferson's Bargain

In August of 1813, Thomas Jefferson sat down at Monticello to answer a letter from a Baltimore miller named Isaac McPherson. The reply has been quoted for two centuries, but only the poetry gets quoted — the line in which ideas are said to spread like fire, expansible over all space, without lessening their density in any point. That line is the opening of the argument. The argument itself is the forgotten half of what Jefferson had to say.

Jefferson told McPherson that ideas, by their nature, belong to humanity. They are not property. They cannot be owned. The exclusive right a patent confers is therefore not the recognition of any natural right. It is a deliberate, narrow, temporary exception carved out of the natural freedom of ideas, created for a specific public purpose. The purpose is disclosure. The inventor receives a temporary monopoly on his invention. In exchange, the public eventually receives the invention itself — on terms better than the public would have had if the inventor had kept the idea secret.

That is the bargain, and it has two halves. The first is the temporary monopoly. The second is the public's reciprocal right, when the monopoly ends, to make, copy, improve, and remanufacture what was invented. The first half has been zealously protected for two hundred years. The second half is the subject of this book. The bargain has been studied. Edward Walterscheid traced the legislative drafting of the Patent Clause and the Patent Act of 1790. Oren Bracha traced the transformation of the Founders' bargain under the pressures of late-nineteenth-century corporate liberalism. Adam Mossoff, in the Cornell Law

Review, reconstructed the natural-rights foundation of early American patent doctrine. The work is scattered across separate books and law reviews, and none of it has been connected to the modern federal regulatory state. That connection is the work this book undertakes.

The Older Lineage

The bargain is older than Jefferson and older than the Constitution. In 1624 the English Parliament passed the Statute of Monopolies, abolishing the Crown's practice of granting monopoly privileges to royal favorites — privileges that had become a notorious instrument of patronage in the late Tudor and early Stuart years. The statute made one exception. New inventions could be granted exclusive rights for fourteen years. After that, the invention belonged to the public.

The fourteen-year term was not arbitrary. It was twice the seven-year apprenticeship English guilds used as the span in which a journeyman became a master. An inventor's monopoly should last roughly as long as it took a new generation of craftsmen to learn the trade. After that, the trade belonged to everyone who had learned it.

That principle was what the American Founders inherited. James Madison, defending the clause in *Federalist* No. 43, observed that the public good and the inventor's claim coincided. The bargain served both parties at once. The bargain had no third constituency, and made no provision for the rent-extracting middleman whose interest lay in preserving the benefit of the monopoly past the patent's constitutional term. Two years after ratification, the Patent Act of 1790 — signed by Washington and drafted in substantial part by Jefferson — granted exclusive rights for the same fourteen years English law had used since the

early seventeenth century. After the fourteen years, the invention belonged to the American people. Article I, Section 8, Clause 8 is the only place in the founding charter where the federal government may grant a private party a monopoly, and the two words “limited Times” are not decorative. They are the bargain. One principle older than either runs through the 1624 statute and the clause descending from it: equality under the law in trade. The Statute ended the grant of permanent commercial advantage to royal favorites; the Patent Clause set a term no inventor’s advantage could outlast. Both rest on the public’s right to compete on equal terms in any trade whose monopoly has run. *The Patent Clause Prong* returns to that premise as a structural argument. Equality under the law has been in the architecture of American patent law since 1624.



The Court Has Said So

Four Supreme Court decisions, spanning forty-four years, set out the doctrine on the post-expiration public domain. In *Scott Paper Co. v. Marcalus Manufacturing Co.* (1945),

the Court rejected the use of a private contract to extend a patent's practical effect past expiration; any attempt by a holder to maintain control after the term ran was contrary to the policy of the patent laws. In *Sears, Roebuck & Co. v. Stiffel Co.* (1964), the Court held that federal patent law occupies the field: when a patent has been invalidated or has expired, the article enters the public domain and is available to anyone, and a state may not use unfair-competition law to recreate the monopoly federal law had ended.

In *Graham v. John Deere Co.* (1966), the Court treated the Patent Clause as both an authorization of congressional power and a limitation on it. Congress could grant patents, but only within the boundaries the clause established — and one of those boundaries was that the rights run for limited times and no longer. And in *Bonito Boats, Inc. v. Thunder Craft Boats, Inc.* (1989), the Court struck down a Florida statute prohibiting the copying of unpatented boat hulls, holding that federal patent law determines not only what is protected during the term but what is free for all to use after it. When an invention is in the public domain, the public has a vested right of access that no state may rescind.

Two centuries. Multiple Supreme Court majorities. A consistent holding: when a patent expires, what comes next belongs to the American people. The Founders meant it. The Court has repeatedly said so.

Where the Bargain Stands Today

Lay out the entire universe of expired-patent commerce in the United States and color each slice by how much of the bargain has actually been honored, and the picture divides

into three regions. The opening chapter charts that division; what follows is what each region contains.

The *blocked* region, roughly 58 percent of the total, consists of industries where federal regulations beginning in the 1970s have made it effectively impossible to manufacture products based on patents that expired long ago: passenger vehicles, light trucks, commercial vehicles, agricultural equipment, construction machinery, small engines, heating and cooling systems, major appliances, and lighting. In each, the core mechanical technology has been public domain for decades, sometimes for over a century. Emissions tiers, safety mandates, and efficiency standards did not repeal the public's right to manufacture. They made the right unusable. Once the regulations were in place, only incumbent manufacturers and their licensed successors could produce compliant versions, and they could charge whatever the market would bear. This breaks the patent bargain on both its halves at once, the inventor's and the public's.

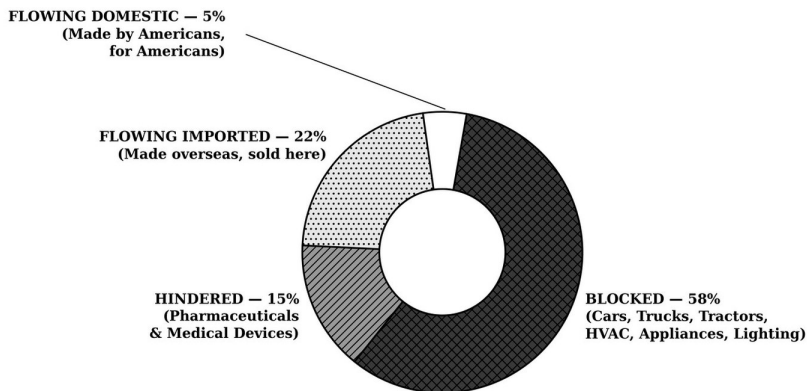
The methodology behind these three figures is set out in the companion article cited in the Endnotes. The audit maps federal regulatory frameworks against United States consumer durable-goods spending by category, weights the result by household consumption, and triangulates it against output data from the BLS Consumer Expenditure Survey, the BEA Industry Economic Accounts, and the Census Annual Survey of Manufactures. *The \$22 Trillion Wealth Transfer* develops the cumulative dollar value of the transfer. These figures are conservative; the true blocked and hindered share may be higher.

The *hindered* region, roughly 15 percent, covers pharmaceuticals and medical devices. The patents on the

underlying chemistry have generally expired on the conventional schedule, but the pathway from expiration to remanufacture is obstructed by approval requirements, licensing-board enforcement, and patent thickets layered around generic formulations. The bargain is not cancelled outright there; it is severely impaired. The chapter *The Hindered (not Blocked) Sector* treats it in detail.

The *flowing* region, roughly 27 percent, covers the categories where the bargain operates roughly as the Founders intended — consumer electronics, hand tools, bicycles, cameras, calculators, watches, sporting goods, furniture, kitchenware. Expired patents enter the public domain on schedule, manufacturers compete, and prices fall. But of those 27 points, only about 5 reach Americans through goods made by American workers. The remaining 22 arrive as imports, because the same regulatory architecture that blocks remanufacturing in the largest tier also raises the cost of domestic manufacturing in the smallest one. Take vehicles. India's Mahindra Thar, built on the long-expired Jeep CJ design, sells new for around twelve thousand dollars against a forty-five-thousand-dollar American counterpart. The prices are evidence, not an argument for importing: they show what Americans should be paying for American-made versions of the same platforms. The vehicle-by-vehicle record is the work of the chapter *Why Wealthy Nations Break the Patent Promise That Poor Nations Honor*. Roughly ninety nations, most of them poorer than the United States, currently honor the bargain in vehicle and equipment markets. Thirty-four wealthy nations, including this one, do not. We are in the smaller and more expensive group by political choice, not by economic necessity.

Of the 27% Flowing tier, only 5 points reach American workers



Four-Tier Breakdown of the Expired-Patent Economy — 58% Blocked / 15% Hindered / 22% Flowing Imported / 5% Flowing Domestic.

The Firearms Exception

One substantial category of expired-patent durable goods has kept its public domain intact. The core patents covering most American firearms — the 1911 pistol, lever-action rifles, bolt-action rifles, the AR-15 platform — have been public domain for decades and in many cases over a century. Americans buy firearms made by American workers in American factories at prices that reflect genuine competition. The reason is not that firearms are technologically simpler than vehicles or appliances. The reason is political: a sustained constituency has, for fifty years, kept federal regulators from applying the car-and-tractor playbook to firearms. When the regulatory wall is absent, the patent bargain delivers what the Founders

wrote it to deliver. Whether a category gets a wall is a political question, not a technological one.

The Public Domain Is a Kit of Parts

The post-expiration public domain has a property the existing patent literature has not articulated. Once a patent expires, the invention does not enter the public domain alone. It enters as a component, available to be combined with every other expired-patent component, in any combination an engineer can devise. That is the combinatorial property, and it is a property of a public domain rather than of any individual patent.

A patent-expired remanufactured vehicle is therefore not a copy of one expired design. It is a freshly engineered product assembled from many: the chassis whose patents expired in the early 2000s, the USB port whose patents expired around 2015, the backup camera whose imaging patents expired years ago, the Bluetooth radio, the touchscreen, the anti-lock braking system whose 1980s patents are public domain, the fuel-injection system whose 1990s patents have expired. None of these components is owned by anyone. The engineer who combines them is producing an assembly of public-domain parts that is itself, in its specific combination, a fresh design. The commons is not a museum of preserved old designs. It is a stockroom of free-to-use parts, available as freely as the alphabet is available to writers.

The Founders Built a Commons

The political scientist Elinor Ostrom, awarded the 2009 Nobel Prize in Economics, spent thirty years studying how human communities actually govern shared resources.

The conventional academic view before her had been that commons necessarily failed — that a resource everyone could use would be used up by everyone. Her work showed that successful commons had governance frameworks with common features: clearly defined boundaries, rules adapted to local conditions, collective-choice arrangements, monitoring, graduated sanctions, and recognition of the right to organize. That framework matches, with eerie precision, the one the Founders wrote into the Patent Clause two centuries earlier. The clause defines bounded rights and adapts them to inventive disclosure. It provides collective choice through legislation, monitoring through patent examination, and graduated remedies through the courts. And it sets a temporal boundary — “limited Times” — beyond which the right passes to the commons. Her synthesis, *Governing the Commons* (1990), arrived two centuries after the clause was written. The commons in question is the public domain of human knowledge. The bounded right is the temporary monopoly. The boundary is the expiration date.

That is the doctrinal foundation. The Constitution authorizes a temporary monopoly. The Supreme Court has affirmed, across two centuries and many majorities, that the post-expiration public domain is a vested right of the American people. Ninety nations currently honor it. What happened to it in the United States — how a doctrine settled in 1813 and repeatedly reaffirmed came to be effectively cancelled across 58 percent of American expired-patent commerce — turns out to involve a specific date: October 26, 1970.

CHAPTER 3

How the Bargain Was Cancelled: The Regulatory Ratchet

The previous chapters told you what the bargain was. This one tells you how it was cancelled. The cancellation was not announced. No statute repealed the second half of the Patent Clause. No agency issued a rule declaring expired patents off-limits to the American public. If you went looking for the moment of cancellation you would not find it — not in the Federal Register, not in the United States Code, not in the briefings of any administration. There is no headline.

What there is instead is a wall. It was built quietly, one statute at a time, beginning around 1970. It does not mention the Patent Clause. It does not describe itself as extending any monopoly. Its declared purposes are clean air, fuel economy, motor vehicle safety, energy efficiency, occupational health, and its declared purposes are real. The wall is real too, and it does something the declared purposes do not require it to do. Each year, a few inches taller. Each decade, much taller. By the time any patent in the affected category expires today, the wall around its underlying technology has already grown so high that the expiration changes nothing on the ground.

The wall is the cancellation. The mechanism by which it grows higher year after year, and by which it cannot — by statutory design — grow shorter, is the regulatory ratchet.

The 2005 Ford F-150

A word first on how long patents last. American patent law treats two kinds differently. Utility patents, covering how an invention works, run twenty years from the filing date; design patents, covering how it looks, run fifteen years from issue. Either way the term is fixed, and when it ends, it ends. One narrow exception exists: patents on pharmaceutical and medical-device inventions can receive a single five-year extension for the years the invention spent awaiting federal approval. It is one-time, statutory, and confined to that category. It reaches no vehicle, appliance, or machine. For the products this book is about, twenty years is the ceiling. No renewal. No second life.

The 2005 Ford F-150 is a useful place to begin. The F-Series has been America's best-selling vehicle every year since 1981, and by 2025 all the patents covering the 2005 model had expired. The twenty-year utility-patent ceiling applies to every engineering feature on the truck — the engine geometry, the drivetrain, the chassis design, the electronics architecture. By 2025 those patents were on the public side of the constitutional line.

Under the Patent Clause as the Supreme Court has read it, that means something specific. After 2025, any American — Ford, a startup in Toledo, a cooperative of independent mechanics in central Texas — has the legal right to build the 2005 F-150 to its original engineering specifications. The right is not a license. It is not a grant. It is the public's residual right, materializing the moment the patent term ends. It is the half of the bargain the public has already paid for. Trademark is a different animal, and the difference is constitutional. Article I, Section 8, Clause 8 grants patent and copyright in the same breath — "To

promote the Progress of Science and useful Arts, by securing for limited Times to Authors and Inventors the exclusive Right to their respective Writings and Discoveries” — and the Founders bounded both with the same two words. It does not authorize trademark at all. The Supreme Court confirmed as much in 1879, striking down Congress’s first attempt to protect trademarks under that clause. Trademark rests instead on the power to regulate interstate commerce, exercised today through the Lanham Act of 1946, and no “limited Times” boundary binds it. So the Ford name, the blue oval, and the F-150 designation do not expire when a Ford patent does, and a remanufacturer must give the truck a different name and badge. The engineered article, yes. The badge, no. The truck can be the truck. It cannot be a Ford.

Now ask what happens when an American actually tries to exercise the right. The 2005 F-150 was certified to Tier 2 emissions standards; Tier 3 has since been phased in at roughly half of what Tier 2 permitted, and the 2005 catalytic converter, fuel-injection calibration, and on-board diagnostics were engineered for the older envelope. The truck was certified to the fuel-economy standard then in force; the 2005 SuperCrew with the 5.4-liter V8 was rated at sixteen miles per gallon combined, and the current light-truck target sits far above that. The engine displacement, gear ratios, aerodynamic profile, and curb weight together produce that number, and they are inseparable from the design. And the truck lacks equipment the Federal Motor Vehicle Safety Standards now require — electronic stability control and tire-pressure monitoring added in 2007, backup cameras after 2018, automatic emergency braking now.

A skeptical reader might note that several of those mandated components have patents drifting into the public

domain themselves. A remanufacturer in 2027 could lawfully integrate a backup camera built from expired-patent parts, owing royalties to no one. The combinatorial property is real. It does not break the wall, because the wall does not require that a vehicle simply have a backup camera. It requires that any vehicle manufactured for initial sale today be certified, on a current-year platform, to current-year safety, emissions, and fuel-economy standards, all together. The certification is the wall. Not the components.

Nor is the F-150 an oddity. The same demonstration works for any pickup, sedan, tractor, combine, semi-trailer, furnace, air conditioner, commercial heating unit, or major household appliance. Of all American consumer durable-goods spending, the audit classifies roughly fifty-eight percent as Blocked: in the patent-expired domain on paper, locked behind the regulatory wall in practice. Another fifteen percent is Hindered, partly blocked and partly flowing. Twenty-seven percent is Flowing — but only five of those twenty-seven points reach American workers, the other twenty-two arriving from overseas, where the foreign manufacturer enjoys exactly the bargain the American manufacturer has been denied. The category-level audit producing these three numbers is documented in the published methodology article cited in the Endnotes.

The result is that no one in the United States can lawfully manufacture and sell the 2005 F-150 today. Not a startup. Not a cooperative. Not a tribal nation building for its own roads. Not Ford itself. The patents have expired. The right exists. The right cannot be exercised, because exercising it would require ignoring rules that did not exist when the truck was originally certified. And this is not because anyone at EPA, or NHTSA, or the Department of Energy decided that expired-patent F-150s should be banned. No

one decided that. Every standard applies to new motor vehicles, and a remanufactured 2005 F-150 is, in regulatory terms, a new motor vehicle. Each standard is facially neutral, written without reference to patents and without intent to extend any monopoly. Aggregated, they cancel the second half of the Patent Clause for the entire vehicle category. Precision matters about what is constitutional here and what is not. EPA has constitutional authority to set emissions standards, NHTSA to set safety standards, the Department of Energy to set efficiency standards. None of those grants, taken individually, raises a Patent Clause question. The Patent Clause does not say the federal government may not regulate vehicles. It says monopolies must be for limited times. The constitutional question is what happens when those grants, exercised one agency at a time over fifty-five years, produce a structure under which no patent in the category ever functionally expires.

The Mechanism Has a Name

A ratchet is a one-way device. It turns in one direction and is locked against turning in the other. The metaphor has to be stated with care, because a careless version is false and the precise version is where the constitutional argument lives. The careless version says federal standards only ever tighten. They do not. Fuel-economy standards in particular have lurched up and down with each administration — tightened under Obama, loosened by the 2020 SAFE Rule, tightened again and beyond the old baseline under Biden, then in 2025 loosened more drastically than ever, with the civil penalty for missing the standard set to zero. A reader watching that alone would see not a ratchet but a seesaw.

The seesaw is the tell, not the refutation. Fuel-economy standards swing because they are held in place by nothing stronger than the policy of whoever currently runs the agency. That is a political gear, and political gears turn both ways. Three federal frameworks are the ratchet's principal gears: EPA emissions standards under the Clean Air Act, NHTSA's Federal Motor Vehicle Safety Standards, and the Department of Energy's appliance and equipment efficiency standards under the Energy Policy and Conservation Act. *The Wall Itself* walks all seventeen. Three are enough to demonstrate the mechanism. The one that matters here is the gear bolted down by statute so that it cannot turn backward no matter who wins the election — 42 U.S.C. § 6295(o)(1).

Consider what did not happen in 2025 — the year a single party controlled the White House and both chambers and made deregulation its banner. The Department of Energy's anti-backsliding statute, which forbids any administration from weakening an appliance-efficiency standard once set, was not repealed. It was not amended. It still stands exactly as it stood before. An administration that zeroed out the fuel-economy penalty with a stroke of the reconciliation pen, proposed rolling back the next tier of tailpipe standards, and nullified three freshly finalized efficiency rules the moment they issued could not loosen a single established appliance standard by ordinary rulemaking — because the statute forbids it, and at least one federal court has read the statute to mean what it says. To get around the lock, the administration had to argue that fully rescinding a rule is not the same as amending it, a theory now bound for litigation, and its allies in Congress had to write a bill creating petition-based exceptions to the lock rather than repealing the lock itself. The lock is so well built that its opponents legislate around it instead of at it.

Two Patterns: The Blockade and the Withdrawal

The ratchet produces two distinct patterns, depending on whether the public was ever able to exercise its post-expiration right in the first place.

The first is the blockade. The wall is built around the technology before the relevant patents expire, so that by the time they do, the wall is already there. The public's right comes into existence into a regulatory environment that already forbids it. The 2005 F-150 is a blockade case: the patents expired in 2025, and the public has had no manufacturing window of any length. Most of what is at stake in this book is blockade-pattern injury — tractors, pickup trucks, sedans, semi-trailers, combines, residential furnaces, commercial heating and cooling, industrial pumps, generators. The blockade is harder for an ordinary American to feel as a loss, because the loss is of a window that never opened. You do not see the F-150 you cannot buy. You see the F-150 you can buy, and you see its price.

The second is the withdrawal, and it is a sharper case. The product was already in the public domain. Americans built it, sold it, and bought it for decades, in a competitive market, at prices the market produced. Then the wall was erected around it after the fact, and the product was withdrawn from a domain it had already been part of. The ordinary incandescent bulb is the canonical case: its patents expired more than a century ago, Americans manufactured it continuously from 1879 onward, and federal efficiency rules issued between 2007 and 2023 progressively withdrew it from manufacturability. *Healthy Lighting and the Anti-Backsliding Trap* develops that case in full.

The distinction is useful to litigators. The blockade case asks whether the federal government may, through regulation, render a future patent expiration economically meaningless before it occurs. The withdrawal case asks whether it may take back a vested public right that had already been exercised. The blockade is the larger problem in dollar terms. The withdrawal is the stronger doctrinal posture, because the right at issue was not theoretical — it had been exercised continuously for over a century, and the precedent on vested rights is firmer than the precedent on hypothetical ones.

Why the Ratchet Is Uniquely Hostile to the Patent Clause

The Patent Clause does not exempt expired-patent products from laws of general applicability, and it does not promise the public-domain manufacturer a regulation-free environment. The constitutional problem is narrower: it is the way the ratchet is built. In the ordinary case a regulation is enacted, imposes a cost, and may render some designs uneconomic — and then a different administration arrives, and the rule may be tightened, loosened, replaced, or withdrawn. An expired-patent design uneconomic under one set of rules may become economic under the next. The bounded monopoly remains bounded. That is not what the ratchet does. Its defining feature is that the regulatory environment does not rise and fall. It only rises.

The structure converts the time-limited monopoly into one whose commercial benefit never ends — not by extending the patent term on paper. The patent expires on its statutory schedule. The holder's exclusive rights end. The Patent and Trademark Office's records correctly show the

patent as expired. None of the formal language of patent law has been touched. What has been touched is the second half of the bargain, the half that begins the day the patent expires and that the regulatory architecture renders economically unusable from that day forward.

That is the constitutional problem in compact form. A monopoly is what the holder enjoys when no one else may legally produce the article. That is what an unexpired patent provides — exclusive legal access, secured by infringement law. It is also what the ratchet provides for an expired patent — exclusive economic access, secured by the impossibility of any new entrant clearing the wall. The mechanism of exclusion is different. The result is the same.

A skeptic might object that the comparison is uneven, since Ford loses its exclusive legal access on the same day the public's right begins. The objection has the shape of a fair point and misses the substance. The Constitution is not concerned with the formal symmetry of expiration. It is concerned with what the public was promised, and what the public was promised was competition. When the patents on the basic scientific calculator expired in the 1990s, prices collapsed by more than ninety-five percent and never returned to their monopoly level. Ford's patents expire on schedule. Ford's monopoly pricing does not. The wall preserves the second when the Constitution intended only the first.

The ratchet does not amend the Patent Clause. It does something more elegant. It leaves the language intact and erases the operative effect. The clause continues to read what it has always read. The patent term continues to be twenty years. The patent expires on schedule. The expiration accomplishes nothing the public can use. The

clause becomes scenery. What matters is the structural shape: the wall is real, it grows in one direction, the growth is held in place by statute, and the statute makes the growth permanent against every actor below the constitutional level. Whether the wall stands against the Patent Clause's plain text is the question the doctrinal chapters answer, and *The Filable Case* explains why that question is now ripe — the Supreme Court retains the authority to ask it.

The Keystone

Seventeen federal frameworks participate in building the wall, and most are one-way only as a matter of practice — political habit, agency culture, the resistance of constituencies that benefit. Practice can change. But one of the seventeen is permanent by statute. One was written, debated, voted on, and signed into federal law specifically to ensure that the rule it produces cannot be loosened by any future administration. That provision is 42 U.S.C. § 6295(o)(1), added by the National Appliance Energy Conservation Act of 1987. It is the single act of Congress that converts the regulatory ratchet from a policy convention into a permanent feature of statutory law.

The story of how it became law is worth telling. The Senate passed it on February 17, 1987, by a vote of 89 to 6. The House passed it without a recorded individual vote — by voice vote, on March 3. President Reagan signed it on March 17. The single statutory provision that holds the ratchet in place across the entire energy-using product universe entered federal law without a single member of the House of Representatives being asked to register an individual yes or no. The next chapter is that story.

CHAPTER 4

The Anti-Backsliding Provision: How 42 U.S.C. § 6295(o)(1) Locked the Ratchet

The previous chapter described the architecture by which the post-expiration patent commons was cancelled — statutes, court decisions, and agency rules, each adding a layer to a wall almost no one set out to build. That wall has a keystone. One provision locks the rest in place. One provision that, by design, cannot be removed by any administration acting alone.

It sits at 42 U.S.C. § 6295(o)(1). Energy lawyers and the Department of Energy itself call it the anti-backsliding provision, though the label is later than the statute. It was enacted in 1987, in a bill called the National Appliance Energy Conservation Act. The vote was not close. Eighty-nine senators voted yes. Six voted no. The House passed it without a recorded vote at all. The President signed it without a public ceremony. It is the most consequential statute almost no one has heard of.

The Vote

The bill had failed the year before because President Reagan killed it. On November 1, 1986, he pocket-vetoed the prior version, H.R. 5465. His veto memorandum was unusually direct: the bill would set minimum efficiency standards for home appliances without regard to technological feasibility or economic justification, would intrude unduly on the free market, would deny consumers the opportunity to purchase lower-cost appliances, would

intrude substantially on traditional state responsibilities, and would mandate a complicated series of nineteen rulemakings over twenty years for fifty-two subcategories of appliances. And it would, he warned, hit low-income consumers hardest.

S. 83 was the same bill, repackaged and reintroduced by the new chairman of the Senate Energy and Natural Resources Committee. The Reagan administration knew it. On February 4, 1987 — thirteen days before the vote — the Office of Management and Budget issued a Statement of Administration Policy saying the President's senior advisors would recommend a veto if the bill reached him, then repeated, almost verbatim, the four objections from the veto memorandum.

Thirteen days later, the Senate voted 89 to 6 to send it to him. Of the forty-six Republicans then sitting, thirty-eight stood with the bill and against their own President's stated position, and one was absent. Six stood with the President: Phil Gramm of Texas, a former economics professor who offered the principal opposing amendment; James McClure of Idaho, who had chaired the Energy Committee until five weeks earlier; Steven Symms of Idaho; Jesse Helms of North Carolina; Don Nickles of Oklahoma; and Malcolm Wallop of Wyoming. Six men, all Republicans, all from Western or Southern states.

Two weeks later the House took up the bill the Senate had passed. It did not vote by roll call. It passed S. 83 on a voice vote. No individual representative is on record in support or opposition; nobody asked for a recorded vote because, after the 89-to-6 Senate result, nobody in House leadership thought one was needed. On March 17, 1987, President Reagan signed it into law. He held no signing ceremony. He issued no signing statement. The bill he had

threatened to veto twice was law, and the keystone of the wall was set in stone.

What Was Said, and What Wasn't

The floor debate filled roughly four pages of the bound Congressional Record. Two of the six dissenters rose to speak. The other four voted no without remarks. Gramm had filed an amendment before debate began, negotiated directly with Bennett Johnston, the bill's sponsor. It softened the rulemaking timeline and left the architecture untouched: it did not weaken a single efficiency standard, did not weaken federal preemption of state law, did not touch the anti-backsliding provision, and introduced no sunset, waiver, or executive off-ramp. Gramm said plainly why. *What I am doing here is offering an amendment which will put the bill into a form where the President's senior advisers will recommend that he sign it.* That is the administration's capitulation in nineteen words, from the most active opponent on the floor. Johnston accepted it with satisfaction on the record, calling it the administration's *retreat in the face of withering fire* — an amendment some had characterized as a fig leaf, he said, but which he would characterize as a sensible compromise. The Senate adopted it without a recorded vote. Gramm then rose to oppose the underlying bill anyway, and his substantive case is what political scientists would later call the bootleggers-and-Baptists analysis, articulated from the Senate floor in 1987 by a senator who voted no. He said the bill was an example of how laws should not be made: two special-interest groups, manufacturers and environmentalists, were the driving elements behind it, and they had worked out a compromise suiting their relative needs. The manufacturers wanted to preempt state standards. Neither

interest, he said, happened to coincide with the interest of the people who do the work and pay the taxes and pull the wagon in this country.

Gramm's second concern was the geographic uniformity problem of national standards, and he gave the warning a name and a place: a retired citizen from Brownsville, Texas would be forced to buy a heating unit that makes sense in Alaska but makes absolutely no sense in Brownsville. It captured the impossibility of designing one furnace that suits a hundred-degree South Texas summer and a forty-below Alaska winter, and tied the impossibility to the consumer most likely to bear its cost. Reagan had warned in general terms about low-income harm. Gramm gave the warning coordinates.

Senator McClure spoke briefly, and his one notable sentence was a philosophical confession: while he still believed the best way to avoid the problems inherent in federally imposed appliance standards was a simple repeal of the existing statute, he recognized that this might not be an option available to him. He voted no anyway.

And in all of it — even in the letter from the Coalition for Federal Appliance Efficiency Standards entered into the Record — no one mentioned patents. No one mentioned limited times. No one observed that the bill, by erecting a permanent federal regulatory wall around appliance design, would eliminate the post-expiration patent commons across the entire household appliance category. The full text of the Senate's February 17, 1987 proceedings runs to roughly one hundred thirty pages of the bound Congressional Record. The word patent appears in none of them.

It is worth pausing on the asymmetry. Reagan had warned in plain language that the bill would force consumers to

spend an extra \$1.4 billion per year on more expensive appliances. Gramm had named the coalition that drove it forward. McClure had stated his preference for repealing the underlying statute outright. The political and economic case against the bill was made, and it was rejected 89 to 6. But the constitutional case was never made. The Patent Clause sat in the Constitution that day exactly as it had for two centuries. And no senator, in either party, on either side of the bill, mentioned any of it. That is the sense in which the clause was forgotten. Not contested. Not overruled. Forgotten.

What the Provision Says

The act runs to thirty-three pages in the Statutes at Large. It covers refrigerators, freezers, room and central air conditioners, water heaters, furnaces, dishwashers, clothes washers and dryers, ranges and ovens, pool heaters, and several other categories. In the middle of all that sits one sentence: the Secretary may not prescribe any amended standard which increases the maximum allowable energy use, or decreases the minimum required energy efficiency, of a covered product.

That is the anti-backsliding provision. In ordinary English: once a federal standard is set, the Secretary cannot ever lower it. Ordinary federal regulatory statutes contain safety valves. The statutes governing air pollutants, spectrum allocation, drug approval, and transportation safety all permit revision in either direction when the evidence warrants. This one permits no such thing — not on a showing of technological substitution, not on a showing of consumer harm, not on a showing of newly available alternatives. It contains no sunset, no waiver for changed circumstance, and no expedited procedure by which

Congress could later authorize relief on application. The one waiver the act does contain runs the other way: states may seek waivers from federal preemption to impose *stricter* standards, never to let the Secretary relax a federal one. A federal court can hear a citizen suit asking the Secretary to issue a stricter standard. It cannot order a weaker one, because the statute denies the Secretary authority to issue one. A regulated party has two options: lobby Congress, or sue. The Secretary can leave it alone. The Secretary can tighten it. The Secretary cannot loosen it. The rest of the subsection only sets the clock — compliance dates, and a fixed schedule on which the Secretary must review each existing standard and decide whether to amend it. And because that one sentence has already foreclosed amendment in any direction but one, “whether to amend” can mean only one thing: whether to tighten.

What the Provision Does

The text does three things. First, it is a ratchet: it permits motion in only one direction, so the long-run trajectory of any covered product is monotonic. The federal floor for refrigerator efficiency, set in 1990, is higher today than it was in 1990. In every covered category, the standard moves in only one direction over time, because there is only one direction it can move.

Second, it binds the executive. In the ordinary architecture of administrative law, when a President is elected on a platform of regulatory relief, the agencies under that President have authority to implement it — to revise rules, grant waivers, relax standards, reverse predecessor decisions on the merits. That authority is what makes the political system responsive to election outcomes. The anti-

backsliding provision removes it for appliance efficiency standards. A President elected on a platform of restoring consumer choice in appliances cannot direct the Secretary of Energy to relax any existing standard. The Secretary has no statutory power to comply. The President can issue executive orders, can fire the Secretary, can replace the Secretary with a more compliant successor — and the standards will still stand, because no Secretary has the authority to lower them.

A reader may object that Presidents routinely decline to enforce laws they dislike, and the Supreme Court blessed that kind of choice in *Heckler v. Chaney* (1985), holding an agency's decision not to bring an enforcement action presumptively committed to its own discretion. But enforcement discretion is the power not to prosecute a violation. It is not the power to repeal the standard. A President could direct the Secretary to stop fining manufacturers who break a standard. The standard would still be law. The banned design would still be unlawful to sell, unlawful to finance, unlawful to stock at retail — and the next administration could resume enforcement the day it took office, without writing a single new rule. Non-enforcement creates legal limbo. It does not reopen the commons.

Third, the provision freezes the regulatory state at the level of stringency that obtained at the moment of greatest political support for stringency. Whatever standard was set when the rulemaking record was strongest, the political winds were favorable, and the activist coalition was best organized — that standard becomes the permanent floor. Subsequent shifts in technology, in consumer preference, in international competition, in cost of living, in patent expiration can never move the floor downward.

That last point connects the chapter back to the project of the book. When a federal performance standard supersedes an expired patent's claims — when a 1995 patent on a furnace mechanism expires in 2015, and a 2009 efficiency standard already prohibits that mechanism's commercial sale — the expired patent's invention does not actually enter the public domain. It cannot be manufactured. It cannot be sold. It cannot be operated commercially. The patent's monopoly has expired. The regulation's prohibition has not, and under the anti-backsliding provision the prohibition cannot be relaxed by any executive action alone, no matter how strong the case for doing so. The wall is high. The wall is also locked.

Why the Provision Is Uniquely Hostile to the Patent Clause

Other regulatory schemes restrict expired-patent designs too. An agency can ban a chemical after the patent on its manufacturing process has expired, ground an aircraft type after the patents on its airframe have expired, withdraw approval of a drug after the patents on its molecule have expired. But in each of those cases the regulation is reversible by executive action. The decisions are reviewable, the standards amendable, the ordinary administrative path running in both directions. A new administration with new priorities can retrace those steps.

The anti-backsliding provision is the one place in the federal regulatory code where that path runs in only one direction. And it covers a domain — household appliances, lighting, plumbing fixtures, motors, distribution transformers — that is unusually rich in expired patents. The basic refrigeration cycle, the basic furnace, the basic

incandescent lamp, the basic flush toilet, the basic single-speed motor, the basic clothes-washer agitator: all mature, off-patent technologies whose fundamental claims expired decades ago. By ordinary patent law they belong in the Expired Patent Commons. Federal regulation bars them from it, and no executive remedy can loosen the bar.

The Patent Clause is the only clause in Article I, Section 8 that qualifies Congress's power with a temporal limitation written into the text. The framers wrote that limit in on purpose. They knew a monopoly, once granted, would be defended forever by its beneficiaries, and that an unbounded grant would defeat the clause's purpose. The anti-backsliding provision is, in operation, a means of converting limited times into permanent times — not by extending the patent itself, which expires on schedule as it always has, but by ensuring that the post-expiration design cannot be brought back to market under any administration. The mechanism is different from an extended patent. The result is the same. This is the constitutional dimension that nobody raised on February 17, 1987. Not the bill's supporters, who wanted energy savings. Not its dissenters, who wanted free markets. Not the President, who wanted, and got, neither. Forgetting the clause was bipartisan.

The Predictable Equilibrium

Why does this configuration persist? Gramm gave the answer on the Senate floor. Two organized interests — appliance manufacturers wanting federal preemption of fifty state regimes, and environmental advocacy groups wanting national efficiency standards — converged on a bill that gave each what it wanted. The diffuse interest of consumers, who would pay for the bill on every appliance

purchase for the rest of their lives, was not represented in the room.

The academic literature names the dynamic too. George Stigler argued in 1971 that regulation is, in the typical case, a service supplied by the state to organized industry rather than imposed on it. Mancur Olson had observed in 1965 that small concentrated interests can bear the cost of organizing politically because each member's share of the gain is large, while large diffuse interests cannot.

Concentrated interests therefore tend to win against diffuse ones, even when the diffuse interests are vastly larger in aggregate.

Bruce Yandle gave the dynamic its name in 1983: bootleggers and Baptists, from the era of dry counties, where prohibition was sustained by a coalition of those who wanted the moral outcome and those who wanted the illicit margins.

Each group was indispensable to the other. The 1987 vote was that configuration exactly. The Baptists were the environmental and energy-conservation groups; the bootleggers were the appliance manufacturers, who had operated under a patchwork of state standards and wanted federal preemption of them. Reagan did not lose to a hostile coalition. He lost to his own party's manufacturers, who had decided federal preemption was worth the price of federal stringency.

The anti-backsliding provision is what such a coalition writes when it is permitted to write its own preferred legal architecture. The coalition's interest is in preventing future deregulation, because deregulation would re-fragment the market and reintroduce the patchwork the manufacturers had paid to escape. A one-way ratchet locks the deal in.

The Baptists are content because the standards stay. The bootleggers are content because the preemption stays. The diffuse consumer interest, being absent, neither gains nor loses politically — only economically, on every appliance purchase, for life. This is not corruption in the criminal sense. It is what political economy predicts when concentrated interests face diffuse ones. It is the predictable equilibrium.

The provision is not bad regulation in the ordinary sense. It imposes no absurd cost on its face, and the standards it freezes are, taken individually, defensible. Refrigerators are more efficient than they used to be. Clothes washers use less water. Each is, considered narrowly, a gain. What the provision is, is bad constitutional architecture. It removes the ordinary executive remedy for regulatory overreach in one domain, forcing every dispute into one of two venues: back to the Congress where the same coalition stands ready to keep the ratchet locked, or into the federal courts, where the question of whether an ordinary statute may convert “limited Times” into permanent times has not yet been litigated. The Constitution’s Patent Clause says “limited Times.” The anti-backsliding provision says, in effect, never.

The keystone was set on March 17, 1987, with no signing ceremony, on a Senate vote of 89 to 6 and a House voice vote with no individual record, against the stated position of the President of the United States, and with no senator on either side mentioning the constitutional clause the bill was about to overwrite. That is how the clause is forgotten.

CHAPTER 5

The Wall Itself: Seventeen Laws That Hold the Patent Bargain Hostage

The previous chapters described the mechanism. This chapter describes the materials, and then shows the mechanism at work in a single function, traced through the public patent record. Seventeen federal frameworks do the load-bearing work. None of them was passed for that purpose. Each can be defended on its own terms — safety, efficiency, environmental protection, workplace protection, public health, aviation airworthiness, maritime safety. The wall is the combination.

The seventeen are not exhaustive. The full Code of Federal Regulations runs to over two hundred titles and tens of thousands of pages, and somewhere in those pages there is a rule for almost every product manufactured in the United States. The seventeen are the statutes whose effect on the post-expiration patent commons is large enough to map. Take any one of them out, and the bargain returns to working order across some segment of expired-patent commerce. Take all seventeen out, and it returns across the whole of it.

They divide by mechanism rather than by agency. Eight apply to motor vehicles, and the foundational one is the Clean Air Act of 1970, at 42 U.S.C. §§ 7401 and following, with the mobile-source provisions at § 7521. It has been amended substantively three times — 1970, 1977, 1990 — and each amendment moved the regulatory floor without ever lowering it. EPA has issued successive

emissions tiers under it: Tier 1 in 1991, Tier 2 in 1999, Tier 3 in 2014, Tier 4 in development, each applying to vehicles manufactured for initial sale. Some frameworks freeze a regulatory floor in place: the Energy Policy and Conservation Act of 1975 with its 1987 amendment. Some require type-certification at every variant of a product — the Federal Motor Vehicle Safety Standards, the Federal Aviation Administration’s airworthiness regime at 49 U.S.C. § 44704, the Coast Guard’s vessel inspection regime under 46 U.S.C., and the international maritime conventions Congress has ratified into domestic law.

Aviation shows the pattern at its sharpest. The patents on the Cessna 172, the most-produced aircraft in history, expired decades ago, and no one may build a new one without a type certificate that takes years and tens of millions of dollars to obtain. That cost amortizes over a production run of several hundred wide-body jets. It does not amortize over a few hundred piston-engine aircraft a year. The result is the American light-aircraft market as it actually stands: a handful of incumbents, prices risen far faster than general inflation, and a fleet whose median age now exceeds fifty years.

Some impose workplace and waste compliance overlays that fall on every domestic manufacturer in any sector: the Occupational Safety and Health Act of 1970, the Resource Conservation and Recovery Act of 1976. One imposes mandatory consumer-product safety standards across a wide range of household durables, through the Consumer Product Safety Act of 1972. The National Technology Transfer and Advancement Act of 1995, Public Law 104-113, instructs every federal agency to use technical standards developed by private voluntary consensus bodies — the Society of Automotive Engineers, ASTM, Underwriters Laboratories, the American National

Standards Institute — in lieu of writing federal standards from scratch. The mechanism is incorporation by reference: the privately drafted standard becomes federal regulatory text. And one framework mirrors and extends the federal architecture at the state level.

The state-level framework is the most diffuse of the seventeen, and the most consequential piece of it is the California Air Resources Board. Its emissions standards have operated since 1967 under a federal preemption-waiver structure: California may set standards stricter than the federal floor, and other states may adopt the California standard in lieu of that floor. More than a dozen states have. So a manufacturer building for nationwide sale must in practice certify to California's standard wherever the vehicle is built. State frameworks raise a federalism complication the federal ones do not. A constitutional ruling reaches federal regulation directly; state regulation is reached through preemption, through the commerce power where state rules unduly burden interstate commerce, and through the same Patent Clause and Takings Clause analyses applied state by state. The Supreme Court addressed the preemption question for the Federal Motor Vehicle Safety Standards in *Geier v. American Honda Motor Co.* (2000).

The inventory here is architectural only: it does not, by itself, make the argument.

The Wall, Traced Patent by Patent

The wall is easy to assert and harder to see. So look at it directly, in one emission-control function, through the one record that cannot be spun: the patents themselves. Every United States patent states on its face when it was filed, when it was granted, when it expired, and which earlier

patents it was built on. String those facts together for a single function and the wall stops being a claim. It becomes a sequence of dated documents, each one handing off to the next, with the public locked out at every step.

The function is catalyst-efficiency monitoring — the on-board diagnostic that decides whether a car's catalytic converter is still working. It compares an oxygen-sensor reading taken before the converter to one taken after it, computes a ratio, and lights the dashboard warning when the ratio drifts past a federally set threshold. A car cannot be sold without it. A California Air Resources Board executive order certifies the 2002 Ford Expedition and Lincoln Navigator with the 5.4-liter V8 and lists the exact hardware that performs the function: dual three-way catalytic converters, dual heated oxygen sensors, exhaust-gas recirculation, and sequential multiport fuel injection, all under the on-board diagnostic rules. That is the certified machine. The monitoring method inside it belongs to Ford, and Ford has held it through not one patent but a relay of them, each replacing the last.

The relay runs nine members deep, and the table below lists every member by patent number, with the dated regulatory steps it tracked. Some members ran their full statutory term and entered the public domain on schedule. Some lapsed early, when their owner stopped paying maintenance fees — ordinary in a competitive market, and by itself proof of nothing. Some are in force today. What matters is the shape: as each member expires, the next has already issued, and the function the public was promised has never once been available to build.

Patent	Fil ed	Gra nted	Status	Role in the chain	Docume nted link
US 5,357,751 (Orzel)	1993	1994	Expired * (2013)	The switch-ratio (SR) ancestor	Named by number in later members
US 5,899,062 (Jerger et al.)	1997	1999	Expired * (2017)	Introduces the arc-length index-ratio (IR) method	Cited on the face of US 10,626,779
US 6,195,986 (Davey et al.)	1999	2001	Lapsed ** (2019)	IR intermediate refinement	Same function, same assignee
US 6,397,583 (Davey et al.)	2001	2002	Lapsed ** (2021)	IR intermediate (flex-fuel)	Text cites US 5,899,062
US 6,532,734 (Nader et al.)	2002	2003	Lapsed ** (2022)	IR intermediate (two-grade catalyst)	Text names US 5,357,751 and US 5,899,062
US 6,550,237 (Adamczyk et al.)	2002	2003	Lapsed ** (2022)	IR intermediate refinement	Same function, same assignee

Patent	Fil ed	Gra nted	Status	Role in the chain	Docume nted link
US 10,626, 779 (Bagna sco et al.)	20 18	2020	Lapsed ** (2024)	Modern IR successor; corrects for sensor aging	Cites US 5,899,06 2; cited by US 12,196,6 40
US 10,947, 910 (Bagna sco et al.)	20 19	2021	In force	Catalyst feedback- control intermediate	Same function, same assignee
US 12,196, 640 (Martin ez Leyva et al.)	20 22	2025	In force	Live successor; oxygen- sensor diagnosis	Cites US 10,626,7 79 on its face

The catalyst-efficiency monitoring relay, patent by patent. Filing and grant dates, expiration status, and the citation links each member states on its own face.

Ye ar	Regulatory step	Effect on the monitored function
19 89	CARB § 1968.1 (OBD II) adopted	Catalyst and oxygen-sensor monitoring required
19 90	LEV I adopted	First Low-Emission Vehicle standards (model years 1994– 2003)
19 91	Federal Tier 1 final rule	Phased in over model years 1994–1997

Year	Regulatory step	Effect on the monitored function
1994	§ 1968.1 and LEV I implementation begins	Monitoring operative on new vehicles
1996	OBD II universal; EPA § 209(b) waiver	Required on essentially all new vehicles
1999	LEV II adopted; federal Tier 2 adopted (Dec. 21)	Tighter NMOG/NO _x (LEV II from model year 2004; Tier 2 2004–2009)
2002	CARB § 1968.2 adopted (codified 2003; model year 2004 onward)	New and tightened light- and medium-duty monitoring requirements
2012	LEV III adopted	Combined NMOG+NO _x , roughly 70 percent tighter (model years 2015–2025)
2014	Federal Tier 3 finalized (Mar. 3)	Phased in over model years 2017–2025

The regulatory steps the relay tracked, 1989–2014.

That is the wall in a single function. The same tracing can be run for emissions aftertreatment, for restraint systems, for compressor control, for anything the federal standards require and the patent record covers.

The Keystone's Placement

Two of the seventeen apply to home appliances and to commercial energy-using equipment. The Energy Policy and Conservation Act of 1975 established the Department

of Energy's authority to set efficiency standards for residential appliances and a long list of other consumer products. The same Act established the Corporate Average Fuel Economy program, counted among the mobile-source frameworks rather than again here.

The 1987 amendment added the anti-backsliding provision at 42 U.S.C. § 6295(o)(1). That single provision is the keystone. The Act without it would be one framework among several; with it, the Act holds the others in place against any administrative attempt to roll them back.

The Architecture, Again

Seventeen frameworks. Eight different federal agencies. State-level extenders. Foreign treaty obligations woven into domestic law. Each piece can be defended on its own merits. None of them was passed to extend the patent monopoly past expiration. The architecture they form does exactly that.

Two things follow. The first is what each piece does, and does not, decide. None of the seventeen frameworks decided that expired-patent vehicles, appliances, consumer durables, aircraft, or vessels should be barred from manufacture in the United States. The decision was never made. It is a side effect of how every rule is written, because every standard applies to new products manufactured for initial sale — and a remanufactured product built on an expired-patent platform is, in regulatory terms, a new product. Each rule is facially neutral, written without reference to patents and without intent to extend any monopoly. Aggregated, the rules cancel the second half of the Patent Clause across the sectors the architecture covers.

The second is the structural placement of the keystone. Sixteen of the seventeen are, in formal terms, capable of relaxation. Congress could amend the Clean Air Act. The Federal Aviation Administration could withdraw an airworthiness standard. Congress could narrow the Consumer Product Safety Act. The Coast Guard could grant categorical exemptions. None of these has happened in any meaningful sense in fifty-five years. One framework, though, is not reversible at all. The anti-backsliding amendment is permanent against every actor below the constitutional level, and the Supreme Court has authority to remove it.

The seventeen-statute architecture would be a serious problem on its own. The anti-backsliding ratchet is what makes it permanent.

Twelve Items Behind the Wall

The inventory translates into the consumer durables behind the wall — twelve ordinary products from the kitchen, the basement, and the garage, each blocked from patent-expired remanufacture by a federal framework named in the inventory or by an overlay layered on top of one. The list is not exhaustive. The list is the household, item by item.

Item	How blocked
Air conditioners	DOE efficiency standards under EPCA (1975) with the NAECA anti-backsliding amendment (1987); central AC standard first effective 1992, raised 2006 and 2023; room AC first effective 1990.

Item	How blocked
Dishwashers	DOE efficiency standards under EPCA/NAECA, first effective 1990; amendments through 2012 capping water and energy use; further amendment 2024.
Freezers	DOE efficiency standards under EPCA/NAECA, first effective 1990; refrigerant-restriction overlay (CFC phaseout, 1995).
Furnaces	DOE efficiency standards under EPCA/NAECA, first effective 1992; 2023 final rule sets 95% AFUE effective 2028.
Heat pumps	DOE efficiency standards under EPCA/NAECA, paired with central air conditioning under SEER and HSPF; refrigerant-restriction overlay (R-22 phaseout 2010–2020; HFC phasedown under the AIM Act of 2020).
Lawn mowers	EPA small spark-ignition engine emissions standards under the Clean Air Act (Phase 1, 1995; Phase 3, 2008), 40 C.F.R. Part 1054; CPSC walk-behind power lawn mower safety standard, 16 C.F.R. Part 1205 (1982).
Ovens	DOE cooking-products efficiency standards under EPCA/NAECA, first effective 1990; multiple amendments since.
Refrigerators	DOE efficiency standards under EPCA/NAECA, first effective 1990; refrigerant-restriction overlay (CFC phaseout, 1995).

Item	How blocked
Snow blowers	EPA small spark-ignition engine emissions standards under the Clean Air Act (Phase 1, 1995; Phase 3, 2008), 40 C.F.R. Part 1054.
Stoves and ranges	DOE cooking-products efficiency standards under EPCA/NAECA, first effective 1990, covering both cooktops and combined ranges.
Table saws	CPSC rulemakings of 2011 and 2017 under the Consumer Product Safety Act, neither finalized; UL 987 and UL 62841-3-1 voluntary standards adopted by retailers through the NTTAA conduit; product-liability exposure on saws sold without flesh-detection technology.
Water heaters	DOE efficiency standards under EPCA/NAECA, first effective 1990; 2015 amendment effectively required heat-pump or condensing technology for residential units above 55 gallons; further amendment 2024.

The cost of this architecture is not borne by the manufacturers, who recover compliance costs in the prices they charge. It is borne by the household that buys the pickup truck, the furnace, the small aircraft. The household pays twice: for the compliance baked into the new product, and for the absence of the remanufactured alternative Jefferson's bargain should have made available at a fraction of that price. The wealth-transfer chapter measures the second payment. The inventory only shows what is doing the transferring.

CHAPTER 6

Why Nobody Noticed: How a Constitutional Violation Hides in Plain Sight

The argument this book has been making is, on its own terms, not a difficult one. The Patent Clause says limited Times. Federal regulators have built a wall that converts those limited times into permanent ones for fifty-eight percent of expired-patent commerce. The Supreme Court has said, across a line of cases running one hundred ninety-five years, that what comes next when a patent expires belongs to the American people. The arithmetic of the harm is straightforward. The constitutional violation is structural. The remedy is available.

So why has no member of Congress raised it? Why has no federal court been asked the question? Why has no presidential administration in fifty-five years acknowledged it? Why has no law professor written the article? Why has no journalist filed the story? The silence is the first thing a reader of this argument has the right to ask about. A reasonable reader, having absorbed the constitutional case, will assume there is a catch. There is no catch. There is, instead, a diagnosis.

One fact precedes the diagnosis. The Patent Clause was written in 1787. The wall was built after 1970. For the 183 years between, no second federal power intruded on the ceiling on limited Times. Patents expired on the calendar, and the design passed to the public with nothing in the way. The Court did meet encroachments on the ceiling in that span, and struck them: from state law in *Sears* and

Compco (1964), from private contract in *Brulotte* (1964). None came from a federal regulatory power. A limit its own government has never intruded on builds no doctrine, no literature, and no vocabulary. When the second federal power arrived, there was no one whose job was to compare the two. That inference is the book's own. The six forces that follow explain why the silence held. The dormancy explains why nobody was watching when it began.

Six forces converge to produce the silence. The first is the corrupt ecosystem of regulatory capture — the loop of lobbyists, legislators, agency staff, and revolving-door employment that processes roughly \$22 trillion of taken value and has every incentive to leave the constitutional question on the floor. The second is the bootleggers-and-Baptists coalition, named by the economist Bruce Yandle in 1983, whose two wings have independent reasons to defend the wall and no reason to challenge it. The third is the institutional blind spot — the categorical absence of the expired patent commons from the operating thinking of the agencies, Congress, every presidential administration of the last fifty-five years, and the academic disciplines that might otherwise have noticed. The fourth is the doctrinal floor: until 2024, even a scholar who privately recognized the violation could not have brought the case in federal court. The fifth is moral acquiescence — the psychological architecture that lets otherwise principled people stay silent when each of the wall's stated purposes seems worthier than the constitutional cost. The sixth is rule-induced behavior — the game-theoretic phenomenon by which individuals inside a system act according to its rules without recognizing the rules as a system, producing outcomes that look coordinated but require no coordinator.

Where do the participants sit, by population? Roughly ninety-five percent sit in Force 6 — going about ordinary life, abiding by rules they did not write, with no professional position from which the constitutional question would have come to their attention. A fraction of a percent occupies Forces 1 and 2, with both the expertise to see the question and the incentive to keep silent. A smaller fraction yet inhabits Forces 4 and 5. These numbers are illustrative. The argument here is not against the ninety-five percent. It is against a system that operates regardless of who staffs it.

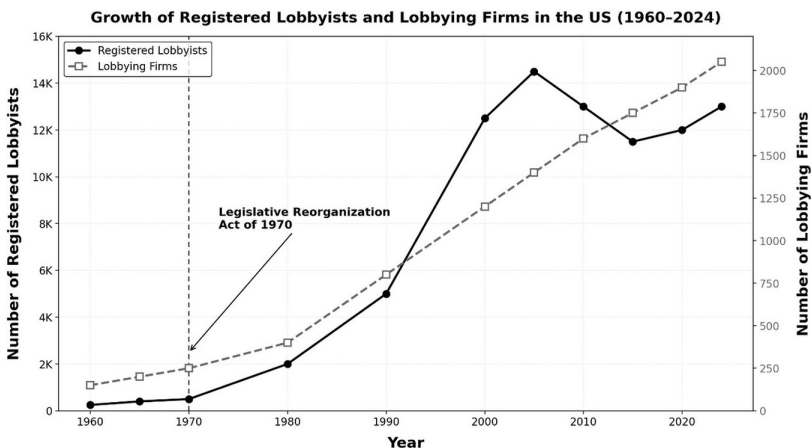
The six forces are independent. Removing any one of them would not break the silence; the others would carry it. That is why the silence has held for fifty-five years. It is also why, when the silence breaks, it will break suddenly. Each of the six forces is contingent. None of them is permanent. What follows walks them in turn.

Force 1. The Corrupt Ecosystem

The opening chapter introduced the 1970 Legislative Reorganization Act (LRA) and the lobbyist-count data that documents what the Act made possible; the chapter, *How the Bargain Was Cancelled: The Regulatory Ratchet*, walks its history in full. The diagnostic point here is narrower. Once the Act passed, the loop closed. The loop is the engine that has produced the silence for fifty-five years, and it requires no conspiracy and no individual bad actor.

Before October of 1970, most committee proceedings and many floor votes were not recorded, and the rooms where bills are written were closed by default. The word lobbyist records the arrangement: they waited in the lobby, kept out of the room, unable to watch their targets vote. Secrecy

had a function. As Senator Bob Packwood put it of the closed sessions, members could vote their conscience there, where no trade association could phone within twelve hours to complain. The 1970 reforms opened those doors in the name of transparency, and the people waiting in the lobby were the ones who walked in first. Every committee vote, floor vote, amendment, and roll call became public, indexed, and trackable — a real-time scorecard of every legislator's behavior on every issue. The Picture Section panels titled *1949 vs. 2013 committee rooms* and *Why they are called lobbyists* show the before and after. The numbers measure the effect: a lobbyist count growing from a few hundred in 1970 to over thirteen thousand by 2024, a lobbying-firm count multiplying fourteenfold between 1971 and 1982. The most rigorous account is D'Angelo and Ranalli's 2019 *Foreign Affairs* essay "The Dark Side of Sunlight," from whose data these figures derive.

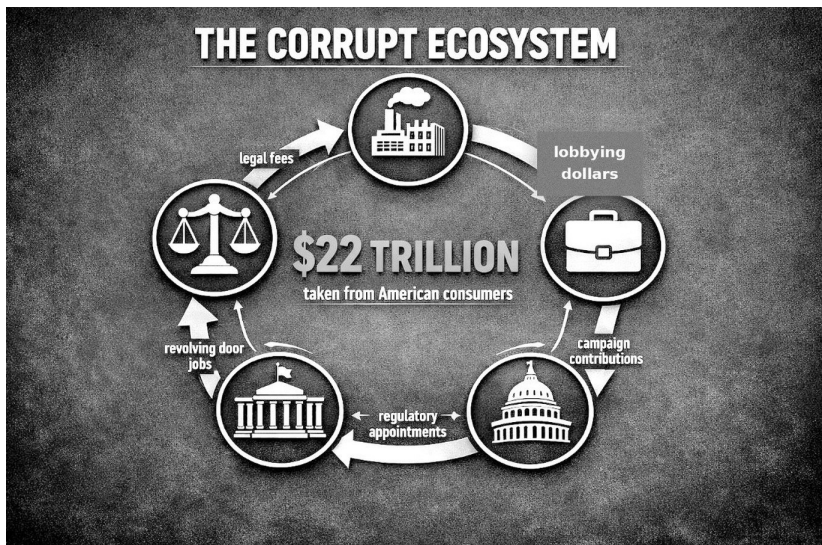


Note: Data for 1960–1990 is estimated based on historical accounts.
Data for 2000–2024 is based on Statista and OpenSecrets reporting.

The scorecard turned a campaign contribution from a general expression of support into a conditional payment. A legislator who voted the wrong way on a tracked vote

could be defunded the next cycle, primaried, or have an opposing candidate funded by the same industry. It does not require corruption in the criminal sense. It requires only the ordinary self-preservation instinct of a legislator whose career depends on funding he does not personally control.

Three other nodes complete the loop. The first is the agency. EPA, NHTSA, and DOE staff their senior policy positions from the regulated industries, because that is where regulatory expertise lives, and a senior official who leaves government typically returns to industry. The “revolving door” is not a metaphor. It is the average career trajectory of the people who write and enforce federal regulations. The second node is the courts: from *Chevron* in 1984 until *Loper Bright* in June of 2024, judicial deference closed the last check on the loop. The third node is the legal-and-consulting economy. Patent attorneys, consulting firms, and trade associations all bill against regulatory complexity, and the professional ecosystem needs only the architecture to remain intricate, which it does, by everyone’s preference.



Now apply the loop to the constitutional question. Suppose a member of Congress, briefed by a thoughtful staff lawyer, decides that the post-1970 wall around expired patents violates Article I, Section 8, Clause 8. The lobbyists see his tracked vote on any bill that would loosen the wall and report it to the trade associations, which decline to fund his next campaign and quietly support a primary challenger. The agencies whose authority he has questioned testify against his bills. His colleagues, none of whom have heard the argument before, worry that supporting him will trigger the same defunding. His bill dies in committee. His career stalls. Five years later he is no longer in Congress, and the question has not advanced. Multiply that scenario by fifty-five years and the result is exactly what the record shows: no member of Congress, in either party, has raised the Patent Clause question against the wall. Not Republican, not Democrat, not progressive, not conservative, not libertarian. The silence is the predictable behavior of a system designed — accidentally or not — to make this particular question structurally unaskable.

The federal government has not passed a budget that reduced the national debt in any single fiscal year since 1957. The opening chapter noted this fact in passing. The system that has not passed a debt-reducing budget in sixty-eight years is the same system that has not raised the Patent Clause question in fifty-five. The two facts share a mechanism. A Congress whose individual members cannot vote against the funding interests that elect them will not raise constitutional questions whose remedies threaten those same interests. That is the corrupt ecosystem.

The Clinton-era surpluses of 1998 through 2001 did not reduce the debt; the reconciliation between operating-

budget surpluses and total outstanding national debt is walked in the chapter, *Hybrid Legislative Voting: How to Stop the Ratchet from Resetting*.

Force 2. Bootleggers and Baptists

Yandle (1983), introduced in the opening chapter, named a pattern now foundational to the public-choice literature. A durable regulatory regime is rarely supported by one coalition. It is supported by two, with opposite motives, who reach the same regulatory conclusion for different reasons. The Baptists support the regulation because it serves their stated values; the bootleggers because it raises the cost of competition. The Baptists provide moral cover. The bootleggers provide political muscle. The result is an architecture harder to dislodge than any single-coalition regulation could ever be.

The Baptists are the environmental groups, public-health organizations, and traffic-safety advocates who have spent fifty-five years pressing for tighter EPA emissions tiers, broader NHTSA safety mandates, and stricter DOE efficiency standards. Their motives are sincere and the stated purposes are real. None of this is in dispute, and the Patent Clause argument does not require it to be. The argument requires only that the constitutional question — whether the architecture the Baptists helped build reconstitutes by regulation the manufacturer’s exclusivity from low-cost remanufactured competition, an exclusivity with no terminus — does not appear inside their operating framework. It does not. “Limited Times” is not part of their working vocabulary, and no career incentive in their world makes it so. Whether restoring remanufacturing would unleash 1970s emissions is taken up in the chapter, *The*

Environmental Question: Why the Patent Bargain Does Not Unleash 1970s Emissions.

The bootleggers are the incumbent manufacturers — the automotive majors, the agricultural-equipment consolidators, the residential-HVAC incumbents, the lighting majors — and the patent attorneys, lobbyists, and trade-association staff who work for them. Their patent attorneys do see the constitutional vulnerability. They have to; Patent Clause doctrine is the foundation of the law in which they built their careers. They simply have no reason to raise it. The wall benefits their clients, and their duty runs to their clients. The silence is a strategy.

The asymmetry is structural, not accidental. The Baptists do not see the constitutional question. The bootleggers see it and suppress it. No party in the coalition has both the expertise to recognize the question and the incentive to raise it.

A grandfather framework — developed in the chapter, *Healthy Lighting and the Anti-Backsliding Trap* — would hold a remanufactured product to the standards in effect during the platform year it mimics. A 2005-platform pickup remanufactured today still meets the 2005 NHTSA crashworthiness standards and the 2005 EPA emissions tier — the standards that produced the outcomes the Baptists campaigned for. In principle, the Baptists could support the framework without abandoning anything they ever fought for. In practice, they have not, because the constitutional argument has never been put in front of them in a form their operating framework can recognize.

Force 3. The Institutional Blind Spot

The expired patent commons does not appear inside the operating thinking of the four institutions where the constitutional question would otherwise have to surface. The agencies that built the wall do not think about patents. The Congress that authorized them does not think about expired-patent doctrine. The presidential administrations that staffed them do not think about Article I, Section 8, Clause 8. The academic disciplines that might have noticed are organized so that each holds only a piece of the question. The blind spot is not stupidity. It is categorical. Each institution sees what its operating framework lets it see, and the commons sits in the gap between the four frameworks.

Begin with the agencies. EPA, NHTSA, and DOE are statutory creations. Their general counsel offices read the Clean Air Act, the Motor Vehicle Safety Act, and the Energy Policy and Conservation Act, and their working questions are the ones those statutes pose. An EPA lawyer asked whether a new emissions tier conflicts with the Patent Clause will not have an answer prepared. The question has not come up. Move to Congress. A senator briefed on an efficiency mandate is briefed by staff from the Hill, the regulated industries, or the policy programs; none of those pipelines includes patent law. The Judiciary Committees, which do have patent-law staff, do not write energy or vehicle-safety legislation. The committees that handle regulation and the committees that handle intellectual property do not talk to each other, and the question lives in the gap between them. The presidency runs through the same pipeline. Presidential agendas are built from coalitions — the climate caucus, the auto-industry caucus, the family-farm caucus — none of which has reason to put the Patent Clause on the agenda. Every

administration in fifty-five years has had constitutional lawyers on staff. None was asked whether the architecture it administered complied with “limited Times.” The question never reached the desk.

Then comes academia, where the blind spot is the most carefully constructed of all. Patent law scholarship is a substantial discipline with its own journals and canon; Walterscheid, Bracha, and Mossoff have written excellent histories of the Clause. But the canon is concerned with active patents — how they are filed, prosecuted, infringed, licensed, and litigated during their term. The post-expiration phase of the bargain receives a sentence in the textbook and not much more.

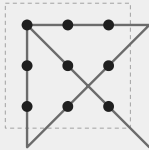
Constitutional law scholarship is the mirror image. Scholars of Article I, Section 8 work on the Commerce Clause, the Spending Clause, the Necessary and Proper Clause, and the Tax and Spend Clause. The Patent Clause is treated as a specialty within intellectual property, not as a piece of Article I structure. The constitutional scholars do not engage with patents because they understand patents to be a patent-law topic. The patent scholars do not engage with the constitutional question because they understand it to be downstream of their working questions. The two disciplines are separated by an academic membrane that keeps the question from crossing between them.

Two outside thinkers help name what is happening. Kuhn (1962) showed that science operates inside paradigms — accepted frameworks that determine which questions count as researchable and which do not. Most academic work is what Kuhn called normal science: question-asking inside the existing frame. Paradigm-shifting questions come, almost always, from outside it. The expired patent

commons is exactly that kind of question. It does not arise within environmental law, administrative law, patent law, or constitutional law as those disciplines currently define their terrain. It arises across the boundaries. De Bono (1967) made the same point with a more popular device. His nine-dot puzzle asks the solver to connect nine dots arranged in a square using four straight lines without lifting the pen. The puzzle is unsolvable as long as the solver assumes — without being told to — that the lines must stay within the square. The solution requires lines that extend beyond the implied perimeter. The puzzle works because solvers impose a constraint that is not in the problem. The institutional blind spot has the same shape. Agencies, Congress, presidents, and academic disciplines have all imposed an unstated constraint — that the regulatory architecture and the patent bargain belong to separate problems — that nothing in the Constitution actually requires.

The Nine-Dot Puzzle

de Bono's illustration of the unstated constraint

The Puzzle	The Solution
 Connect all nine dots using four straight lines without lifting the pen.	 The lines must extend beyond the implied perimeter — a constraint the puzzle never imposed.

"The puzzle works because solvers impose a constraint that is not in the problem."

— from Chapter 6, "Why Nobody Noticed"

The next force takes up the closely related problem of why even a participant who recognized the question would, until very recently, have had no way to bring it.

Force 4. The Doctrinal Floor

A constitutional case has to be filable before it can be filed. The fourth force is different in kind from the first three. Even a participant who saw the question, recognized it as constitutional, and wanted to bring it would, until very recently, have had no procedural and no substantive doctrinal path. The doctrinal floor under the Patent Clause case did not exist for most of the post-1970 period. It was assembled piece by piece across the last fifteen years and was not complete until 2024.

The dominant fact, for forty of those fifty-five years, was *Chevron* deference. From 1984 until June of 2024, *Chevron U.S.A., Inc. v. Natural Resources Defense Council* required federal courts to defer to an agency's reasonable interpretation of an ambiguous statute. A challenge to the wall would have met the agency's claim that its emissions tier was a reasonable reading of its statute, and the court was not free to disagree. The Patent Clause merits would never reach the bench, and every patent attorney advising a potential plaintiff knew it. *Chevron* was the doctrinal door in front of the constitutional question, and it was bolted shut.

The other pieces of the floor were assembled across the same period. The Declaratory Judgment Act of 1934 supplies the procedural vehicle: a court may declare a party's rights in an actual controversy without requiring the party to break the law first. Pre-enforcement standing was not settled in its modern form until 2014. The modern reach of the Takings Clause was not in place until 2012 and 2021. The Major Questions Doctrine was formalized only in 2022. The chapter, *The Filable Case: Why 2026 and Not Before*, walks them in order with full case names and citations.

The floor was not complete until the *Chevron* door came down. In June of 2024, in *Loper Bright Enterprises v. Raimondo*, the Supreme Court overruled *Chevron* and restored to the federal courts the independent judgment required to confront an agency assertion of authority that effectively extends a patent monopoly past expiration. *Loper Bright* did not decide the Patent Clause question. It made the question askable.

For most of the period during which the silence held, this force was decisive. The case was not filable. So it was not filed. So no court was asked. So no answer came. The silence around the Patent Clause is not, primarily, the silence of cowardice or self-interest. It is, primarily, the silence of waiting — until the doctrinal floor was ready, and then for someone to notice it was.

Force 5. Moral Acquiescence

Force 5 covers the smallest population and carries the heaviest moral weight. Its participants saw the constitutional question, understood the harm being inflicted on American households, and declined to raise it. They are a fraction of a fraction: the patent attorney whose private analysis revealed the Patent Clause vulnerability and who argued the client's position anyway; the agency lawyer who sensed that an emissions tier was an indefinite extension of an expired monopoly and defended the tier; the constitutional scholar who suspected the architecture was unconstitutional and chose a different research project. Across fifty-five years and four institutions, they may number no more than a few thousand. They are the hardest case. Five thinkers help name what happens inside a person who recognizes a structural wrong and continues to participate in the system that produces it.

Bandura (2016) cataloged the mechanisms by which a moral self-regulatory system can be selectively disengaged from one activity without dismantling the system as a whole — a framework tested across decades of research on industrial pollution, military atrocities, and corporate fraud. Moral justification: the regulation does serve clean air, however incidentally it also serves an extended monopoly. Euphemistic labeling: one calls the architecture “consumer protection,” not “monopoly extension.” Displacement of responsibility: the agency wrote the rule, the legislator passed the statute, the court would not have heard the case anyway. Each mechanism is rational on its own. Their cumulative effect is to render the constitutional question unspeakable inside the speaker’s own conscience.

Asch (1956) asked subjects to judge which of three lines matched a reference line. Working alone, they answered correctly over ninety-nine percent of the time. Placed among confederates who unanimously gave a wrong answer, roughly a third of responses went with the group, and three quarters of subjects conformed at least once. The experiment was not about lines. It was about how social pressure overrides perception. The patent attorney in a trade-association meeting where the room is uniformly defending a regulatory phase-in schedule is, in Asch’s terms, a minority of one against a unanimous majority. The Asch literature predicts what such a minority will do, and it is what they have done.

Janis (1972) named groupthink — the suppression of dissent and convergence on consensus inside cohesive groups whose members value membership and expertise. Its symptoms include collective rationalization, belief in the group’s inherent morality, direct pressure on dissenters, self-censorship, and self-appointed mindguards who

shield the group from disconfirming information. The American patent bar exhibits each. So does the administrative-law academy, the agency policy staff, and the trade-association legal team. The Patent Clause question, raised inside any of them, would not be answered. It would be deflected — first politely, then by silence, then by exclusion from the next round of cases or invitations.

Kuran (1995) named the most powerful framework for explaining how silence can hold for decades and then break suddenly: preference falsification. Where the public expression of a view carries social cost, those who privately hold it conform in public, and the resulting consensus is illusory. When conditions change — a court decision, a high-profile defection, a moment of permission — each defection lowers the cost of the next, and the cascade runs fast. Kuran's most-cited application was the collapse of Eastern European communism in 1989, which surprised even the intelligence services because the public consensus had appeared stable while private preferences had long since diverged.

The pattern of a profitable truth held quiet for decades and then broken by nothing more than public awareness is not hypothetical. The Progressive Era ran the experiment four times. Sinclair (1906) made the Chicago meatpacking plants visible, and the Pure Food and Drug Act and the Meat Inspection Act followed within months. Tarbell (1904) assembled the documentary record on Standard Oil, and the company was broken up in 1911. Steffens (1904) exposed municipal graft and forced a wave of city-government reform. The tobacco industry concealed for four decades what it privately knew until leaked documents made it public, and the 1998 Master Settlement Agreement arrived quickly. In each case the

trigger was the informational fact itself, not a new coalition and not a new statute. The constituency that had borne the cost needed only to be told. The Patent Clause silence is a Kuran case in slow motion, waiting on the same kind of trigger. *Loper Bright* (2024) is the kind of condition change that lowers the cost of defection, and the people who lead the cascade will be drawn from those who have been privately waiting for permission.

The four social-psychology frameworks name the mechanisms. They do not address the moral cost. For that, the argument turns once to Martin Luther King, Jr., who in his Letter from Birmingham Jail (1963) wrote that the great stumbling block to justice was not the open opponent of justice but the moderate who preferred order to it. The Patent Clause silence is not a civil rights problem, and the book does not claim it is. But King's framework — that recognized injustice met with silence carries a moral cost that cannot be transferred to the system that produced it — is general enough to apply. King's contribution is the reminder that rational silence is still silence. The cost is real even when no individual chose to inflict it, and naming it is the first step toward accepting that the silence cannot continue now that the doctrinal floor is in place.

Those are the first five forces. Every one of them acts on a population close enough to the question to have raised it. The sixth is different. It acts on everyone else, and it asks nothing of them but the ordinary rule-following that ordinary life already requires.

Force 6. Rule-Induced Behavior

Force 6 takes up the vastly larger population whose participation required no recognition at all. Force 5 covered those who knew and stayed silent. Force 6 covers

those who never knew, and never had a position from which knowing was possible — the ninety-five percent: consumers, manufacturers, agency career staff, journalists, lawyers, judges, and ordinary citizens abiding by rules they did not write. The framework that names what happens in that population is game theory, and it has been settled mathematics for over eighty years.

Von Neumann and Morgenstern (1944) founded the discipline in *Theory of Games and Economic Behavior* on a central claim: rational participants optimize against the rules in front of them, not against the rules' origin, intent, or downstream consequences. A consumer chooses the legal vehicle that fits the budget; a manufacturer builds the vehicle that meets the standard; an agency enforces the standard; a judge applies the precedent. None of them needs to want the wall for the wall to stand. The rules aggregate, and the aggregation is the wall.

Schelling (1978) translated the insight into the vocabulary of *Micromotives and Macrobehavior*: aggregate outcomes nobody intended reliably emerge from the sum of individual rule-following. Everyone in a slightly chilled room adjusts the thermostat by one degree, and the room becomes uncomfortably warm — a temperature no individual asked for and no one is responsible for. The consumer buys the legal pickup. The manufacturer builds the truck the EPA tier permits. The EPA enforces the tier the Clean Air Act authorizes. The court defers to the agency. The wall around the expired diesel-engine patent stands, and no participant in that sequence had to want it.

Olson (1965) supplied the mathematics of why concentrated interests defeat dispersed majorities in *The Logic of Collective Action*. A regulation that transfers \$22 trillion from households to incumbent manufacturers

spreads the cost across roughly one hundred thirty million households — about \$3,385 per household per year — while the benefit concentrates in a few hundred firms, where it is worth tens of billions per firm. The firm's lobbyist fights for that one regulation across decades; the household's \$3,385 of incentive is too small, too dispersed, and spread across forty different regulations to organize against. The asymmetry is structural. It requires no one to be malicious.

Stigler (1971) closed the loop with "The Theory of Economic Regulation," published one year after the LRA passed. It predicted what Force 1 documents: regulation is, in the general case, captured by the regulated industry — not because the participants are corrupt but because the economics of group size, information cost, and concentrated benefit make capture the predictable equilibrium. The industry has a permanent stake in the regulation; the dispersed beneficiaries do not. Capture follows. Stigler received the Nobel Prize in Economic Sciences in 1982 in part for the paper. He did not need to be shown the patent case. The patent case is what the framework predicts.

Von Neumann and Morgenstern named the formalism. Schelling named the aggregate dynamics. Olson named the asymmetry. Stigler named the equilibrium. The ninety-five percent are not in the silence because they failed to notice anything. They are in a system whose rules produce the silence regardless of who staffs it. A constitutional violation can persist for fifty-five years inside a population of more than a hundred fifty million people without any of them needing to want it. The argument is against the system, not against the people inside it.

The Silence Breaks

Six forces. Six populations. The corrupt ecosystem is the captured loop of legislators, lobbyists, agency staff, and the legal-and-consulting economy around them. The bootleggers-and-Baptists coalition is the two wings whose independent reasons for defending the wall have removed any political incentive to challenge it. The institutional blind spot is the four institutions whose operating frameworks keep the question off every desk. The doctrinal floor is the set of doctrines that, until 2024, made the case unfileable. Moral acquiescence is the small fraction who saw the question and chose silence. Rule-induced behavior is everyone else.

The argument is not, in any of these forces, against the participants in general. It does not call every legislator corrupt, every trade-association lawyer cynical, every agency career employee captured, or every patent attorney cowardly. But it does not pretend that no one is culpable either. Some legislators are corrupt. Some trade-association lawyers are cynical. Some captured insiders built the wall and have defended it knowing what it does. Both populations exist, and both are morally responsible for what they did. The diagnosis explains the silence without requiring greed or awareness from most participants. It does not excuse the participants from whom greed and awareness are not missing.

What Kuran saw in 1989 was that public consensus and private preference can diverge for a long time without visible motion, then close the gap rapidly when the cost of dissent falls. The Patent Clause silence has been holding under exactly that pattern, and *Loper Bright* (2024) is the kind of condition change that lowers the cost of the first defection. Each of the six forces is contingent. The corrupt

ecosystem is a function of the LRA, which Congress can amend. The bootleggers-and-Baptists coalition is a function of regulatory architecture, which a Supreme Court ruling can dismantle. The institutional blind spot is a function of disciplinary boundaries, which a public argument can cross. The doctrinal floor is in place. The moral acquiescence is, in Kuran's framework, exactly the population that defects first when permission arrives. The rule-induced behavior changes when the rules change. The cascade, when it comes, may be fast. Kuran's framework predicts that it will be.

The next chapter answers the question this one has been circling: if the case is real, who can bring it, and how, without anyone breaking a law?

CHAPTER 7

You No Longer Need to Break a Law to Challenge One

*How at Least Nineteen Plaintiff Categories Can
Challenge Regulatory Barriers
to Expired-Patent Remanufacturing*

On December 1, 1955, Rosa Parks boarded a Montgomery, Alabama, city bus and refused to give up her seat to a white passenger. She was arrested and convicted of violating a state segregation law. Her conviction became the test case the Montgomery Improvement Association used to mount the bus boycott, and her name became one of the most familiar in twentieth-century American history.

In 1955, that was how a citizen challenged an unjust law. You broke the law. You were arrested. You stood trial. And from the witness stand or the appellate brief, you argued that the law itself was unconstitutional. The Constitution did not get litigated except by people willing to go to jail for their convictions.

That changed in 2007. In *Massachusetts v. EPA*, twelve states and several local governments sued the Environmental Protection Agency in federal court, alleging that the agency had unlawfully refused to regulate greenhouse-gas emissions from new motor vehicles. No state attorney general had violated any statute. No state had been arrested. No state had stood trial as a criminal defendant. The states filed a civil complaint, supported by affidavits and expert reports, asking the federal courts to

declare the agency's inaction unlawful and to order corrective action. They won.

That was the inflection point. Across roughly a quarter-century of decisions, the Supreme Court has rebuilt the procedural geometry of constitutional litigation in the United States. A would-be plaintiff who believes a federal statute, regulation, or agency action is unconstitutional no longer has to break the law to find out. Four doors into the federal courthouse have been opened. Each admits a different class of plaintiff, and each has been walked through by plaintiffs who broke no law and who won.



The purpose here is narrow. The patent-bargain case is filable. It is not waiting for a trucker willing to go to jail. It is not waiting for a remanufacturer willing to be enjoined. It is waiting for one plaintiff, in one of at least nineteen categories, to walk through one of the four doors recent Supreme Court doctrine has opened wide. A plaintiff category is a population of would-be plaintiffs sharing a common doctrinal posture, a common factual injury, and a common operational point of contact — usually a trade

association, a public-officer organization, or an advocacy coalition. The case needs one plaintiff. Nineteen categories mean nineteen separately viable paths to the courthouse.

The First Door: Sovereign Plaintiffs

Massachusetts v. EPA did more than let a state sue. It established that a state suing to protect its citizens and its territory is entitled to what the Court called *special solicitude* in the standing analysis — a lower threshold than a private plaintiff faces, grounded in the state's status as a sovereign that surrendered certain powers to the federal government at the founding and retains an interest in how those powers are exercised.

State attorneys general are the most obvious holders of that door. Every state purchases fleet vehicles, equipment, and supplies at prices the wall inflates, which supplies a direct proprietary injury; every state may sue on behalf of its citizens for a diffuse economic harm too small for any single household to litigate; and every state has a sovereign interest in ensuring that federal regulation does not nullify a constitutional provision written to benefit its economy. Three independent theories, any one of which survives a motion to dismiss.

Tribal nations hold the same door and arguably a stronger claim to it. Tribal sovereignty is not a courtesy extended by statute. It predates the Constitution, and the government-to-government relationship between a tribe and the United States may support the same *special solicitude* the Court extended to a state. Three hundred forty-five federally recognized tribal nations sit in the contiguous states, each a sovereign decision-maker, none able to veto what another decides. And tribal plaintiffs have walked this

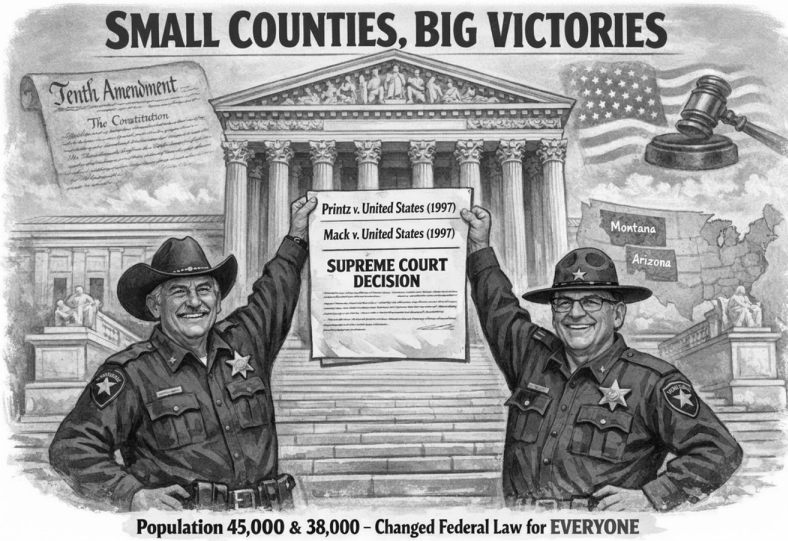
ground before, without breaking any law. In *United States v. Sioux Nation of Indians* (1980), the Supreme Court held 8-1 that an 1877 Act of Congress had effected a Fifth Amendment Takings Clause violation against the Sioux Nation and ordered just compensation of \$17.1 million plus interest from 1877 — an award now past a billion dollars, which the Sioux have refused, wanting the Black Hills rather than the money. That is a tribal plaintiff winning on the same clause that carries this book's second prong. In *Menominee Tribe of Indians v. United States* (1968), the Court held 6-2 that treaty hunting and fishing rights survived the Menominee Termination Act, because Congress had not extinguished them in clear and unambiguous language — a clear-statement rule that bears directly on whether an agency may cancel a constitutional right by implication. And *Cobell v. Salazar*, filed in 1996 by Elouise Cobell of the Blackfeet Nation for roughly 500,000 tribal members, ran fifteen years and settled in 2010 for \$3.4 billion. None of those plaintiffs broke a law to file.

And the door is not reserved for large sovereigns. In *Printz v. United States* (1997), two county sheriffs from counties of a few tens of thousands challenged a federal statute and won at the Supreme Court. Jay Printz of Ravalli County, Montana, and Richard Mack of Graham County, Arizona, filed before the Brady Act provision took effect against them, and the Court held 5-4, through Justice Scalia, that Washington may not commandeer local officers to run a federal program. A sheriff's department is a direct purchaser of vehicles, and the wall inflates the price of every patrol car the county buys. The path from a small county to the United States Reports has been walked before.

The Second Door: Direct and Associational Plaintiffs

The second door admits the parties a regulation actually falls on. In *Loper Bright Enterprises v. Raimondo* (2024), four family-owned herring-fishing businesses challenged a rule that forced them to pay about seven hundred ten dollars a day for an at-sea monitor. They broke no law. They sued as the parties the rule had injured, and they won. A year earlier, in *Sackett v. Environmental Protection Agency*, a single landowning couple did the same.

Associations hold the same door on their members' behalf. Under *Hunt v. Washington State Apple Advertising Commission* (1977), an association has standing when its members would have standing in their own right, when the interests at stake are germane to the organization's purpose, and when the claim does not require each member to take part. A declaratory-judgment action that the regulatory wall is unconstitutional as applied to patent-expired designs satisfies all three: every member pays the inflated price, challenging the regulations is squarely within the mission, and a declaration of unconstitutionality requires no member to testify to anything. An association has already won this way. In *American Hospital Association v. Becerra* (2022), the hospitals' trade body challenged a 28.5-percent cut in Medicare reimbursement that the Department of Health and Human Services had imposed without the survey the statute required. The Supreme Court read the statute itself, gave the agency no deference, and reversed 9-0. SCOTUSblog headlined the day's coverage as an opinion that shuns *Chevron*. Two years later *Loper Bright* overruled it outright.



The Third Door: Pre-Enforcement Review

The third door is the one that most directly replaces the Rosa Parks model. The Declaratory Judgment Act of 1934 permits a plaintiff to ask a federal court to declare rights before any enforcement action has been taken, and *Aetna Life Insurance Co. v. Haworth* (1937) settled that the Act clears Article III: a live controversy between adverse parties is justiciable even where neither has yet sued the other or breached any duty. The question after that has always been how imminent the threat must be.

MedImmune v. Genentech (2007) answered it, and the answer matters here more than anywhere else in the chapter. A party paying under protest — complying with a requirement while contesting its validity — does not forfeit the right to challenge it. The plaintiff need not stop paying, breach, and invite suit to reach the merits. Before that ruling, compliance was treated as concession, which is what this book calls *the trap inside the fix*. Every plaintiff category this book identifies is paying the compliance

costs it would challenge. Under *MedImmune*, every one of them may challenge those costs while continuing to pay them.

Susan B. Anthony List v. Driehaus (2014) supplies the companion rule for a plaintiff not yet paying. A credible threat of enforcement is enough; a plaintiff who intends to engage in conduct arguably proscribed by a statute, and who faces a substantial threat of enforcement, has standing without waiting to be prosecuted. A remanufacturer who intends to build a patent-expired platform and would face certain enforcement does not have to build it first.

The Fourth Door: What 2024 Changed

Two decisions in a single term rebuilt the ground under all of this. *Loper Bright Enterprises v. Raimondo* overruled forty years of judicial deference to agency interpretations of ambiguous statutes. A court now decides for itself whether an agency's reading is correct. For a case arguing that an aggregate of agency actions has cancelled a constitutional clause, that change is not incidental. It is the difference between a case that survives a motion to dismiss and one that does not.

Corner Post v. Board of Governors, decided the same term, held that the six-year limitations period on a challenge to a federal rule runs from when the plaintiff is injured rather than from when the rule was promulgated. A rule issued in 1987 is therefore challengeable by a business that first felt its effect last year. The wall was built between 1970 and 1987. Under the old rule, most of it was unchallengeable by anyone who had not sued decades ago. Under *Corner Post*, it is challengeable now.

The Declaratory Judgment Act in US Constitutional Law

A federal court may declare the rights of any party in a case of actual controversy, without requiring anyone to first break the law they seek to challenge.

28 U.S.C. § 2201 · Enacted 1934.

Confirmed: *MedImmune v. Genentech*, 549 U.S. 118 (2007)

The Doors Are Open

Three doors, and the ground beneath them. Sovereign plaintiffs with *special solicitude*. Directly regulated parties and the associations that represent them. Pre-enforcement review for anyone paying under protest or facing a credible threat. And beneath all three, a doctrinal landscape remade in 2024, under which the courts owe the agencies no deference and the clock does not bar the claim. That last change admits no one on its own. It decides what happens to those already inside.

None of these doors requires anyone to break a law. None requires a plaintiff to be arrested, prosecuted, or enjoined. Each has been walked through, in the last two decades, by plaintiffs who won. The case this book describes is not waiting on a doctrine that has yet to be developed. It is not waiting for a Rosa Parks. It is waiting for a tribal chairman, a state attorney general, an elected sheriff, a hospital association, a trucker association, or any one of fourteen other plaintiff categories to decide to file.

CHAPTER 8

The \$22 Trillion Wealth Transfer

What the Forgetting Costs America — The Largest Hidden Wealth Transfer in American History

Three thousand three hundred and eighty-five dollars. That is the figure this chapter is here to defend. It is the per-household per-year cost of forgetting the second half of the Patent Clause — what the average American family pays, every year, in surcharges built into the price of a pickup truck, a refrigerator, a tractor, a furnace, a snow blower, an air conditioner, and several dozen other categories of equipment whose underlying patents expired ten or twenty or fifty or one hundred years ago. The household never sees the surcharge as a separate line item. It pays it anyway.

What does three thousand three hundred and eighty-five dollars buy? At a community college, much of a year's tuition, and a full year of tuition and fees in the lowest-cost states. At a grocery store, ten weeks of food for a family of four. At the dentist, three crowns. At the mechanic, a transmission rebuild. At the emergency room, a single visit on a high-deductible plan. None of that money is being lit on fire. It is going somewhere.

Multiplied across roughly one hundred and thirty million American households, and across the fifty-five years since the regulatory architecture first took shape, the cumulative cost is approximately twenty-two trillion dollars. That figure is conservative. The true figure is higher.

Two Models, One Conclusion

A figure this large invites skepticism, and it should. Who calculated this, with what data, by what method, and how do we know the calculation is sound?

Two independent calculations, working from different data sources and using different methods, produced figures of \$15.73 trillion and \$28 trillion. The lower figure was a deliberate floor that excluded entire categories of cost. The higher was a best estimate that included them. The midpoint between a stated floor and a best estimate is, by construction, more conservative than the best estimate alone. Twenty-two trillion dollars, the rounded midpoint, is the figure this book uses.

Both numbers are worth showing in the open, because each rests on a base a reader can check. The higher calculation began by totalling American consumer spending from 1970 to 2025 across four Bureau of Economic Analysis categories — motor vehicles and parts; household durables; recreational vehicles and other durables; and nonresidential equipment, which takes in aircraft, farm tractors, factory machinery, and utility turbines. Nominal spending across those four categories over fifty-five years came to \$48.4 trillion, which a present-value calculation at three percent brings to \$81.1 trillion in 2025 dollars. Against that base it applied a savings rate of 25 to 40 percent, the range observed in international markets where patent-expired remanufacture is permitted. Thirty-four percent of \$81.1 trillion is the \$28 trillion figure. The \$81.1 trillion is not a claim about what the wall cost. It is the size of the spending the savings rate is taken against, and it is the reason a rate that sounds modest produces a total that does not.

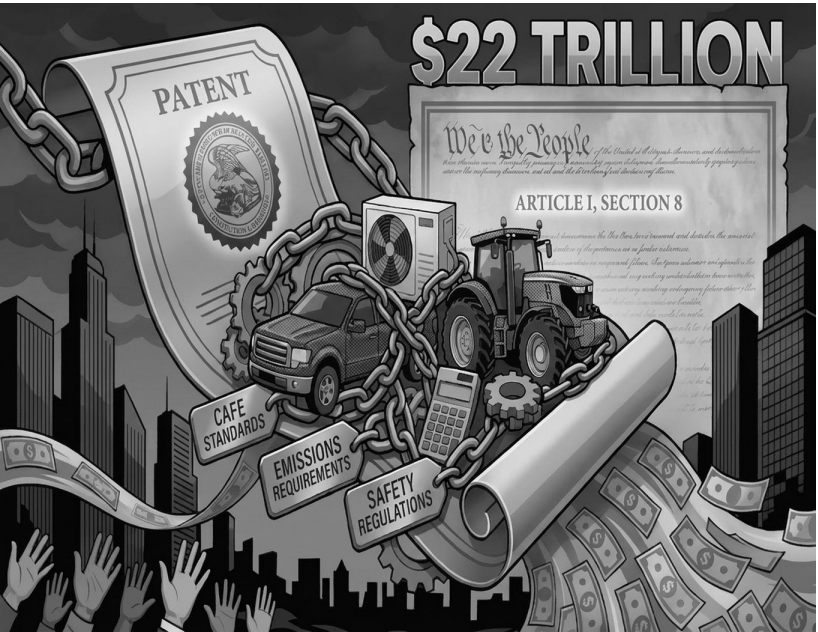
The lower calculation justified its savings rate the same way, from prices rather than from theory. India, Iran, Pakistan, Kenya, Ghana, and China each build and sell new vehicles on American mechanical designs whose patents expired ten or twenty years ago, and those vehicles sell for between \$6,000 and \$15,000 where the American equivalents, blocked from domestic remanufacture, sell for between \$25,000 and \$49,800. The implied savings rate is 50 to 88 percent. The floor calculation declined to use it, applying 40 to 60 percent instead, and only to the categories where remade-versus-new competition would actually arrive.

Two Independent Estimates of the Cumulative Wealth Transfer

	Anthropic’s Claude	xAI’s Grok
Data source	BLS Consumer Expenditure Survey	BEA Industry Economic Accounts via FRED
Scope	60+ product categories behind the regulatory wall	Motor vehicles, household durables, recreational vehicles, nonresidential equipment
Time frame	1975–2025	1970–2025
Method	Direct savings for remake buyers plus 8–15% competitive pressure on remaining buyers	34% midpoint savings rate (range 25–40%) applied to \$81.1T present-value base
Result	\$15.73 trillion	\$28 trillion
Author’s characterization	“The floor, not the ceiling”	Best estimate, market-adjusted
Per household per year	\$2,420	approx. \$4,300

	Anthropic's Claude	xAI's Grok
Headline figure used in this book	\$22 trillion cumulative \$3,385 per household per year (midpoint of the two estimates)	

The two calculations side by side. Different data, different scope, different method — and a spread that brackets the figure this book uses.



What the Floor Excludes

The floor calculation excludes seven categories of cost, each of which would justify a separate book. Together, the published methodology states, they bring the plausible true total to a range of \$30 to \$45 trillion over the same fifty-year period — between \$4,600 and \$6,900 per household per year.

The exclusions do not dilute the case; they strengthen it. A floor figure is one whose author has shown the reader, in

advance, every place where the real number is larger than the figure on the page. The reader is invited to see the figure as the bottom of a range rather than the middle of one, and the constitutional argument does not require the bolder number. It matters for a second reason too. When the wall comes down, the excluded categories are exactly where the restoration is felt: cheaper construction equipment lowers the cost of housing, cheaper farm equipment the cost of food, cheaper aviation and marine equipment the cost of transportation.

The table below names all seven. One deserves a word more: the innovation tax, the products never developed because regulatory barriers made market entry impossible. Certifying a new mechanical design under post-1970 standards runs between \$50 million and \$300 million per model, so only a firm with existing model-line revenue can afford the attempt. What the economy did not get from the small manufacturer, the regional cooperative, and the inventor with a better idea is unmeasurable by its nature. It is not zero.

Category	Foregone savings, 1975–2025
Counted in the floor figure	
Transportation (cars, trucks, RVs, motorcycles)	\$11.13 trillion
Agricultural and construction equipment	\$1.46 trillion
Residential appliances (HVAC, water heaters, washers, dryers)	\$1.26 trillion
Commercial aviation and rail	\$0.83 trillion
Commercial and industrial equipment	\$0.48 trillion
General aviation and marine (private aviation, boats)	\$0.12 trillion
Lawn and garden equipment	\$0.12 trillion

Category	Foregone savings, 1975–2025
Industrial machinery (CNC, robots, presses)	\$0.11 trillion
Medical equipment (MRI, CT, X-ray, ultrasound)	\$0.09 trillion
Building systems (elevators, chillers)	\$0.07 trillion
Other categories (telecom, food service)	\$0.06 trillion
Floor total	\$15.73 trillion
Excluded from the floor figure	
Compounding economic effects (lost business formation, diminished entrepreneurship)	not priced
Indirect costs through the rest of the economy (housing, food, transportation ripple)	not priced
Time value of money (present-value calculation)	not priced
Specialized industrial equipment (mining, oil and gas, semiconductor, pharmaceutical production, textile, packaging)	not priced
Commercial vehicles beyond those counted (delivery vans, buses, specialized trucks)	not priced
Building materials and systems affected by energy codes (windows, insulation, roofing)	not priced
The innovation tax (products never developed because of regulatory barriers)	not priced
Plausible value of excluded categories	\$15–30 trillion
Plausible true total over 50 years	\$30–45 trillion

What the floor counts, and what it leaves out, category by category.

The Patent Royalty Hidden in the Price

Every figure here measures the regulatory wall in aggregate. One cost inside that aggregate can be isolated and watched on its own: the patent royalty. A royalty is the price of a monopoly. While a patent is in force, every

maker who uses the design pays it, and the payment is built into the price the buyer pays at the counter. When the patent's limited term ends, the royalty ends with it. Nothing else about the product need change for its price to fall. The design is the same. The function is the same. The only thing that leaves the price is the royalty the public no longer owes.

This is the one variable the rest of the book's comparisons cannot cleanly separate. The comparison across roughly ninety countries carries differences in labor and regulation. The comparison across a century carries inflation and changing features. The royalty, isolated, carries none of that. It is simply a cost that was in the price, and then was not. The HP-12C financial calculator — same design, same case, same function, now on a faster processor — sells today for over sixty percent less than its 1981 price in actual dollars, and over ninety percent less once inflation is accounted for. A race bicycle built entirely on expired-patent technology sells for over sixty percent less than its still-patented equivalent. In each case the product is as good or better. Only the royalty left. Scaled across the hundreds of patented components in a vehicle, that vanished royalty is part of what the \$22 trillion measures.

The Income-to-Price Collapse

The figures measure a national aggregate. The collapse they measure has a household face. In 1970, the median American household income could buy a new pickup truck for 34.4 percent of one year's earnings. In 2025, the same purchase consumes 54.3 percent. The truck did not get more expensive in absolute terms only. It got more expensive relative to what the household earns to pay for it.

THE AFFORDABLE AMERICAN SEDAN

1960 – 2020



CHEVROLET IMPALA

1958–2020

#1 Best-Selling Car 1970
35% of annual income



FORD GALAXIE 500

1962–1974

#2 Best-Selling Car 1970
38% of annual income



PLYMOUTH FURY

1956–1978

#3 Best-Selling Car 1970
36% of annual income

2025: Ford sells ZERO passenger sedans.

Regulations didn't just raise the price.
They killed the car.

1970: Car = 42% of income
3 affordable sedans available

2025: Car = 70% of income
0 affordable sedans available

Post-1970 federal regulations priced the affordable American family car out of existence. @RaleighMartin/Articles

The \$3,385 figure is the same money, measured three ways. Here it is a national aggregate. At the dealership it is a household line item: a new pickup that should cost \$25,000 in real dollars costs \$72,495, and the \$47,495 gap is approximately half federal regulatory compliance and half incumbent-manufacturer rent extraction made possible by it — the wall plus what the wall lets the protected firms charge on top of the wall. At the truck dealership the same arithmetic prices the owner-operator's rig: a pre-2000 Class 8 highway tractor that should cost \$80,000 as a faithful remake costs \$172,900 new. And in a county budget it is a procurement injury: that same gap multiplied across forty patrol vehicles becomes a \$1.9 million annual line item, and multiplied across roughly thirty-two hundred American counties it becomes injury enough to give a sheriff, a tribal transportation officer, or a state attorney general standing to bring a constitutional case in federal court. Every price in this book, unless a year is stated, is in 2025 dollars.

The Floor Has a Comparator

The size of the figure is the size of the constitutional violation. A wealth transfer of \$22 trillion over fifty-five years is, on any honest accounting, the largest hidden transfer of wealth from American consumers to American producers in the history of the country. It exceeds the combined market capitalization of the S&P 500. It exceeds the total federal debt accumulated from the founding of the Republic until the late 2010s. It exceeds the cumulative cost of every American war, every entitlement program, and every disaster-relief appropriation since 1970, combined.

The figure is not, on its own, an argument that anything is unconstitutional. Large transfers can be lawful, and the argument that this one is not is made elsewhere. What the figure establishes is the magnitude of what the constitutional question is about. A reader can disagree about the doctrine. The reader cannot reasonably disagree that this is a constitutional question worth answering.

One thing the figures lack is a comparator. A reader confronting them for the first time has reasonable grounds to ask whether the problem really is as large as the calculation says, and whether the savings would materialize if the wall came down. That comparator exists, and it is not hypothetical. Roughly ninety countries already permit what the wall forbids, and the next chapter is what their prices show.

CHAPTER 9

Why Wealthy Nations Break the Patent Promise That Poor Nations Honor

*Ninety Countries Prove the Bargain —
Thirty-Four Wealthy Nations Refuse It*

The Natural Experiment

The cleanest way to test the patent bargain is to find a place where it has not been broken. Most of the world is that place. Roughly ninety countries permit the new manufacture of vehicles, farm equipment, construction equipment, and household appliances built on designs whose underlying patents have expired. The buyer in those countries pays a fraction of what an American pays for the same kind of machine — a new machine, built this year on a generation-old design.

Roughly thirty-four wealthy nations do not permit it: the United States, the European Union's twenty-seven member states, the United Kingdom, Japan, South Korea, Canada, Australia, and New Zealand. Inside those countries a new patent-expired vehicle cannot be found on any showroom floor at any price. This is the international comparison the book has been pointing at since Chapter 1. The ninety countries that permit it are the experiment. The thirty-four that do not are the control group. The dependent variable is the price.

The bargain is not an American peculiarity. The United States Constitution carries the most explicit version —

Article I, Section 8, Clause 8, with its “limited Times” — but the same logic appears in the constitutions of Brazil (1988), Mexico (1917), and the Philippines (1987). TRIPS, the trade agreement signed by 166 nations including the United States, settles the global patent term at twenty years from filing. The expiration is universal. The honoring of the expiration is not.



The Mahindra Thar

The single cleanest illustration of the gap is the Mahindra Thar. It traces its mechanical lineage to the Willys-Overland Jeep CJ, whose patents, filed across the 1940s, 1950s, and 1960s, have all been in the public domain for at least three decades. The current Thar — turbocharged diesel, four-wheel drive, anti-lock brakes, electronic stability control — is the lineal descendant of that platform, redesigned in 2020 to current Indian crash and emissions rules.

The licensing record is documented and unbroken. CJ kits arrived in Bombay in 1947, Mahindra acquired the CJ-3B

manufacturing license in 1954, and the license persisted across every successor owner of the Jeep brand, from Willys-Overland to Stellantis today. Every CJ-era utility patent, moreover, had lapsed by the late 1980s. Mahindra's right to keep selling the Thar rests on two pillars. The American buyer's inability to purchase its domestic counterpart rests on neither.

The Thar sells, in the 2024 and 2025 model years, for between roughly \$10,500 and \$18,500. The American descendant of the same lineage is the Jeep Wrangler, built in Toledo, Ohio, by Stellantis, on the same public-domain patents. The 2024 Wrangler Sport two-door started at \$33,890 and the four-door Sahara, the trim most Americans buy, at \$48,895; with options the typical buyer paid \$45,000 to \$55,000. Same lineage. Same expired patents. Same modernization. Two and a half to four times the price.

The difference is not the engineering, which is mature and proven in both cases. The difference is the wall. The Thar is built to the Indian safety and emissions standards in effect in the year of its manufacture — real standards, and ones the platform can carry without ceasing to be that platform. The Wrangler is built to the standards the federal regulatory ratchet has constructed since 1970. The American buyer pays for the platform once, and pays a second time for every regulatory tier added since the platform's patents expired. The Indian buyer pays once.

The Roster of Working Markets

The Mahindra Thar is not unique. It is representative. The table below sets out six more, built on expired Toyota, Peugeot, and Jeep platforms and selling for \$7,500 to \$20,000 — among them ZX Auto's pickups, which China

exports to more than ninety countries, and Iran Khodro's sedans, built under a Peugeot 405 license held since 1991. Pakistan's Buffalo Auto, not in the table, builds compact pickups on expired Toyota and Suzuki platforms for \$6,000 to \$10,000.

The pattern is broader than vehicles. A 50-horsepower diesel tractor sells in India for roughly \$6,000 against a \$54,000 American equivalent — a nine-to-one gap. A compact sedan the developing-country buyer purchases new for roughly \$9,000 sells in comparable European Union markets for roughly \$35,000; a mid-size pickup at roughly \$13,000 there sells in the EU for roughly \$48,000. The detailed industry numbers are documented in the chapter *The \$72,495 Pickup Truck* and in *The Trucker's Burden*.

Brand and Model	Western/U.S. Pattern	Foreign Price (USD)	U.S. Equivalent (USD)
Mahindra Thar	Jeep CJ / Wrangler	≈ \$10,500–18,500	≈ \$45,000 (Wrangler 4-door)
ZX Auto Grand Tiger	1997–2004 Toyota Hilux Pickup	≈ \$13,000–18,000	≈ \$35,000 (Tacoma)
ZX Auto Terralord	1997–2004 Toyota Tacoma	≈ \$13,000–18,000	≈ \$35,000 (Tacoma)
Toyota Hilux Champ	Mature Hilux Pickup	≈ \$12,000–15,000	≈ \$35,000 (Tacoma)
Mahindra Bolero	Jeep CJ lineage, modern diesel	≈ \$8,400–10,400	(no current U.S. equivalent)
Iran Khodro Soren Plus	Peugeot 405	≈ \$7,500–9,900	≈ \$22,000–25,000 (compact sedan class)
Karsan Jeep	Jeep TJ Wrangler	≈ \$20,000	≈ \$45,000 (Wrangler 4-door)

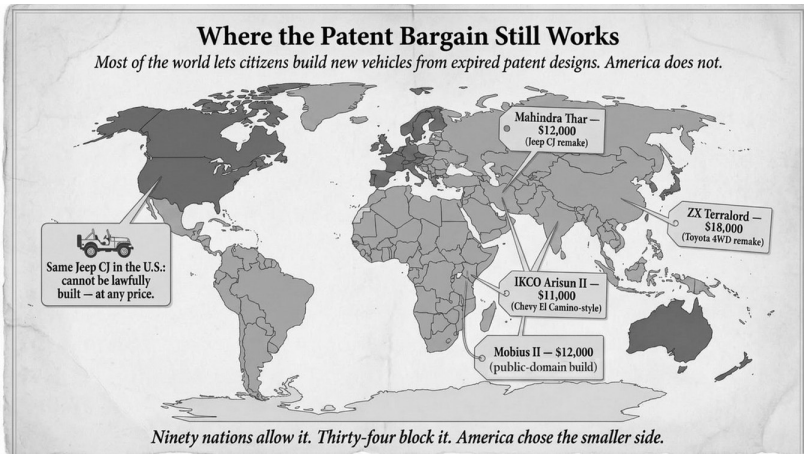
Foreign-market prices reflect 2024–2026 model-year retail in the country of manufacture; U.S. equivalents reflect 2024 manufacturer suggested retail prices for the trims Americans most commonly buy. The Soren Plus figure, distorted by sanctions and currency controls, is at the free-market exchange rate.

The Thirty-Four Countries That Block It

The thirty-four countries that wall their citizens out of these markets share two features. The first is wealth: every country on the blocking list has a per-capita gross domestic product comfortably above the global median and incumbent manufacturers that can afford the lobbying that keeps a regulatory ratchet tightening one notch at a time.

The second is administrative architecture. Every country on the list has built, since 1970, an agency-rulemaking apparatus at some distance from its legislature — the American EPA, NHTSA, and DOE; the European Union’s Whole Vehicle Type Approval framework; the equivalents in the United Kingdom, Japan, South Korea, Canada, and Australia — each an agency with the power to set, tighten, and never relax the floor every new vehicle must meet.

The countries that honor the bargain have safety, emissions, and certification regimes too.



The difference is not that those regimes time-shift; India, like the United States, fixes compliance to the date of manufacture. The difference lies in what the requirements demand. Theirs are additive rules — things a patent-expired platform can carry and still be that platform. The American requirements have become replacement rules: they cannot be met by adding to the expired design, only by replacing it, so that what must change to comply is the very thing the expired patent was supposed to release. (The chapter *The Patent Clause Prong* calls the second kind constitutive; the two names describe the same rule.) Where the rules stay additive, the bargain needs no remedy; the remedy this book asks for becomes necessary once they have not.

The contrast is the bargain Jefferson described being honored most faithfully in the countries the United States routinely lectures about intellectual property law. India, the country American trade negotiators warn about for piracy of pharmaceutical formulations, is also the country building the Thar at one-third the American price. China, the country warned about for copying active patents, is also the country whose ZX Auto exports patent-expired pickups to ninety-plus countries — the post-expiration commons

the United States Constitution mandates and the United States denies its own citizens. The comparison is uncomfortable, and it is meant to be.

The wall does not violate the formal terms of TRIPS or any other international agreement. It violates the bargain itself. In the United States it also violates the Constitution: Article I, Section 8, Clause 8 limits the federal monopoly to limited Times, and the post-1970 regulatory ratchet has converted those limited times into perpetual times by other means. The international comparison is the empirical proof that the constitutional argument matters. The bargain, when honored, produces a price the American buyer would recognize as an answer to the cost-of-living crisis; when forgotten, the price the buyer is paying now.

Why the Comparison Holds

The objection comes immediately: that the foreign-built vehicles in the table above are unsafe, low-quality, or environmentally inferior to anything an American buyer would want. The objection is partly correct and entirely beside the point. Partly correct, because a Thar certified to Indian standards is not engineered to the latest FMVSS or EPA Tier 3 standard. Beside the point, because the argument has never been that the foreign vehicle is the same as the current American vehicle.

The reference comparison is not the Thar against the 2024 Wrangler. It is the Thar against the Wrangler the American buyer cannot purchase: a 2024 vehicle built in Toledo on the same patent-expired platform, certified to the standards in effect when the platform was first released to market, sold for \$13,000 to \$20,000 instead of \$45,000. That vehicle does not exist on any American showroom

floor. The comparison being drawn is one the American market does not permit the buyer to make.

These vehicles are not death traps. The Thar carries hundreds of thousands of Indian families every day. In November 2020 Global NCAP awarded the 2020 Thar four stars for adult and child occupant protection; in November 2024 Bharat NCAP awarded the redesigned Thar ROXX five stars for both — the first body-on-frame SUV to achieve that result. That is a current-year crash certification on a platform whose patents expired half a century ago.

A Note on What the Book Does Not Advocate

The book is not advocating the importation of foreign-built vehicles, tractors, construction equipment, or anything else. The remedy this book argues for is American.

Chapter 1 set out the framing. The constitutional remedy operates through two judicial mechanisms working together: a grandfather exemption that permits any product whose underlying patents expired some years before a regulation's effective date — twenty years is the figure I have suggested elsewhere — to be built to the safety and emissions standards in effect at the time of the patent's filing; and independent engineering certification, modeled on Underwriters Laboratories, to verify that new manufacture under the exemption meets those original standards. A vehicle carries hundreds of patents filed across many years; the standard is the one in effect when the fully assembled vehicle first reached the market, not any single patent's filing date. Both flow from the constitutional holding itself and require no act of Congress.

The third element is import protection. The judicial remedy unleashes a domestic manufacturing revival in the patent-expired platforms the wall has been blocking, which must be protected from foreign producers undercutting American workers. The protection comes from Congress or the President under existing Article I trade authority — tariffs, import quotas, or both. The third leg is therefore not a remanufacturing protection. It is a domestic-manufacturing protection. Remove any one leg of the three, and the whole structure topples.

The international evidence in this chapter is not a recommendation. It is a measurement. It tells the American reader what the household budget would look like if the bargain were honored on American soil. The Mahindra Thar is not the goal. The American Thar, built in Toledo and sold at the price the bargain produces, is the goal.

What the Comparison Says About Restoration

The thirty-four-nation pattern is not destiny. It is a policy choice. It is datable. It is constitutionally vulnerable. Against actors below the constitutional level, the wall is permanent by statutory design; against the Supreme Court, applying the original meaning of Article I, Section 8, Clause 8, it is not, because the Court can order what no agency, Congress, or White House has the political room to order: that “limited Times” must mean a limited time after which the American public regains the ability to build and to remanufacture. The standing question is taken up in the plaintiff chapters; the remedies in Part VII.

The international comparison settles a different question: whether the bargain works. It works in India, Pakistan, Iran, Kenya, Ghana, Vietnam, Mexico, the Philippines,

Bolivia, Turkey — in roughly ninety countries on five continents. That an Indian family of four can put itself inside a new four-wheel-drive vehicle for \$12,000, and an American family of four cannot, is not a story about relative wealth. It is a story about which country walled off the commons the international patent regime treats as public, and which did not. The wealthier country has chosen the worse outcome for its own citizens.

The next chapter takes up the objection that follows: if the patent-expired vehicle could be built in Toledo, would it not roll back the air-quality and safety gains since 1970? That is the subject of *The Environmental Question*.

CHAPTER 10

The Environmental Question

*Why Restoring the Patent Bargain Does Not Unleash
1970s Emissions —
and Why the Constitution Already Has the Answer*

By now the reader has begun to picture what an unblocked patent commons would look like. A 2005 Ford F-150 platform remanufactured today at \$25,000 instead of \$72,495. A 1999 Peterbilt 379 platform semi at \$80,000 instead of \$172,900. A 1999 tractor reborn at a fraction of the price of a modern combine. A residential furnace on a proven 2005 platform at \$3,000 installed instead of \$18,000.

And then comes the question. The reader who voted for environmental candidates in every election since 1992, who has spent years watching the air and water of America become measurably cleaner — that reader looks at the photograph of a freshly built 1999-platform tractor and feels a chill. It is honest worry. Are we about to undo fifty years of environmental progress for a one-third reduction in the cost of living?

This chapter is for that reader. It is for the senator weighing an amicus brief who shares the worry, and for the Sierra Club attorney who, having read the chapter *You No Longer Need to Break a Law to Challenge One*, must decide whether the argument deserves environmental opposition or environmental support. The worry, taken at face value, overstates what the Patent Clause remedy does, and the existing constitutional architecture already provides a clean answer to the residual risk the remedy creates. It does not ask the environmentalist reader to

compromise. It asks him to look at what is actually being proposed.

The Worry, Stated Honestly

Let me state the worry the way someone holding it would state it.

“You are arguing that the Patent Clause restores the public’s right to remanufacture products whose patents have expired. Most of those products — pickup trucks, tractors, furnaces, light bulbs — sit inside categories the Environmental Protection Agency has spent fifty years regulating. The 1970 Clean Air Act eliminated leaded gasoline, mandated catalytic converters, and cut photochemical smog in American cities. The 1975 Energy Policy and Conservation Act doubled passenger-vehicle fuel economy.

“And you are now arguing that all of it can be set aside by a single court ruling, because the patents expired twenty or more years ago. The tractor you want back has no particulate filter. The truck predates the 2007 EPA tier that mandated diesel exhaust fluid. The light bulb is the incandescent Congress phased out for a reason. Your remedy hands the American buyer a 1970-grade emissions profile, a 1975-grade fuel-economy profile, and a 1990-grade safety profile, all in the name of price. That is the deliberate dismantling of fifty years of environmental and public-health policy for a one-third reduction in the price of consumer durables.”

That is a fair statement, and the sections that follow take it seriously.

What the Remedy Actually Does

The worry assumes a static picture of the patent-expired technology base — as if remanufacturing a 2005 F-150 today means producing it with 2005 emissions controls, 2005 fuel economy, and 2005 safety equipment. That is not what the remedy compels, and it is not what the market would produce. The set of expired patents is not a single fixed technology level. It is a continuously expanding kit of components, each with its own expiration date, and the remanufacturer is not building a 2005 vehicle from 2005 parts. He is building from any combination of expired patents available to him today.

Consider what is in that kit. The catalytic converter was refined into the three-way catalyst introduced commercially in 1975; its core patents expired in the mid-1990s, and it has been public-domain technology for thirty years. Electronic fuel injection, which replaced the carburetor across the 1980s, saw its core patents expire across the 1990s. The same holds for modern oxygen sensors, engine-control units, crankcase-ventilation systems, and onboard diagnostics — all long out of patent.

And the fuel itself is regulated independently of the vehicle. Federal rules govern what refiners may sell, and they apply to all gasoline sold at retail regardless of what burns it. A remanufactured 1970-platform vehicle running on the only gasoline a buyer can legally purchase in 2026 is running on unleaded fuel by definition. The lead phase-out was a fuel rule under 40 C.F.R. § 80, not a vehicle-manufacturing rule. The remedy does not touch it.

The remanufactured patent-expired vehicle, in other words, is not a 1970 vehicle. It is a freshly engineered product on a 2005-era chassis with the emissions-control

technology of the late 1990s and early 2000s, and its emissions profile is roughly 1995 to 2005 emissions. By any environmental metric that is a vast improvement over the 1970 baseline the worry imagines; the largest per-vehicle reductions since the Clean Air Act came from technology that was fully public-domain by the mid-2000s. The marginal environmental cost of a remanufactured fleet is not zero. It is not catastrophic. It is the cost of operating a lightly older fleet, and it is a fraction of what “undoing fifty years of progress” implies. The remedy unlocks late-1990s engineering at late-1990s prices, not 1970s engineering at 1970s prices.

Why Markets and Manufacturers Will Not Race to the Bottom

Suppose a remanufacturer tooled a factory for 1970-vintage vehicles in their original configuration — no catalytic converter, no oxygen sensor, no electronic engine controls. What would the market for that product look like?

The fuel market would not cooperate. No leaded gasoline is sold at retail in the United States, and no refiner will produce it. The 1970 engine without a catalytic converter does not run cleaner on unleaded fuel; it runs hot, fouls, and produces unburned hydrocarbons.

State inspection would not cooperate. Inspection regimes are matters of state law, untouched by the federal remedy. California will still require passing emissions results to register a vehicle; Massachusetts will still require a working catalytic converter. The 1970-emissions vehicle is unsalable in roughly two-thirds of state markets on the day it leaves the factory.

The brand, labor, and financing markets would not cooperate. Ford is not going to be the company that builds a truck without a catalytic converter. American manufacturing labor is deeply attached to post-Clean-Air-Act standards. And tooling a vehicle line costs three to five hundred million dollars, supplied by banks and pension funds whose screening regimes would refuse a product visibly worse than its competitors.

The market-discipline argument is not that no bad actor is possible. It is that the bad-actor strategy is a poor business plan. The remanufactured pickup that wins commercially carries the most advanced emissions technology then in the public domain; the one that loses fails inspection, fails the brand test, and fails the financing test. The market sorts the two within the first two product cycles, in favor of the cleaner version. American firearms manufacturers, in a category where the regulatory architecture stayed light, do not produce visibly defective products in pursuit of marginal savings. The market punishes everything else.

One further step is open to the political branches, and it answers the emissions objection at its root. The remanufacturer's kit of expired patents grows by one year with every passing year. A favorable ruling is unlikely before 2030, and by then the cleanest emissions technology free of patent will be the integrated systems certified around 2010. That is cleaner than most of the fleet already lawful on American roads. It is not, however, as clean as a new vehicle certified to the current year's standard. The gap is real, and the book does not hide it.

Congress can close that gap without touching the constitutional remedy. After a favorable ruling, it could offer an incentive to remanufacturers — not to consumers — to fit the most advanced emissions-control technology

that has entered the public domain. The incentive attaches to the producer, so no consumer is subsidized. The floor rises on its own as the public domain advances. The public is not asked to underwrite a buyer's choice; the remanufacturer is given a reason to install what the public already owns.

What form should the incentive take? Government has three blunt tools for steering private behavior, and none fits. An ordinary deduction is too weak, since a business already deducts what it spends on parts. A tax credit is powerful but expensive. A civil penalty is coercive. A fourth tool fits, one the author proposed in *ShareDebate International* in 1994: the multi-value tax deduction. The remanufacturer that fits the newest patent-expired emissions technology would deduct two or three dollars as a business expense for every dollar spent. It costs the Treasury a fraction of what a credit costs, and it makes the clean choice cheaper than the dirty one at the moment of purchase.

The instrument is not hypothetical. Several governments already run it for research spending. China lets a company deduct two hundred percent of qualifying research spending and made the policy permanent in 2023. Singapore allows four hundred percent on the first four hundred thousand dollars. Malaysia has run a double deduction for decades. The United Kingdom grants research-intensive small companies an additional eighty-six percent. India allowed two hundred percent before phasing the figure down to one hundred in 2020. Those statutes also answer the obvious question about the multiplier: India applied different multipliers to different categories of research, and Singapore sets a separate cap for each qualifying activity. The multiplier is a dial. Other legislatures have already learned to turn it. The United

States has a version of its own: since 1976, a corporation donating inventory for the care of the ill or the needy may deduct more than the goods cost, capped at twice basis. What is new is the purpose.

The Constitutional Safety Valve: The Fifth Amendment

Suppose, however, that some genuinely dangerous category emerges that the market and social mechanisms above fail to address — a product whose external costs exceed what democratic society will tolerate and which survives the market disciplines in volumes that matter. What is the constitutional answer?

The answer is the Fifth Amendment, and it has been on the books since 1791. The Takings Clause states that “private property” shall not “be taken for public use, without just compensation.” The Supreme Court has applied it to physical takings (*Loretto v. Teleprompter*, 1982), regulatory takings (*Pennsylvania Coal v. Mahon*, 1922), per se total takings (*Lucas v. South Carolina Coastal Council*, 1992), temporary physical occupations (*Cedar Point Nursery v. Hassid*, 2021), appropriation of personal property (*Horne v. Department of Agriculture*, 2015), and appropriation of property value beyond what is owed (*Tyler v. Hennepin County*, 2023).

A successful ruling restores the public’s right to remanufacture products built on expired patents. That right is a property right of the American public, affirmed by *Bonito Boats v. Thunder Craft* (1989) and *Kimble v. Marvel* (2015). If a future federal regulation extinguishes it for a particular category, that regulation is a taking, and the Fifth Amendment requires just compensation.

The democratic implication is meaningful. The current architecture extinguishes the public's remanufacturing right without compensation. The remedy does not weaken that regulatory power. It conditions it. The federal government may still regulate any category of remanufactured product into commercial unviability — but it must pay the public for doing so. This is not a weaker form of environmental regulation. It is environmental regulation with a price tag, and the price tag is the Founders' contribution to environmental policy, written into the Bill of Rights two hundred and thirty-five years before the Clean Air Act.

Suppose the Sierra Club or the Environmental Defense Fund determines that a category of remanufactured product creates emergency-grade public harm. The path is open. Petition Congress to extinguish that category of remanufacturing right. Accept that the Treasury will pay the affected manufacturers the fair market value of the right being taken. Then let the political process decide, in public, whether the regulation passes the cost-benefit test. That is the system the Constitution actually establishes.

The Friedman Concern: When Compensation Becomes a Subsidy

That would be sufficient if the safety valve were costless. It is not. Milton Friedman's enduring observation — *if you want less of something, do not subsidize it* — applies here too. The Takings Clause, by requiring compensation, creates a marginal subsidy for the very behavior the regulation is meant to discourage. A speculative remanufacturer may reason that tooling a factory creates an asset whose value includes both the operating profits during the pre-regulation window and the just-

compensation payment after the regulation arrives. At the margin, more such projects get built.

The concern is real. It is also constrained, by doctrine and by market reality. The just-compensation measure is the fair market value of the property at the time of the taking, established in *United States v. 50 Acres of Land* (1984); the remanufacturer who tools a factory to attract a taking receives the depreciated value of the tooling and inventory, not the strategic value of the buyout. The investment-backed-expectations doctrine of *Penn Central Transportation Co. v. New York City* (1978) weakens his claim further, because expectations formed with the taking in mind are not reasonable. And he still has to operate during the pre-regulation window, under the market disciplines described above. In expectation, the strategic-builder business plan loses money.

Gaming is therefore individually constrained. But at the aggregate level, across many manufacturers and product categories, the backstop produces some additional remanufacturing in categories society would prefer not to see. The amount is bounded, and it is also positive. Friedman's principle says only that subsidies produce more of the subsidized behavior, and that is true here.

Two answers. The first is to accept the residual subsidy as the price of the architecture's other virtues: the Takings Clause makes the cost of regulation visible and aligns the regulator's political incentives with the financial costs of the regulations it imposes. The second is to close the residual subsidy with a structural change. The Constitution is amendable, and a properly drafted amendment could reach the marginal subsidy directly.

The Sunset Amendment: Phase 2 Structural Insurance

What follows is one possible draft. It is offered in the spirit that the chapter *Hybrid Legislative Voting* offers its reform: as structural insurance, available if the Phase 1 remedy proves to need it, not required for the Phase 1 remedy to operate.

Proposed Amendment to the Constitution of the United States. The right to remanufacture an invention whose patents have entered the public domain shall remain unrestricted for twenty years from the date of first lawful remanufacture by any party, after which Congress may by law restrict or prohibit further remanufacture without obligation under the Fifth Amendment to provide just compensation, provided that any such restriction shall apply only prospectively and shall not impair products already manufactured or contracts already entered into.

The amendment does three things. It preserves the remedy at full strength for twenty years from the date a category enters lawful production. After that window it authorizes Congress to restrict the category without compensation, prospectively only. And it fixes the window as a constitutional fact, giving every potential remanufacturer clear notice of a product line's long-run exposure. The speculative builder has no backstop in year twenty-one and beyond, so the expected value of his plan drops to the operating-period profits alone. The Friedman concern is closed at the long-run boundary while the remedy keeps its full force during the window where the public benefit is greatest.

The political case is neither right nor left. A populist-right reader sees a structural end to the regulatory ratchet

without unbounded long-run subsidy. A populist-left reader sees a clean mechanism for long-run environmental regulation that does not pay manufacturers for the privilege of cleaning the air. A libertarian reader sees in it a constitutional rule that respects property rights for a defined term and then transfers full regulatory authority to Congress without procedural ambiguity. A coalition running from the Sierra Club to the Federalist Society could plausibly support it, the way the coalition behind the 27th Amendment ran from Gregory Watson at the University of Texas to senators on both sides of the aisle.

The amendment is not necessary for the Phase 1 remedy. It answers the Friedman-grade question of whether the country eventually wants the marginal subsidy removed.

A More Democratic Arrangement Than the One We Have

The worry was that the remedy undoes fifty years of environmental progress. The answer has come in five steps. The worry overstates what the remedy does. Market and social mechanisms address most of the residual risk. The Fifth Amendment provides a constitutional safety valve. The marginal subsidy that safety valve creates is constrained by doctrine and by market reality. And a Sunset Amendment can close even that, if the country decides it should.

Set the two architectures side by side. Under the current one — the post-1970 regulatory ratchet — the federal government extinguishes the public's remanufacturing rights by regulation, without compensation, and permanently. The cost is borne entirely by the buyer of the regulated product, in the price markup created by the

artificial extension of the patent monopoly. The regulator who imposes the cost faces no fiscal accountability for it. The asymmetry runs one way. The wall holds a remanufactured 2005 F-150 to 2026 standards. The electric vehicle those same rules favor carries costs of its own — a battery fire that vents hydrogen fluoride, a spent battery with no recycling system yet built to receive it — and no regulation prices either. The political incentives of the regulator and the financial incentives of the regulated industry are aligned against the buyer.

Under the proposed one — the Patent Clause remedy plus the Takings Clause safety valve, with the Sunset Amendment as long-run insurance — the public retains its remanufacturing right by default. Federal regulation that extinguishes it requires compensation paid out of the Treasury, so the cost becomes visible on the federal balance sheet. Environmental groups retain their full authority to advocate for any regulation they believe the country should pay for, and the regulatory architecture's environmental work continues — on terms the Constitution actually authorizes.

The reader who began worried that the remedy was a deregulatory smuggle should now be able to see that it is not. The environmentalist reader who concludes that this architecture is worse than the one we have is invited to make that case in the political arena. But the argument does not require the reader's agreement. It requires only the recognition that the remedy is not what the worry stated it was. It asks the country not to want less environmental protection but to pay for the protection it wants, visibly, out of the Treasury, rather than invisibly, out of the price of a furnace.

The argument continues in the next chapter, with the historical witnesses to what the patent bargain looked like when it worked.

CHAPTER 11

When the Bargain Worked: Pre-1900 American Prosperity

*Three Saddles That Crossed a Continent, Nine Industries, —
and the Frontier the Patent Bargain Built*

By 1900, an American rider could cover seventy-five miles in a working day, and on the relay system the Pony Express ran in 1860 and 1861 a rider could cover one hundred. A century earlier, the same rider — on the saddle his English grandfather had brought across the Atlantic — would have done well to make twenty-five.

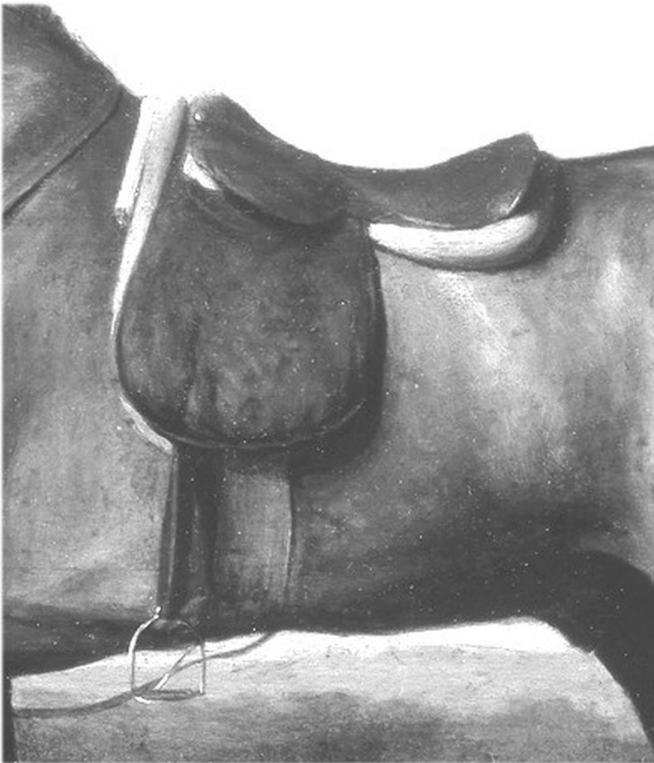
The improvement is real and documented, in U.S. Army field-trial records, in cattle-drive logbooks from Texas to Montana, and in the operational reports of the firm that ran the Pony Express. It is also a story about what the American patent system did, and did not do, in the saddleries of the nineteenth century. The most consequential American saddles of that century were unpatented at the design level. The component improvements that made them work were patented, in dozens of separate filings, and entered the public domain on schedule. Those two halves together are the first piece of evidence that the patent bargain worked.

This chapter runs in two registers. Saddles are the human story: it opens with the saddle the colonial American rider inherited from England, then turns to the three working American saddles of the nineteenth century that the patent bargain made possible. Each followed the same trajectory — patented improvements flowing into a public-domain

integrated design, and the design itself diffusing across the continent on terms ordinary Americans could afford. The second register is broader still: nine other product categories from the late eighteenth and nineteenth centuries that share nothing with saddles except the structure of their commercial history. The inventory is the pattern proof. The pattern is the argument.

Saddles, the Human Story

The English Inheritance



Detail of a hunter's saddle from George Stubbs's A Saddled Bay Hunter (1786), Denver Art Museum (Berger Collection, 2019.10). The horse belonged to Arthur Annesley, the 9th Viscount Valentia. The low pommel and cantle, flat seat, and long stirrup leather are the

*diagnostic features of the English hunting saddle
Americans imported during the colonial and early federal
periods.*

What the colonial American rider had inherited from England, by the time of the Revolution, was the English hunting saddle. George Stubbs's 1786 painting *A Saddled Bay Hunter* shows the type clearly: a wooden tree of beech or similar hardwood covered in vegetable-tanned leather, a low pommel at the front and a low cantle at the rear, a flat and lightly padded seat between them, a single girth running under the horse's belly, and a stirrup leather hanging from a bar set roughly under the rider's knee.

The saddle was excellent for what its name described. A gentleman could spend a day in the field at the trot and canter, jumping hedges and ditches in pursuit of foxhounds, and come off the horse in something better than ruin. For long-distance travel it was poorly suited. The flat, lightly padded seat that lets a foxhunter rise easily over fences puts concentrated pressure on the rider's seat bones over a multi-day journey. The low cantle that clears the pelvis on a jump removes the lower-back support a distance rider wants. The narrow weight-bearing panels that make the saddle light put the same load on a smaller patch of the horse's back — fine for one day's hunt, punishing on consecutive days. A colonial American rider could comfortably cover twenty to thirty miles a day on this equipment, and beyond that fatigue and horse discomfort cut the day short. This was the saddle the country inherited. The country would build something different.

The three saddles that carried the American rider across the continent — the Model 1904 McClellan, the Visalia stock saddle, and the Pony Express mochila — were not patented as integrated designs.

The Model 1904 McClellan Cavalry Saddle

Captain George B. McClellan submitted his design to the War Department in 1857 and never filed a patent; the Army adopted it in 1859, and its open-tree, split-seat construction of poplar or beech under shrunk rawhide weighed eleven to thirteen pounds against the thirty to forty of a civilian stock saddle. It has been manufactured continuously by multiple American companies, available to any buyer, for one hundred and sixty-five years.



U.S. Army Model 1904 McClellan cavalry saddle, manufactured under government contract by the Rock Island Arsenal. National Museum of American History, Smithsonian Institution. Captain George B. McClellan designed the original 1859 pattern after a study tour of European cavalry equipment, and never patented it. The design served the U.S. Army through the Civil War, the Indian Wars, the Spanish-American War, both World Wars, and into the Korean War era.

The Visalia Stock Saddle

The Visalia Stock Saddle Company opened in 1869 in Visalia, California, founded by three Mexican-American saddlemakers — Juan Martarell, Alsario Herrera, and Ricardo Mattley — and developed the center-fire slick-fork California tree that stockmen across the West adopted.



DOUBLE D RANCH.

Antique Visalia Stock Saddle, c. 1890–1905, with the company's early maker's mark. Double D Ranch collection. The Visalia Stock Saddle Company was founded in 1869 in Visalia, California, by Mexican-American saddlemakers Juan Martarell, Alsalia Herrera, and Ricardo Mattley, and produced the saddle that

vaqueros and stockmen across the western territories regarded as the finest working ranch saddle of the period.

The Pony Express Mochila Saddle

The Pony Express mochila was a flat leather cover with four locking corner pockets draped over a stripped-down California saddle of eight to ten pounds, and no patent was ever filed on it. None of the three named designers held a saddle patent on what the rider was actually using.



Pony Express saddle with mochila, c. 1860, Buffalo Bill Center of the West, Cody, Wyoming (Accession 1.69.16).

The mochila was a removable leather cover with four corner pockets for mail, designed to be transferred from one horse to another in seconds at relay stations every ten to fifteen miles. No patent was filed on the design.

The American Patent Record on Saddles

What the rider was actually using, however, was filled with expired patented components.

The record behind those components runs back to the founding of the Patent Office. Thomas Stickney filed Patent X-187 in 1797, among the very first American saddle patents under the Patent Act of 1790; Gosker and Degenhart filed a pack-saddle improvement, Patent 39,285, in 1863.

The most striking entry belongs to Davis, a Buffalo Soldier of the Tenth Cavalry stationed at Fort Assiniboine in northern Montana, who patented an improvement to the saddle he had spent his working life in.

What the nineteenth-century record shows is not a sequence of breakthrough integrated designs by famous names. It is a steady stream of patented components — rigging systems, tree construction, stirrup mountings, girths, horn attachments, spring seats — flowing into designs that were public from the day they were built.

The McClellan absorbed patented components across its production history, and they arrived in sequence. Theodore Wint, a Fourth Cavalry officer, patented an adjustable saddle for cavalry use in 1885 (Patent 320,522). Davis's spring-seat patent followed in 1896, feeding into officer's variants without ever displacing the standard tree. Samuel S. Adams patented a McClellan saddle-tree in 1914 (Patent 1,111,817), its side bars built of stitched layers of leather containing spring-steel strips, for firmness and rigidity without chafing the horse. Claude Brookins Mills patented an improved stirrup-leather mounting in 1923 (Patent 1,461,282).

The Visalia's center-fire rigging tradition absorbed its components on the same schedule. Rufus A. Lane patented a saddle girth with mounted take-up in 1892 (Patent 481,180). Frank O. Hicks patented a unified seat cover in 1900 (Patent 662,564), and Milton Randolph

Hunter an improved saddle-horn attachment in 1901 (Patent 671,765). Charles A. Harrington of Elko, Nevada, patented an adjustable rigging-plate in 1912 (Patent 1,040,316), and the Hamley brothers of Pendleton, Oregon — with Frederick J. McMonies — a single-piece cable rigging system in 1913 (Patent 1,055,410). Each addressed a specific component problem — chafing, fit, durability, adjustability — and each entered the public domain on the schedule the Patent Act set, where competing California saddleries were free to incorporate it without royalty. Swell-fork rigging, saddle-tree, and bucking-roll patents appeared steadily across the same span, and the public-domain saddles that crossed the continent absorbed every one of them as the original terms ran out.

This is the patent bargain at its purest. The federal government granted limited monopolies on saddle components — improvements in rigging, in tree construction, in stirrup mounting, in spring seats — and when each expired, the improvement entered the integrated working saddles that ordinary Americans, ordinary cowboys, and ordinary cavalrymen actually rode. No federal agency stood between the patented improvement and the public-domain saddle that absorbed it. The system ran exactly as written, and the rider in 1900 covered three to four times the distance his great-grandfather had, on equipment whose underlying design belonged to the American people.

The Inventory: Nine Categories, One Pattern

The second register is broader. Nine product categories span the late eighteenth and nineteenth centuries and share nothing in common except the structure of their

commercial history. Each was patented in the early American republic. Each was held under monopoly during the patent term. Each entered the public domain on a specific, recorded date. And each followed the trajectory the saddle components followed: a real-terms price collapse within a generation of expiration, and a marketplace in which the design became available to ordinary buyers at a fraction of what the patent term had charged.

Eight of the nine are set out below. The ninth, the dandy horse — the proto-bicycle patented in 1818 by Karl von Drais — follows the same curve. Across all of them, real-terms prices fell between fifty and sixty-nine percent within a generation of expiration.

Invention (Inventor, Filed / Expired)	During-Patent Price	Post-Expiration Trajectory	Real-Term s Drop
Nail-making machine (Jacob Perkins, 1795 / 1809)	~\$6 per pound (hand-forged nails)	~\$3.20/lb by 1830s; ~\$1.50 by 1860s; ~\$0.80 by 1880s with machine-cut production; ~\$0.30 by 1890s with mass production	69.4 %
Wooden shelf clock (Eli Terry, 1816 / 1830)	~\$150 (complete clock)	~\$80 by 1840s; ~\$50 by 1860s; ~\$30 by 1890s with mass production	64.4 %
Screw-cutting machine (David Wilkinson, 1798 / 1812)	~\$4 per pound (screws)	~\$2.50/lb by 1830s; ~\$1.20 by 1860s; ~\$0.60 by 1880s with automated threading	64.2 %
Cast-iron plow (Charles Newbold, 1797 / 1811)	~\$350 (complete plow)	~\$200 by 1830s; ~\$150 by 1860s; ~\$100 by 1880s as steel plows competed but cast iron remained common	57.1 %

Invention (Inventor, Filed / Expired)	During-Patent Price	Post-Expiration Trajectory	Real-Term s Drop
Cotton gin (Eli Whitney, 1794 / 1808)	~\$1,900 (service-based)	~\$1,200 by 1830s; ~\$800 by 1860s; ~\$500 by 1880s as mass-produced gins entered the market	56.1 %
High-pressure steam engine (Oliver Evans, 1804 / 1818)	~\$12,000 (industrial engine)	~\$8,000 by 1830s; ~\$5,000 by 1860s; ~\$3,000 by 1880s with design refinements	55.6 %
Potash production process (Samuel Hopkins, 1790 / 1804)	~\$12 per bushel	~\$8 by 1830s; ~\$5 by 1860s; ~\$3 by 1880s as chemical alternatives emerged	55.6 %
Cotton carding machine (Amos Whittemore, 1797 / 1811)	~\$0.30 per pound	~\$0.20/lb by 1830s; ~\$0.15 by 1860s; ~\$0.10 by 1880s with factory scaling	50.0 %

Why the Pattern Held Before 1970

The nineteenth-century United States produced this pattern not by accident but because the federal government had built nothing that could prevent it. The Patent Act of 1790 gave the Patent Office a single function: examine applications, grant valid patents for fourteen years — later seventeen, ultimately twenty — and let them enter the public domain when the term ran out. No parallel agency was authorized to take over regulation of the post-expiration product. The Constitution authorized a temporary monopoly. The Patent Act administered it. After it ended, the design belonged to the public, and Congress had not authorized any further federal intervention in what the public did with it.

That structural absence is what made the pattern possible. The cast-iron plow could be manufactured by competing firms after Newbold's patent expired because no apparatus existed by which a competing manufacturer could be told he could not. The McClellan saddle could be reproduced by competing saddleries because no federal agency had authority to set design standards a remanufactured McClellan would have to meet. The high-pressure steam engine could be built across the industrial Northeast because no agency had jurisdiction over what came after Oliver Evans's patent expired.

The Supreme Court did not need to intervene to protect this state of affairs, because nothing was challenging it. Across the nineteenth century the Court heard patent cases on validity, on infringement during the term, on the boundary between state and federal authority — but not on whether the federal government could prevent the public from making and selling expired-patent designs, because the federal government did not try. When the Court did eventually speak to the post-expiration question, it spoke clearly.

Scott Paper Co. v. Marcalus Manufacturing Co. (1945) rejected the use of a private contract to extend the practical effect of a patent past its expiration. *Sears, Roebuck & Co. v. Stiffel Co.* (1964) held that a state could not use unfair-competition law to do what the patent had been forbidden to do — prevent public copying of an expired or invalid design. *Graham v. John Deere Co.* (1966) returned to the Patent Clause itself, including the Founders' insistence that the limited-term grant served the public's eventual right to the disclosed invention. *Bonito Boats, Inc. v. Thunder Craft Boats, Inc.* (1989) struck down a Florida statute that prohibited the use of a direct molding process to copy unpatented boat hulls. The four cases,

decided across forty-four years, said the same thing. When a patent expires, what comes next belongs to the American people, and no other branch of government may take it back.

Proof of Concept

The pre-1900 American pattern is the proof of concept for what the Patent Clause was designed to deliver, and for what restoration of the bargain produces in practice. It is one thing to argue in the abstract that a limited-term monopoly serves the public good because the commons that follows transfers the technology into common use at sharply reduced cost. It is another to read the historical record and find that this is exactly what happened, nine times over, in an American century when no federal apparatus existed to stop it.

The proof is measurable in what a rider could do in a day. A colonial American on the English hunting saddle covered twenty to thirty miles in eight to ten hours. A cavalry trooper on a McClellan covered twenty-five to thirty as a standard pace and forty to fifty in a forced march. A Pony Express rider covered seventy-five to one hundred on the relay. Across roughly a hundred and twenty-five years the typical American daily ride rose from twenty or thirty miles to forty or fifty in cavalry use and seventy-five to a hundred in mail relay, and the improvement was not horse breeding alone, nor better roads alone, nor accumulated horsemanship alone. It was equipment, built on components that had entered the public domain on the Patent Act's schedule. The trajectory ran out around 1925, when the registered motor-vehicle population passed the combined population of horses and mules.

The pattern did not end because the technology changed. It ended because, beginning around 1970, the federal government built the apparatus the nineteenth century had lacked.

CHAPTER 12

The Bargain Still Works: Bicycles, Calculators, and the Political Warning of Firearms

*Three Living Sectors, One Pattern —
and the Politics That Decides Which Categories Get a
Wall*

The previous chapter closed in 1925, the year the registered motor-vehicle population of the United States passed the combined population of horses and mules. After that, the saddle stopped being a serious answer to how Americans got across the country. The pattern that the saddle, the cast-iron plow, the cotton gin, and the high-pressure steam engine had documented did not stop. It continued. It is still continuing today.

Three sectors show it: the bicycle, the pocket calculator, and the firearm. None faces the closed federal compliance wall that the post-1970 automobile, tractor, household appliance, and incandescent bulb face. Each is therefore living evidence of what the Patent Clause delivers when nothing is built to prevent it. Real-terms prices fall. Competition broadens. The post-expiration design enters ordinary American life on terms ordinary Americans can afford.

The thesis travels through all three sections. When the federal regulatory wall is absent or light, the patent bargain delivers exactly what the Founders wrote it to deliver. What determines whether a category gets a wall is not the technology, the safety profile, or the public interest. It is

purely political. The bicycle, the calculator, and the firearm escaped the wall for reasons that have nothing to do with whether the underlying technology could have been regulated, and everything to do with whether the constituencies that make regulation politically possible were organized to demand it.

Bicycles: The Public-Domain Boom Continues

The lineage runs from Karl von Drais's two-wheeled steerable running machine, patented in Baden in 1818, briefly monopolized, then copied freely on both sides of the Atlantic when the term ended. The first decisive American filing in the lineage was Pierre Lallement's pedal velocipede patent, U.S. Patent 59,915, granted in 1866. Two centuries of inventions followed the same path — the diamond frame, pneumatic tires, ball bearings, roller chains, derailleurs, disc brakes — each entering the public domain through ordinary expiration, each absorbed by competing manufacturers. The labor-hours required to buy a bicycle fell 95.5 percent. A machine that cost six months of wages in 1890 costs three hours of labor today.

Every component in the modern bicycle arrived by that route, and each entered the public domain on the schedule the Patent Act set:

Steerable two-wheel frame — Drais, 1818

Pedal-crank drive — Lallement, 1866

Wire-spoked wheels — Meyer, 1870s

Tangential spoke lacing — Starley, 1870s

Ball bearings — Hughes, 1877

Roller chain — Renold, 1880

Diamond frame with chain drive — freely available from 1885

Pneumatic tires — Dunlop, 1888

Freewheel coasting — Van Anden, 1869

Coaster brake — Townsend, 1898, expired 1924

Slant-parallelogram derailleur — SunTour, 1964, expired 1984

Cliplless pedals — Look, 1984; Shimano SPD, 1990

Integrated brake and shift levers — Shimano STI, c. 1990

Suspension forks — RockShox, 1989

V-brakes — Shimano, 1990s

Mechanical disc brakes — mid-1990s

Some of that list was invented abroad. Starley's diamond frame, Renold's roller chain, Dunlop's pneumatic tire — British. Drais's running machine — German. Meyer's wire-spoked wheel and Look's clipless pedal — French. SunTour's slant-parallelogram derailleur and Shimano's integrated brake and shift levers — Japanese. It made no difference. A patent is good only in the country that grants it. A foreign inventor who wanted an American monopoly had to come to Washington and take the American bargain: a limited term, and then the public domain. One who never filed here held nothing here at all, and American manufacturers were free from the first day.

Renold's roller chain came out of Manchester, and imported chain cost more than ordinary buyers could pay. On Christmas Eve of 1890, three men in Indianapolis put up five thousand dollars, rented rooms above a tinner's shop, and began making it themselves with four machines and four workers. Making the chain in Indiana is what brought the bicycle within reach of ordinary Americans.

Within fifteen years that shop supplied two-thirds of the bicycle chain in the country. Two of its dealers, brothers who kept a bicycle shop in Dayton, ordered seven special chains in 1903 and flew them at Kitty Hawk.

Nobody paid a royalty. No federal agency reviewed any of it, because none existed that could.

Each generation of components became free on schedule. The integrated bicycles that absorbed them were free from the start. Compare the \$72,495 pickup truck: the same accumulation of expired patents sits underneath it, and none of it is available to anyone.

What that pattern produced is one of the best-documented affordability records in American consumer history. Drawing on Sears Roebuck catalogs, Bureau of Labor Statistics wage data, and contemporary retail pricing, the Human Progress Institute calculates that the time price of a bicycle has fallen ninety-five and a half percent since 1910, when a Sears catalog bicycle cost \$11.95 against an average blue-collar wage of eighteen cents an hour — sixty-six hours of labor then, about three hours now.

The pattern is still visible in real time. In 2005 the Trek Madone 5.2 was the dream bike of serious American road cyclists: carbon-fiber frame, Shimano Ultegra nine-speed groupset, caliper rim brakes, 19.2 pounds, and a price between \$3,400 and \$3,600 — roughly \$5,400 in 2026 currency. In 2026 an American can buy the SAVA Warwind V3, a full-carbon road racing bicycle built entirely from expired-patent technologies: Toray T800 frame and fork, Shimano Sora eighteen-speed groupset, Tektro V-brakes, 19.6 pounds, certified by the Union Cycliste Internationale for competitive racing. Eleven hundred dollars. Same category, same level of performance, roughly an eighty percent real-terms price collapse in

twenty-one years, and no federal agency involved in any of it.

In an 1896 interview with Nellie Bly in the New York World, Susan B. Anthony said the bicycle had done more to emancipate women than anything else in the world. What Anthony saw at the price point of her own time, the contemporary American buyer is still seeing.

BICYCLEBLUEBOOK.COM

Value Guide

Buy

Sell

2005 Trek Madone 5.2

2005

2008

2009

2010

2011

2012

2013

Not eligible for trade in. Learn more.





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Home /

V3 Carbon Road Bike 18S US

★★★★☆ 4.6 (145 reviews)

~~\$1,499.00~~ \$1,099.00 Save \$400.00
Tax included. Shipping calculated at checkout.

SIZE

47cm

51cm

54cm

56cm

COLOR

Black

White

Where the Safety Improvement Came From

The bicycle's safety record is not a paradox to be explained away. Child cyclist deaths fell 88 percent over forty-eight years with no federal product-design regulation, outperforming the heavily regulated automobile's 62 percent improvement per vehicle-mile over the same period. That improvement is the visible product of market-driven innovation on expired-patent technologies that no agency mandated and no ratchet produced.

Mechanical and hydraulic disc brakes replaced rim brakes across the 1990s and 2000s, providing superior stopping power in rain, mud, snow, and cold where rim brakes lose effectiveness. No federal agency mandated the change. Manufacturers adopted disc brakes because consumers wanted predictable braking in every condition, and expired automotive disc-brake patents made the technology affordable to adapt. High-brightness LED lighting replaced the dim incandescent lamps that had shaken loose on the bicycles of the 1970s and 1980s, making cyclists vastly

more visible at night. The federal reflector mandate persists unchanged from the early 1980s; cyclists who wanted to be seen bought their way past that minimum once the underlying LED patents expired and competing manufacturers began producing systems. The mandate did not move. The market moved.

The rise in adult cyclist deaths over the same period is not a failure of the bicycle. It is a failure of the road and of the vehicles sharing it, and both were reshaped by federal regulatory choices made after 1970. The Corporate Average Fuel Economy standards of 1975 treated passenger cars and light trucks as separate categories with sharply different fuel-economy requirements. The loophole rewarded automakers for building truck-frame vehicles in place of the station wagon. The Insurance Institute for Highway Safety has measured the result at the roadside: the average American passenger vehicle is roughly four inches wider, ten inches longer, eight inches taller, and a thousand pounds heavier than its early-1990s equivalent. Vehicles with hoods above forty inches are forty-five percent more likely to kill a pedestrian than vehicles with hoods at thirty inches or below. On the largest pickups sold today, the hood clears the head of a child on a bicycle.

The road changed with them. Federal Housing Administration lending standards financed postwar suburban development, required wide rights-of-way, and imposed no sidewalk requirement. Developers responded predictably. By 2002 a third of Americans reported no sidewalks in their communities. In 1969, forty-eight percent of American children walked or bicycled to school; by 2017, ten percent did. Adult cyclist deaths rose from 212 in 1975 to 1,041 in 2023, and 1989 was the first year adults outnumbered children among American cyclists

killed by motor vehicles. The bicycle did not change. The road changed, and the vehicles changed. The same federal apparatus that keeps the expired-patent pickup off the market turned the family station wagon into the truck beside which today's cyclist rides.



The pre-1970 American residential street, left, beside its post-1970 equivalent, right. The contrast — sidewalks present and then erased, modest sedans at child eye level replaced by pickup hoods above the child's head, porch life replaced by absence — is what the Corporate Average Fuel Economy light-truck loophole and Federal Housing Administration lending standards produced in combination.

Calculators and the F-150 Contrast

In 1981 Hewlett-Packard launched the HP 12C, a financial calculator with Reverse Polish entry, a small numeric display, and a battery that ran for years on a single set of cells. The introductory price was one hundred and fifty dollars — five hundred thirty dollars and thirty-six cents in 2025 currency. By 2011 the same thirty-year-old design sold for roughly eighty dollars. By 2025, still in production and still in the same physical case, it retailed for forty-nine

dollars and ninety-nine cents. The real-terms drop is approximately 90.6 percent across forty-four years: the same product, in the same packaging, on the same retail shelf, at one-tenth the inflation-adjusted price.

The patents covering the original design expired by the early 2000s, and public-domain clones followed. Competing manufacturers reproduce the keypad, the function set, and the Reverse Polish entry system, and they compete on the post-expiration commons exactly as the safety bicycle's manufacturers competed after the Lallement patent ran out. Calculator manufacturing operates under ordinary product-liability law and consumer-electronics standards, but under no federal product-approval mandate, no emissions test, and no fuel-economy floor. Set that against the pickup truck built in the same forty-four years, whose real price rose while its function did not, and the difference is not technology. It is the wall.

Firearms and the Political Warning

The American firearms industry has the longest continuous record of expired-patent competition of any category in this book. Samuel Colt's revolving-cylinder handgun, patented in 1836, sold as a Colt Paterson for roughly fifty dollars in 1840s currency — months of wages for an ordinary laborer. The core Colt patents expired by the late 1850s. Smith & Wesson, who had been waiting for that moment, entered within a year. Remington followed. Within a decade the one dominant manufacturer became dozens of competitors, and prices fell. By the Civil War, soldiers on both sides were privately purchasing revolvers that existed only because the Colt monopoly had ended on the schedule the Patent Clause fixed.

The pattern repeated across every major firearm innovation of the nineteenth century. The Spencer Repeating Rifle opened the lever-action era. The Winchester Model 1873 was built on accumulated expired-patent technology — lever actions, tubular magazines, metallic-cartridge chambers. Pump-action shotgun patents from the 1880s and 1890s ran their term, and Winchester, Remington, and Ithaca then competed on the same fundamental mechanism, which became the universal American farm and home-defense weapon at prices the patented original had never approached.

The reason firearms are different is not that they are technologically simpler than vehicles or appliances. The reason is political. A sustained constituency has, for fifty years, prevented federal regulators from applying the car-and-tractor playbook to firearms. That is the warning inside the good news. Whether a category gets a wall is decided by organized political attention, not by the merits — which means the bicycle and the calculator hold their commons on the same terms, and hold it only as long as no one organizes to take it.

Three Sectors, One Pattern, One Political Warning

Three sectors. One pattern. Prices fall in real terms, competition broadens, and safety improves without a federal design mandate. Each is what the post-1970 automobile, tractor, appliance, and light bulb would look like if the wall had not been built around them.

The warning is that none of the three is protected by anything structural. They are protected by the absence of a coalition organized to wall them off. Firearms have a

constituency that has fought that fight for half a century and won it. Bicycles and calculators have no such constituency; they have simply never been worth the trouble. The Patent Clause was written to make the commons a matter of law rather than a matter of political luck. In three sectors it still is. In fifty-eight percent of the economy it is not.

CHAPTER 13

The \$72,495 Pickup Truck: Cars, Trucks, and the Household Budget

*Cars, Trucks, and Eight Major Appliances —
the Wall's Bill at the American Kitchen Table*

Set a calculator beside a checkbook and add up what an American family spends in a year, and three line items dominate. The first is the part of the roof this book leaves aside — mortgage or rent, property tax, homeowner's insurance, and the building itself. None of these is a patented article, so none of them can enter the public domain. The second is health care, and the chapter *The Hindered (not Blocked) Sector* — treats the pharmaceutical and medical-device piece of that bill at length. The third line item is the family's transportation and the durable goods that make a household function — the truck or the car parked in the driveway, the refrigerator running in the kitchen, the furnace cycling in the basement, the water heater warming the morning shower, the washer and dryer running through the week's laundry. The third line item is the subject of this chapter.

The below discusses the Blocked fifty-eight percent — the category where the federal regulatory wall built between 1970 and 1987 is highest, most consequential to the household's monthly budget, and most invisible to the household paying the bill. It walks through the wall's cost in three movements: what an ordinary American family pays today for a new pickup truck, what a hundred years of federal regulation has done to the share of a working household's income consumed by a basic family car, and

what the same architecture has done to the eight major appliances on the American kitchen floor.

The \$72,495 Pickup Truck

The 2025 Ford F-150 Lariat 4×4 SuperCrew costs \$72,495 to drive off a typical American Ford dealer's lot. That figure is the out-the-door total — base manufacturer's suggested retail price (MSRP), destination charge, sales tax, title and license fees, and the dealer documentation fee. It is not a luxury figure. Ford's Platinum and Raptor trims start at \$74,905 and \$79,005 in base MSRP alone. The Lariat is the mid-tier family truck Ford itself positions as the volume sweet spot.

The prior chapter walked through the base F-150 XL: \$4,235 nominal in 1976, about \$23,860 in 2025 dollars, against \$37,250 for the 2025 XL — a fifty-nine percent real-terms increase in the bare work-truck floor. The Lariat at \$72,495 is what happens above that floor when a household that needs a family-capable truck walks into the showroom. Ford can charge nearly double the base-trim price for the family truck because no patent-expired alternative competes against it on the lot.

The should-cost baseline for that truck is \$25,000 in 2025 dollars — the wealth-transfer chapter's price for an equivalent family truck without the post-1970 regulatory architecture. The figure is conservative. It accommodates air conditioning, a heater, a sound system, an automatic transmission, a four-door cab. It requires only that the truck be built around the mechanical core — V6 engine, ten-speed automatic, ladder frame, leaf-spring rear suspension, disc-brake stack — that has been in the public domain for decades. The \$72,495 actual price minus the \$25,000 baseline gives \$47,495: what the typical family

pays per vehicle for the regulatory architecture built around the truck it drives.

The post-1976 trajectory does not climb in a straight line. In 1976 the F-150's base MSRP was \$23,860 in 2025 dollars. In 1991 it was \$24,048 — essentially flat for fifteen years, while the wall was being built. By 2001 it had climbed to \$31,756, as the Clean Air Act Amendments of 1990 and the second-generation Corporate Average Fuel Economy (CAFE) standards began to compound. By 2016, \$35,831. By 2025, \$37,250. The wall held, then started to push, then kept pushing.

A second figure puts the household's experience in context. The average transaction price for a new vehicle sold in the United States in June 2023 was \$48,808, per Kelley Blue Book and Cox Automotive. The 2025 American working man's median income is \$67,964. An average new vehicle cost more than seventy percent of a working man's annual income before tax. The household financing a new truck is not pricing in luxury; it is pricing in the regulatory layer.

A third figure is a sanity check on the direct-compliance component. Conrad and Graham (2021), aggregating Office of Management and Budget data across 1967 through 2015, found the cumulative per-vehicle federal regulatory cost — safety, emissions, and fuel economy combined — to be about \$8,200 in 2025 dollars, or fifteen to twenty percent of the price the consumer pays. Sperling and his coauthors at the University of California, Davis, established the floor in 2004: federal safety and emissions regulation added about \$11 in 1967 dollars — \$106 today — to a 1967 vehicle. The Conrad-and-Graham figure is the direct-compliance floor of the wall premium. The \$47,495 is what happens above that floor when the architecture

creates the price ceiling, and the incumbent pricing power, that the wall makes profitable in the absence of patent-expired competition.

Congress is aware of the constraint. The Low Volume Motor Vehicle Manufacturers Act allows specialty manufacturers to sell replicas of designs at least twenty-five years old without full current-year compliance — and caps production at three hundred twenty-five vehicles per manufacturer per year, nationwide. The exception's existence is evidence that Congress understands the constraint; its size is evidence of what the constraint produces when Congress leaves it in place. The injury is the latest chapter in a hundred-year story.

A Hundred Years of Cars

In 1925 Henry Ford sold the Model T Runabout for \$260 nominal — about \$4,776 in 2025 dollars. The Bureau of Labor Statistics, in Bulletin No. 852, reports the average annual wage for full-time non-agricultural workers in 1923 as \$1,428, roughly \$26,800 in 2025 dollars. The Runabout cost about eighteen percent of that wage — substantial but reachable. Ford's 1925 tagline was *Within the Means of Millions*. It was not marketing language. It was a description.

Move forward to 1969 and the Ford Falcon, Ford's basic family sedan of that year. It was larger and more comfortable than the Runabout. Its price in real terms was higher and it consumed a larger share of the working man's income, but the increase was driven by features the buyer wanted, not by a wall the buyer could not see. That is the market doing what it is supposed to do, with General Motors, Chrysler, and the early Japanese imports keeping the price disciplined.

Ford's cheapest current offering is the 2025 Escape, a compact crossover at a base MSRP of \$28,150. Against the working man's median income of \$67,964, the Escape costs 41.4 percent of a year's earnings — more than double the eighteen percent the Runabout cost against the period's average wage. The income share consumed by Ford's cheapest car has more than doubled across a hundred years. That doubling is the household-budget expression of the constitutional violation this book traces.

Almost the entire post-1970 layer of that doubling traces to federal regulation, not consumer demand, and the two studies above settle the point. The pre-1970 layer — principally the Motor Vehicle Safety Act of 1966 and the Air Quality Act of 1967 — cost about a hundred dollars per vehicle in today's money. The post-1970 layer — the Clean Air Act Amendments of 1970, the Energy Policy and Conservation Act of 1975 — produced the wall. More than ninety-nine percent of the federal regulatory burden a 2025 vehicle carries was installed in the fifty-five years since 1970.

The wall did not require corruption from most participants. Most regulators who drafted the 1970 amendments acted from genuine belief in the public-health rationale. Most members of Congress who voted for the 1975 Act voted under the structural incentives the regulatory-ratchet chapter documented. At the same time, some legislators knew exactly what the architecture would do for incumbent manufacturers and supported it because of those effects. Some executives understood it would protect their pricing power against patent-expired competition, and welcomed it. Both populations exist. Both are morally responsible. The argument notes both without resting on either.

What it rests on is the architecture itself. The Low Volume Motor Vehicle Manufacturers Act gave the National Highway Traffic Safety Administration and the Environmental Protection Agency one year to issue implementing rules. Six years passed. The DeLorean Motor Company of Texas, which had championed the legislation specifically to build new DMC-12s on the long-expired patents of the 1981 design, lost its suppliers and its engine-certification path during the paralysis, and placed the project on indefinite hold in October 2022. Superformance sells Shelby Cobra replicas under the same three-hundred-twenty-five-vehicle ceiling and prices the few it sells at luxury levels, because the ceiling makes low-volume affordability impossible. The patent-expired commons exists on paper for the Cobra and the DeLorean. It is unreachable in practice.

The International Mirror

The technological prerequisites for an affordable patent-expired vehicle exist, and the proof is on sale. In 2025 Mahindra sells the Thar 3-door — a modernized successor to the Jeep CJ-5 and CJ-7 with four-wheel disc brakes, anti-lock braking, and a four-star Global NCAP crash rating — for \$10,500 to \$18,500 in India. Its Bolero, a CJ-lineage vehicle on a modern Mahindra diesel, sells for \$8,400 to \$10,400. Maruti Suzuki's Alto K10, descended from the 1979 Suzuki Alto, sells for \$3,900 to \$5,700. Iran Khodro's Soren Plus, built around the patent-expired Peugeot 405 platform it has manufactured under license since 1991, sells for roughly \$7,500 to \$9,900 at the free-market exchange rate. The mechanical bones of all four are public-domain technology.

The same expired-patent Jeep CJ that Mahindra sells new in India for \$12,000 would cost an order of magnitude more to certify for sale in the United States, and no American manufacturer has attempted it. The wall is American, not technological. The same mechanical core that supports a \$12,000 Indian SUV sits underneath the \$72,495 Lariat.

The Eight Appliances on the American Kitchen Floor

Eight major appliances make an American kitchen and laundry room functional: refrigerator, dishwasher, washer and dryer, water heater, furnace, central air conditioning, oven and range, and microwave. Each descends from a design whose core patents expired decades ago. None is available to the American household in a patent-expired configuration in 2025.

The keystone statute is the National Appliance Energy Conservation Act of 1987 (NAECA), which directed the Department of Energy (DOE) to set efficiency standards on a continuously tightening schedule across more than sixty product categories. The anti-backsliding provision at 42 U.S.C. § 6295(o)(1) means that once a standard is set, it cannot be relaxed. It can only tighten. Every patent-expired appliance design met the DOE standard on the date the standard was set. The standard has tightened since. The design is therefore unsalable as a new product, no matter how long ago its patents expired, no matter how reliably it performed in service.

The Association of Home Appliance Manufacturers (AHAM) spent \$1,104,386 on federal lobbying in 2019, per Senate LD-2 filings. It is the appliance-industry parallel to the automotive lobbying the regulatory-ratchet chapter

documented. The same two registers apply here as in the auto sector: most participants acted under structural incentives, and some insiders built and maintain the architecture knowing what it does to patent-expired competition.

The household experience is documented. Ben Lieberman's March 2025 *City Journal* article "Home Appliance Regulations Are Out of Control" surveys the DOE regime. The most recent standard for residential electric water heaters will add a DOE-estimated \$953 to the upfront cost of every new model when it takes effect in 2029. The current dishwasher standard already produces wash cycles of two hours or more for a normal load — roughly twice the time of pre-standards models — and the response was to tighten the dishwasher standard again.

Each of the eight categories has the same story. Refrigerators on the patent-expired vapor-compression cycle ran twenty to thirty years. The washing machine built around the patent-expired direct-drive transmission and mechanical timer, the kind a household repaired with a thirty-dollar belt and a forty-five-minute service call, is not lawfully manufacturable for sale. The water heater built around a patent-expired tank, a single thermostat, and a sacrificial anode no longer meets a floor that tightened in 2015 and again in 2024. The furnace built around a patent-expired gas valve and steel heat exchanger is not lawfully manufacturable. Neither is the single-stage air conditioner, the gas-burner range, or the magnetron microwave. At every line the patent-expired design is what the household would buy if it could, and at every line the wall stands between the household and the design.

In January 2025 the Trump administration issued Executive Order 14154, *Unleashing American Energy*,

and a companion Presidential Memorandum, directing agencies to review, postpone, or withdraw appliance-efficiency mandates. Seven DOE standards were postponed that February. But the order and the memorandum are reversible by the next administration in a way a Supreme Court ruling is not. The anti-backsliding provision stands untouched. The path back through the executive branch closes the moment a future president decides to close it. The path back through the Supreme Court does not close. A Supreme Court ruling cannot be reversed by the next election.

The eight are not the whole household. The chapter *The Wall Itself* lists twelve durables behind the wall, including the lawn mower, the snow blower, the freezer, the heat pump, and the table saw — the wall does not stop at the kitchen door.

The household pays the appliance premium in pieces. A refrigerator costs more than it should every twelve years. A water heater every ten. A furnace every twenty. A microwave every nine. It never sees the figure as a single line item, because it pays fifty dollars and four hundred dollars at a time, across the years.

The Household Sum and the Constitutional Reframe

The wealth-transfer chapter cashed the headline figures at the national level: \$22 trillion in cumulative wealth transfer since 1970, and \$3,385 per household per year. At household altitude the figures are arithmetic. The \$47,495 per-vehicle premium amortizes across a ten-to-twelve-year ownership cycle at roughly \$4,000 to \$4,750 a year — and that is the vehicle alone. Add the appliance premium

across the eight categories on their replacement cycles, and the per-household line item comes into focus: a four-figure number every household sends every year to the manufacturers of the durable goods it cannot avoid buying.

The household is paying for the architecture. The architecture is the wall. The wall is unconstitutional.

The core mechanical patents underlying the truck on the driveway and the eight appliances on the kitchen floor — from the V6 engine and the ladder frame to the vapor-compression cycle and the magnetron — expired decades ago. *Bonito Boats v. Thunder Craft* (1989) and *Kimble v. Marvel Entertainment* (2015) hold the anti-re-protection rule: when a patent expires, the design enters the public domain, and the expired monopoly may not be revived. The right to remanufacture on that foundation — free of the patent monopoly, though not immune from all regulation — is the inference this book defends. The architecture built between 1970 and 1987 — the Clean Air Act Amendments of 1970, the Energy Policy and Conservation Act of 1975, and NAECA — does not formally cancel that right. It blocks every practical exercise of it, without repealing the Patent Clause, without announcing the trade, without asking the household to vote on it. The household has been paying the difference for fifty-five years.

About 58 percent of the products in American commerce are blocked from remanufacturing after patent expiration. The pickup truck on the driveway and the eight appliances on the kitchen floor are the worked example.

The grocery-bill chapter that follows turns to the bill the household pays at the grocery store, the downstream consequence of what the same architecture has done to the American farm.

CHAPTER 14

How Prohibited Expired Patents Double Your Grocery Bill

A forty-four-horsepower Mahindra tractor sells in India for roughly \$7,300. The same company sells a forty-one-horsepower Mahindra in the United States for \$27,992. A combine harvester rated for one hundred horsepower costs an Indian farmer between \$21,000 and \$24,700. The American counterpart costs about \$646,000. These numbers are not typos.

The objection arrives immediately, and it is fair. The two Mahindra tractors are not identical machines. It does not move the argument. Even on a generous feature-adjusted estimate that adds four-wheel drive and the closest match on amenities, the India-bought Mahindra would still cost \$9,000 to \$10,500. The mechanical gap is small. The price gap is enormous.

Some of that spread reflects genuine regulatory overhead — engineering, documentation, compliance labor, legal risk. Much of it reflects what happens in a protected marketplace, where regulation makes it difficult for low-cost competitors to discipline incumbent prices after patent rights expire. What an American farmer pays is the combined cost of compliance overhead and a structure that shelters producers from the competition expired patents were supposed to restore.

The question here is narrow. What would American food cost if farm equipment had followed the bicycle's path instead of the car's?

Before 1970, Tractors Were Not Regulated Like Cars

The first thing to notice about the American tractor is what it was not. Before 1970 the tractor was not born into the federal highway apparatus that captured the passenger automobile. It was an off-road machine first and a road user only incidentally, and the federal government's pre-1970 relationship with it ran mostly to youth labor rules. It did not extend to what could be manufactured and sold.

The National Highway Traffic Safety Administration (NHTSA) confirmed this in its 2016 rulemaking on agricultural equipment lighting, acknowledging that it had not historically regulated most agricultural equipment because such equipment was not designed primarily for public road use. That is the critical legal distinction. Tractors did not begin on the automobile's regulatory path. They began, functionally, on the bicycle's. Every federal layer added after 1970 is therefore visible as an accretion.

The Three Regulatory Layers

The capture of the tractor did not happen in one legislative act. It happened in stages, which is what makes it harder to see and easier to dismiss.

Layer One. The Occupational Safety and Health Administration (OSHA) became effective in 1971. By 1976 its agricultural rollover rule, 29 CFR 1928.51, required rollover protective structures and seatbelts on tractors manufactured after October 25, 1976, when operated by employees. The rule was targeted, not total. It turned on employee use, not every private owner, and small farms with fewer than ten employees were exempt from routine

enforcement. But it established the principle that for a federally relevant class of tractor use, the new machine had to embody design features earlier generations had not. Once that principle exists, later layers are easier to justify.

Layer Two. The Environmental Protection Agency (EPA) nonroad diesel regime is where the ratchet truly bit. EPA's nonroad rules expressly cover farm tractors and agricultural equipment. Tier 1 was finalized in 1994, Tiers 2 and 3 in 1998, Tier 4 in 2004. A new tractor could no longer be a durable machine with a proven layout and a reliable engine. It had to be a compliance platform — catalytic converters, diesel particulate filters, selective catalytic reduction, and the diagnostic electronics to monitor all of it.

Here is the asymmetry that matters constitutionally. *Old tractors may stay in service. Pre-standard engines may be rebuilt without emissions restrictions. But a company that wants to sell a newly manufactured tractor must cross the modern emissions gate.* EPA is not saying yesterday's machine is unlawful. It is saying, more subtly, that yesterday's machine may remain in use, but if you want to make a fresh one for today's buyer, you must satisfy requirements written decades later. That is the structure that preserves patent expiration on paper while destroying its practical value.

Layer Three. In 2016, NHTSA adopted 49 CFR Part 562, implementing the lighting and marking provisions Congress mandated through the Moving Ahead for Progress in the 21st Century Act (MAP-21). Standing alone, the rule would not be decisive. Placed beside OSHA's rollover requirement and EPA's emissions framework, the pattern crystallizes. Tractors are not subject to one sweeping federal tractor code. They are

fenced in by layers — workplace safety for covered employees, emissions certification for new powertrains, road-use standards for new equipment — each narrowing the space for simple, low-cost remanufacture of patent-expired designs.

If the law recognized a grandfathered remanufacture pathway for patent-expired tractor designs older than twenty years, the public could finally obtain the benefit the Constitution contemplates: competitive production of formerly patented machines after the patent term ends.

The Price Gap

The combine harvester gap shows the architecture in full. A John Deere W70 sold in India is a one-hundred-horsepower self-propelled wheeled combine priced at about \$21,000 to \$24,700, with a maximum capacity of 2.5 acres per hour. The closest current American Deere, the S7 600, is listed at about \$646,000 and, with a twenty- to thirty-foot platform, harvests seventy-eight to one hundred sixteen acres a day.

So the American machine costs twenty-six to thirty times more up front while delivering three to five times the one-day output. On cost per unit of daily harvesting capacity rather than sticker price alone, the Indian Deere comes out roughly six to ten times cheaper. Taking six as the conservative floor, a productivity-equivalent combine built on patent-expired technology should cost on the order of \$108,000 rather than \$646,000.

These are not isolated price distortions. Mahindra, Sonalika, and TAFE export tractors and farm equipment to between eighty and one hundred fifty countries. Sonalika alone produces more than 120,000 units a year. They

build machines meeting ISO 9001 and multiple regional safety certifications. Their products cannot enter the American market at their natural prices because of the same architecture that blocks every other category of patent-expired durable goods.

The Bicycle Thought Experiment

Here the bicycle comparison stops being an analogy and becomes a projection. No one regulates which bicycle a manufacturer may produce, and the market delivers both affordability and continuous safety improvement without a federal gatekeeper. We do not have to guess what the same freedom would do for tractors, because we can look at the countries where it already happens. Across forty-three countries individually named on the public sales-network maps of ZX Auto and Mahindra alone, patent-expired vehicles and farm equipment are legally sold and in many cases manufactured. A Kenyan farmer buying a Mahindra tractor pays a fraction of what a Kentucky farmer pays for a machine of similar capability. He is not getting a worse machine. He is getting one whose price reflects the actual cost of manufacturing a proven design rather than the cost of satisfying a regulatory architecture written decades after that design matured.

From the Dealership to the Dinner Table

Tractors and combine harvesters are not discretionary luxuries. They are the capital backbone of food production, and when federal regulation inflates the cost of manufacturing them, the inflation does not stop at the dealership. United States farm production expenditures hit roughly \$500 billion in 2023, per the Department of Agriculture — a two-year increase of 28 percent, the

sharpest on record, with farm machinery and equipment purchases alone running about \$19.4 billion. When a combine costs \$646,000 instead of \$108,000, the farmer amortizes that gap across every bushel of grain. The consumer pays for it at the grocery store.

The Grocery Cost Projection

Suppose the constitutional case succeeds, and patent-expired designs become legal to remanufacture across every sector. The requirements are minimal. A licensed engineering firm certifies the product as an authentic reproduction of the expired-patent design, and the labeling discloses which standards it meets and which it does not. The buyer knows what he is getting.

The cost reduction would not be confined to tractors and combines. It would reach every manufactured input in the food supply chain — the combine that harvests, the eighteen-wheeler that hauls, the refrigeration unit that preserves, the machinery that mills and packages.

Follow the food dollar. Per the Department of Agriculture's Food Dollar Series, the farm share of every retail dollar Americans spend on food is 15.9 cents, and the marketing share is 84.1 cents. Americans spent \$2.58 trillion on food in 2024. The food dollar runs roughly 45 to 60 percent manufactured-goods-dependent when every input is traced honestly, and observed international price ratios in that category run from 3:1 to 20:1.

In a competitive economy where remanufacturing exists across every sector, grocery prices fall 35 to 50 percent once the market fully adjusts. On \$2.58 trillion in annual food spending, that is \$903 billion to \$1.29 trillion per year. A family spending \$1,000 a month on groceries would

spend \$500 to \$650 instead — \$4,200 to \$6,000 a year back in the family's pocket, not from a subsidy and not from a tax cut, but from the competition the Constitution was designed to deliver.

The Patent Question Inside the Tractor

The core mechanical patents underlying the modern agricultural tractor — three-point hitches, hydraulic remotes, power take-off geometries, the diesel engine architectures used in pre-emissions farm equipment, transmission designs reaching back to the 1960s — expired decades ago. *Bonito Boats v. Thunder Craft* (1989) and *Kimble v. Marvel Entertainment* (2015) hold the anti-re-protection rule: when a patent expires — and the Constitution grants it for “limited Times” only — the design enters the public domain, and the expired monopoly may not be revived. The right to remanufacture on that foundation — free of the patent monopoly, though not immune from all regulation — is the inference this book defends.

The federal architecture just walked through — OSHA's rollover rule, EPA's nonroad emissions tiers, NHTSA's lighting and marking requirements — does not formally cancel that right. It blocks every practical exercise of it. A farm-equipment company that wants to manufacture a new copy of a 1975 John Deere 4430, certified by an independent engineering authority as a faithful reproduction, currently has no lawful path to do so.

The tractor did not follow the bicycle's path. It was pulled, later and more slowly than the automobile but just as effectively, into a regulatory architecture that grandfathers old machines while blocking new, affordable remanufacturing of patent-expired proven designs.

The Model T Bracket

Consider what the automobile's own history shows. When Henry Ford introduced the Model T in 1908 at \$825 — roughly \$18,000 today — a working family could buy one for about four months of wages. By 1925 mass production had driven the price to \$260, about \$4,500 today. No federal agency told Ford what he could manufacture, and competition and standardization did what they always do when left alone. A century later a 2026 Ford Explorer starts at roughly \$38,000 — eight to ten months of median household income. The car costs roughly twice as much in labor-hours as it did a hundred years ago, despite a century of manufacturing improvement that should have driven the price the other way. The Model T was on the bicycle path. The Explorer is on the regulatory-ratchet path. The difference is not technology. It is law. The regulatory ratchet did not stop with cars. It reached into the barn.

How Farmers Help Restore the Back Half of the Bargain

The constitutional path is the durable one, and who files the case is the work of the plaintiff chapters. Nineteen categories are ranked there, and the independent family farmer represented through his association is one of them. The strongest candidates are organizations whose missions and memberships already align with the injury — the National Farmers Union, with more than 200,000 family farms across thirty-three state divisions, among them. What follows a ruling is the work of the policy chapters. The family farmer reading these pages will see the equipment line of his operating budget redrawn there.

All that is missing is the association president or board member who decides to be the name on the complaint.

Farm bankruptcies were up 46 percent in 2025, and equipment costs are a central driver. About 58 percent of the products available in American commerce are blocked from remanufacturing after patent expiration. The grocery bill is the worked example.

CHAPTER 15

The Trucker's Burden: A Constitutional Brief for the American Cab

You are somewhere west of Cheyenne at one in the morning, hauling a time-sensitive load, when the dashboard lights up. The diesel particulate filter wants attention. You cannot ignore it. Ignore it, and the truck's computer cuts your engine power, then shuts you down on the shoulder of Interstate 80 in the dark. You pull off. You wait two and a half hours for a parked regeneration cycle to complete. Your delivery window is gone. Your dispatcher is calling. And the five-thousand-dollar filter that caused all of this exists on your truck because a federal regulation says it must.

The argument is often misread. Counsel for one regional trucking association, asked to evaluate an early summary of this brief, thought the position was that “government regulation of a product must stop after twenty years.” A court would reject that argument. So would the author. That is not the argument.

This chapter does not argue that the Constitution forbids diesel particulate filters. The filter is a useful device when fitted to a new engine architecture designed around it. The argument is narrower. The Constitution forbids the federal government from forcing a filter — or a cooled exhaust gas recirculation loop, or a selective catalytic reduction tank — onto a newly manufactured truck built on engine technology no longer protected by patent, if the patent expired before 2000. The pre-2000 engines that powered

American freight before the regulatory cascade began — the Caterpillar 3406E and the first-year C-15 that replaced it in 1999, the Cummins N14, the Detroit Series 60 — ran without aftertreatment, without cooled recirculation, without exhaust fluid, proved themselves over millions of miles, and now belong, by the express terms of the Patent Clause, to the American public. Federal regulations have erased the promise that placed them there. The owner-operator pays the bill.

The architecture that captures the cab is the same architecture that captures the tractor, the appliance, the light bulb, and the family pickup. The cab is one worked example. The chapter is the trucker's. The remedy is everyone's.

What the Argument Is Actually About

The argument is a grandfathering argument. Federal law already recognizes the principle. A 1999 Detroit Series 60 drives legally on most American roads today, because the Clean Air Act vests EPA with the power to set emission standards only for “new motor vehicles or new motor vehicle engines,” 42 U.S.C. § 7521(a)(1), measured at the time of manufacture. Vehicles older than twenty-five years are exempt from current safety standards under 49 U.S.C. § 30112(b)(9), and the Low Volume Motor Vehicle Manufacturers Act permits replicas of designs at least twenty-five years old. Federal regulation has always treated old designs and new designs differently. The constitutional question is what happens when a manufacturer wants to make a new copy of an old design.

The Patent Clause answers that question. Article I, Section 8, Clause 8 authorizes Congress to grant exclusive rights only “for limited Times.” When those times

end, the design enters the public domain, and the monopoly's benefits may not be extended past expiration. Regulation can still apply to a remanufactured product, but it must apply equally to the original manufacturer and the remanufacturer — the same standards, measured at the same point in the design's life. That equality under the law is what grandfathering delivers. An authentically remanufactured pre-2000 truck, certified as a faithful reproduction by an independent engineering authority, should receive the grandfathering every truck of the same vintage already receives: the standards under which the engine was first lawfully certified, not 2026 standards that transform the 1999 design into a different product.

The \$5,000 Filter

A diesel particulate filter is a stainless-steel cylinder holding a ceramic honeycomb substrate. Exhaust passes through the porous walls and soot is trapped — up to ninety-eight percent of particulate matter. When enough soot accumulates it must be burned off by heating the substrate past a thousand degrees Fahrenheit. If the truck is moving fast enough the system does this passively. If it cannot, it forces an active regeneration. If that fails, it parks the truck.

The 2007 federal mandate required not just the filter but the full aftertreatment stack: selective catalytic reduction injecting exhaust fluid to neutralize nitrogen oxides, diesel oxidation catalysts, the sensors monitoring each stage, and the diagnostic integration tying everything to the engine control unit. The hardware is the smaller part of the burden. The emission-control components run roughly three to five thousand dollars on a new Class 8 tractor, and the price the buyer ultimately pays is higher once

integration, sensors, the fluid tank and dosing equipment, diagnostics, and dealer markup are added.

The rule had a genuine public-health basis, and the argument concedes it plainly. The International Agency for Research on Cancer classifies diesel exhaust as a Group 1 carcinogen, and the Environmental Protection Agency estimated the 2007 rule would prevent eighty-three hundred premature deaths a year at full implementation. The organizations that pressed for it acted in good faith on the science. They did not ask whether the Constitution permitted the method.

The heavier cost recurs. Industry research puts ongoing aftertreatment maintenance at thirteen percent of total operating maintenance across the life of the vehicle. A filter replacement runs two to eight thousand dollars; forced regenerations, cleanings, sensor and catalyst failures, and the exhaust fluid consumed at every fill add thousands more — well into the tens of thousands over a truck's working life. When EPA finalized the 2007 rule it told the public the incremental cost would be \$1,990 to \$3,230 per truck, declining to \$1,170 to \$1,870 by 2012. Those projections covered the filter substrate, not the full stack the standard ultimately required, and not the cost of operating it. The realized figures ran the other way. For an owner-operator with one truck and a mortgage, those numbers are not abstractions.

The Engines Were Already Clean

Pre-2000 trucks were not belching black soot. By 1999, new heavy-duty diesels were eighty-three percent cleaner on particulate matter than the 1988 federal baseline and ran at half the nitrogen-oxide output of a decade earlier — entirely through engine design. No aftertreatment. No

filters. No exhaust fluid. The black-smoke trucks people associate with diesel were a pre-1994 phenomenon; the real cleanup ran from 1991 to 1994, met through electronic fuel injection, optimized combustion chambers, better turbocharging, and low-sulfur diesel. The table traces the progression.

Year	PM Standard	Reduction from Baseline	How Manufacturers Met It
Pre-1988	No limit	Baseline (~0.6 typical)	Nothing required
1988	0.60 g/bhp-hr	First limit (matched existing)	No changes needed
1991	0.25 g/bhp-hr	58% reduction	Electronic controls, injection timing
1994	0.10 g/bhp-hr	83% reduction	Electronic fuel injection, low-sulfur diesel
1998–2006	0.10 g/bhp-hr	Unchanged for 12 years	Continued engine refinement
2007	0.01 g/bhp-hr	98% reduction	DPF required — engine design alone cannot reach this

Heavy-duty diesel particulate standards, 1988–2007. The 1994 limit cut particulate emissions by eighty-three percent from the 1988 federal baseline through engine design alone (and approximately ninety percent from the unregulated pre-1988 in-use level). The 2007 standard demanded the final cleanup — a step that engine design could not reach, and that forced a diesel particulate filter onto every new truck.

The standard held at 0.10 grams for twelve years because engine design had reached its practical limit. Then, on December 20, 2000, the Clinton administration signed the rule that changed everything. Phased in with the 2007 model year, it demanded another ninety percent reduction, from 0.10 to 0.01 grams — a step engine design could not reach, and that bolted a filter onto every new truck built from 2007 onward.

The particulate rule was only half of the cascade. The 1997 NOx rule, 62 Fed. Reg. 54694, and the 1998 consent

decrees against the six major engine makers pushed every manufacturer toward cooled exhaust gas recirculation — a system that shortens engine life, increases fuel consumption, and adds points of failure to a previously mechanical engine. The 2010 NOx standard then added selective catalytic reduction and diesel exhaust fluid on top of the filter. A 1999 truck pre-dates every layer of that cascade, and the Clean Air Act's plain text holds its 1999 certification in place for the life of the engine. That is what was lost. That is the truck this chapter argues should be lawful to remanufacture today.

Imagine an America Where...

... energy and safety laws recognized the public domain, expired patents received grandfather exemptions, and remanufacturers only had to show independent engineering certification for the year-standard they meet.

-
- A newly remanufactured 1999 Cat C-15 Peterbilt 379 rolls off a domestic remanufacturing line for ~\$80K — no DPF.
 - No more 1 AM strandings on I-80 because a \$5,000 filter demanded attention.
 - No more \$45,268 penalties for delete kits — the original design needs no kit to delete.
 - Owner-operators choose: pay \$170K for latest tech, or \$80K for proven pre-2000 reliability.
 - Competition drives ALL truck prices down — the generic drug effect.
 - CARB's DPF mandate for pre-2000 patent-expired truck designs eliminated in all 18 states via the Supremacy Clause.
-

The Constitution already promises this. Someone just has to ask.

What the Public Lost

Walk the loss in three steps. First, the patent on a 1999 Detroit Series 60 engine design expires, and Bonito Boats

(1989) and *Kimble v. Marvel* (2015) promise the public the unrestricted right to make or use the design as against federal patent law and any state law trying to recreate the monopoly. Second, the agencies require any newly manufactured engine to meet current standards — a filter, a catalytic reduction system, a fluid tank, cooled recirculation, and the diagnostic architecture the 1999 design was never engineered to carry. Third, compliance does not merely raise the price of remanufacturing the 1999 engine. It abolishes it. What emerges is a different engine. The public-domain right is not burdened. It is destroyed.

The Supreme Court has named this kind of destruction. *Ruckelshaus v. Monsanto* (1984) held that an agency's use of a submitter's proprietary data was a Fifth Amendment taking, and that intangible intellectual property is protected property under the Takings Clause. *Lucas v. South Carolina Coastal Council* (1992) held that regulations eliminating all economically viable use of property are per se takings, with no balancing required. The public's right to remanufacture a 1999 Series 60 is intangible intellectual property of the kind *Ruckelshaus* protects, and the regulatory cancellation of that right eliminates every economically viable use. No litigant has yet asked a federal court to apply those cases to the regulatory cancellation of an expired patent.

What Americans Could Build

The trucks this chapter argues should be lawful to remanufacture are American. They are the platforms American workers built in American shops, drove for American carriers, and parked at American truck stops for thirty years. The Peterbilt 379 with a Caterpillar C15 was

the legendary million-mile rig of the 1990s and early 2000s — sleeper cab, eighteen-speed transmission, four hundred seventy-five to five hundred fifty horsepower. The Kenworth W900 with a Cummins N14 was its twin in the mechanical-simplicity school, serviceable by any independent mechanic with hand tools. The Detroit Series 60 powered Freightliner, Western Star, and other platforms for two decades. All three engines, and the chassis designed around them, are now out of patent or close behind.

An authentic remanufactured 1999 Peterbilt 379 with a new Caterpillar 3406E or first-year C-15 certified to 1999 emission standards would carry a price of roughly eighty thousand dollars, against a new Class 8 tractor well past a hundred and seventy thousand. The saving on acquisition is half the picture. The other half is what the truck would not carry: no filter, no catalytic reduction system, no fluid tank, no cooled recirculation loop, and none of the recurring maintenance that follows them.

The International Proof

The comparison is not hypothetical anywhere else. Heavy trucks built on long-expired powertrain designs sell new, today, at a fraction of American prices in markets that never built the wall. A new Sinotruk HOWO 6×4 tractor, 371 to 420 horsepower, sells for \$28,000 to \$35,000 with no filter, no catalytic reduction system, and no exhaust fluid. The closest American comparison, a 2026 Kenworth T680 day cab in the same configuration, lists at \$161,850. The American truck is the better truck on horsepower, brakes, and electronics; it is not five times the truck. Those figures are evidence of what the market clears at when the wall is gone. They are not a recommendation to import a

foreign truck — there is not one Chinese-built heavy truck on an American road, and no administration is likely to permit one. The trucks called for here are American, built by American workers, on platforms that already belong to the American public.

A 2025 T680 sleeper runs about \$172,900 and a heavy-spec 2026 T800 reaches \$284,900: an American owner-operator pays five to eight times what a trucker in Ghana, Nigeria, or Indonesia pays for a truck that does the same job. The proposal here is a Peterbilt 379 with a Caterpillar C15 built in America at the price the developing world pays for its serviceable equivalent. The market would do it. The wall blocks the result.



Brand-new Sinotruk HOWO 6×4 tractor, 371–420 HP, Euro II — \$28,000–\$35,000 FOB China. No DPF, no SCR, no DEF. Exported to one hundred fifty-plus countries.



New 2026 Kenworth T680 day cab — \$161,850. Full DPF/SCR/DEF aftertreatment stack required by EPA.

Bootleggers and Baptists

The bootleggers-and-Baptists pattern the opening chapter named has a specific cast here. The Baptists are the public-health organizations already credited above. Two layers of beneficiaries, intentional or not, appear opposite them. The first is the emission-controls manufacturers — the Manufacturers of Emission Controls Association (MECA), whose members include Corning, Johnson Matthey, BASF Catalysts, and Umicore. The mandate created that category, and every Class 8 truck sold since carries roughly three to five thousand dollars in MECA-member components. None of this is an accusation; the pattern simply fits Yandle's framework. The second layer is the truck original-equipment manufacturers — Paccar, Daimler, Volvo and Mack, and Navistar — reported as

having fought the 2007 rule and lost. Once final, the mandate gave them a higher barrier to entry, a captive service-and-parts aftermarket, and an architecture under which no firm could lawfully build a new Class 8 truck on expired-patent engine technology. For fifty-five years the only parties with a voice in the EPA rulemaking room have been the parties who profit from the wall. The owner-operator, who pays for it, has had no seat at the table.

Three Paths, One That Holds

Three paths exist for removing the wall. Two are open and active. The third has never been opened. Only the third is durable.

The EPA path. In 2025 EPA Administrator Lee Zeldin announced reconsideration of the 2022 Heavy-Duty NOx rule and issued guidance ending the five-mile-per-hour DEF derate that had been stranding truckers on the shoulder, and in February 2026 the agency announced rulemaking to eliminate DEF derates permanently. Real relief — and none of it bound by anything beyond administrative discretion. The next administrator can rescind every guidance.

The legislative path. Senator Cynthia Lummis introduced the Diesel Truck Liberation Act, S. 3007, in October 2025, with a House companion from Representative Mike Collins in March 2026. It is the strongest political effort the trucker constituency has ever mounted against the wall. What Congress writes today, the next Congress can repeal.

The constitutional path. A federal court ruling on the Patent Clause and the Takings Clause binds every future administration, every federal agency, and every state through the Supremacy Clause. And it produces a benefit

the other two cannot: by opening domestic remanufacturing of pre-2000 platforms, it puts a price floor under the new-truck market. The owner-operator who keeps buying brand-new trucks still pays less, because manufacturers must compete against the remanufactured alternative — the same dynamic by which brand-name drug prices fall by thirty to eighty percent when generics enter. It also does what neither of the others can for the shops that would build the trucks: a ruling that cannot be rescinded is the only footing on which a remanufacturer will finance a production line.

Three Paths — Only One Holds

Comparing the three available routes for removing the regulatory wall from heavy-duty diesel.

CATEGORY	EPA GUIDANCE	LEGISLATION (S. 3007)	PATENT CLAUSE RULING
Permanence	Reversed by next administrator	Repealed by next Congress	Binds every future administration
Scope	Only what EPA can reach	Only what the bill specifies	Every patent-expired design, all agencies
Authority	No constitutional authority	Statutory authority only	Supreme law of the land
CARB / States	Does not bind states	Federal only	Supremacy Clause binds all states
Status	Active — partial relief, reversible	Introduced Oct 2025, in committee	Never raised — prime opportunity

Everything gained by EPA guidance or legislation can be undone by the next administration. A Patent Clause ruling cannot be reversed.

Three paths to restore the patent bargain. EPA guidance can reach what the Clean Air Act lets it reach, and the next administration can take back. Legislation can reach what its drafters specify, and the next Congress can repeal. A Patent Clause ruling reaches every patent-expired design and every agency that touched it, and cannot be reversed by any administration.

Not Just the Long-Haul Driver

The long-haul owner-operator is the clearest face of the injury, but not the only one. The logger, the grain hauler, and the dump-truck owner buy the same over-priced engine. Five trades carry the burden compounded — a distinct federal regulatory load the dry-van driver never touches, with the diesel wall paid on top.

The fuel hauler. Roughly sixteen thousand fuel-dealer firms, most of them multi-generation family companies, move product through their own trucks. The cargo is hazardous material: a tank above one hundred nineteen gallons triggers federal hazmat and tanker-endorsement requirements, a DOT-406 construction standard, and a TSA background check. The wall lands on him twice.

The milk hauler. Nearly every drop of America's milk leaves the farm on a truck that must run every day or every other day, under FDA sanitary-transportation rules and state bulk-milk permits — a vehicle that can never sit idle waiting for a cheaper option to become legal.

The refuse hauler. Of roughly twenty-one thousand waste-collection firms, only a few dozen are national chains; the rest own their fleets outright under EPA disposal rules and state permitting. *The septic and vacuum operator* — some seven thousand three hundred owner-run firms — carries EPA disposal rules and state manifest requirements as the price of doing the work at all. *The concrete mixer* — some three thousand seven hundred ready-mix firms, the trucks frequently owner-driven — lives under axle-limit regulation and the ninety-minute clock of a perishable cargo. Each meets the same wall: the same expired patents held out of remanufacture, the same inflated price on the only new truck he may buy.

The Cab and the Constitution

There are roughly three and a half million truck drivers in the United States. Between five hundred eighty-seven thousand and nine hundred twenty-three thousand are owner-operators. The injury reaches more Americans than any other single plaintiff category the book identifies. The trucker who paid eight thousand dollars for a filter last Tuesday does not need a spreadsheet.

Other voices have begun to describe what the war on the trucker looks like from inside the cab. Gord Magill's *End of the Road: Inside the War on Truckers* (Creed & Culture, 2026) is the firsthand account of how the once-iconic occupation has been ground down over two decades, in a voice the constitutional literature has not heard. During the drafting of this book he put the aspiration on the other side of that war in plain terms: what the owner-operator wants is not a used truck but a newly manufactured one built on the pre-2000 platform — the rig he would buy today if it were lawful to make. That denial, rather than the modern truck he already owns, is the injury this chapter describes.

Return to the shoulder of Interstate 80. The driver waiting on a regeneration cycle at one in the morning is waiting because a federal rule requires a device his engine was never designed to carry, on a design the Constitution says belongs to him. The patent expired on schedule. The right came into existence on schedule. The right cannot be exercised. The wall is permanent against every actor below the constitutional level. The Supreme Court is not below that level.

The trucker also has a reason not to wait. The same wall that prices his new rig at \$170,000 is the head start the driverless platforms run on, and the chapter *The Last*

Human Driver works the arithmetic: a ruling in 2030 puts the remanufactured truck well under the autonomous truck's cost per mile, and a ruling in 2033 puts it barely under. A case filed this year lands in that window. A case filed in five lands after it.

Who files the case is the work of the plaintiff chapters, where the owner-operator trucker — the category this chapter has documented, the category the foreword writer represents — is one of nineteen. The wall is removable. The remedy is constitutional. It is waiting for the trucking-association president, the OOIDA board member, or the owner-operator who reads this chapter, recognizes the opportunity, and picks up the phone.

CHAPTER 16

Healthy Lighting and the Anti-Backsliding Trap

The aisle is forty feet long. Every bulb on it is an LED.

The American with photosensitive epilepsy walks the length of it twice. He is looking for the warm filament glow he grew up with — the one his neurologist has told him to use, because LED flicker can trigger his seizures. He finds nothing labeled for residential use. Specialty and rough-service incandescents are tucked at the bottom of one end-cap display, six or seven dollars apiece. The packaging says they are for ovens, garage door openers, and industrial machinery. The packaging does not say they are for him.

The bulb he is looking for is not a new invention. It is the descendant of a design whose foundational patents had expired by the end of the nineteenth century. From 1879 to 2022 it sat on every hardware-store shelf in the United States. American factories made it. Households bought it for less than a dollar. His grandmother bought it. He bought it as a young adult.

He cannot buy it now.

The question this chapter asks is narrow. Edison's foundational filament patents had expired by the mid-1890s. Why is the descendant of that bulb effectively illegal to manufacture for general residential use in 2026, while the descendants of the LED patents — patents that have not all yet expired — are the only thing on the shelf?

What the Argument Is Actually About

The argument is not that the Constitution forbids energy-efficient lighting. It does not. It requires no American household to buy an incandescent bulb, and the household that prefers an LED is free to buy one after every remedy this book proposes is in place.

The narrower claim is this. The federal government has every constitutional authority to set efficiency standards for newly invented lighting technology. It may regulate LEDs, fluorescents, and the next lighting technology not yet invented. What it may not do is use those standards as instruments for retroactively withdrawing a public-domain technology — one whose patents expired more than a century ago, one that Americans manufactured and bought freely for a hundred forty-three years — from the residential market, on behalf of the patent-protected competitors that benefit from the withdrawal.

The constitutional question is not about lighting. It is about whether the “limited Times” the Patent Clause guarantees can be quietly extended by regulation after the time has run out.

The Cost Imposed

The cost shows up in three places. The first is at the store, where a bulb that retailed for less than a dollar in 2022 has been replaced on residential shelves by an LED retailing for several. The per-bulb comparison cuts against the argument for households without medical light-sensitivity, because the LED has a longer rated life and lower operating cost. The Department of Energy projected national annualized consumer savings of roughly \$2.3 billion and \$1.2 billion for the 2022 and 2024 rules. The

argument does not contest that arithmetic for the household that can use an LED.

The second is at the bulb the LED cannot replace. For the American with photosensitive epilepsy whose neurologist forbids flicker, for the patient with a circadian-rhythm disorder prescribed warm-spectrum light, for the patient using a UVB lamp in approved psoriasis therapy, for the person whose retinal sensitivity makes blue-shifted LED output intolerable, the arithmetic does not apply. Savings projected on the assumption of LED substitution are not available to a household for whom substitution is medically contraindicated. The injury to those populations does not appear in the regulatory impact analysis because the analysis did not measure it.

The third is at the disappearance of an option. Americans who had relied for generations on the warm-spectrum light of an incandescent bulb are no longer free to choose it. The relevant cost is the constitutional one: the withdrawal of an exercised public-domain right, without compensation, on behalf of the patent-protected competitors that benefit from the rule.

What the Public Lost

Three doctrinal anchors, briefly. In *Bonito Boats v. Thunder Craft Boats* (1989), the Supreme Court reaffirmed that when a patent expires, the federal patent laws create a federal right to copy and to use the design. In *Ruckelshaus v. Monsanto* (1984), the Court held that intangible intellectual property — including data, formulas, and the legal right to use them — is property protected by the Takings Clause. In *Lucas v. South Carolina Coastal Council* (1992), the Court held that a regulation eliminating all economically viable use of a property right is a per se

taking. Each is treated at length in the chapter *The Takings Clause Prong*.

What separates the lighting injury from every other modern injury in this book is the temporal shape of the loss. The blockade pattern captures most expired-patent durable goods — tractors, pickup trucks, residential heating and cooling, semi-trailers. The federal regulatory wall was built around the technology before the patents expired, so that by the time the patents crossed the public-domain line, the wall was already there. A government defendant can argue, plausibly, that no public right ever vested.

The withdrawal pattern is different. It is what happens when the federal government withdraws a public-domain technology from a market in which the public has already been exercising the right to manufacture, freely and competitively, for generations. The ordinary incandescent bulb is the canonical case. The Department of Energy's own 2020 Lighting Market Characterization documented approximately 819 million incandescent lamps still installed in U.S. residential sockets — roughly twelve percent of the entire residential lighting stock at the moment the federal machinery moved to withdraw their manufacturing future. The right was exercised, in plain commerce, for a hundred forty-three years. What the 2022 backstop rule did was take it away.

The pattern does not stop with the incandescent. The compact fluorescent that replaced it is itself public-domain technology. General Electric's engineer Edward Hammer bent a fluorescent tube into a spiral in 1976, and the company patented the work; those foundational patents expired years ago. The April 2024 rule raised the efficiency threshold to a level the compact fluorescent cannot meet, effective in 2028. A second expired-patent design is now

scheduled to leave the shelf. The LED that survives both thresholds is the one whose patents have not all expired, and a newer patented successor already waits. The regulatory wall does not fall when a patent expires. It moves to the next design the public has come to own.

The Bulbs Were Already What They Needed to Be

That ordinary incandescent bulbs are wasteful is a policy argument, not a constitutional one. Policy arguments may justify federal regulation of newly invented technologies. They have never been held sufficient to justify the retroactive withdrawal of a century-old public-domain technology from the market. The horse-drawn carriage is less efficient than the automobile, and the federal government did not outlaw the carriage. The acoustic guitar is less efficient than the electric at filling a stadium, and the federal government did not outlaw the acoustic guitar. The 2022 general service lamp rules did not conclude that the incandescent bulb endangered consumers, or that it was unsafe. They concluded it was less efficient than the LED. That is a preference the federal government may articulate for new lighting technology. It is not a preference that may, consistent with Article I, Section 8, Clause 8, be applied retroactively to a design the public already owned. The bulbs that served the American household for a hundred forty-three years were what they needed to be.

Bootleggers and Baptists

The framework comes from the economist Bruce Yandle, who named it in 1983. A regulatory regime is at its most durable when it is supported simultaneously by a

moralizing constituency that believes in the regulation's stated purpose and a profit-taking constituency that profits from the regulation's suppression of competition. The Baptists win the moral argument. The bootleggers win the market.

The 2007 Energy Independence and Security Act was passed by such a coalition. Environmental and energy-efficiency advocates wanted national standards in good faith. The patent-protected lighting incumbents supported the architecture for reasons more closely tied to their balance sheets. Not every advocate at that table understood what the standards would do to the residential incandescent market. Some did and supported them anyway.

The structural fact is what matters constitutionally. Signify, the successor to Philips Lighting, holds more than twenty thousand patents and reports annual research spending on the order of a quarter billion euros, the great majority of it on solid-state lighting. Its market position depends on patented LED architectures whose royalty streams expired-patent incandescent manufacture would substitute against. The same is true of GE Lighting, Osram, Cree, and Acuity. The constitutional injury is not that any particular company lobbied for any particular rule. It is that the architecture, taken as a whole, removed the unpatentable competitor from the residential market in favor of patent-protected manufacturers whose business model depends on its absence.

The Supreme Court's proportionality jurisprudence makes this configuration constitutionally relevant. *Nollan v. California Coastal Commission* (1987) requires that a regulatory condition bear an essential nexus to the harm the regulation claims to address; *Dolan v. City of Tigard*

(1994) requires that the burden be roughly proportional to that harm; and *Sheetz v. County of El Dorado* (2024) reaffirmed both and extended them to legislatively imposed conditions. A lumens-per-watt threshold appears facially neutral. But when its practical effect is to eliminate the one technology that cannot be patented, and the alternatives that survive are owned by patent-protected manufacturers, the proportionality question becomes more honest.

The International Picture, Acknowledged Thin

One candor point belongs here, because the international evidence in this chapter is weaker than elsewhere in the book. For most expired-patent durable goods the comparison is dramatic: India sells a four-wheel-drive Mahindra tractor for about \$9,000 against an American equivalent above \$27,000, and combine harvesters retailing abroad for \$24,000 retail here above \$600,000. The lighting evidence is thinner. The European Union's incandescent phase-out preceded the American one by several years and most wealthy nations have followed similar trajectories, so the continuous-manufacture comparison that runs against ninety countries for tractors and trucks does not run as cleanly for the residential bulb. The argument rests its weight on the American historical record instead — a hundred forty-three years of open manufacture, ended by rule.

Three Paths, One That Holds

The executive path is closed by the anti-backsliding provision at 42 U.S.C. § 6295(o)(1), and the closure has been tested twice. On January 20, 2025, President Trump signed Executive Order 14154, Unleashing American

Energy, which referenced consumer choice in light bulbs. Within days the trade press confirmed that the order could not override the statutory lock. On July 2, 2026, the Department issued a proposed rulemaking its own announcement described as ending home-appliance mandates; the proposal changed only the Department's internal process for setting future standards. It did not, and could not, repeal the standard already in force.

The legislative path is closed by politics. On May 1, 2025, Senator Mike Lee of Utah — the sitting chairman of the Senate Committee on Energy and Natural Resources — introduced the Liberating Incandescent Technology Act with five cosponsors. Twelve days later a House companion followed. The bills repeal the statutory authority under which the 2022 and 2024 lamp rules were issued. Neither bill has received a hearing. Neither has been marked up. The sitting chairman of the Senate committee of jurisdiction has not been able to move his own bill out of his own committee.

The constitutional path is the one this book is built around. After a successful ruling, American manufacturers certified by an independent engineering authority build incandescent bulbs at the prices and to the specifications that served Americans for a century. The bulb returns to roughly a dollar at the hardware store, and specialty and rough-service incandescents become a market segment rather than the regulatory loophole that survived the residential withdrawal. One honest complication remains: since 2022 the productive capacity for residential incandescent manufacture has been substantially dismantled, and a bare repeal of the rule would not restart it. That is part of why the remedy is constitutional rather than legislative. A ruling does more than remove a rule. It establishes the framework under which a remanufactured

product built on a public-domain platform is certified to the engineering standards in effect during the platform year the product mimics. The patent commons becomes a buildable thing rather than a paper right.

The Lamp and the Constitution

Return to the American with photosensitive epilepsy who opened the chapter. He stands in a hardware-store aisle in 2026 and cannot buy the bulb his neurologist has told him to use. Americans manufactured and bought it freely from the year his great-grandfather was born to the year he turned forty-three. The federal government, between 2007 and 2024, withdrew it from the residential market on behalf of the patent-protected competitors that benefit from the withdrawal. He has done nothing to forfeit the right his grandmother exercised every time she replaced a bulb in her kitchen. The Constitution still says what it said when she did.

The Patent Clause has been on the page for two hundred and thirty-eight years. The withdrawal pattern is younger than most Americans. The wall is permanent against every actor below the constitutional level. The Supreme Court is not below that level.

CHAPTER 17

The Surveillance Premium

The showroom floor is bright. The car parked under the lights is gleaming. The buyer has come in prepared to ask for three things. He wants a manual transmission. He wants a large screen that will run the navigation app on his phone without a connected-services subscription. He wants a vehicle that will not transmit his driving data to insurance companies.

The dealership owner is an old friend, and he is patient. The manual transmission is not offered in this model, in any trim, and inventory checks across three nearby dealerships confirm no match within four hundred miles. The two of them find out together that fewer than thirty of the roughly three hundred new-car models on the U.S. market in the 2026 model year offer a manual at all. The screen runs Apple CarPlay from the buyer's phone; the salesman is happy to demonstrate. The third request takes longer. The event data recorder is installed in every new vehicle on the lot. So is the driver-attention camera in the rearview-mirror housing on the higher trims. So is the connected-telematics package, which on this brand transmits driving-behavior data to the manufacturer by default. The buyer can decline some downstream uses by adjusting settings after purchase. The salesman cannot confirm which uses, or for how long.

The buyer drives home in the fifteen-year-old sedan he arrived in. It has a manual transmission. It has no screen. It has no telematics. None of the three things he wants in a new vehicle is buildable, by any U.S. manufacturer, into any vehicle on any showroom floor in the United States in

2026. The reason none of them is buildable is the subject of this chapter.

What the Argument Is Actually About

The argument is not that the Constitution forbids safety technology. It does not forbid backup cameras, automatic emergency braking, driver-attention monitoring, event data recorders, or telematics broadcast, and it requires no household to refuse any of them. The federal government has every constitutional authority to set safety standards for newly invented automotive technology. What it may not do is use those mandates as instruments for blocking the public's right to manufacture vehicles from patents that expired long before any of the mandates were written, on behalf of the new-vehicle manufacturers and the downstream data brokers that benefit from the blocking.

The injury named here is not primarily economic. It is the involuntary conscription of the owner's property — the car the owner paid for, parks at home, and drives on the public road — into a continuous data-collection network operated by, and for the benefit of, parties the buyer never chose. Something was signed at delivery, and that is the point. He can decline the paperwork. He cannot decline the hardware, because no new vehicle is sold without it.

The mechanical platform of the modern American passenger vehicle crossed the public-domain line decades ago: the internal combustion engine, the four-wheel suspension, the disc brake, the automatic transmission, the manual transmission alongside it. The foundational electronic architecture crossed it more recently and no less completely. USB, Bluetooth, capacitive touchscreens, and the imaging sensors backup cameras run on are all buildable from patents now in the public domain. The

components mandated for surveillance, by contrast, apply only to newly manufactured vehicles, by the statutory architecture's own design. That asymmetry is the seam this argument pries open.

The Four Federal Mandates

The surveillance architecture rests on four federal pillars. Title 49 of the Code of Federal Regulations, Part 563, published in 2006, defines what an event data recorder must record; the National Highway Traffic Safety Administration withdrew a proposed installation mandate in 2019 because voluntary adoption had already reached substantially universal coverage. Federal Motor Vehicle Safety Standard No. 111 completed its rear-visibility camera phase-in on May 1, 2018. Standard No. 127 mandates forward and pedestrian automatic emergency braking on every light vehicle manufactured on or after September 1, 2029, requiring forward-looking sensor arrays and the computational architecture to interpret them. And Section 24220 of the Infrastructure Investment and Jobs Act directs the agency to require advanced impaired-driving prevention technology; the statutory deadline was missed, the rulemaking is ongoing, and the implementation form is not yet settled. The direction of motion is.

Each of those rules applies only to newly manufactured vehicles. None applies to a vehicle remanufactured from patent-expired technology and certified under the framework the chapter *The Policy Framework After a Successful Ruling* develops. The federal compliance burden attaches at the point of new manufacture. The remanufactured vehicle sits outside it.

What the Surveillance Produces

The architecture mandated for safety has been put to a second use. In March 2024, Kashmir Hill of the New York Times documented that General Motors, Honda, Hyundai, Kia, Subaru, and Mitsubishi had been transmitting driving-behavior data from their connected vehicles to two data brokers, LexisNexis Risk Solutions and Verisk. The brokers compiled the transmissions into individual driver-risk reports and sold them to insurance companies, which used them to raise premiums.

The data fields were granular: trip dates and times, distance driven, counts of hard-braking events, rapid accelerations, speeding events, and GPS location traces. On General Motors vehicles the collection ran through a feature called Smart Driver, presented at delivery as a gamified driver-feedback experience and tucked into a consent flow most buyers did not read with care. The per-vehicle payments were small — twenty-six cents from Honda, sixty-one from Hyundai. The surplus was not at the broker payment. It was downstream, at the insurance-premium markup the reports enabled.

General Motors discontinued the arrangements in March 2024. In January 2025 the Federal Trade Commission announced a proposed order barring the company and OnStar from disclosing precise geolocation and driving-behavior data to consumer reporting agencies for five years, finalized in January 2026. Four things about what that did and did not accomplish. It did not require removal of the collection infrastructure, which remains installed. It regulated the downstream sale, not the upstream collection. Of the carriers Hill named, only General Motors faced enforcement. And the five-year term expires in 2031.

What the surveillance architecture produces, in plain terms, is this. You are paying for a prosecuting attorney to have live witnesses in your car twenty-four hours a day. The witnesses are the federally mandated sensors. The prosecuting attorney is whichever insurer, civil litigant, or law enforcement officer eventually requests the data.

The Same Architecture in the Trucking Cab

On the working side of the road the same architecture exists in a more intense form. On December 16, 2015 the Federal Motor Carrier Safety Administration published a final rule at 49 C.F.R. Part 395 requiring most commercial trucks to install an Electronic Logging Device — a federally mandated digital monitor recording hours of service, location, engine state, and motion — with compliance required by December 2017. The Owner-Operator Independent Drivers Association sued, arguing the device amounted to warrantless continuous surveillance of roughly three and a half million American drivers. The Seventh Circuit rejected the challenge in 2016, holding that the industry's status as pervasively regulated permits the inspection the rule requires. The Supreme Court denied review.

Federal regulation supplied the legal floor; commercial vendors built the stack above it. Karen Levy's *Data Driven: Truckers, Technology, and the New Workplace Surveillance* (Princeton, 2022) documents what the device produces, including the Omnitracs Driver Retention Model, which analyzes hours-of-service data to predict which drivers are likely to quit. Driver-facing cameras now sit on top of that floor. A 2023 American Transportation Research Institute study of roughly 2,100 commercial drivers reported a driver-facing-camera approval rating of

2.24 on a zero-to-ten scale, with female drivers rating the cameras' privacy protection twenty-four percent lower than male drivers.

And the safety case advertised for the mandate has not survived contact with the data. Working with the RigDig commercial-trucking database, *Overdrive* reported in 2022 that truck-involved crashes rose from 504,214 in the four years before the mandate to 565,043 in the four years after — an increase of more than sixty thousand. The Fourth Amendment frame has lost. The Patent Clause and Takings Clause prongs this book develops have not been argued.

The Doctrinal Foothold

Three holdings, briefly. In *Ruckelshaus v. Monsanto Co.* (1984), the Supreme Court held that intangible intellectual property — including data, formulas, and the legal right to use information of commercial value — is property protected by the Takings Clause. The public-domain right to manufacture a vehicle from expired patents is exactly that kind of property. In *Cedar Point Nursery v. Hassid* (2021), the Court held that a government-compelled physical intrusion onto private property is a per se taking. The compelled installation of a mandated data recorder, sensor array, camera, and telematics broadcast is the modern automotive analog. In *Sheetz v. County of El Dorado* (2024), the Court held that regulatory exactions must bear a proportional relationship to the government's stated interest. A mandate justified by traffic safety is one thing. A mandate that produces a continuous data stream available to insurers, civil litigants, and law enforcement is another.

The Remanufactured Exit

Component by component, the electronic architecture of a modern passenger vehicle is buildable from patents now in the public domain. The foundational Intel-assigned patents establishing the Universal Serial Bus protocol were filed in October 1994 and expired between 2014 and 2018, and the device-class specifications for video and audio output are published royalty-free. The Ericsson-assigned patents covering the Bluetooth piconet architecture, filed in 1997, expired between 2017 and 2018. Samuel Hurst's electrographic-touchscreen patent expired decades ago, and the multi-touch patent underlying most contemporary implementations expired in 2019. The foundational imaging-sensor patents expired in the 1990s, and the CMOS active-pixel patents that succeeded them have also expired. A remanufactured vehicle's dashboard screen, fed by a cable from the buyer's phone, can display the navigation application the buyer chose — with no licensed infotainment stack between the buyer and the map.

General Motors is moving further in the same direction. In October 2025 the company confirmed it will phase out support for both Apple CarPlay and Android Auto across its entire lineup in favor of a proprietary system, with full transition expected by the 2028 model year. On a General Motors vehicle, the buyer who declines to surrender control of the dashboard will have no remaining alternative on the licensed-software market. On a remanufactured vehicle built from the patent-expired components named above, the buyer needs no licensed-software stack at all.

The Manual Transmission: Regulatory Death, Not Market Death

The manual transmission is the cleanest single example of regulatory death rather than market death. The Environmental Protection Agency's Automotive Trends Report documents the trajectory. In 1980, manual transmissions accounted for 34.6 percent of newly manufactured American light-duty vehicles. By 1990, 22.2 percent. By 2000, 9.7. By 2010, 3.8. By 2020, 1.1. By 2024, 0.9.

The market explanation — that American buyers simply stopped wanting manuals — does not survive contact with the data. Enthusiast forums, dealership surveys, and manufacturer market research consistently report a stable minority who prefer the manual and will pay a premium for it. Corporate Average Fuel Economy compliance favors automatics with adaptive shift logic optimized for the test cycle, dual-clutch automatics, and continuously variable transmissions with no fixed gear ratios at all. Each manual variant of a model line requires a separate emissions certification, a separate fuel-economy test, and a separate crash-test program, and the fixed cost spreads across the manual's smaller volume. The unit economics push the manufacturer to drop the option.

The buyer who wants the option finds that the option has been removed by the test cycle, not by his neighbors. A remanufactured vehicle built on a 1980s or 1990s platform restores it to a buyer who has been told for thirty years that it is unavailable. The deeper question — what is lost when the relationship between the driver and the machine is mediated by a computer that cannot be turned off — belongs to the chapter *The Last Human Driver*.

The Car and the Constitution

Return to the buyer in the showroom. He wanted three things: a manual transmission, a screen that runs the navigation he chose, and a vehicle that does not transmit his driving data to insurance companies. He left and drove home in his fifteen-year-old sedan.

The argument is not against safety. It is not against backup cameras. It is not against drivers who choose to share their data with their insurer. It is against a federal regulatory wall that has converted what the Patent Clause guarantees — the public's right to manufacture vehicles from expired patents — into a perpetual monopoly extended on behalf of the manufacturers and data brokers that profit from its absence. The wall is fifty-two years old. The Patent Clause has been on the page for two hundred and thirty-eight. The wall is permanent against every actor below the constitutional level. The Supreme Court is not below that level.

CHAPTER 18

The Last Human Driver

Three Scenes

The yard is small: two trucks, a gravel lot, a single-wide office trailer. The owner-operator who runs it has come out to meet a man he has known for twenty years. The two drove team once, long-haul, splitting the wheel on runs that paid better than either could earn alone. The visitor is not driving today. He arrived in a rented sedan, in a pressed shirt, carrying a tablet. He works now for an autonomous-freight company, in a role the company calls remote support specialist. When his own truck grew too expensive to replace, this was the job on offer, and he took it.

He has come to make his friend an offer too. The platform will haul the owner-operator's freight on its own driverless trucks for a flat fee per mile, lower than what it now costs to keep a human in the cab. The arrangement has a name: Driver-as-a-Service. The owner-operator asks a question the tablet has no slide for. If the driving is the cheap part, why is the truck the expensive part? Why does a new rig cost more than his first house did?

Three hundred miles away, a woman sits in a holding lot outside an airport waiting for the app to send her a fare. She has carried passengers for nine years, and the car she is sitting in is hers. The loan on it has four years left to run. She watches a car leave the terminal with a passenger in the back seat and no one in the front. More of those appear every month. She suspects, correctly, that the driverless ride is not actually cheaper to provide than

the ride she provides. It is only priced as if it were. What she cannot work out from the holding lot is why the car she had to buy to compete cost as much as it did.

Closer to home, a father takes his sixteen-year-old to an empty parking lot on a Sunday morning, the way his own father once took him. The lot is empty because the stores are closed. The family car is fifteen years old. The teenager stalls it once, laughs, and gets it moving. The father is glad and uneasy at the same time. He is teaching a skill he loves, and he is not certain his child will be able to afford to use it. A new car now sells, on average, for close to fifty thousand dollars, and the cheap, simple models he learned on have nearly vanished from the showroom. He is not against the future. He would like his child to keep the choice he had: to own a car, and to drive it.

What the Argument Is Actually About

The argument is not that the Constitution forbids the autonomous truck. It forbids no company building one, and it asks no shipper to refuse driverless freight and no rider to refuse a robotaxi. The federal government may set safety and emissions standards, and it may regulate the autonomous vehicle as the new technology it is. What it may not do is use that power to price the human driver's lawful alternative off the road — the patent-expired truck, the patent-expired car — until the autonomous platform is not the better choice but the only affordable one.

The autonomous vehicle is not a different machine. It is the surveillance-equipped vehicle with the human removed. The connected-telematics layer described in the chapter *The Surveillance Premium* is the same layer an autonomous fleet uses to dispatch its trucks, monitor them, and update them. Federal mandates did not build

the autonomous vehicle. They normalized the platform it runs on, and they made every new vehicle carry that platform. The human trucker who bought a compliant new rig helped pay to make standard the machine now offered as his replacement.

The Per-Mile Economics

Begin with the truck. A new Class 8 sleeper tractor costs between \$170,000 and \$195,000 in 2026, and a working truck covers roughly 750,000 miles before it leaves line-haul service. Spread across those miles, the truck alone costs about twenty-three to twenty-six cents a mile. The American Transportation Research Institute puts the fully loaded operating cost of a tractor-trailer — fuel, maintenance, tires, tolls, insurance, the truck, and the driver — at \$2.26 a mile in its 2025 analysis. Of that, the driver's wages and benefits come to roughly ninety-nine and a half cents.

Now change one thing. Suppose the owner-operator could buy, new, an American-remanufactured truck built on a pre-2000 mechanical platform whose patents expired decades ago — a 1999 Peterbilt 379 rebuilt around a new Caterpillar 3406E certified to the 1999 emission standards, or its Kenworth W900 equivalent with the Cummins N14. The projected price is roughly eighty thousand dollars. That figure is a projection, and it is offered as one; no such market exists yet, because the regulatory wall does not permit it. Spread across 750,000 miles, it comes to roughly eleven cents.

The remanufactured truck removes twelve to fifteen cents a mile of capital cost. It also sheds the maintenance overhead of the post-2000 emissions hardware the older platform never carried — the diesel particulate filter, the

selective catalytic reduction system, the diesel exhaust fluid, the cooled exhaust gas recirculation. On the capital saving alone, the fully loaded cost falls from \$2.26 to roughly \$2.11 a mile. The driver keeps his ninety-nine-cent wage. That wage is not the inefficiency in the number. It is his living, and it is the point.

The Cost Calculation With Remanufacturing Patent-Expired Trucks

Cost per Mile	Operation	Comment
\$2.26	Start	ATRI 2025 fully loaded cost — fuel, maintenance, tires, tolls, insurance, truck, driver
\$0.23–\$0.26	Of which	Capital cost of a new \$170,000 to \$195,000 tractor spread over 750,000 miles
\$0.11	Substitute	Remanufactured pre-2000 platform, projected \$80,000, over the same 750,000 miles
\$0.12–\$0.15	Less	Capital saving from the substitution
\$2.11	Total	Fully loaded cost, remanufactured truck — capital saving alone; the emissions-hardware maintenance it also sheds (DPF, SCR, DEF, cooled EGR) is not counted here
\$0.995	Memo	The driver’s wages and benefits per mile, carried unchanged in both totals — not the inefficiency in the number

What the Autonomous Platform Actually Costs

The autonomous truck is sold on a single number: the fee per mile. Aurora, running freight on Texas interstates without a driver in the cab, charges carriers a Driver-as-a-Service fee in the range of eighty-five cents a mile. Set beside the human trucker's \$2.11, eighty-five cents looks decisive. It is not a cost. It is a price.

The cost of operating an autonomous truck is the sum of four things the fee is meant to cover: the sensor and computing stack, which Goldman Sachs Research puts at \$125,000 to \$150,000 per truck; the remote-operations center from which specialists watch the driverless fleet, increasingly from overseas; insurance, which runs well above the premium on a human-driven truck because the claims record is thin and the liability of a crash with no driver is unsettled law; and the platform's margin.

A fifth cost sits inside the first and is worth naming separately. Every component of that stack — the lidar, the radar, the compute, the software — is under live patent, and every truck built carries the royalty. The mechanical platform the human trucker drives shed its royalties decades ago when its patents expired. The autonomous truck is the one vehicle on the road whose cost structure the Patent Clause has not yet begun to relieve.

Add them. In May 2026, Goldman Sachs Research put the all-in operating cost of an autonomous Class 8 truck, before corporate overhead, at roughly \$8.56 a mile — more than three times the cost of a human-driven truck and more than four times the remanufactured truck described here. The eighty-five cents Aurora charges does not begin to cover the \$8.56 it costs Aurora. The gap is

paid by investors funding the platforms below cost to take the market before the bill arrives. A price held under cost by outside capital is a price that must, in time, rise to meet it.

The Cost Calculation For The Autonomous Platform

Cost	Per Unit	Operation	Comment
\$125,000—\$150,000	Truck	Add	Sensor and computing stack (Goldman Sachs Research)
Not disclosed	Mile	Add	Remote-operations center; specialists monitoring the fleet, increasingly from overseas
Not disclosed	Mile	Add	Insurance, well above the human-driven premium — thin claims record, unsettled liability
Not disclosed	Truck	Add	Patent royalties on the stack: lidar, radar, compute, software, all under live patent
Not disclosed	Mile	Add	Platform margin
\$8.56	Mile	Total	Goldman Sachs Research, May 2026 — all-in operating cost before corporate overhead
\$0.85	Mile	Less	Aurora's Driver-as-a-Service fee — the price, not the cost

Cost	Per Unit	Operation	Comment
\$7.71	Mile	Gap	Paid by investors funding below cost; a price held under cost must eventually rise to meet it

The 2028 Crossover

The most serious objection to this argument arrives on a schedule, and it will not be dodged. Goldman Sachs Research projects that the cost of autonomous trucking will fall below the cost of a human-driven truck around 2028, as sensor and computing prices drop, and continue falling toward roughly \$2.03 a mile by 2035. The arithmetic of the projection is not disputed here. Its premise is examined below.

But read what it is measured against. Goldman pairs its falling autonomous figure against the conventional new truck, and puts that truck’s cost at \$2.55 a mile — higher than the \$2.26 used here. The crossover the analysts foresee is a crossover with that truck. It is not a crossover with the remanufactured truck, running near \$2.11 a mile with no emissions overhead and no investor subsidy beneath it.

Read also when the remedy arrives, and what the projection assumes. A case filed now and argued through the courts reaches a Supreme Court ruling somewhere between 2030 and 2033. By the earlier date, on a straight line between Goldman’s own endpoints, the autonomous truck costs about \$2.40 a mile; by the later date, about

\$2.18. The remanufactured truck enters at \$2.11, before the emissions-hardware maintenance it sheds is counted. It undercuts the platform on either date. The margin is wide in 2030 and thin in 2033, and the thin margin is the case for filing now rather than later.

The projection also assumes something a ruling on either date takes away. Sensor and computing costs fall on learning curves, and learning curves run on volume. Goldman's falling line presumes a growing autonomous fleet, accumulating the miles and the orders that push down the price of lidar and compute, bought with investor capital at \$8.56 a mile against an 85-cent fee. Put the remanufactured truck on the road in 2030, and the carrier who can buy a new American rig for \$80,000 has no reason to lease a driverless platform. The freight the platforms need to reach scale goes to human drivers instead. Volume that does not arrive does not lower cost. Capital that sees no path to a price above cost does not renew. A ruling in 2033 checks a fleet that has had five more years to grow; a ruling in 2030 checks one that has barely begun. Forecasting is not a hard science, and this forecast rests on the wall staying up.

Two limits belong in the same breath. Bankruptcy ends companies, not technology. A failed platform's patents would pass out of the estate, and its code would wait for costs to fall. And costs would keep falling on other industries' volume: the lidar and compute in a phone, a drone, or a state-funded fleet abroad. So the claim here is not that the autonomous truck disappears. It is that the remedy takes away the head start the wall gave it, and leaves it to beat a fairly priced human driver on cost, in the open, on its own money. That is the contest the Constitution permits. The one it forbids is the contest fixed by pricing the human off the road first.

The Safety Premise

The crossover projection rests on a premise it does not itself test: that by the time the cost lines meet, the autonomous platform will be operationally safe enough for deployment at the scale assumed. From federal crash data, the human driver in the United States is roughly 99.999819 percent crash-free on a per-mile basis — about one crash every five hundred fifty thousand miles. An autonomous vehicle would have to clear that bar merely to match him. The bar is set by the driver the technology proposes to replace.

Missy Cummings directs the Mason Autonomy and Robotics Center at George Mason University and is a former senior safety advisor to the National Highway Traffic Safety Administration. She has named a malfunction the industry has not solved. Phantom braking is the unexplained activation of an autonomous vehicle's emergency-stop systems; shadows, weather, and animal silhouettes can trigger it, because the software treats them as hazards. She has told the New York Times that no solution is on the horizon and that the problem will not be addressed soon, because nobody wants to admit it is happening. Its consequences are likely sharper in a Class 8 tractor-trailer at highway speed. The argument does not rest on that doubt — the per-mile economics work on their own — but a wall built to make the human alternative unaffordable should not be tightened in reliance on a safety claim that has not been proved.

The American Remanufacturing Geography

Where would these trucks come from? From American shops, built by American workers, on platforms Americans

already know: the Kenworth W900, the Peterbilt 359 and 379, the Freightliner Classic. The engines that power them — the Detroit Diesel Series 60, the Cummins N14 — rest on foundational patents now in the public domain. The legal core of the machine is free for anyone to build.

The grandfather exemption this book proposes is keyed, in general, to the patent calendar. For trucks the line is drawn tighter and fixed: the remanufactured semi is a pre-2000 platform, built to pre-2000 standards. A pre-2000 engine escapes the entire post-2000 emissions architecture — cooled exhaust-gas recirculation in 2004, the diesel particulate filter in 2007, selective catalytic reduction and diesel exhaust fluid in 2010 — and the capital and maintenance cost that follows. It is also, by long report from the people who drive them, the truck working drivers most want. The line is not drawn by this book. It is drawn by the trucker, and the book follows it.

One state regulation must be named here and answered in the same breath. California's Truck and Bus Regulation bars trucks with pre-2010 engines from operating on California roads, and roughly seventeen other states have adopted some or all of California's heavy-duty rules. Read alone, that looks like a permanent ceiling on the remedy. It is not. It is a wall that stands only until the Supreme Court rules. The Court has already held, in *Sears v. Stiffel* and *Bonito Boats v. Thunder Craft*, that a state may not re-monopolize what federal patent policy keeps free for the public. A state regulation whose effect is to forbid the patent-expired vehicle is the *Bonito Boats* situation in a new sector. Once the Court rules for the forgotten half of the Patent Clause — the half that begins when the "limited Times" the Constitution grants have run — the Supremacy Clause forecloses any state from re-cancelling the commons the ruling protects.

The Robotaxi, the Rideshare Driver, and the Family Car

The wall does not stop at the loading dock. Consider the second American, the independent driver who carries passengers for a living. The machine built to replace her is already on the road. The pitch is the trucking pitch again, and it rests on the same confusion of a price with a cost. A full-cost study from the Massachusetts Institute of Technology found that a conventional human-driven vehicle costs about seventy-two cents a mile to operate, while a robotaxi — once licensing, insurance, cleaning, remote oversight, and the low utilization of the fleet are counted — costs between \$1.58 and \$6.01. The robotaxi is not cheaper than the human driver. It is two to eight times more expensive. Published robotaxi fares sit inside that range.

Now the third American, the family that wants only to own a car and drive it. The affordable American car is disappearing. The average new vehicle now sells for roughly \$49,000, and the number of new models priced under \$25,000 has fallen from about a dozen in 2012 to four. By one industry analysis, a third of American households can no longer afford any new vehicle at all, and the share of new-car buyers earning under \$100,000 has fallen from half the market to barely a third in six years. A trade analysis of the collapse names the causes without flinching: tariff protection, the industry's pivot to high-margin trucks and sport-utility vehicles, and a regulatory structure that punishes automakers for building small, cheap cars.

The Move to Outlaw Human Driving

A further step follows, and it is no longer hypothetical. A current of advocacy now argues that human driving should not merely be priced down but prohibited — that manual driving will be outlawed on public roads, the human permitted to operate a vehicle only on a closed track, as a sport. A peer-reviewed paper in *Science and Engineering Ethics* argues the case directly: once autonomous cars are meaningfully safer than human drivers, manually driven cars should for moral reasons be banned from ordinary traffic. The reasoning deserves to be stated fairly. If the machine is safer, the person who insists on driving imposes a measurable risk on everyone around him, and the law restrains such risks all the time.

This chapter does not resolve that argument. It raises it, because the reader asked to invest in a remanufactured drivable car deserves to know the question is coming. What can be done here is to mark the constitutional ground the question would have to cross. Suppose the Court holds that the patent-expired vehicle may lawfully be built. A right to build a car that may not lawfully be driven is not a right at all. It is the same cancellation reached by a longer road. A regulation that permits the manufacture and forbids the operation would re-take, in its second move, the very property the ruling protected in its first. The move to outlaw human driving is real, it is documented, and it is challengeable. It is not a settled future. It is a contested one, and the Constitution gives the human driver ground to contest it from.

One asymmetry in the ban proposal deserves naming. What its advocates ask is that once the machine is safer than the person, the person must yield the road. Run their argument backward. Every mile the machine drove on a

public street before it could claim that superiority was a mile driven under a standard they would now apply to the human alone — and it was driven among drivers who were never asked. The newcomer was excused from the rule while it was losing. The rule is urged on the incumbent the moment the newcomer starts winning. Equality under the law is the principle this book has pressed at every turn, and it does not bend because the party seeking the exemption is a machine.

Who Benefits

All this expensive is not an accident. It has beneficiaries. The autonomous-platform companies benefit directly: every dollar the wall adds to the price of a human-driven truck is a dollar of headroom beneath their subsidized fee. The largest carriers benefit too, having the scale to absorb compliance cost and the lobbying presence to shape the rules that impose it. The wall does not require bad faith from every participant to stand; ordinary commercial self-interest, lawfully pursued, is enough to hold it up. But some who built it understood exactly what it would do, and they bear responsibility for having built it. The mechanism does not need their intent. That does not absolve the intent that was there.

And the human cost has a face. A driverless truck is not a truck with no human attached. It is a truck whose human has been moved off the road and into an office. A former driver sits at a desk before a bank of monitors, watching live camera feeds from many driverless trucks at once and taking control of any one of them over a remote link when it meets something its software cannot handle. The arithmetic is the whole point of the job. One driver in one cab gives way to one specialist watching a fleet. The ratio

is roughly six trucks to one specialist today; the platforms' own investor materials project a single specialist eventually overseeing as many as a hundred. Every step up that ratio is more drivers without work. The visitor in the pressed shirt on the gravel lot has that job.

The Bridge to the Foreword

This book carries a foreword by Gord Magill, a working trucker of three decades and the author of *End of the Road: Inside the War on Truckers*. His chapter on autonomous freight, "Truckzilla: Invasion of the RoboTrucks," documents this war from inside the cab — Aurora's first driver-out run on Interstate 45, the remote-support-specialist scheme and its one-per-hundred ambition, and the congressional bill written to clear the robot's road. Magill's book documents the war. This chapter supplies the mechanism for winning it. Roughly three and a half million professional truck drivers work in the United States, some nine hundred thousand of them independent owner-operators who own the truck they drive. The patent remedy is the concrete economic instrument by which those drivers keep it.

The Trucker, the Driver, and the Constitution

The last human driver is not one person. The phrase names a category, and three people stand inside it: the owner-operator who wants to keep his own truck, the independent driver who carries passengers for a living, and the ordinary motorist who wants to own a car and drive it. Not one of the three asks to be shielded from the future. Each asks only for the bargain the Constitution already wrote — that when the patents expire, the machine belongs to the public, and the public may build it, buy it,

and drive it. The wall is permanent against every actor below the constitutional level. It is not permanent against the Supreme Court, which stands above it.

CHAPTER 19

The Hindered (not Blocked) Sector: Pharmaceuticals and Medical Devices

This book is about the Blocked fifty-eight percent of the patent-bargain pie — vehicles, tractors, heating and cooling, appliances, and lighting, where federal manufacturing mandates make remanufacturing of twenty-year-old expired-patent platforms legally impossible. That sector has had no book-length constitutional treatment until now. The Hindered fifteen percent — pharmaceuticals and medical devices — is a different matter. Other authors have been mapping it for half a century. A short acknowledgment is owed, both as a courtesy to those who did the work and as a signpost for readers who want to follow the pattern into a sector this book does not address at length.

The defining feature of the Hindered sector is that the wall is porous rather than statutory. A generic version of an off-patent drug is legal to manufacture. What throttles its use is Food and Drug Administration labeling, off-label prescribing risk, medical-board discipline, pharmacy-board refusals, hospital-formulary exclusion, and the ordinary tools of pharmaceutical patent gaming — evergreening, pay-for-delay agreements, citizen petitions, and Orange Book listings. The harm pattern matches the durable-goods pattern. The public is denied affordable use of designs whose patents have expired. The enforcement mechanism is administrative and professional rather than statutory. The wall stands. It just stands by different means.

Half a Century of Documentation

The COVID-era literature is the most visible recent layer. Robert F. Kennedy Jr.'s *The Real Anthony Fauci* (Skyhorse, 2021) is the genre's commercial flagship, and its scope reaches back to the AIDS-era suppression of bactrim, dapson, fluconazole, and ribavirin in favor of patented AZT. Pierre Kory and Jenna McCarthy's *The War on Ivermectin* (2023) names the conflict in its subtitle as a war on a generic medicine. Peter McCullough and John Leake also wrote *Overcoming the COVID Darkness*, Justus Hope wrote *Ivermectin for the World*, and Steve Kirsch's *Doctored Results* treats the trial record itself. John Leake and Peter McCullough's *The Courage to Face COVID-19* (2022) documents the early-treatment protocol and the complex that suppressed it.

Behind the COVID books sits a much older cancer literature that established the pattern long before hydroxychloroquine and ivermectin. Suzanne Somers's *Knockout* (2009) reached number one on the New York Times list. G. Edward Griffin's *World Without Cancer* (1974) is the foundational text on laetrile suppression. Ralph W. Moss's *The Cancer Industry* (1980) is the Sloan-Kettering insider account, and Daniel Haley's *Politics in Healing* (2000) walks through twelve documented cases. Paul E. Marik's *Cancer Care* (second edition, 2024) is the current standard, covering ivermectin, mebendazole, fenbendazole, doxycycline, curcumin, metformin, and propranolol. One legal title deserves separate mention: Richard A. Jaffe's *Galileo's Lawyer* (2008), a defense attorney's account of three decades of litigation against state medical boards and the FDA, is the closest thing in this literature to a procedural manual.

The medical-device side has its own literature, anchored in the right-to-repair movement. The U.S. Public Interest Research Group, iFixit, and the International Association of Medical Equipment Remarketers and Servicers have documented how original equipment manufacturers extend monopoly power past patent expiration through copyright, trade secrets, and post-sale contractual controls on service manuals, diagnostic codes, and replacement parts. The FDA's own 2018 report on third-party servicing concluded that independent servicers are critical to American healthcare and that adverse events from independent repair run under one percent of reported incidents. The debate runs on antitrust grounds, right-to-repair grounds, and FDA-guidance grounds. It does not run on Patent Clause grounds.

How the Hindrance Works

The catalog of patent-expired and non-patentable items in this sector is large. Some are pharmaceuticals, some devices, some procedures using the patient's own tissue. The hindrance comes from many angles — FDA labeling, the FDA Office of Criminal Investigations, the Department of Justice, state medical boards, state pharmacy boards, state attorneys general, Medicare coverage denial, and private-insurer coverage denial. Often more than one acts at once. Each item in the catalog has a favorable peer-reviewed literature and a contrary one, and the Endnotes cite the principal papers on both sides for each. This chapter takes no position on clinical efficacy. Its subject is the mechanism by which an off-patent or unpatentable option is kept from the patient, whatever the merits of the option. Four examples show the range.

Hydroxychloroquine is an antimalarial first patented in the 1950s, off-patent for decades, and on the World Health Organization Essential Medicines List. During the 2020 emergency, multiple state medical boards issued advisories against off-label prescription, multiple state pharmacy boards refused to fill prescriptions, and the FDA revoked its Emergency Use Authorization in June of that year. The drug's status as a generic was unchanged. Its availability for the indication patients sought collapsed anyway. Ivermectin, discovered in the 1970s and recognized with a Nobel Prize in 2015, followed the same pattern: medical-board advisories, pharmacy refusals, and an FDA consumer advisory whose social-media posting included the line that humans are not horses.

Platelet-rich plasma is the patient's own blood, separated and re-injected. By definition it is not patentable. A Medicare local coverage determination excludes it for tendinopathy, osteoarthritis, rotator-cuff tears, plantar fasciopathy, and chronic low-back pain as not medically reasonable and necessary; the major private insurers follow Medicare's lead. Patients pay five hundred to two thousand dollars per injection out of pocket. The hindrance is a coverage decision. The procedure is legal and routine. Prolotherapy, a dextrose-injection technique whose active ingredient is medical-grade sugar in saline, is excluded on the same grounds. The substance has no patent. The procedure has no patent. The bill goes to the patient.

A word on what that experience was. The author has had all three procedures — autologous stem-cell injection, platelet-rich plasma, and dextrose prolotherapy — and found PRP the most cost-beneficial of the three. The occasion was a complex vertical and lateral tear of the left meniscus at sixty-two, for which conventional arthroscopy was recommended — a course that, for patients past fifty,

carries a documented rate of conversion to knee replacement within a few years. He took the injections instead, needed them for three or four years, and has section-hiked the Pacific Crest Trail in the twelve years since. The knee behaves as though nothing happened. One patient is not evidence, and this chapter takes no position on clinical efficacy. What the case illustrates is the mechanism: not one of the three is patented, all three were paid for out of pocket, and the coverage decision that made that necessary is the hindrance this chapter describes.

Methylene blue is the textbook capture case in the catalog. Synthesized in 1876 and FDA-approved for methemoglobinemia, it has seen a recent off-label surge for cognitive and mitochondrial use. In the 2014–15 ProveyBlue episode, Provepharm, the maker of the approved branded formulation, secured an eight-month near-monopoly on the American market by petition, and the resulting litigation is reported as *Akorn v. Provepharm*. The molecule has been off patent for more than a century. What produced the monopoly was the regulatory pathway, not the patent.

Three further items fill out the range. Ozone therapy runs against 21 C.F.R. § 801.415, which classifies ozone as a toxic gas with no useful medical application; the FDA has prosecuted clinic operators since 1991, and state medical boards have suspended licenses, most famously Robert Atkins's in New York. Royal Raymond Rife's electromagnetic frequency devices, built in the 1920s and 1930s, have never been FDA-approved, and the hindrance there is multi-agency — the FDA Office of Criminal Investigations has prosecuted operators since the 1990s and the Postal Inspection Service has acted on mail-fraud grounds. Homeopathic preparations are mostly

unpatentable, being extreme dilutions of nonproprietary substances, and the FDA's 2017 enforcement-policy revision tightened its stance on products bearing disease claims while the Federal Trade Commission acted separately on advertising.

The most thoroughly documented item is the orthopedic stem-cell procedure that Christopher J. Centeno and John R. Schultz developed in Colorado beginning in 2005, using the patient's own bone-marrow-derived mesenchymal stem cells. The FDA argued the cultured-cell version was a drug under the Federal Food, Drug, and Cosmetic Act. The physicians disagreed and fought the case. *United States v. Regenerative Sciences* ran from a 2008 warning letter through a 2010 federal complaint, a 2012 permanent injunction, and a 2014 affirmance by the D.C. Circuit — the federal court directly below the Supreme Court. The court ruled the cultured cells were a drug. The underlying commercial logic is this chapter's broader theme. Biological material that comes from the patient's own body cannot be patented, cannot be mass-produced, and cannot be sold by a pharmaceutical company at scale — which is precisely why the regulatory apparatus treats it the way it does.

The Constitutional Gap

The gap is the same across all of it. No published source argues that post-expiration FDA, medical-board, pharmacy-board, or Medicare-coverage hindrance of off-patent drugs constitutes a violation of the Patent Clause or the Takings Clause. The medical-device right-to-repair literature is similarly anchored in statutory and regulatory grounds. One academic paper makes an inferential constitutional link. The connection has not been pressed in

court. The Patent Clause has not been deployed as a lens through which post-expiration regulatory control of either drugs or devices is examined.

Considered together, the books named here and the catalog above are the work of half a century of physicians, attorneys, journalists, and patients. They have established the regulatory mechanics, the financial incentives, and the casualty count. What none of them does — what none of them set out to do — is name the harm as a violation of the Patent Clause's limited Times guarantee. The constitutional frame this book brings to the durable-goods sector applies without strain here as well. A public-domain asset, paid for in advance with the disclosure-for-monopoly bargain, is being denied to the public after the monopoly has ended.

One consequence follows, and it bears on the chapters ahead. A ruling for the Blocked sector would not stop at the Blocked sector. The holding that a federal regulatory architecture may not convert an expired patent into a permanent monopoly is stated at the level of the Patent Clause, not at the level of trucks and appliances, and it would be available to any plaintiff whose injury fits the same shape. The hindered sector's injury fits it. What would follow is a second wave of suits, brought by different plaintiffs against different agencies, testing whether labeling policy and coverage denial do to an expired drug patent what manufacturing mandates do to an expired engine patent. Those suits are not this book's case, and the plaintiffs the following chapters recruit are not drawn from this sector. But a reader deciding whether the argument is worth carrying should know how far a favorable ruling would reach.

Why This Book Belongs on the Hindered Sector's Shelf

A reader who came to this book from the medical fight may reasonably ask what a litigation strategy aimed at energy, safety, and emissions agencies has to do with a doctor disciplined for prescribing an off-patent drug. The answer is that the strategy is a template, and the template does not care which agency built the wall. Both prongs of this book's argument run on the same two facts: a design whose patent has expired belongs to the public, and a federal regulatory architecture that keeps the public from using it has taken something the Constitution says the public owns. Neither fact changes when the design is a molecule rather than an engine, or when the barrier is a labeling rule, a pharmacy-board refusal, or a coverage denial rather than an emissions standard. What changes is the defendant.

That is why this book can serve the hindered sector as a catalyst rather than a brief. The plaintiffs recruited in the chapters that follow will sue the EPA, NHTSA, and the Department of Energy, because the durable-goods wall is statutory and the injury is on a ledger. A sibling movement, drawn from the physicians, patients, and attorneys whose half-century of documentation this chapter has just surveyed, could take the same two prongs and aim them where its own wall stands: at the Food and Drug Administration, at the state licensing boards, at the pharmacy boards, and at the coverage policies that price an unpatentable procedure out of reach. That case will be harder in places. Manufacturing a generic is legal, so the taking is of the right to use rather than the right to make, and a state board is a state actor with defenses a federal agency lacks. But the doctrine is the same doctrine, and

the first favorable ruling in the durable-goods case would be the strongest citation the sibling movement could hold. A public fight over the regulatory wall around trucks and tractors will draw attention that the wall around medicine has never had. The hindered sector should be ready to use it.

The constitutional lens is what remains to be added here. This book supplies the lens. Carrying it into a courtroom is a job for another plaintiff, and perhaps another book.

CHAPTER 20

Sovereign and Public-Fiduciary Plaintiffs

The Cleanest Injury in the Book

A household that pays \$72,495 for a pickup truck has been injured. The injury is real, and earlier chapters have measured it. In a federal courtroom, though, that injury is nearly useless. It is shared by every household that buys a vehicle, in roughly the same way. That means it belongs to no one in particular, and a court cannot hear it.

This is the wall that standing doctrine builds. A constitutional wrong does not correct itself. Someone has to carry it into a court, and the law is particular about who that someone may be. It is not enough to have been harmed. The plaintiff must have been harmed in a way the court can see as his own.

Seven plaintiffs clear that bar with room to spare, and they share a single feature. Every one of them is a government — a state, a tribe, or an arm of one — and every government buys things. It buys patrol cars and school buses. It buys fire engines and transit coaches. It buys road graders and ambulances. It buys them in a market the regulatory wall has rigged, and it buys them year after year, out of a budget that is a matter of public record. When the wall inflates the price, the overpayment lands on a named line in a named ledger. That is the difference between an injury a court can hear and one it cannot.

Call it the procurement-injury theory. It is the cleanest claim in the book. The harm to a county that overpays for

its patrol fleet is not larger than the harm to the household that overpays for its pickup. It is the same harm. What makes it useful is that it is particular. It has a name on it, a figure attached to it, and a federal regulation it can be traced back to. A household grievance multiplied by a hundred million households is still a household grievance. A county budget is a plaintiff.

The scale of the wrong is not in dispute here; earlier chapters established it. A regulatory wall built since 1970 has transferred something on the order of \$22 trillion from American households, roughly \$3,385 a year for the average household. The question here is narrower. Of all the parties that figure has injured, which ones can take it to court? Seven of them can.

One boundary belongs at the start. The question here is whether these plaintiffs can sue, not why any would want to. Motive has its own chapter, *Why You Want to Be the Plaintiff*; standing is only the legal right to be heard.

The Bar, and the Four Ways Through It

The bar is *Lujan v. Defenders of Wildlife* (1992). A federal plaintiff must show an injury in fact that is concrete and particularized, a causal line from the challenged action to that injury, and a remedy the court can grant. *Clapper v. Amnesty International USA* (2013) polices the first of those: an injury must be concrete and imminent, not a fear of harm that may come later. The generalized grievance — a harm shared by the whole public in substantially the same way — fails there.

Four routes clear that bar, and the first two belong here. Sovereign standing belongs to a state or a tribe suing to protect its territory, its citizens, or its sovereign interests.

Public-fiduciary procurement standing belongs to a government body that buys the regulated machine out of a public budget. The other two — occupational-injury standing and its associational form — belong to the chapter *Direct and Associational Plaintiff Categories*.

Sovereign standing carries an advantage no private plaintiff has. In *Massachusetts v. EPA* (2007), the Supreme Court held that a state suing to protect its citizens and its territory is entitled to special solicitude in the standing analysis. It rests on the state's status as a sovereign that surrendered certain powers to the federal government at the founding and retains an interest in how those powers are exercised.

The Seven Categories

State attorneys general hold three independent theories at once. Every state purchases fleet vehicles, equipment, and supplies at prices the wall inflates, which is a direct proprietary injury. Every state may sue *parens patriae* — as parent of its people — on behalf of citizens bearing a diffuse economic harm too small for any single household to litigate. *Alfred L. Snapp & Son, Inc. v. Puerto Rico* (1982) is the controlling authority. A sovereign's quasi-sovereign interest includes the health and well-being, physical and economic, of its residents in general. And every state has a sovereign interest in ensuring that federal regulation does not nullify a constitutional provision written to benefit its economy. Any one of the three survives a motion to dismiss. The office also has a well-worn history of multi-state litigation. Tobacco and opioids were both launching points for the attorneys general who led them.

Indian tribes hold the same three grounds, a stronger claim to the sovereign one, and a fourth ground no other plaintiff has. Tribal sovereignty predates the Constitution, and the government-to-government relationship may support the same special solicitude the Court extended to a state. The fourth is the federal trust responsibility — a duty, recognized across more than a century of federal Indian law, to act in the interest of tribal nations. A regime that inflates the price of every vehicle a tribal government buys sits uneasily beside that duty. The doctrine need not be settled to make the point; it is one more reason a tribal plaintiff's standing is secure. A tribal government buys vehicles, equipment, and machinery at the wall's inflated price, and may sue on behalf of members who pay the same \$3,385 a year every American household pays. That burden falls hardest on rural and reservation economies, where a blocked-tier pickup consumes a far larger share of household income than in a wealthy suburb.

The irony is one no other plaintiff can claim. In Guatemala, Maya families buy new vehicles built on patent-expired designs for around \$9,000. In Kenya, Maasai communities buy them for around \$18,000. Indigenous people in the world's poorer nations have the benefit of the patent bargain. Indigenous people in the nation that wrote the Patent Clause — the nation whose Constitution grants a patent for “limited Times” and then owes the design to the public — do not.

The recruitment facts sit alongside the doctrinal ones. Three hundred forty-five federally recognized tribal nations exist in the contiguous states, distributed across ten federal circuits. That distribution maximizes the chance of a circuit split, the single most reliable path to Supreme Court review. Each is a sovereign decision-maker, and no central authority can veto what one of them decides. Tribal

nations are among the most active litigants against the federal government in the country, and the Native American Rights Fund carries such cases at no cost to the tribe. The case needs one tribal council willing to pass a resolution.

State governors hold the proprietary and *parens patriae* grounds through the executive branch, and control the procurement apparatus directly. A governor does not litigate, though: one who wants this case filed directs the state's attorney general to file it.

Sheriffs in elected-sheriff states hold the cleanest procurement injury of all. A sheriff's department is a direct purchaser of vehicles, and the wall inflates the price of every patrol car the county buys. The posture is proven: in *Printz v. United States* (1997), two sheriffs from small counties sued the federal government and won. *Printz* decided a Tenth Amendment question and said nothing about the Patent Clause; it is cited here for the posture, not the doctrine.

That posture depends on the sheriff being elected, and it does not hold everywhere. Alaska has no county sheriff at all. Connecticut abolished the office in 2000. Delaware's sheriffs serve civil process but run no law-enforcement fleet, and so have no procurement injury to claim. Hawaii and Rhode Island appoint theirs, and an appointed sheriff can be removed by the authority that appointed him — he lacks the political independence a multi-year constitutional challenge demands. Forty-five states qualify, 3,091 jurisdictions among them — counties, Louisiana parishes, and Virginia's independent cities. The precision is doctrinal: the independence that makes the elected sheriff a clean plaintiff is exactly what the appointed sheriff lacks.

School districts buy bus fleets on replacement cycles out of budgets that are public record, and the procurement line that eats a gifted-and-talented program is documented in board minutes. Public transit authorities replace coaches that now run from roughly half a million dollars into seven figures apiece. Volunteer fire departments in counties too small to fill a tax base buy engines the wall has priced up from something near \$250,000 to something near \$750,000.

One distinction belongs on the fire departments. Those organized as special districts or county departments are governments, and they hold this standing. Those organized as 501(c) charitable corporations do not, or hold a weaker version of it, because the injury runs to a charity rather than to a public budget.

Why the Procurement Injury Is the Strongest

Three features make it strong. The injury is documented before the case is filed, in procurement contracts and budget line items no plaintiff has to reconstruct, and it recurs every budget year — the concrete, imminent injury *Clapper* requires. The causal line is short: the federal regulation makes the lower-priced machine impossible to build new, and the buyer pays the difference. And the remedy is a declaratory judgment, which a court has full authority to issue and which requires no individual accounting of anyone's loss.

A skeptic will say no government can prove that injury without auditing its whole budget. Standing does not require an audit. It requires a plausible, concrete injury, and a population-scaled estimate with a disclosed method supplies one. The plaintiff's own procurement records settle the precise figure later.

Why Your Category Is Not Here

A reader whose own government body is not among the seven is owed a direct answer, and the answer is doctrinal rather than dismissive. The seven are the categories where the procurement injury is largest, best documented, and most clearly traceable to a federal rule. The rule that generated the list is simple: a public body qualifies when it buys regulated durable equipment at wall-inflated prices out of a public budget, and the overpayment appears on a documented line. A municipal government, a port authority, a state university system, a public hospital district — any body that meets the rule belongs on the list whether or not it appears above.

Some bodies will not meet it. The mayor of a council-manager city does not control procurement; the injured party is the city, and the city is the proper plaintiff. A charitable fire company has no public budget to point to. Neither is left without a role. A voice that cannot clear the bar as a plaintiff can still be heard as an amicus, and the effort is strongest when every name on the docket clears it and every other supporter arrives on the brief.

The regulatory wall has held against every actor below the constitutional level — against the household, against the would-be manufacturer, against the consumer with a generalized grievance and no way into court. It was built to hold against them. What it was never built to withstand is a plaintiff with standing and a court with the authority to look over the top of it. The Supreme Court sits above the wall. The only thing still missing is the name on the complaint.

CHAPTER 21

Direct and Associational Plaintiff Categories

The Business Is the Machine

The companion chapter set out the federal courthouse door and the test for passing through it. It named four ways a plaintiff can clear the generalized-grievance bar of *Lujan v. Defenders of Wildlife*. The first two, sovereign standing and public-fiduciary procurement standing, belong there. Here are the other two: occupational-injury standing, which belongs to a plaintiff whose trade the wall directly taxes, and the associational form of that injury, which a trade body carries on behalf of its members.

The eleven plaintiff categories described here share a single feature. For each of them, the regulated machine is not one cost of doing business among many. It is the business. A laundromat is a room of washing machines. An earth-moving contractor sells the use of an excavator by the hour. A restaurant is a kitchen of regulated equipment with tables in front of it. When a post-1970 federal rule inflates the price of that machine, it does not inflate an overhead line. It taxes the enterprise at its center. That is why these eleven categories can sue. The injury is not diffuse. It is specific, it is large, and it lands on an identifiable owner. The lead exemplar of the posture is the independent trucker, argued in full in the chapter *The Trucker's Burden*.

Two Routes to the Courthouse

A direct individual plaintiff and a trade association reach the federal courthouse by two different routes. Both are well worn, and in most fields the campaign should prepare both.

The direct route is simpler. An individual who buys the regulated machine — the laundromat owner, the landscaper, the earth-moving contractor, the rideshare driver — sues in his own name. He must show the three things every federal plaintiff must show: an injury in fact, a causal line from the federal rule to that injury, and a remedy the court can grant. When the machine is the business, all three are easy. The injury is the inflated purchase price he actually paid. The cause is the regulation that made the lower-priced machine impossible to build new. The remedy is a declaratory judgment a court has full authority to issue.

The associational route runs through *Hunt v. Washington State Apple Advertising Commission* (1977). A trade association may sue on behalf of its members when three conditions are met. At least one member must have standing in his own right. The interests must be germane to the association's purpose. And neither the claim nor the relief may require individual members to participate. The first two prongs are satisfied without difficulty by every trade body named here: each has members who are direct purchasers, and each exists by its own charter to lower its members' costs. The third prong does the work, and a government brief will press it hardest. A suit for money damages, measured member by member, can founder there, because proving each member's loss draws every member into the case. A suit for a declaratory judgment does not. It asks for a single ruling of law that applies to all

members at once. That is why the remedy this book seeks is declaratory rather than compensatory. It is what keeps the strongest plaintiff structure available.

One consequence orders what follows. A direct individual plaintiff escapes the third prong of *Hunt* entirely, because no membership roll exists for a defendant to exploit. Direct therefore ranks above pure-associational, and the strongest filing pairs the two: an owner who bought the machine, and an association that lends its name and its weight.

A second factor orders them as well: the statutory hook. The more precise the rule that governs the machine, the shorter the causal line a plaintiff must draw. Four hooks recur. The cleanest is 42 U.S.C. § 6295(o)(1), which names clothes washers as a covered product. EPA Part 1039, the Tier 4 nonroad diesel standard, governs earth-moving, construction, and farm equipment. EPA Part 1054 governs small spark-ignition engines. And 10 CFR Part 431 governs restaurants, hotels, and much of the hospital plant.

The Eleven Categories

Aside from truckers, discussed in their own chapter, earth-moving equipment owners next hold the cleanest direct-individual posture. The contractor buys the bulldozer, the excavator, and the loader outright under EPA Part 1039, and the foundational excavator patents — undercarriage, swing, hydraulic circuit — expired between the early 1990s and the mid-2010s. The price gap is the widest in the book. The rigging contractor's mobile crane runs on the same platform and stands on the same terms.

Healthy-lighting medical groups carry the strongest merits posture in the book. Edison's filament patents expired around 1900 and the bulb was made and bought freely for 110 years, until the 2007 Energy Independence and Security Act and the Department of Energy rules of 2022 and 2024 withdrew it from sale. The bodies here are health and science nonprofits: the Epilepsy Foundation, DarkSky International, the American Academy of Environmental Medicine, the BrightFocus Foundation, and the National Psoriasis Foundation. Their boards would join an amicus brief sooner than lead.

Hospitals are the highest-sympathy plaintiff and hold the largest per-institution exposure — sterilizers, laundries, HVAC, food service — nearly all under 10 CFR Part 431. The American Hospital Association speaks for roughly 5,000 of some 6,000 hospitals, but its policy relationships run through the funding agencies and it will not lead. State associations are more open.

Farming associations buy tractors and combines under the same Part 1039 hook, at multiples of the Indian price for comparable machines. The farm groups already fought Tier 4 in the rulemaking arena and lost, which is the doctrinal point: a Patent Clause challenge is not a re-run of that fight but an argument the rulemaking forum could not reach. The American Farm Bureau Federation would be receptive; the sustainable-agriculture and family-farm coalitions are mixed.

Coin-laundry and laundromat operators hold the cleanest statutory hook in the economy. Section 6295(o)(1) names clothes washers by name, and a laundromat is, in plain physical fact, a room of those covered washers. No gap sits between the regulated equipment and the business model for a government brief to pry open, and a full retool

is a six-figure capital event. The Coin Laundry Association is the trade body for an industry of roughly 30,000 storefronts — genuine but resource-limited, and free of political baggage in either direction.

Restaurant owner associations offer the broadest network and the least dramatic per-unit injury. Roughly a dozen categories of kitchen equipment fall under 10 CFR Part 431, so no single machine is a smoking gun and the case is carried by a worked build-out well into six figures. The National Restaurant Association represents more than 380,000 locations; its disposition is favorable but its attention is on labor and tax.

Construction contractor associations broaden earth-moving from a direct posture to an associational one under the same Part 1039 hook, adding institutional weight rather than new doctrine: the Associated General Contractors of America at roughly 28,000 firms, the Associated Builders and Contractors at roughly 23,000, and the National Association of Home Builders at some 140,000.

Hotel and lodging operators replicate a per-property exposure across roughly 60,000 properties, much of it under Part 431. This is a tighter case than the laundromat's: the hook is diffuse rather than concentrated in one named product, and the cost disappears into a nightly room rate instead of appearing to the public as a price. The American Hotel and Lodging Association represents more than 30,000 members.

Apartment and multi-family building owners face compliant HVAC and water-heater requirements at every unit turnover. One nuance must be handled honestly, because a government brief will raise it: the cost arguably passes through to tenants in rent. That is an argument about

damages, not a defect in standing. The injury in fact is the capital outlay the owner makes at purchase. The National Apartment Association, with roughly 95,000 member companies, and the National Multifamily Housing Council have pressed for years to have housing-cost rules reviewed.

Rideshare and taxicab drivers hold the purest direct posture of all. The driver buys, in her own name and on her own loan, the regulated vehicle that is her entire business. Well over 1.7 million work in the United States. The difficulty is not standing but organization: the field has no unified national trade body, which makes the direct-individual filing the natural vehicle here.

Landscaping and grounds-maintenance operators buy the commercial mower, blower, and chainsaw under EPA Part 1054. The per-unit cost is the smallest in the list, and the category should be sized accordingly rather than inflated. What gives it value is a track record: the National Association of Landscape Professionals, speaking for roughly 100,000 professionals, has actively opposed small-engine emissions rules and state gasoline-equipment bans.

Why Your Category Is Not Here

A reader whose own trade is not among the eleven is owed a direct answer, and the answer is doctrinal. A plaintiff must clear the *Lujan* bar by showing a particularized injury rather than a harm shared by the whole public in roughly the same way. Two groups that feel the wall acutely cannot clear it.

The first is the individual car buyer. A household that paid far too much for a pickup truck has suffered a real loss, but

one shared by tens of millions of households in substantially the same way. *Lujan* is explicit that multiplying a generalized grievance by the size of the country does not convert it into the particular injury the Constitution requires. The household's injury is genuine. It is simply not, standing alone, a justiciable one.

The second is the commercial fleet buyer, and the car-rental industry is the clearest example. A rental company buys well over a million vehicles a year and bears a large absolute regulatory premium. Its standing problem is structural. It buys the same consumer good every household buys, and it recovers the premium through its daily rates, so a government brief will argue it has not been injured in the relevant sense. A framing cost compounds it. A multinational rental corporation suing over its fleet-purchase bill reads as corporate rent-seeking rather than as vindication of the public domain.

Neither group is turned away. Each is redirected. A party that cannot stand on the docket can still file an amicus brief and supply industry data the named plaintiffs cannot generate on their own; the car-rental industry is a valuable amicus and a valuable data source.

The author has proposed — though, as of this writing, not yet formed — a coordinating nonprofit, provisionally named Expired Patent Commons, to organize that participation. Nothing here depends on it. The exclusion test is not there to shrink the campaign. It is there to make it legible.

The Plaintiff List Is Not Closed

The eleven are the named plaintiffs, not the universe of them. The rule that generated the list is simple. A trade

association qualifies when its members must buy or operate regulated durable equipment at wall-inflated prices, and when those injuries are concrete, traceable, and redressable by declaratory judgment. Name the machine, name the regulation, and the category writes itself.

Cousin postures sit just offshore. The independent tug and barge operator owns his vessel outright, built around a marine diesel governed by EPA's Part 1042 rather than the nonroad Part 1039, and the American Waterways Operators could carry the associational claim under *Hunt*. The short-line railroad runs switcher locomotives whose platforms long predate EPA's Part 1033, and the American Short Line and Regional Railroad Association could do the same. The commercial fishing fleet sits in the same Part 1042 orbit. Each is one machine, one regulation, and one association away from a complaint.

A reader who finds his own trade missing should not read the omission as a ruling. The named categories are exhibits, not boundaries. Every industry that recognizes itself in the pattern strengthens the docket, one particularized injury at a time.

That is the standing question answered, eleven times over and with its boundaries marked. The wall built by post-1970 regulation has never been the seamless thing it appears to be. It holds against the household and the generalized grievance. It does not hold against a plaintiff with a particularized injury and a court with the authority to look over the top of it. Eleven such plaintiffs have just been named.

CHAPTER 22

Why You Want to Be the Plaintiff: The Personal, Political, and Institutional Case for Filing

*The standing chapters proved the case can be brought.
This chapter is about why you would want to be the one
who brings it.*

The standing chapters set out to answer one question, and they answered it. Could the case be brought? Was there, somewhere in the country, a plaintiff who could walk into a federal courthouse, satisfy the court that the dispute was real, and be heard on the merits? The answer was yes, nineteen times over. Nineteen categories of plaintiff, each with an injury a court will recognize and a fact pattern built to win.

The two prong chapters set out the arguments that plaintiff would carry into court: the Patent Clause prong and the Takings Clause prong, the two suspenders, either one sufficient on its own.

Every door has been checked, and every door opens. And yet not one of those doors has anyone standing in it.

That is the gap to close. Standing settles whether a plaintiff may file. It does not settle whether anyone will. A right to sue is worth exactly nothing until someone exercises it, and this case has not been filed. The seat marked for the named plaintiff has been built, measured, and proven sound. It is empty. What follows is about the person who fills it.

The Most-Discussed Lawsuit in America

Whoever files this case becomes the named plaintiff in the most-discussed federal constitutional lawsuit in a generation. That is not a hope. It is a structural fact, and it follows from three things already established.

The first is the size of the number. The wall this case challenges has moved twenty-two trillion dollars from American households since 1970 — roughly three thousand three hundred eighty-five dollars from every household, every year. A lawsuit with that figure attached is a front-page lawsuit on the day it is filed. The second is the novelty. No federal court has ever been asked this question, so the case arrives with no prior coverage to compete against and no settled narrative to displace. The third is the shape of the docket. A constitutional case of this kind is not a single news event. It is filed, briefed, argued, decided, appealed, argued again, and taken up by the Supreme Court, the plaintiff's name on the caption at every stage.

The case becomes shorthand: the way the country says *West Virginia v. EPA* to mean a decision that reset a field, it will say the name of the plaintiff who filed this one. And a court ruling is not an administration's policy. Agency guidance can be rewritten by the next administrator and an act of Congress repealed by the next Congress; a constitutional ruling binds every future administration and, through the Supremacy Clause, every state.

The Semi-Truck Owner-Operator Who Files

The likeliest first plaintiff is the independent owner-operator — the trucker who owns the truck he drives rather than driving a carrier's. Two facts put him there.

The first is the depth of the injury. Of every plaintiff category this book identifies, the owner-operator carries the heaviest documented per-person harm on the thinnest margin. A modern Class 8 truck arrives with a federally mandated aftertreatment system built into a price the buyer cannot decline. The hardware is the smaller cost. The larger cost recurs: filter cleanings, forced regenerations, component failures, a replacement filter running into the thousands, and a fluid bill that never ends — well into five figures across the life of the truck. The engine designs that ran without any of it, proven over millions of miles and their patents long expired, are designs federal regulation forbids anyone to build new. A new Class 8 highway tractor runs well past a hundred and seventy thousand dollars; a faithfully remanufactured pre-2000 semi built on those designs would sell for something closer to eighty thousand. The owner-operator pays that gap every time he replaces a truck, for a design the Constitution already says is his.

The second fact is the clarity of the story. His injury needs no economist to explain it. It is a man with one truck and a mortgage on it, stranded on an interstate at one in the morning by a filter that costs thousands to replace. That is a story the country already half-knows and already resents on his behalf. Two further burdens press on him. The driver-facing camera puts him under recorded surveillance for every shift, in a space that is also his bed. The driverless truck advances toward removing him from the cab altogether. Neither is an injury this case can remedy. Both are why he has the most reason to file now.

He can reach the courthouse by either of two routes. The first is associational. Under *Hunt v. Washington State Apple Advertising Commission* (1977), an association has standing when its members would have standing in their

own right, when the interests are germane to its purpose, and when the claim does not require each member to take part. A declaratory-judgment action against the wall satisfies all three. The second route is direct. In *Loper Bright Enterprises v. Raimondo* (2024), four family-owned herring-fishing businesses challenged a federal rule that forced them to pay about seven hundred ten dollars a day for an at-sea monitor. They broke no law, sued as the parties the rule had injured, and won at the Supreme Court. A year earlier, in *Sackett v. Environmental Protection Agency*, a single landowning couple did the same. One trucker, one fact pattern, is enough.

The Tribe That Files

The second-likeliest first plaintiff is a federally recognized tribal nation, and the reason is a feature of standing law no other plaintiff in the book can claim. Standing can be satisfied in more than one way. A plaintiff can show a direct injury to its own purse. A government plaintiff can sue to protect the economic welfare of the people it represents — *parens patriae*, the standing a state invoked successfully in *Massachusetts v. Environmental Protection Agency* (2007). And a sovereign can sue to vindicate its sovereign interests. Most plaintiffs hold one of these. A few hold two. A tribal nation holds all three at once.

It holds the proprietary ground because a tribal government buys vehicles, equipment, and machinery and pays the wall's inflated price on every purchase. It holds the *parens patriae* ground because it may sue on behalf of its members, who pay what every American household pays. That burden falls hardest on rural and reservation economies, where a blocked-tier pickup consumes a far larger share of household income than in a wealthy

suburb. And it holds the sovereign ground because tribal sovereignty is not a courtesy extended by statute. It predates the Constitution. A plaintiff standing on three independent grounds is a plaintiff a court finds it hard to turn away at the threshold. Knock out one ground and two remain.

The recruitment fact sits alongside the doctrinal one. Three hundred forty-five federally recognized tribal nations exist. Each is a sovereign decision-maker; no central authority speaks for them, and none can veto what one decides. Tribal nations are among the most active litigants against the federal government in the country, and the Native American Rights Fund carries such cases at no cost to the tribe. In one wave beginning in 2001, about a hundred and twenty tribes brought trust-mismanagement suits against the United States; the government settled with more than a hundred for roughly \$3.7 billion. The disposition to file is there, and the infrastructure to file for free is built. The case needs one tribal council willing to pass a resolution. And a tribe that files is not joining a lawsuit. It is becoming the named sovereign in the highest-stakes constitutional litigation in the country, on the most favorable possible ground: a tribe asking a federal court to honor a promise the Constitution made and the government broke.

The Case That Costs the Plaintiff Nothing

One objection stops most plaintiffs before any question of upside arises, and the answer to it is better than most plaintiffs expect. The objection is cost. Constitutional litigation carried from a district court to the Supreme Court is expensive — millions in lawyers, expert witnesses, research, and the years a case of first impression

demands. A county, a tribe, an association, a small business looks at that number and stops. That is the responsible thing to do with the members' money, or the taxpayers', and it is also a misreading of how the case would be funded.

The plaintiff would, in all likelihood, pay nothing. The United States has a class of public-interest law firms that exist to carry constitutional cases at no charge to the client. They are not legal-aid offices. They are litigation shops with appellate specialists, deep benches, and decades of Supreme Court experience, chosen cases for constitutional significance, and several of them argued the very decisions this book's argument is built on. A first-impression Patent Clause and Takings Clause challenge, with a clean fact pattern and a twenty-two-trillion-dollar figure attached, is precisely the kind of case those firms were built to take.

The division of labor is simple. The plaintiff supplies a name on the caption and a documented fact pattern — the procurement records, the invoices, the proof of what the wall has cost. The firm supplies the complaint, the briefing, the expert economists, the oral advocacy, the appeal, and the years. The plaintiff is the client, not the litigator. He needs to be injured, documented, and willing to be named. One narrower worry deserves a straight answer. A losing plaintiff in federal litigation can sometimes be ordered to pay the government's costs. That risk is small in a case raising a genuine question of first impression, and where a public-interest firm carries the case the exposure runs to the firm, not to the plaintiff.

Cost is one half of what a plaintiff weighs. The other half is risk. A favorable ruling lifts the wall from patent-expired designs and opens the Expired Patent Commons. A partial

ruling lifts it from some categories and not others. An unfavorable ruling in one circuit changes nothing the plaintiff faced before filing — the wall stood before the suit and stands after it. Every other circuit remains free to be asked the same question. That is the heart of the strategy. The case is meant to be filed in more than one circuit, by more than one plaintiff. The more circuits that hear it, the greater the chance two decide it differently, and a circuit split is the single strongest reason the Supreme Court takes a case at all. No outcome leaves the plaintiff worse off for having filed. The downside has a floor, and the floor is the situation that already exists.

The Asymmetric Bet

Why the decision to file has a floor under its downside

WIN	PARTIAL WIN	LOSS
A federal court strikes the regulatory wall as applied to patent-expired designs. The Expired Patent Commons opens. Domestic remanufacturing becomes lawful, and competition pulls prices down across the category.	A court lifts the wall for some categories and not others. The blocked share of the economy shrinks. Pressure builds on what remains — and on the other circuits still free to be asked the same question.	A court in one circuit upholds the wall. Nothing the plaintiff faced before filing is made worse. Other circuits remain open, and the loss becomes the petition that can carry the question upward. And even a loss puts the forgotten half of the Patent Clause before a public that has never heard of it.
The wall comes down.	The wall is narrowed.	The wall's full impact is revealed
No outcome leaves the plaintiff worse off than before filing. The Supreme Court remains the forum where the question is finally resolved.		

The decision to file, seen as a wager.

The Same Logic, Seventeen More Doors

The owner-operator and the tribal nation are the two likeliest first plaintiffs, not the only ones. The structural upside belongs to every category the standing chapters established. What changes is the particular shape of the reward.

For a state attorney general the reward has a well-worn name: a career. Multi-state litigation has launched the office before — tobacco, opioids — and those who led became governors and federal judges. For an elected sheriff the precedent is *Printz* itself: two sheriffs from small counties challenged a federal statute and won at the Supreme Court, and their names are in the casebooks. For the medical and circadian-health constituencies the reward is scientific as much as public. The case moves an under-recognized body of evidence toward recognition, in the most visible forum the country has.

For the direct and associational categories — the farming associations, the restaurant operators, the construction contractors — the reward is a conversion. These are organizations that already do political work. They lobby, comment, testify, issue statements. That work is abstract, diffuse, and easy to ignore. This case offers something their ordinary advocacy cannot: a narrow, concrete claim with a measurable dollar figure attached and a court at the end of it. A farming association does not have to persuade a legislature to care about equipment prices. It has to document what its members overpay and put that documentation in front of a judge.

Across all of them the division of labor never changes. The plaintiff brings the injury, the documentation, and the name. The public-interest firm brings the law. And every one of these plaintiffs is filing the same case — which means a win for any one of them is a win for all of them, and for every American household behind them. Nineteen categories. Nineteen open doors. The case needs one plaintiff to walk through one of them.

The Brief the Allies Were Waiting For

One more group remains to account for, and it is not a plaintiff category. The case belongs, by every marker of principle, to the constitutional conservatives — the cohort that has spent its public life arguing that the administrative state has slipped its constitutional moorings. The case this book describes is that argument made concrete: a single, named, justiciable violation, with the Supreme Court's own precedents behind it. It is, on its face, their case.

A second body of allies sits in American legal scholarship: the public-domain movement, the law professors and intellectual-property researchers who have spent their careers defending one principle — that knowledge, once its term of legal protection has run, belongs to everyone. That movement has worked mostly on the copyright side, fighting term extensions that keep old books, films, and songs out of the public's hands. But the principle is not a copyright principle. It is what the Constitution states in two words — limited Times — and those words are a single clause, reaching patents and copyrights in the same breath. The reversionary interest these scholars defend for an expired copyright is the identical interest the wall cancels for an expired patent. What they supply is what no plaintiff and no senator can: the doctrinal framework, the

amicus briefs and law-review articles that turn what a court might read as a regulatory complaint into a recognized constitutional question.

So why has none of them brought it? The easy answer is that they failed to see it, and the easy answer is wrong. Every figure in that cohort entered public life after 1970 — the year the Legislative Reorganization Act remade Congress and made the regulatory ratchet politically possible. A legislator who took office in 1985, or 2005, or 2015, entered a Washington in which the post-1970 settlement was simply the furniture of the world. The pre-1970 condition is not part of anyone's working political memory. It is history, and history has to be assembled before it can be argued from. That is the work this book did. The constitutional conservatives were not standing idle in front of an obvious case. They were working without the brief. The brief had not been written.

The Name Not Yet Attached

Stand back and see what has and has not been claimed. Nothing here says filing is easy, or that winning is certain. The claim is narrower and, for the right plaintiff, decisive. The cost of filing is, in practice, nothing. The standing is real and proven nineteen times over. And the upside — the platform, the durability, the place in the country's law — is larger than anything the plaintiff could reach otherwise. For every category named here the arithmetic comes out the same way. Filing is the best decision available.

Consider what every constitutional case of this scale has in common. *Printz*. *Loper Bright*. *Sheetz*. Each is known by a name, and the name is a person or a place that was simply a plaintiff who decided to file. The name was ordinary until the case made it permanent. That is how constitutional law

is built: not by scholars, and not by the government, but by a named plaintiff willing to be the one.

This case has no name yet. It has a doctrine worked out across two prongs, nineteen categories of plaintiff with standing, a remedy a federal court can order, and a body of precedent already assembled. The one thing it lacks is a name on the caption, because it has not been filed. The seat is still empty. It does not have to stay empty. The reader who recognizes his own injury here — owner-operator, council member, association president, elected official — will find contact pages for the plaintiff categories, and for the proposed nonprofit the author calls the Expired Patent Commons, at

https://marti124.github.io/Patent_Letters/.

The case is built. It is waiting for the plaintiff.

CHAPTER 23

The Filable Case: Why 2026 and Not Before

The case could not have been won until now. Now it can.

On the morning of June 28, 2024, a forty-year-old obstacle to the argument of this book disappeared.

The obstacle had a name. It was called *Chevron* deference, after the 1984 Supreme Court decision that created it, *Chevron U.S.A. Inc. v. Natural Resources Defense Council*. For four decades, *Chevron* required a federal court reviewing a federal agency's reading of an ambiguous statute to defer to the agency — to accept the agency's interpretation, even one the court would not have chosen on its own, so long as that interpretation was reasonable. The doctrine governed thousands of cases. It shaped how every regulatory lawsuit in the country was argued. And it stood squarely across the path of the case this book describes. On June 28, 2024, the Supreme Court decided *Loper Bright Enterprises v. Raimondo* and overruled it.

The patent-bargain case is a question of first impression. No federal court has ruled on it, because no plaintiff has filed it. A fair reader is owed an obvious question. If the wall went up beginning in 1970, and the harm has compounded for fifty-five years, why has the case sat unfiled for half a century? The answer is not that the lawyers were asleep. The answer is that, for most of those fifty-five years, the case could not have been filed in a form a federal court could rule on and a plaintiff could win. The tools did not exist. They exist now. As a complete set, they have existed

for less than two years. Draw the inference. The long silence is not evidence that the case is weak. It is evidence that the case was, for most of its history, unfileable — and a violation against which no one can successfully file is one scholars do not develop, journalists do not investigate, and litigators do not pursue. The floor was missing. Then, piece by piece, it was laid.

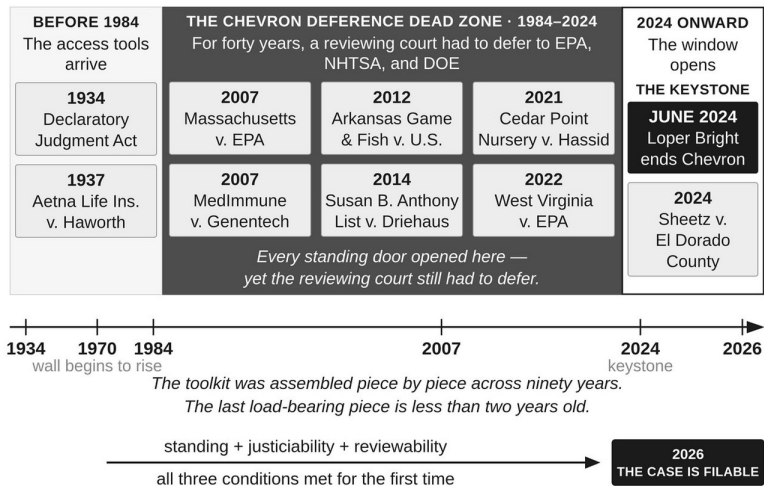
The First Clock: The Doctrinal Toolkit

A plaintiff needs a door into the courthouse before anything else. The doors were built one at a time across ninety years. The Declaratory Judgment Act of 1934 let a party ask a court to declare rights before enforcement. *Hunt v. Washington State Apple Advertising Commission* (1977) established associational standing. *Massachusetts v. EPA* (2007) established that a sovereign suing to protect its citizens receives special solicitude in the standing analysis. *MedImmune v. Genentech* (2007) held that a party paying under protest does not forfeit the right to challenge what it pays. *Susan B. Anthony List v. Driehaus* (2014) held that a credible threat of enforcement suffices without waiting to be prosecuted.

Take any one of those away and a class of plaintiff loses its route. Assembled, they mean the patent-bargain case does not require anyone to break a law, be arrested, or be enjoined. The access clock finished striking in 2014. One objection belongs here. A defender of the wall may say the question is political, not for the courts. It fails. Courts have tested patents against the Constitution's "limited Times" requirement since the founding era, and the standard is concrete: does the regulation make it impossible to remanufacture a patent-expired design?

THE ASSEMBLING OF THE FILABLE CASE

Why the patent-bargain case became filable in 2026 — and not before



Standing-door mechanics are developed in the chapter on challenging a law without breaking it; the merits of each prong are developed in the two chapters that follow.

The assembling of the filable case — how the doctrinal toolkit came together between 1934 and 2024. The access doors took ninety years to build; the reviewability keystone was set less than two years before this book went to press.

The Keystone: June 28, 2024

Suppose the access clock had read the right hour earlier. Suppose a plaintiff in 2015 had every standing door open to her. She still could not have won, because the second clock — reviewability — had not yet struck.

Recall what the case asks a court to decide. It asks whether the Environmental Protection Agency, the National Highway Traffic Safety Administration, and the Department of Energy, in writing the rules that close around patent-expired designs, stayed within the authority Congress actually gave them. The agencies say they did. They point to statutory language — the Clean Air Act’s authority over new motor vehicles, the efficiency

mandates, the safety standards — and argue that the language reaches any vehicle manufactured for sale, a remanufactured patent-expired design included. The plaintiff argues the language cannot bear that weight. That is a dispute about the meaning of a statute.

Under *Chevron*, that dispute had a thumb on the scale. From 1984 to 2024, a federal court facing an ambiguous statute and a competing agency interpretation was required, at the decisive step, to defer to the agency. The court did not ask what the statute meant. It asked only whether the agency's reading was reasonable — and an agency defending forty years of accreted rulemaking is well practiced at making its reading look reasonable. For four decades, the case would have been argued in a courtroom where the agencies began the day ahead.

That is what June 28, 2024 changed. A federal court reviewing an agency's interpretation now reads the statute itself, using the ordinary tools of legal interpretation, and reaches its own conclusion. The agency's view is entitled to respect to the extent it is persuasive — what *Skidmore v. Swift & Co.* (1944) called *Skidmore* weight — but it is not entitled to deference. The thumb came off the scale. This is not yet the merits; whether the agencies in fact exceeded their authority is the argument of the chapter that follows. What June 28, 2024 settled is narrower and prior: the court is now free to decide that question for itself. A companion development sharpened the same edge. In *West Virginia v. EPA* (2022) the Court gave its fullest statement of the major-questions doctrine. An agency claiming power over a matter of vast economic and political significance must point to clear authorization from Congress, not to the ambiguous margins of an old statute. Closing the public's access to expired patents across most of the durable-goods economy is such a matter. Together

with the end of deference it forms the reviewability floor, poured in the 2020s.

Corner Post v. Board of Governors, decided the same term, supplies the companion rule. The six-year limitations period on a challenge to a federal rule runs from when the plaintiff is injured rather than from when the rule was promulgated. The wall was built between 1970 and 1987. Under the old rule, most of it was unchallengeable by anyone who had not sued decades ago. Under *Corner Post*, a business first injured last year may challenge a rule issued in 1987.

The Second and Third Clocks

The injury had to age. A constitutional argument that a regulatory architecture has cancelled a clause of the Constitution is not made by pointing at one rule. It is made by pointing at an accumulation, and the accumulation needed decades to reach a scale a court would recognize as more than a policy complaint. In 1985 the wall was fifteen years old and the transfer was a fraction of what it is now. Fifty-five years produced \$22 trillion, and a figure of that size is what makes the case a constitutional question rather than a grievance.

The evidence had to come in. The international natural experiment — roughly ninety countries permitting what the wall forbids, thirty-four blocking it — is a live comparison only because those markets developed and were documented. The historical price collapses, the nine pre-1900 categories, the bicycle and the calculator, all had to be assembled as record. A court asked to believe that prices would fall if the wall came down needs more than assertion. It needs a benchmark, and the benchmark is now in hand. India's Mahindra Thar, built on the long-

expired Jeep CJ platform, sells for about twelve thousand dollars against a Jeep Wrangler's forty-five thousand. The safety objection meets a documented answer: Global NCAP rated the Thar four stars for adult and child occupant protection in 2020, and Bharat NCAP rated the Thar ROXX five stars in 2024. The objection that it is a new vehicle rather than a remanufactured one was settled by the party with every reason to dispute it. When a Mahindra-built Roxor on the same platform was assembled in Michigan, Jeep's owner went to the International Trade Commission to stop it — and sued on trade dress, the grille and the boxy shape, not the mechanical design. It could not: an expired patent cannot be infringed.

What 2026 Adds That 2024 Did Not

A careful reader will press the chapter's own logic. One further 2024 decision belongs on the calendar, its work for the second prong. In *Sheetz v. County of El Dorado* the Court held that the Takings Clause applies to burdens imposed by general legislation, not only case by case. That floor, like the structural one, was finished in the 2020s. So a careful reader will press: if the keystone was set on June 28, 2024, and *Sheetz* came down the same year, the case became filable in 2024. Why does the title say 2026?

Because filable and filable-with-the-full-advantage are not the same thing, and two developments separate them. The first is that doctrine settles by being applied. A Supreme Court decision is announced on one day; its reach is worked out over the years that follow, as the lower federal courts apply it to real cases. In the time since *Loper Bright*, the federal courts have begun that work — testing its edges, building the body of decisions that tells a litigant

how the new rule behaves in practice. A plaintiff filing in 2026 files into a doctrine that has been used, not merely announced.

The second is that the plaintiff architecture is now in place. Nineteen categories of plaintiff, ranked, many paired with the practical tools to reach the organizations that can carry the case. That architecture is the product of work done across the past year. A case is filable, in the practical sense, not when the doctrine permits it in the abstract but when an identified plaintiff with standing can be put in front of a court. The year 2026 is the first in which both halves are true together.

None of this makes 2026 a magic date. The case did not become filable at the stroke of a particular midnight, and a plaintiff who files in 2027 or 2028 files the same sound case. The claim is narrower and more honest. It is that 2026 is the first year in which every clock reads its hour at once: the access doctrine assembled, the reviewability keystone set and tested, the injury aged into measurable scale, the redressability evidence in hand, and the plaintiff architecture built. One practical clock remains, uncounted among the three but worth naming. The cost of living is the dominant subject of American public life. A constitutional case that addresses it directly will find a plaintiff, and an audience, in a way it would not have a decade ago. That is not a doctrine. It is a description of the moment — and the moment is now.

The Door Is Open

Three clocks. The doctrinal clock, whose hands are the access toolkit assembled between 1934 and 2014 and the reviewability keystone set in 2024. The injury clock, which needed fifty-five years to age the harm into measurable,

doctrinally significant scale. The evidence clock, which needed the international natural experiment and the historical price collapses to come in as documented fact. For most of the period since 1970, at least one of those clocks read the wrong hour. In 2026, for the first time, all three read midnight together.

That is what this chapter set out to establish, and all it set out to establish. The question has been the courthouse-access question: can the patent-bargain case be brought? The answer is yes. It can be brought by a single plaintiff, with standing, in a single federal courtroom, without anyone breaking a law. Whether the plaintiff who walks through that door wins is the question the two prong chapters answer. And the door does not close. The doctrine that makes the case filable is not a window that may shut as quietly as it opened. *Loper Bright* is the law. The Declaratory Judgment Act is ninety years old and unthreatened, and the standing doors of 2007 and 2014 are settled precedent. A toolkit, once assembled, does not disassemble. What remains is not a doctrinal question but a human one. Which plaintiff, in which of the nineteen categories, picks a date and files?

CHAPTER 24

The Patent Clause Prong: “Limited Times” as Constitutive Language

The patent expires on schedule. The protection was never supposed to outlive it.

The filable-case chapter left the plaintiff standing inside the courtroom. The door she came through — standing, justiciability, the end of the deference that once bound the court — is behind her. Whether she could get inside is settled. What follows takes up the first question she faces now that she has.

It is a question about power. Prong one of the patent-bargain case does not ask whether the regulatory wall is unfair, or whether it is expensive, or whether it is unwise. Those are real questions, and the diagnostic chapters have answered them. They are not this question. The question here is narrower and harder: did the agencies that built the wall ever have the power to build it?

The book rests its case on two independent constitutional grounds. It calls them the two suspenders, because either one alone is enough to hold the argument up. The argument here is the first: the Patent Clause prong. It is a structural argument — an argument about the limits of an enumerated federal power. The chapter that follows argues the second, the Takings Clause prong, which a litigant can win even if a court rejects the Patent Clause argument made here, or never reaches it at all, deciding the case on Takings grounds alone. The two do not depend on each other. That independence is the reason for having two.

The structural argument begins with a single observation, and the observation is the title of this book. The Patent Clause has two halves. The first half has been remembered, litigated, and enforced for two centuries. The second half was forgotten. Not repealed — forgotten. The regulatory state was built across fifty-five years without the second half of the Patent Clause in view. What follows is about what the Constitution says when someone finally holds it up to the light.

The Only Monopoly the Constitution Allows

Start with the clause itself. Article I, Section 8, Clause 8 gives Congress the power “to promote the Progress of Science and useful Arts, by securing for limited Times to Authors and Inventors the exclusive Right to their respective Writings and Discoveries.”

Read it for what it authorizes. It authorizes Congress to grant an inventor an exclusive right — a monopoly — on what he has invented. Then read the rest of the Constitution for anything comparable. There is nothing. The Patent Clause is the only place in the entire founding charter where the federal government is given the power to hand a private party a monopoly. Nowhere else in the document does that power appear. The clause is also called the Intellectual Property Clause, because the same sentence provides for copyrights, likewise granted only for limited Times. This book is not concerned with copyright.

That is not an accident of drafting. The generation that wrote the Constitution distrusted monopoly. They had watched the English Crown sell exclusive privileges to court favorites, and they had read the long English fight to abolish the practice. They feared a monopoly the way they feared a standing army, because both put concentrated

power where a free people could not easily reach it. They allowed the federal government one monopoly power, and one only.

Notice the verb. The clause does not say Congress may permit an exclusive right, or recognize one, or allow inventors to keep what they have made. It says Congress may secure it. And there was, in 1787, no patent office to do any securing. The word looks past the paper to the thing behind it: the force of the general government, pledged to stand between the inventor and anyone who would copy him. The First Congress read it exactly so. When it wrote the Patent Act of 1790, it directed that letters patent issue “in the name of the United States,” and it armed the right with a federal lawsuit — damages set by a jury, the infringing articles forfeited, all recoverable in the circuit courts of the United States. To secure a right, the Founders’ generation understood, is to put the nation’s coercive power behind it. Madison had said as much in defending the clause: the States could not separately make effectual provision for inventors’ rights, so the national government would. Securing was never a passive permission. It was an act of force.

Hold that word against the wall. The general government may secure an inventor’s exclusive right — but only for limited Times. That is the whole of the grant, force and limit in one sentence. Now ask what the regulatory ratchet does with the government’s securing force. It does not touch the patent. It does not let the inventor rebuild his own expired design. It does one thing: it ensures that no one else may rebuild it either. And that single prohibition delivers the entire practical benefit the exclusive right ever gave — freedom from a cheaper copy of the design — with no terminus.

Note who receives it. Where the original maker survives, the benefit runs to him, though he need not lift a finger: his protection is complete the moment everyone else is barred, and it does not matter that he cannot remanufacture either. Where the original maker is long gone, the benefit does not lapse — it passes to whoever holds the live patents on the technology that replaced the barred design. So the force the clause pledged for a bounded period is now applied without end. The general government, through the ratchet, keeps securing one party's exclusivity against all comers, decades after the sentence that authorized the securing commanded it to stop.

Edison's incandescent patents expired a century ago, and the securing force that once stood behind them now stands behind the LED and fluorescent patents that supplanted the bulb his patents released. The force outlived the bargain that justified it. That is the deeper defect: the general government's securing power, cut loose from any inventor's limited term, still running — now protecting parties who invented nothing that was taken from the public. This is why critics who argue over what the patent statute does and does not permit have aimed at the wrong target. The limit does not live in the statute. It lives in the clause, above the statute — a cap on how long the general government's force may secure one party's exclusivity against everyone else. The ratchet runs straight through that cap.

And the Founders did not hand even that one power over whole. They bounded it. The grant comes with words of limit built into the same sentence. The power to grant the monopoly and the limit on the monopoly are not in two different clauses. They are not even in two different sentences. They are the same sentence. Congress may

secure exclusive rights, and it may secure them for limited Times, and the second phrase is the price of the first.

This is the frame the rest of the argument builds on. The Patent Clause is a power that arrives with its own boundary already attached. You cannot pick up the power and leave the boundary behind. The boundary is not a separate rule about how to use the power. It is part of the description of the power itself. A grant of exclusive rights for limited Times is the only patent power the Constitution created. A grant of exclusive rights with no limit is a different thing — and it is a thing the Constitution gave nobody.

“Limited Times” Is Constitutive, Not Decorative

Everything in the structural argument turns on a single question about the words. When the Constitution says “for limited Times,” is it stating a boundary on the power it grants, or is it offering Congress a recommendation about how to use a power it grants in full?

Call the second reading the decorative reading. On it, “limited Times” is hortatory. The Constitution grants Congress the patent power, and then adds, in the same breath, a piece of advice: keep the terms limited. Good advice, on this reading, but advice — the kind of language that expresses a preference without fixing an outer edge. If “limited Times” is decorative, then the patent power is large and the words are trim.

Call the first reading the constitutive reading. On it, “limited Times” is part of what the power is. It does not describe how to use the patent power well. It describes the patent power’s outer edge. The power Congress received is the

power to secure exclusive rights for limited Times — and a grant of exclusive rights that runs without a terminus is not a lawful use of that power, because it is not that power at all. It is outside the four corners of the clause. This book takes the constitutive reading, and so has the Supreme Court.

The clearest statement is in *Graham v. John Deere Co.* (1966). *Graham* was, on its surface, a case about when an invention is novel enough to be patented at all. But the Court used the occasion to say what kind of provision the Patent Clause is. The clause, the Court explained, is both a grant of power and a limitation upon the exercise of that power. Congress may legislate within it, but Congress may not overreach the restraints imposed by the stated constitutional purpose. The constitutional question, the Court said, is not answered merely because Congress has acted; the question is whether Congress has stayed within the limits the clause itself sets. That is the constitutive reading, in the Court's own words. The limit is not advice. It is a wall around the power.

Two older cases sharpen what kind of thing a patent is, and therefore what kind of thing its expiration is. In *James v. Campbell* (1882), the Court held that a patent is property — that when the federal government itself uses a patented invention, it has taken private property and must pay for it. A patent is not a license the government lends and may quietly call back. It is property, with the legal solidity the word carries. That holding has stood, unbroken, for more than a century and forty years.

In *McCormick Harvesting Machine Co. v. Aultman* (1898), the Court drew the line that matters most here. Once a patent has been issued, the Court held, the Patent Office — the very agency that issued it — has no power to

revoke, cancel, or annul it. The patent has become the property of the patentee, and the only authority that may set it aside is a court. An executive agency cannot. *Florida Prepaid Postsecondary Education Expense Board v. College Savings Bank* (1999) confirmed the point for the modern Court: patents are a species of property. What the Fifth Amendment requires when government cancels one belongs to the Takings Clause prong.

Hold those two cases together and turn them around. If a patent is property, and if not even the agency that grants a patent may cancel a live one by its own act, then consider what the regulatory wall does. It is an agency act. It does not cancel a live patent. It does something stranger: it reaches an expired patent, and by regulation it restores the exclusivity that expiration was supposed to end. If an agency may not revoke the patent right while it lives, the inference this argument draws — and it is the argument's own step, not the Court's — is that an agency may not resurrect its commercial substance after it dies.

What “limited Times” Stands For

The previous section settled what kind of word “limited Times” is. The next question is what the words stand for. A constitutional command is not a mood; it is a rule with content. So what is the content?

The content is equality. The Constitution authorizes a single departure from the rule that the federal government may not hand a private party a permanent commercial advantage. That departure is the patent. And the departure is allowed only for limited Times. Once the time runs, the departure is supposed to end. The equal commercial footing the rest of the Constitution presumes is supposed to govern once again. The inventor and the

future remanufacturer are supposed to stand on the same ground. Neither is supposed to hold a permanent shield against the other. That is the equality the Clause was written to keep.

The cases on the post-expiration public domain — *Sears*, *Bonito Boats*, *Kimble* — describe that equality in their reasoning, in words the Court has treated as the settled background of patent law. They use different words for it. They say at expiration the invention “passes to the public,” that the patent-expired article “may be made and sold by whoever chooses to do so,” and that what is in the public domain is “free for all to use.”

Read the Clause once more with that purpose in view. Congress may secure exclusive rights for limited Times. The first phrase grants a temporary monopoly. The second phrase ends it. What lies on the other side of the second phrase is not a void; it is a particular state of affairs the Founders required. The patent-expired invention belongs to no private party more than any other. The market for it is open. No participant holds permanent commercial advantage. That is what limited Times stands for, and that is the equality the regulatory wall destroys.

This is not the equality the Fourteenth Amendment imposes on the States. Nor is it the parallel equality the Fifth Amendment’s Due Process Clause imposes on the federal government — the equality the Supreme Court recognized in *Bolling v. Sharpe* in 1954. That equality covers many things, gets rational-basis review in the economic context, and would not by itself defeat the wall in court. This is something narrower and stronger: the Patent Clause’s own internal equality command, encoded in the words of the Clause itself, binding on Congress regardless of how a court rates the wall under rational basis. The

Patent Clause does not ask whether a federal law treating remanufacturers unequally is rational. It asks whether Congress preserved the equal commercial footing the Clause requires at the end of the patent term. When the answer is no, the question is settled. The wall is outside the power the Clause grants, because the power the Clause grants is the power to confer exclusivity for limited Times and not the power to confer exclusivity that survives the time in another name.

What No Federal Power May Undo

A grant of power in one clause does not authorize Congress to nullify what another clause commands.

The Patent Clause does two things, not one. It grants a power, and it fixes an outcome. The grant is limited to “limited Times.” The outcome is that at the end of those times, the exclusive right ends and the invention passes to the public. A statute enacted under the Commerce Clause, or the Necessary and Proper Clause, or any other enumerated power, may regulate a great many things. What it may not do is produce, by another route, the very outcome the Patent Clause forbids: an exclusivity without end.

The objection writes itself, and it deserves a straight answer. The agencies were not exercising the patent power. They were exercising the commerce power, delegated through the Clean Air Act and its companions. True. The commerce power is real, read as broadly as the Supreme Court read it in *Wickard v. Filburn* (1942). Congress may also arm an agency with rulemaking authority; the Court has permitted such delegation since *J. W. Hampton, Jr., & Co. v. United States* (1928), so long as Congress lays down an intelligible principle. This

argument disputes neither power. Both are also beside the point. The question is not which power the agencies thought they were using. The question is whether the result lies inside any power they had. Congress cannot do under the Commerce Clause what the Patent Clause forbids it to do under the patent power, because the Patent Clause's limit is a limit on the federal government, not a limit on one drawer of its filing cabinet. A settled rule reinforces the point: when a specific constitutional provision and a general one bear on the same act, the specific governs. The commerce power is general. The "limited Times" boundary is specific. The Supreme Court applied that rule in *Railway Labor Executives' Association v. Gibbons* (1982), holding that Congress may not use the commerce power to escape the uniformity the Bankruptcy Clause requires. *Gibbons* is a bankruptcy case. Carrying its rule to Clause 8 is this argument's step, not the Court's.

The Expired Patent Commons

Return to the sentence. Congress may secure exclusive rights for limited Times. The first half of that bargain is the exclusive right, and the argument has spent its pages so far on the first half. Now the second.

What happens at the limited Time's end? The exclusive right stops. That much is uncontested. But the Constitution's purpose is not served by the mere cessation of a legal right. It is served by what the cessation is supposed to produce: a public able to make, use, copy, and improve the invention. Call that condition the Expired Patent Commons — not a metaphor, but the place of actual public use the bargain was written to open. A commons is not a void where a right used to be. The village commons was land the villagers walked onto and

grazed. This one is defined by use, not by paper: a patent that has expired on paper, but that no one may lawfully build, has not entered it.

The Supreme Court has affirmed the public half as plainly, and as long, as it has affirmed the inventor's. *Scott Paper Co. v. Marcalus Manufacturing Co.* (1945) rejected the use of a private contract to extend a patent's practical effect past expiration. *Sears, Roebuck & Co. v. Stiffel Co.* (1964) held that when a patent expires the article enters the public domain and is available to anyone, and that a state may not use unfair-competition law to recreate the monopoly federal law had ended. *Bonito Boats, Inc. v. Thunder Craft Boats, Inc.* (1989) held that federal patent law determines not only what is protected during the term but what is "free for all to use" after it, and that the public holds a vested right of access no state may rescind.

The Court grounded that rule in the Clause itself: "free exploitation of ideas will be the rule, to which the protection of a federal patent is the exception." *Kimble v. Marvel Entertainment* (2015) restated the rule for the modern Court: when the patent expires, the right to make and use the invention passes to the public — free of the patent monopoly, though not immune from all regulation. One question a careful reader will press: what does the Clause release? What was patented. A patent is granted on a claim, and the claim covers the integrated invention, not its parts list. A complex product -- a vehicle, a tractor, an appliance -- is an aggregation of separately claimed inventions, each expiring on its own schedule. The whole becomes the public's when the last of them enters the commons.

That last qualification is the honest one, and the argument does not hide from it. *Kimble* does not say an expired-

patent product is exempt from law. It says the patent monopoly is over. A remanufactured product must still meet the rules that apply to it. The question this book presses is which rules those are — the ones in force when the design was certified, or the ones written decades after. That is the difference between a rule an expired design can carry and a rule that forbids the expired design as such. That history is also what makes “forgotten” the right word, and the only honest one. The doctrine was never overruled. No decision ever held the public’s half a mistake. The cases still stand. The law did not change; the institutions stopped acting as though it existed.

The Patent Expires; the Wall Does Not

Here is the mechanism, and it must be stated precisely, because imprecision at this step invites an easy and false rebuttal.

The patent expires. It expires on schedule — twenty years from filing, under the term in force today, and sooner when maintenance fees lapse. Federal regulation does not touch the patent term. It does not add a year to it. The agencies that built the wall have no power over patents at all, and they have not tried to claim one. On paper, exactly as the Constitution promises, the exclusive right ends. Anyone who says this book argues that regulation creates a patent that never expires has misread it. The patent is not perpetual. The patent ends.

What does not end is something else, and naming it correctly is the whole of the matter. What does not end is the original manufacturer’s protection from low-cost remanufactured competition. The patent gave him two things while it lived: a legal monopoly, and the commercial

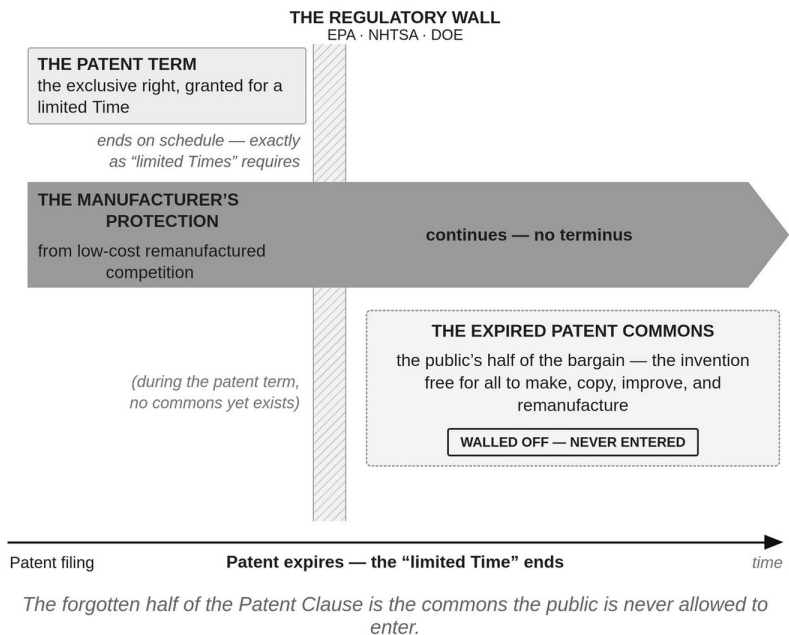
benefit of that monopoly — freedom from anyone selling a cheaper copy. Expiration is supposed to take back both.

It does take back the first: the legal monopoly is gone. But the regulatory wall hands the second one back. And it adds a third thing the patent itself never gave him: extra pricing power on his still-patent-active products today, because the competitor who would have appeared the day his patent expired never appears at all.

It does so without ever mentioning patents. Beginning around 1970, three families of federal regulation closed around the durable goods of American life — the Environmental Protection Agency's emissions tiers, the National Highway Traffic Safety Administration's safety mandates, and the Department of Energy's efficiency standards. Each requires that a product offered for sale meet the current year's standard. And the law grants no grandfather exemption for remanufactured goods. So no one may lawfully build a patent-expired design today — not even the original manufacturer — because a product built to that older design no longer meets the standard the current year demands.

This is not a theory about a world that never existed. Before 1970 a durable good whose every underlying patent had expired could be rebuilt by anyone who chose to, subject only to the trademark and trade-dress law that protects the maker's name. The engine remanufacturers, the parts-interchange trades, and the machine-tool rebuilders were an entire independent industry operating on exactly that principle. What changed after 1970 was not the Constitution and not the patents. It was the wall.

The Patent Term Ends; the Regulatory Wall Does Not
WHAT THE PATENT CLAUSE PROMISED — AND WHAT THE WALL PRODUCES



The Grandfather Denied the Public, Kept for the Owner's Invention

A government brief will answer that the rules treat everyone alike. Nobody may build a non-compliant vehicle — not the original manufacturer, not a remanufacturer, not anyone. Equal treatment, evenly applied.

But look closer, and the equal treatment dissolves. The federal safety standards bind by the date a vehicle is manufactured, not the date it is driven. A vehicle built to the standards in force on its manufacture date stays lawful to operate and resell for as long as it survives. It is never forced to retrofit to a later standard. So a vehicle built on a mature, backup-camera-era platform — the kind the market would most want to remanufacture — remains lawful on the road for its full life, years after the patents on

its design have expired. The grandfather exemption outlives the patent. The original manufacturer's customer holds a legal article the regulation will never touch.

Now set beside it the identical design, remanufactured by a member of the public after a favorable ruling. The patents are the same expired patents. The design is the same design. Built today, the remanufactured article is never dirtier than the original it replaces — the very vehicle, of its own generation, the law already permits to be driven, sold, and registered. In the ordinary case it is cleaner, because the remanufacturer may fit any emissions technology that has since entered the public domain. And yet the law that permits the original to remain on the road forbids the public from ever building that design at all.



That is the asymmetry the equality command reaches. The exemption runs with the unit rather than with the first owner, which is why it survives resale — and which is also why it cannot be explained as a personal accommodation to a buyer who relied on the old rule. It is a standing legal permission attached to an article of a certain vintage. The

public is denied the same permission for the same vintage, on the same design, after the patent that once justified the difference has expired. That is the inventor's half of the broken bargain. The public's half is the other face of the same wall, and the precise word for it is prevention. Nothing is seized from the public's hands; the public never gets the invention into its hands at all. The wall does not raid the commons. It keeps the commons from ever encompassing the invention. The right attaches however the patent ends, by full term or early lapse, because the Constitution conditions it on the patent's end and not on its length.

The reader has already met the price of all this. A pickup truck that should cost twenty-five thousand dollars in real terms costs \$72,495. The gap between the two numbers is, in large part, the cash value of an exclusivity that the Patent Clause said should have ended and the wall did not let end.

The Power Congress Never Had to Give

One more step completes the structural argument, and it is the step that reaches Congress rather than the agencies. The agencies answer that they were authorized: Congress told them to set standards for products offered for sale, and a remanufactured vehicle is such a product. There are two answers to that, and they work at different depths. Either one defeats the position.

The first stays inside the statute, and the filable-case chapter has already set the ground. *Loper Bright Enterprises v. Raimondo* (2024) ended *Chevron U.S.A., Inc. v. Natural Resources Defense Council, Inc.* (1984) deference, so the court now reads “new motor vehicle” for itself, and *West Virginia v. Environmental Protection*

Agency (2022) supplies the major-questions doctrine, which requires clear congressional authorization. That chapter stopped there, at reviewability. Here the merits follow. Holding the public out of the Expired Patent Commons moves twenty-two trillion dollars from households to incumbents, and no statute grants that in clear words. “New motor vehicle” was written to catch new sources of pollution, not to cancel the public’s half of the patent bargain.

The second answer goes deeper, and no reading of any statute reaches it. If the wall exceeds the power the Patent Clause allows, a defender will say the fault lies with Congress, which wrote the statutes, and that the remedy is legislative.

The answer is that Congress could not have granted what it did not hold. An agency’s authority is delegated authority, and no delegation can carry more power than the delegating body possessed. If Congress may not, under any enumerated power, produce an exclusivity without end, then Congress could not authorize an agency to produce one either. Whether the agencies’ statutes are read narrowly or generously, the wall fails, because the result is outside the power.

Answering the Objections

A merits argument is only as strong as its answer to the best case against it. The defenders of the wall are able lawyers, and they have arguments. Here are the strongest, each stated as its proponent would state it, and then answered. The chapter *Objections Considered* takes up the wider field.

The first objection. The agencies do not regulate patents. They regulate emissions, crash safety, and fuel economy. The Environmental Protection Agency has never issued a rule that mentions a patent. The patent-bargain case, the objection runs, is an attack on environmental and safety regulation dressed up in the language of the Patent Clause, and a court should see through the costume.

The answer is that the Constitution looks at what a government action does, not at what the agency files it under. This is a settled habit of constitutional analysis, not a special rule invented for this case.

A tax that falls only on newspapers is a First Amendment problem even though it is filed as a revenue measure. A zoning rule that exists to keep a disfavored congregation out of town is a religious-liberty problem even though it is filed as land use. The label does not control. The effect does. And the argument does not ask a court to strike down emissions standards, safety standards, or efficiency standards. It asks for a ruling on one narrow question: whether those standards may be applied to bar the manufacture of a design whose patents have expired.

The second objection. Nothing stops anyone from building a patent-expired design that complies with today's standards. The public domain is intact; the public simply has to meet the same rules everyone meets.

The answer is that compliance with today's standards is not a burden on the expired design. It is the abolition of it.

A 1999 engine architecture cannot carry a 2010 aftertreatment stack and remain that architecture; what emerges is a different engine. The distinction the argument rests on is between a rule an expired design can simply carry — a labeling requirement, a warranty rule, a

disposal rule — and a rule that forbids the expired design as such.

The first is additive and raises no constitutional question. The second is constitutive, and it is what this argument is about.

The third objection. A ruling for the plaintiff would unleash unsafe and dirty products on the American public.

The answer is the remedy itself. The argument does not ask for deregulation. It asks that a remanufactured product built on a public-domain platform be held to the federal standards in force during the platform year it reproduces, certified by an independent engineering body, and labeled so the buyer knows exactly what he is getting. The standards do not vanish. They time-shift, and only for the remake. New designs still meet today's standards, and the wall is untouched for them. Agencies retain audit, recall, penalty, and enforcement authority in every lane.

The Desired SCOTUS Ruling Time-Shifts Regulations for Remanufacturing; the Commons Opens

THE PRINCIPLE

The regulatory standards for the original invention should be the same for a patent-expired remake — equal justice (standing) under the Patent Clause. A remake can happen if it need only meet the original standards.

THE PROBLEM TODAY

Current standards have ratcheted up and make it impossible to remanufacture the expired-patent item. The remedy time-shifts the rule back to the original's standard — the public's remake need meet only what the original met.

THE REGULATORY STANDARD

A grandfather exemption, earned by independent engineering certification, time-shifts the remanufactured design to the standards in force when its original was certified.

THE PATENT TERM

the exclusive right, granted for a limited Time

the patent still ends on schedule — it was never what blocked the public

THE MANUFACTURER'S PROTECTION
from low-cost remanufactured competition

THE CERTIFICATION GATE

A remanufactured patent-expired design passes the gate only if it contains nothing still under patent and meets the federal safety and emissions standards of its certified platform year — confirmed by independent engineering certification.

the patent-expired invention

this protection ends.

Low-cost remanufactured competition is lawful again.

THE EXPIRED-PATENT COMMONS

Now open — the public may make, copy, improve, and remanufacture.

Held to the standards the original met — only the brand name and logo stay with the original maker.

the wall still holds for newly designed vehicles — modern rules are untouched

patent filing Patent expires — the “limited Time” ends time →

Simplified: the patent is shown expiring in the present, with remanufacturing desired now.

The standard's applicability is time-shifted to the original's certified year — the wall stands, but for the remake it speaks the standards of that year.

The First Suspender Holds

Stand back from the argument and see its shape.

The Constitution created exactly one federal power to grant a private monopoly, and it created that power with a boundary inside it: exclusive rights for limited Times, the grant and the limit in a single sentence. The Supreme

Court has read the boundary as constitutive — a real edge on the power, not advice about using it — for as long as there has been a Court reading it. At the limited Time's end, the invention is meant to pass into the Expired Patent Commons, the place of actual public use, and the Court has affirmed that public half of the bargain as plainly, and as long, as it has affirmed the inventor's.

The post-1970 regulatory wall lets the patent expire on schedule and then, without ever naming a patent, hands the original manufacturer back his protection from low-cost remanufactured competition with no terminus, and holds the public out of the Expired Patent Commons that should have opened. It produces, by regulation, an exclusivity without end. That is the single thing the Patent Clause permits no one in the federal government to produce — not the agencies, and not the Congress that made them.

That is the first suspender. It holds on its own. A plaintiff who proves it needs nothing else: the wall, as applied to patent-expired designs, exceeds the constitutional power of the government that built it, and a federal court can say so and order it lifted.

Be plain about what kind of remedy that is. The wall has stood for fifty-five years, and against the agencies that built it the wall is effectively permanent — the ratchet was designed to keep it so. But the agencies are not the last word. A court stands above them, and the remedies available to a court are set out in the chapter *How a Remake Is Built and Certified*. The wall is permanent only until a plaintiff asks the right court the right question. The whole of this chapter is that question, written out.

And the plaintiff need not stake everything on this single argument. The Patent Clause prong is the first of two independent grounds, and the second does not depend on

a single word of the first. A ruling that “limited Times” is decorative after all would have to run against two distinct lines of Supreme Court doctrine at once: the cases reading the Clause itself as a binding limit on Congress — *Graham*, *James v. Campbell*, *McCormick*, and *Florida Prepaid* — and the cases enforcing the public’s right to the expired invention — *Scott Paper*, *Sears*, *Bonito Boats*, and *Kimble*.

CHAPTER 25

The Takings Clause Prong: Property the Government Cannot Cancel for Free

*The patent expires on schedule. What it leaves to the
public is property —
and property cannot be taken without paying for it.*

The plaintiff has made her first argument. She has told the court that the agencies that built the regulatory wall never had the power to build it — that “limited Times” is a boundary on an enumerated federal power, and that an exclusivity without end lies outside it. That is the Patent Clause prong, the first of the two suspenders.

Now she turns to the second argument, and she begins it with a concession that costs her nothing. Suppose, she tells the court, that the first argument fails — that the court holds “limited Times” is decorative after all and that the agencies’ statutes authorize every brick of the wall. That ruling would be a sweeping reversal of the eight Supreme Court holdings that read “limited Times” as constitutive language. She does not expect it. The likelier scenario is quieter: the Supreme Court grants cert on both prongs, rules for her on Takings, and writes that it need not reach the structural argument. Either way — a Patent Clause loss or a Patent Clause silence — the work falls to the second prong. The second argument is built to do it.

That is the discipline of the two suspenders. They are not one argument stated twice. They are two arguments that fail and succeed independently. Even if the wall was within

federal power to build, what the wall did to the public's property is a taking. The Fifth Amendment does not allow that without compensation.

The Second Suspender

Begin with the words. The Fifth Amendment closes with a short clause: nor shall private property be taken for public use, without just compensation.

The clause is not a grant of power. It is a limit on power. It assumes the government will sometimes take property — for a road, a fort, a courthouse — and it does not forbid the taking. It attaches a price. The government may take private property, but it must pay for what it takes.

That structure matters here, and in a way that will not be obvious until the end. The Patent Clause prong ends with the wall removed: a power exceeded is a power a court can confine. The Takings Clause prong does not open by promising the same ending. It opens by naming a price. Where it arrives, and why it arrives at the same place the first prong does, is the subject of the final pages.

Four things must be established, in order. First, that what the regulatory wall destroyed is property — not a privilege, not a hope, not a loose figure of speech, but property in the constitutional sense. Second, that the wall's destruction of it is a taking — that a permanent regulatory cancellation of a vested property right is the kind of government action the Takings Clause governs. Third, that the clause reaches this property at all. The Fifth Amendment protects *private* property, and the thing the wall destroys carries the word *public* in its name. The objection is obvious and the chapter meets it: the right of access is held by every American individually, so the maker who built a business

on the expired incandescent bulb lost something as private as any acre when the federal efficiency rule took his product off the market. Fourth, what follows: what a court does when it finds a taking of this kind, and why the answer leads where the book's remedy leads.

One reminder before the argument starts. The previous chapter named the public's half of the patent bargain: the Expired Patent Commons. The argument takes that term as established. What it adds is a single word. The word is property.

Patents Are Property; So Is the Commons

Start with the half of this that no one disputes. A patent is property. In *James v. Campbell* (1882), the Court held that the government's use of a patented invention is a taking of private property — that the inventor must be paid, exactly as the owner of a seized acre must be paid. A patent is not a license the government lends and may quietly recall. More than a century later, in *Florida Prepaid Postsecondary Education Expense Board v. College Savings Bank* (1999), the Court said it again in plain words: patents are a species of property. The structural chapter needed only that a patent has a boundary. The takings argument needs the other fact the same case supplies, because the Takings Clause speaks only to property, and the first thing a takings argument must do is put property on the table.

But the inventor's patent is not what the regulatory wall destroys. The wall does not touch a live patent. The patent expires on schedule, untouched, exactly as the Constitution promises. What the wall destroys is on the other side of the expiration date. It destroys the public's half. And here the argument must do real work, because

the public's half is not as obviously property as the inventor's patent. It must be shown to be property, and the argument shows it three ways.

The first is the nature of the thing itself. When a patent expires, the right that comes into being is not vague. It is the right to make, use, and sell the invention — the same right the inventor held, now held by everyone. *Kimble v. Marvel Entertainment* (2015) describes that moment: at expiration, the unrestricted right to make and use the invention passes to the public — unrestricted, that is, by the patent monopoly, with the invention now governed by whatever generally applicable law would govern any maker of any other thing.

The patent's expiration ends the inventor's exclusion of others; it does not enlarge the regulatory regime the invention faces. What was lawful to make under patent is lawful to make once the patent has lapsed. The same legal rule applies at both ends of the term. Read that as a property lawyer reads it. A right to make, use, and sell a defined thing is the classic bundle of rights the word property names.

The public's right of access to an expired invention is not a different kind of thing from the inventor's patent. It is the same bundle, after it has changed hands from one owner to many. And the Court has used that word itself: in *Singer Manufacturing Co. v. June Manufacturing Co.* (1896), it held that at expiration the right to make the thing formerly covered by the patent becomes public property.

It said the same in substance in *Scott Paper* (1945) and *Bonito Boats* (1989). Those were patent, trademark, and assignor-estoppel cases, not takings cases, and none held that the public's interest is property the Fifth Amendment protects — that step is this argument's, not theirs. An

estoppel case, in plain terms, is one in which a court stops a party from taking a position that contradicts one it already profited from. Assignor estoppel is the patent-law version: an inventor who sells his patent and pockets the price is not later heard to argue that the patent he sold was invalid all along. But the noun is the Court's own. When the monopoly ends, the Court does not say the right vanishes. It says the right becomes the public's.

The second is a case about a different kind of intangible property, and it is the case this prong leans on hardest. In *Ruckelshaus v. Monsanto Co.* (1984), the Court considered trade secrets — commercially valuable information a company keeps confidential. A trade secret is intangible. It is not land, not a machine, not a raisin. The Court held that it is nonetheless property protected by the Takings Clause, and that when a federal agency's regulation appropriates the value of a trade secret, the agency has taken property and may owe compensation. The lesson is precise. Property, for Takings Clause purposes, is not confined to things a person can stand on or hold. An intangible right of defined economic value, recognized by law, is property — and an agency does not escape the Takings Clause merely because what it appropriated had no physical form. The public's right to practice an expired invention is exactly such a right: intangible, defined, economically real, recognized by two centuries of patent law.

The third is the most recent, and it answers the move the wall's defenders will reach for. They will say the public's right of access is not property because the government may simply decline to recognize it — that what the government never had to grant, it may withhold. In *Tyler v. Hennepin County* (2023), the Court closed that door. The county had seized a home for unpaid taxes, sold it, and

kept the entire proceeds, including the surplus far above the tax debt, arguing that state law said the county got to keep it. The Court rejected the argument. A government cannot evade the Takings Clause by passing a law that declares a historically recognized property right to be no property at all. The label the government applies does not control. What controls is whether the interest has long been recognized as property in fact. The public's right to an expired invention has been recognized since the first patent statute of 1790. A regulation that declares it worthless does not make it so.

Three cases, one conclusion. The public's right to access and utilize the Expired Patent Commons is property. It is the inventor's bundle, passed to the public. The Takings Clause has its subject. The argument can proceed to the verb.

When Regulation Goes Too Far

The verb is "taken." The wall's defenders have an obvious reply. The government did not take anything; it passed regulations. Nobody confiscated a building or seized a crop. Governments regulate property all the time without buying it — zoning, building codes, speed limits — and no one calls those takings.

It would be the end of the argument, except that the Supreme Court rejected exactly this reasoning more than a century ago. In *Pennsylvania Coal Co. v. Mahon* (1922), a Pennsylvania statute forbade coal mining that would cause the surface above to subside, even where a coal company had sold the surface rights while expressly keeping the right to mine the coal beneath. The State said it was merely regulating for safety. Justice Holmes gave the answer that founded an entire branch of constitutional

law: while property may be regulated to a certain extent, if regulation goes too far it will be recognized as a taking. Regulation is not a safe harbor. A regulation can take. The only question is how far it has gone.

The Court has since marked the field with two roads. The first is a balancing test, set out in *Penn Central Transportation Co. v. New York City* (1978), weighing the economic impact of the regulation, the degree to which it interferes with reasonable investment-backed expectations, and the character of the government action. *Penn Central* is the general test, the one that applies when a regulation diminishes the value of property without destroying it.

The second road is shorter and steeper. In *Lucas v. South Carolina Coastal Council* (1992), a coastal statute barred a landowner from building anything at all on two beachfront lots he had bought to develop. The Court held that when a regulation denies an owner all economically beneficial use of his property, the regulation is a taking without any balancing at all. The Court's words are worth having exactly: when an owner is "called upon to sacrifice all economically beneficial uses in the name of the common good, that is, to leave his property economically idle, he has suffered a taking." That is the rule on which this argument turns. It is a taking *per se*, because total deprivation is different in kind from partial diminution. A regulation that leaves an owner ninety percent of his property's value is weighed under *Penn Central*. A regulation that leaves him nothing is a taking on its face.

The *per se* road did not start with *Lucas*. It started a decade earlier with *Loretto v. Teleprompter Manhattan CATV Corp.* (1982), where a New York statute required landlords to allow cable operators to install a junction box

and a few feet of cable on the outside of a building, with compensation set at one dollar. The Court held that a permanent physical occupation of property is a taking per se, regardless of economic impact or the strength of the public interest. The hardware was small. The lost rent was effectively zero. The public benefit was real. None of it mattered. Lucas added the total-economic-wipeout branch in 1992; *Cedar Point Nursery v. Hassid* (2021) extended the physical-occupation branch to forced access for limited hours; *Horne v. Department of Agriculture* (2015) applied the per se rule to appropriation of personal property. The per se category is not one rule. It is a family of categorical rules, and the family has grown across forty years rather than shrunk.

The argument rides the Lucas branch. The wall does not physically occupy anything in the *Loretto* sense; the regulator does not install cable boxes or send organizers onto private land. What the wall does is eliminate all economically beneficial use of the right that passes to the public when a patent expires. The *Loretto* branch is named because the per se architecture is wider than the one branch this argument walks, and a constitutional argument is stronger when it shows the architecture it is part of rather than only the part it needs.

Now apply the two roads, and the wall's situation is not close. The wall does not diminish the public's right of access. It does not reduce it by some percentage and leave the rest. An invention has entered the Expired Patent Commons only when an American is legally able to actually rebuild it. The wall makes that legally impossible — not difficult, impossible. No American may lawfully manufacture the patent-expired design, because the design cannot meet the current year's standard and the law grants no grandfather exemption for remanufactured

goods. The right to practice the invention is not worth less under the wall. It is worth nothing. What is wiped out is the right to remanufacture the whole durable good once the totality of its patents has entered the public domain — not a parts list whose components the public already had, and not any single assembly in isolation, but the public-domain vehicle itself.

That is a Lucas total deprivation. The public's property in the expired invention is not diminished by the wall; it is wiped out. Under Lucas, that is a taking per se, and the analysis could stop there.

It does not have to stop there, because the wall is also a taking under the gentler Penn Central test, and a plaintiff is wise to win on both roads rather than one. The economic impact of the wall on the public's right is total, which Penn Central weighs heavily. The interference with investment-backed expectations is as complete as interference gets: the public's expectation of receiving the invention at expiration is not a private investor's hope but a two-century constitutional promise. And the character of the government action is the final Penn Central factor — the subject of the next section, because the character of this particular government action is the wall's deepest vulnerability of all.

The Government Cannot Escape by Calling It Regulation

Effect controls, not label. A government action is what it does, not what the agency files it under. That principle did structural work in the Patent Clause prong. It does takings work here, and the takings cases state it in their own voice.

Begin with *Horne v. Department of Agriculture* (2015). A New Deal–era marketing order required raisin farmers to turn over a percentage of their crop to a federal committee each year. The government argued this was not a taking but a regulation of commerce — a condition of participating in the raisin market, and farmers who did not like it could grow something else. The Court rejected the argument in full. Requiring the farmers to hand over their raisins was a physical appropriation, and a physical appropriation is a taking *per se*. The government could not escape that conclusion by calling it a regulation, by routing it through a committee, or by characterizing the lost raisins as a contribution to an orderly market. The Court looked at what the order did. It took raisins. That it was filed as agricultural regulation did not change what it was.

The wall is filed as emissions regulation, safety regulation, efficiency regulation. Horne says the filing does not control. What controls is the operation. The operation of the wall is to take from the public the right to practice an expired invention and to hand the original manufacturer back his protection from low-cost remanufactured competition. A government action that does that is not rescued by the cabinet it is filed in.

Three more cases close the remaining exits. In *Cedar Point Nursery v. Hassid* (2021), a California regulation gave union organizers a right to enter an agricultural employer’s property for a set number of hours. The Court held that a regulation granting a right of physical access is a *per se* taking even if the access is intermittent and even if the owner keeps every other use of the land. The parallel is not exact — the wall grants no one a right to walk onto land — but the principle is the one that matters: when the government takes a right of access and reallocates it, the Takings Clause is engaged. In *Arkansas Game & Fish*

Commission v. United States (2012), the Court considered government-induced flooding and rejected the argument that a temporary, recurring intrusion belongs to some other, non-takings category. Whether a taking has occurred turns on the effect of the government action on the property, not on whether the action fits a tidy label.

And in *Sheetz v. County of El Dorado* (2024), the Court answered the most technical objection of the family. A county had imposed a traffic-impact fee set by a general fee schedule its legislature had enacted, rather than by an official's case-by-case decision, and argued that the exactions doctrine of *Nollan* and *Dolan* reaches only individualized administrative conditions. The Court rejected the distinction unanimously. The Takings Clause does not contain a legislative exception. A burden on property is subject to takings scrutiny whether it arrives by an administrator's order or by a statute of general application. The wall is built of general rules. After *Sheetz*, that is no defense. A taking accomplished wholesale is still a taking.

Put the four together. *Horne*: the label "regulation" does not rescue an appropriation. *Cedar Point*: the appropriation of a right of access is read strictly. *Arkansas Game & Fish*: the effect on the property controls, not the formal category. *Sheetz*: general legislation enjoys no exemption. Every escape route from the word "taken" is, by recent and repeated Supreme Court decision, closed.

The Inequality the Wall Imposes

One more strand belongs to the takings argument. The wall does not cancel the public's vested right of access in isolation. It cancels that right while leaving other vested rights of the same constitutional class in place. The

asymmetry is the inequality, and it sharpens the claim in a way the Penn Central factors recognize.

Consider what the wall does not cancel. The owner of an old building lawfully built under earlier standards is not required to tear it down because the code has been ratcheted upward. The owner of a firearm lawfully purchased under earlier standards is not required to surrender it. The holder of a professional license issued under earlier standards is not required to retake the examination. American regulatory practice grandfathers vested rights as a matter of course. The wall is the unusual case. The vested right at the end of the patent term is the one vested right of its constitutional class that gets no grandfather.

The Court has met this shape of asymmetry before, under the Equal Protection Clause rather than the Takings Clause. In *Allegheny Pittsburgh Coal Co. v. County Commission of Webster County* (1989), a county taxed recently acquired coal properties at full current value while leaving long-held properties at decades-old assessments. Same property class, vastly unequal burden. The Court struck the practice down unanimously. Similarly situated property, treated unequally, by a distinction tied to when the activity occurred, is a constitutional shape the Court has already rejected. The remanufacturer of an expired patent and the original manufacturer of the same now-expired patent are similarly situated. The original was held to the standards of his patent's issuance year. The remanufacturer is held to ratcheted current-year standards.

One bridge is needed, because *Allegheny Pittsburgh* policed a county and the wall is federal. The Court supplied that bridge in *Bolling v. Sharpe* (1954) and reaffirmed it in

Adarand Constructors, Inc. v. Peña (1995): the Fifth Amendment's Due Process Clause imposes on the federal government an equal-treatment requirement equivalent to the one the Fourteenth Amendment imposes on the states. Those cases concerned racial segregation and federal contracting, not patent-expired goods; the extension to this asymmetry is the argument's, not theirs. The principle they establish is not. One limit belongs with the citation: in *Nordlinger v. Hahn* (1992) the Court sustained an acquisition-value tax scheme and confined *Allegheny Pittsburgh* to classifications no plausible policy could explain. The wall's vintage line has no such explanation, and on that ground the citation survives.

The inequality sharpens the second Penn Central factor — the character of the government action. A regulation that treats similarly situated property unequally on the basis of when the activity occurs is government action of a kind the Court has identified and disfavored. It is not the ordinary background hum of economic regulation. It is structural singling-out.

Answering the Objections

The first objection. A right of access is not property. It is a liberty, or an expectancy, or a hope. Until an American actually builds the patent-expired design, he holds nothing a court can call property — and the wall makes sure he never builds it, so the property never comes into being.

The answer has two parts. The objection proves too much: by its logic the inventor's patent would not be property either, until he had actually manufactured something, and *James v. Campbell* and *Florida Prepaid* hold flatly that the patent is property from the moment it issues, exercised or not. Property is the right to do a thing, not the doing of it.

The second part is Tyler. A right does not fail to be property merely because a recent law refuses to honor it. The public's right of access vested by operation of two centuries of patent law. The wall is the recent law refusing to honor it, and a government may not sidestep the Takings Clause by disavowing traditional property interests in what it wishes to appropriate. The objection has the order of events backward. The property exists, and the wall cancels it.

The second objection. Takings claims belong to identifiable owners. "The public" is not a person. A right held by everyone is, for takings purposes, a right held by no one.

The answer is that the plaintiff is not "the public" in the abstract. The plaintiff is a particular party with a particular, concrete stake in entering the Expired Patent Commons. A sheriff whose county must replace its vehicle fleet, barred from buying the remanufactured patent-expired trucks that competition would produce. A farmer barred from the remanufactured tractor. A remanufacturer ready to build — a factory, a workforce, and no lawful design to build. Each is an identifiable holder of the right of access, injured in a way no one else is injured. The right belongs to the public, but it belongs to each member of the public individually, and any one of them whose stake is concrete enough to satisfy standing is a proper takings plaintiff. The Constitution does not protect this property less because it is widely held. It protects each holder.

A further answer comes from a corner of the Court's own law where the property taken was never individually held at all. Tribal land held in common — owned by no member in fee, titled to the people as a body — is nonetheless property the Fifth Amendment protects when the federal

government takes it. In *United States v. Sioux Nation of Indians* (1980), the Court held that Congress's 1877 seizure of the Black Hills was a taking requiring just compensation, though the land belonged to the Sioux collectively. Half a century earlier, in *Shoshone Tribe v. United States* (1937), Justice Cardozo put the principle in four words that fit the wall exactly: "Spoliation is not management." The step the argument asks a court to take is the neighboring one. If a bounded people's common property is compensable when taken, the public's common right to the Expired Patent Commons — an interest the Court has itself called property since *Singer* — is of the same character, and its taking is of the same kind. The argument does not claim the tribal cases held as much; they concerned Indian lands under treaty. It claims only that they establish the governing principle — that property need not be privately and individually titled to fall within the Just Compensation Clause. The plaintiff who stands as an individual holder needs no such extension; the reader who doubts that route is offered this one.

The third objection. All regulation reduces some property values, and the Constitution cannot mean to make every regulation a compensable taking, or government could not function.

The answer is the distinction already drawn. Most regulation diminishes; the wall eliminates. A regulation that trims a property's value is weighed under *Penn Central* and often survives. A regulation that destroys all economically beneficial use is a taking per se under *Lucas*. And there is a second answer: the wall is not general regulation in the relevant sense. A genuine general regulation binds everyone alike and hands no one a competitive gift. The wall regulates the patent-expired design in a way that only the original incumbent can

satisfy, and in the same motion hands that incumbent his freedom from competition back. That is a transfer.

What Is Owed: Compensation, or the Wall

One objection remains, and it is the most important, because it is not really an attack on the argument. It is a question about what the argument wins.

Suppose the plaintiff is right. Suppose the public's right of access is property, and suppose the wall takes it. The Takings Clause still does not say the government may not take property. It says the government must pay for what it takes. So the plaintiff who proves a taking does not win the removal of the wall. She wins a check. The wall stays up, the Treasury cuts a payment, and the wall is still standing — now with a price tag, but no less in place.

This objection states the law correctly as far as it goes, and it stops one step too soon. The argument says so plainly rather than pretend otherwise. *First English Evangelical Lutheran Church v. County of Los Angeles* (1987) confirmed the point: the Takings Clause does not prohibit the government from taking property; it requires compensation when the government does. A finding that the wall is a taking does not, by itself and automatically, order the wall torn down. It establishes that the public is owed.

But follow what being owed actually means, because the objection stops one step too soon. What is owed is just compensation for the property taken, and the property taken is the public's right of access to the patent-expired durable goods of American life, across the better part of the durable-goods economy. On the conservative figure this book uses throughout, the wall has moved twenty-two

trillion dollars from American households to incumbent manufacturers, and it goes on moving roughly three thousand three hundred eighty-five dollars from the median household every year it stands. A compensation judgment for a taking of that property is a judgment on that scale.

No government will pay it. That is not cynicism; it is arithmetic. A standing obligation to compensate the public, every year, for the continuing denial of the Expired Patent Commons is an obligation no Congress can fund and no Treasury can carry. So a court's finding that the wall is a taking does not present the government with a bill it will quietly pay. It presents the government with a choice, and the choice has only two doors. Door one: keep the wall, and pay just compensation at the scale the taking is measured, and keep paying, year after year, for as long as the wall stands. Door two: lift the wall from patent-expired designs, open the Expired Patent Commons, and owe nothing going forward, because once the public has its access there is no longer a taking to compensate.

Set the two doors next to each other and the outcome is not in doubt. The Takings prong does not order the wall down. It makes the wall too expensive to keep standing. And the instrument the government reaches for when it walks through door two is the one the remedy chapter has already designed: the grandfather-exemption framework, under which a remanufactured patent-expired design is held to the federal standards of its certified platform year. That framework is not a defeat for the Takings plaintiff. It is door two. It is how a government that has been told it owes trillions stops the meter.

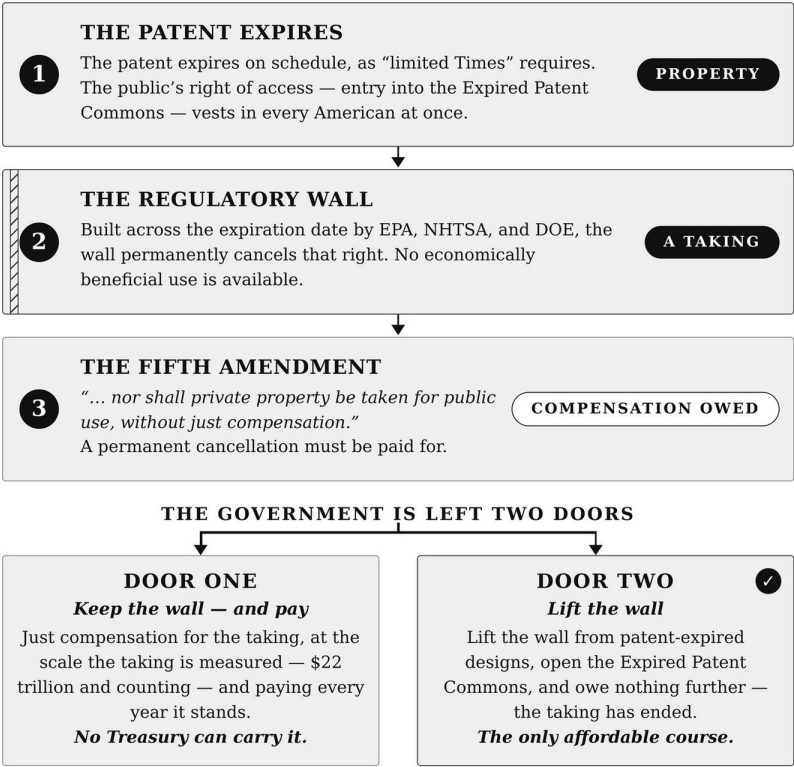
So the plaintiff does not plead for a check. She pleads in the alternative, the way good lawyers always have. She

asks the court to declare that the wall, as applied to patent-expired designs, effects a taking of the public's property without just compensation. And she asks the court, having declared the taking, to enjoin the wall unless and until the government provides the compensation the Constitution requires. The court names the violation. The political branches face the two doors. And the door they can afford to walk through is the one that lifts the wall.

This is why the two prongs, which travel by such different roads, arrive at the same place. The Patent Clause prong reaches the wall's removal directly: a power exceeded is a power a court confines, and the wall comes down because no one ever had the authority to build it. The Takings Clause prong reaches the wall's removal by a second road: the wall may stand only if the government pays a price it cannot pay. One road runs through the limits of federal power. The other runs through the cost of ignoring them. Both end with the Expired Patent Commons open.

What the Fifth Amendment Names

THE PUBLIC’S PROPERTY, TAKEN — AND THE PRICE OF
KEEPING THE WALL



A permanent cancellation of a vested property right is a taking. The Fifth Amendment leaves two doors — pay just compensation, or lift the wall — and only one is affordable.

And it should be said, against any reading of this as a counsel of despair: the actor in all of it is a court. The wall has stood for fifty-five years, and against the agencies that built it the wall has held. But the agencies are not the last word. A federal court can find the taking. A federal court can declare what is owed. The wall’s permanence has always run only as far as the level of the government that built it, and a court sits above that level.

The Second Suspender Holds

Stand back from the argument and see its shape. The Fifth Amendment protects property against being taken without just compensation. A patent is property; the Court has said so since 1882, and said so again in 1999. And the public's half of the patent bargain — the right of access that vests when the patent expires — is property too. It is the inventor's bundle of rights, passed at expiration from one owner to the whole public. It is intangible, and the Court has held that intangible rights of defined economic value are property all the same. It is old, recognized since the first patent statute of 1790, and the Court has held that a recent regulation cannot define a long-recognized property right out of existence.

The post-1970 regulatory wall takes that property. It does not diminish the public's right of access; it eliminates it, wholly, leaving no economically beneficial use behind — and a regulation that destroys all beneficial use is a taking per se. The wall cannot escape by being filed as emissions or safety or efficiency regulation, because the Constitution looks at what a government action does and not at the cabinet it is filed in. It cannot escape by being built of general rules rather than individual orders, because the Takings Clause has no legislative exception.

What the Constitution requires as payment is just compensation, measured against the scale of what was taken — and the scale is twenty-two trillion dollars and counting. That is a price no government will pay to keep a wall. So the wall comes down, not because the Takings Clause orders it down, but because the Takings Clause makes keeping it cost more than any Treasury can bear.

That is the second suspender. It holds on its own, and it holds independently of the first. That independence was the point of building two. A plaintiff who argued the Patent Clause prong and lost — or whose Patent Clause prong a court never reached, deciding instead on Takings grounds alone — would walk out of that courtroom with this argument still unspent. Nothing here borrows a single load-bearing premise from the chapter before it. The structural prong asks whether the government had the power. This prong grants the power and asks what the government owes. Lose the first, and the second is still standing. That is what two suspenders are for.

Both suspenders are now in place. The chapters that follow turn from whether the wall can be brought down to what is built once it is. A litigant may win on either one alone, and the wall, for all its fifty-five years, has never yet been asked to survive either.

CHAPTER 26

How a Remake Is Built and Certified

A favorable ruling reopens the commons. The skeptic and the court ask the same practical question. Reopened how? How is a remanufactured good built? How is it certified as safe and compliant, without rebuilding the wall and without waving regulation away?

The answer is at the level of mechanism: how a faithful remake is built, and how that build is certified. The point is not deregulation. It is parity. The public receives the expired design on the same regulatory timeline the original occupied, while genuinely new designs remain fully subject to current standards.

Design First, Then Build

A remake begins as a model, not a physical object. The remanufacturer assembles the design in a three-dimensional computer model, simulated, tested, and revised on a screen before any metal is cut. This is not exotic; it is how modern vehicles already come into being. The order matters for certification: the model is what gets certified, and the physical build is confirmed against it afterward.

What Counts as a Faithful Remake

A faithful remake is not any product inspired by an old design, nor a modern product wearing a vintage nameplate. It is a build whose material design elements

trace to the expired platform. Deviations are limited to necessary manufacturing substitutions, ordinary replacement materials, or separately lawful improvements depending on no still-active patent.

The rule is binary. A product is either a certified platform-year remake or a new design, and a new design meets current standards. A regulated feature may be materially changed — architecture, crash structure, emissions system, drivetrain integration, safety system, energy-consuming design. If the change leaves the product no longer the same platform, it leaves the commons for the current-year system.

A remake may also carry equipment the original did not. A backup camera, a tire-pressure monitor, a better lamp — the platform carries the addition and remains that platform. The test is not whether a change touches a regulated system but whether the design survives it. Such additions face the same limit as every deviation: they must depend on no still-active patent. A requirement the expired design can carry is additive; one that forbids the design as such is constitutive, and a remanufacturer's own improvements sit on the additive side. That boundary keeps the remedy from becoming a loophole. The claim protects the public's right to build the expired design. It gives no manufacturer the right to hide a new design inside an old label.

Mix and Match Is the Method

A remake is an assembly. The original was itself an assembly of separately invented parts, each patented on its own schedule, and the remake reassembles the integrated invention as it was disclosed and lawfully sold. The constitutional basis for treating that as the public's right is the aggregation principle argued in *The Patent*

Clause Prong: the whole becomes the public's when the totality of its relevant patents has entered the public domain.

That licenses no copying of anything still protected. A remake must contain no still-patented component unless the remanufacturer holds a license or uses an independently lawful substitute. Nor may it use protected trademarks or trade dress in a way that confuses buyers about source. The public-domain right is a patent right, not a trademark defense and not a license to infringe an unexpired patent. What the remake reassembles is certified against the federal standards in force when the original platform was certified, not against today's. The standard is time-shifted, not removed.

What Qualifies for the Commons

Not every expired design belongs to the commons this book defends. One rule sets the boundary: a design enters when it was sold lawfully to the public and the invention has passed into the public domain. How the patent ended does not matter. A full twenty-year term and an early abandonment for unpaid maintenance fees end in the same place. The Constitution draws no line between them, and neither does this remedy.

One thing is excluded, and only one. A product pulled from the market while under patent because it was unsafe never entered the commons. It was a failed design, and its passage into the public domain changes nothing, there being nothing legitimate to inherit. The test is not how long the patent lasted but whether the design lawfully served the public and was never banned as unsafe.

It must be defined this way, for a reason bearing on how the wall could otherwise survive a ruling. Suppose eligibility turned on a patent running its full term. The agencies could then empty the commons by tightening the ratchet until no manufacturer carried one to term, and early abandonment would become the wall's escape hatch. The public's right cannot depend on how long a private party pays maintenance fees. It depends on two facts: lawful sale, and a now-public invention.

Self-Certification Is Already the American System

The United States does not pre-approve most vehicles for safety before sale. It relies on self-certification: the manufacturer certifies that the vehicle meets every applicable federal safety standard and stands behind it. For emissions, the Environmental Protection Agency issues certificates of conformity on submitted data. Certification is already part of the American system, and the means are already flexible — physical testing, or simulation and engineering analysis where those demonstrate compliance. Independent firms do this work today, against federal standards. The proposal invents no new concept.

How a Remake Is Certified

Every remade good needs certification, whatever blocked it — emissions, energy, or safety alone. The structure is the same for all three, and it does two jobs. It documents that the remake qualifies for the platform-year treatment a ruling restores. It affirms that the remake meets the federal rules of the year its design maps to.

The mechanism has four parts. The remanufacturer builds a three-dimensional model. The model is mapped to the original certified platform year. An independent engineering certifier affirms it meets the federal safety, emissions, and energy rules of that year. The physical build is confirmed against the certified model. A remade light bulb uses the same structure in miniature. The simplest blocked good and the most complex use one mechanism, scaled to the product.



The Independent Certification Layer

The independent certification layer is what makes the remedy administrable. The certifier must be independent of the remanufacturer’s production decision, professionally qualified, subject to record-retention duties, and exposed to legal consequences for false certification. It may be paid by the remanufacturer, as compliance professionals often are, but its duty runs to the legal standard, not to the client’s preferred outcome.

It answers three questions. Is this a faithful remake of the expired platform? Does the build contain no still-patented

component unless independently licensed? Does the model satisfy the federal safety, emissions, and energy standards of the platform year? If any answer is no, the product receives no platform-year treatment. That gives courts and agencies a manageable line: the court identifies the constitutional class, the agencies administer the technical rules, the certifier supplies the record, and the remanufacturer bears the risk of error.

Equal Treatment, Not Exemption

The remake is not exempt from certification. It submits to it. It is not exempt from safety law; it is mapped to the safety law the original met. It is not exempt from emissions or energy law; it is mapped to the rules of the certified platform year. That is the equality principle made operational. The original earned lawful entry under the rules of its year; the public, inheriting the expired design, earns the same opportunity under the same rules — not a privilege, and not a loss of regulation.

Why the Remake Floor Is Higher Than the Original's Was

Does the proposal lower the standard at the point of certification? It does not. It raises the floor in five ways.

First, the original deal was self-certification. The manufacturer attached its own label to its own product, attesting to the federal safety standards in force on the date of manufacture. No agency reviewed most vehicles before sale. The remake adds an independent engineering layer at the threshold.

Second, the proposal is structurally limited to expired-patent designs lawfully sold. It reaches no failed product banned as unsafe, no design with still-active patents in the relevant components, and no newly designed product.

Third, the rule prevents gaming. A pre-standard design that cannot be mapped to an applicable federal platform-year standard receives no certification under this remedy for that standard. A modernized product departing materially from the expired platform is a new design under current standards.

Fourth, market review adds discipline after certification. Insurance underwriting, fleet purchasing, crash-test ratings, consumer reviews, wholesale grading, and ordinary buyer inspection all operate on remakes as on other products. They are not substitutes for the regulatory floor but checks on top of it.

Fifth, the tort system reaches remakes as it reaches originals. A remanufacturer whose vehicle injures someone faces the products-liability exposure any manufacturer faces today. The lawsuit is not a substitute for the regulatory floor. It is the backstop the floor sits on.

One suspicion remains: that the remake commons competes with the patent system. It does not. A remanufacturer who wants to ship a genuinely new improvement — a safer restraint, a cleaner aftertreatment — takes it through the patent system, as any inventor would. The patent system rewards novelty; the remake commons restores continuity. They are the two halves of the bargain the Constitution wrote: exclusive rights for a time, public access at the end.

The Question Tort Law Leaves Open

One question remains, and candor requires naming it. Tort law lets a person injured by a product sue the maker for damages — the private lawsuit, separate from government regulation. It runs through a different door: the state courthouse, not the federal agency.

A favorable ruling time-shifts the federal wall, making the remake lawful to build and sell, certified to its platform year. It does not tell a state what its tort law must hold. Federal safety standards, as a rule, do not displace state tort law. The savings clause of the federal motor-vehicle statute, 49 U.S.C. § 30103(e), says plainly that meeting a federal standard does not exempt anyone from common-law liability. That clause is not absolute. In *Geier v. American Honda Motor Co.* (2000) the Supreme Court held that it does not bar ordinary conflict preemption, and found a state claim preempted where it collided with a specific federal standard. Absent such a conflict, state suits survive.

For the remanufacturer, federal certification is strong evidence. A design built and certified to a recognized federal standard is powerful proof it was not defective, admissible in any courtroom in the country. But it is evidence, not immunity, and states differ on the weight they give it: some a full defense, some a rebuttable presumption, some one factor among many, and at least one no defense at all.

This is not a flaw unique to remakes. The tort system judges era-relative products every day — the restored vintage car, the rebuilt industrial press, the refurbished appliance — and already knows how to ask what a maker reasonably owed given the standards of the time. The

remedy opens the commons. It does not dictate the verdict in every courtroom that follows, nor should a single ruling try.

Who Can Order This Remedy

The remedy does not require the Court to write a new regulatory code. It requires the agencies to apply existing standards to a constitutionally required certification date for a narrow class. That is work they do daily, administering model-year rules, conformity rules, grandfather provisions, and compliance testing. The judgment identifies the class: faithful remakes of fully patent-expired designs containing no still-patented component and passing independent certification to the standards of the platform year.

Technical implementation remains with the agencies. They decide forms, records, testing protocols, audits, recall obligations, and enforcement, subject to the constitutional boundary. The Court supplies the rule of law: the public may not be denied the expired design by moving the compliance year beyond the timeline the patentee itself enjoyed. That division respects institutional roles.

The certification test does not change with the body that orders it. Only who applies it changes. The Supreme Court may adopt platform-year parity directly, with independent certification as its condition — the book's inference, not a holding any court has entered. It may instead find the violation and remand. Congress may codify it by statute. The agencies may take the mechanics, though candor requires naming the worry: they built the wall, and an agency asked to set the year that unbuilds it has an interest in the status quo. An independent certification layer answers that. Or the Court may fashion its own

remedy, curing the same injury in a way that still lets the public build the expired design.

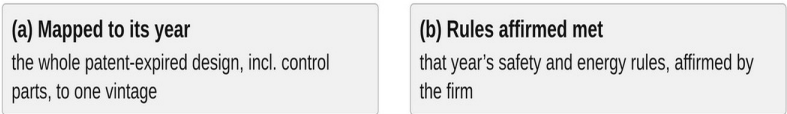
The paths differ in who decides. They converge on what is decided. Because the mechanism is stable across deciders, the constitutional holding does not wait on the Court's choice of remedy.

The Two-Step Certification

EVERY REMADE GOOD USES THIS — BULB TO SEMI



TWO FINDINGS



Whatever the reason a good was blocked — emissions, energy, or safety — its remake is certified the same way. A remake light bulb uses exactly this, and nothing more.

Closing

Certification keeps remade goods safe and compliant. The wall never did that work. Safety, emissions, and energy regulation do that work, during the patent term and after, for originals and remakes alike.

The wall did something else. It barred the public from the commons after the bargain was complete, holding the expired design to a standard that rose every year — one the design was never allowed to meet while the patent kept

the public out. Strike the wall, and the commons operates again under honest certification, equal to what the original met and more carefully documented. The Court can strike it as applied to faithful remakes and require platform-year treatment; the agencies apply existing tools. The Court cannot direct Congress to spend, and this remedy does not need it to. What Congress may add — tariffs, import restrictions, a statutory remake program — is for the political branches.

CHAPTER 27

Objections Considered

Every argument worth making attracts objections. What follows takes the strongest objections this book has drawn since previews of the argument began circulating in 2025 and 2026 — from economists, lawyers, libertarian friends, and populist skeptics — and answers them in the form they were raised. The correspondents are anonymized at their request, but each objection is reproduced in its strongest form rather than its weakest, because a book that ducks the strongest version of an objection has not earned the right to be read past it.

Seventeen objections follow. They are not the easiest ones. The first nine and the twelfth through the seventeenth are answered in full. The tenth and eleventh are objections this book takes seriously enough to have given them their own chapters; the entries below point the reader to them.

1. Regulation Has Always Reached Patents

The patent has never been immune from regulation. Why does expiration change anything?

This is a strong objection, and the answer begins by conceding what is correct in it. A patent has never conferred immunity from regulation. The Supreme Court said so in 1878 in *Patterson v. Kentucky*, rejecting a patentee's claim that his federal patent put his improved burning oil beyond a state safety-inspection statute. The patent secured the inventor's exclusive right against private competitors. It left the product subject to police-

power regulation aimed at public safety. *Webber v. Virginia* reaffirmed the principle two years later, and *Aronson v. Quick Point Pencil Co.* restated it a century on. A patented drug must still pass the Food and Drug Administration. A patented vehicle must still meet federal crash standards.

This book does not argue otherwise. The argument is structural, not absolute. The patent expires on schedule. So does the original manufacturer's obligation to meet the federal safety, emissions, and efficiency standards in force on its date of manufacture. Both endings were written into the system the Constitution authorized. What this book identifies is a regulatory architecture that ends neither.

Look at what the remanufacturer is actually asking for. He is not asking to escape regulation. He is asking for the same regulation: to build to the federal standards in force on the date the original was manufactured. Those are the standards the original passed when it rolled off the line, and the ones it still bears as a lawful vehicle on the road. The certification is by an independent engineering firm rather than his own staff, a more rigorous floor than the original's self-certification ever had to clear. Equal treatment is not exemption. The current architecture does the opposite. The original was tested against the standards in force when it entered the market. The remake is held to a moving standard ratcheted upward on the incumbent product cycle, and denied any goalpost it can ever reach. That is the inequality, and it is not a metaphor.

The Supreme Court has been precise about the bargain's second half. *Kimble* (2015) states that when a patent expires, the unrestricted right to make or use the article passes to the public. *Sears* (1964) held that an article on which the patent has expired is in the public domain and

may be made and sold by whoever chooses to do so. Made and sold — the language of a substantive right, not of a residual freedom from infringement liability. The wall makes that right unusable in practice while preserving its bare legal recognition on paper.

Consider a case this book loses. In 1971 the National Highway Traffic Safety Administration bans a patented in-dash coffee-maker as unsafe, and the patentee loses his challenge — reasonable safety regulation, entirely consistent with *Patterson*. The patent expires in 1987. In 1995 a remanufacturer wants the coffee-maker back. He cannot have it. He loses on takings under *Penn Central* and *Lucas*, and he loses on the structural argument too, because the 1971 rule preserved no incumbent's market position, tracked no patent-expiration cycle, and Congress could repeal it tomorrow.

The book's argument attaches only where three conditions are jointly satisfied. First, the underlying technology was once patented, so the two-half bargain was genuinely in play. Second, the regulation has the effect of preserving the incumbent's market position past expiration, rather than imposing a public-safety floor that would have been imposed regardless of patent status. Third, the regulatory mechanism is statutorily designed to be irreversible — the anti-backsliding provision at 42 U.S.C. § 6295 in the efficiency context, New Source Review in the emissions context, the motor-vehicle-safety ratchet at NHTSA.

The third condition is misread most often. A rising floor is not the complaint; a standard that rises can also fall, and the possibility of revision is what makes it a standard rather than a boundary. Anti-backsliding removes the revision by statute, as the chapter *The Anti-Backsliding Provision* sets out. A floor with the return path removed is what converts a

moving standard into a permanent exclusion of the expired design. A safety ban on a coffee-maker meets the first condition and neither of the others.

The first condition does more work than it appears. A never-patented or open-source design does not enter the Expired Patent Commons, because no patent bargain was struck for it. Such a design may be perfectly free to make under the general public domain. What it lacks is the reciprocal claim the Patent Clause creates only where an inventor invoked patent protection and the term then ran out. The wood-burning stove that was never patented draws no Patent Clause claim against a later ban. The wood-burning stove whose patent expired in 1985 does.

The book's actual case satisfies all three. The diesel-engine patents underlying every Class 8 truck expired decades ago, and the emissions tiers surrounding them ratchet in lockstep with patent-expiration cycles rather than with any independent safety floor. The same holds for the internal-combustion patents underlying every American light vehicle. The incandescent-bulb patents expired more than a century ago, and the federal efficiency rule that withdrew the bulb finalized in 2023.

That is the heart of the answer. The book's case is narrower than the strongest objections to it would suggest, and it is stronger. Its quarrel is not with safety regulation. It is with regulation engineered to work as an enduring private benefit at public expense — regulation that imposes on the remanufacturer an unequal burden the original never faced, whichever enumerated power Congress nominally invokes.

2. The Carpet Case Proves Too Little

Your carpet case proves too little. A safety rule the old design can simply carry does not extend any monopoly, so where is the constitutional injury?

A federal appeals lawyer posed this objection in its cleanest form. Picture a levitating-carpet company that patents its design in 1930, expecting exclusivity until 1950. In 1940 a safety law requires fore-and-aft blinking navigation lights. Every carpet built after 1940 must carry them. The company complies by bolting the lights on and keeps its exclusive right until 1950. After that a competitor may build the same 1930 design, lights included. Both makers are free. No monopoly outlasts the term. Every step of that is correct, and it is conceded without reservation.

The objection's force depends entirely on one feature of the rule it chose. Blinking lights are something you add to a design that otherwise stays intact; the 1930 carpet is still the 1930 carpet, now wearing lights. The chapter *The Patent Clause Prong* draws the line this turns on. A rule the expired design can carry is additive, extends no monopoly, and draws no complaint from this book. A rule that cannot be satisfied by supplementing the design, but only by replacing it, is constitutive — a modern emissions tier is not a lamp clamped onto an older engine, and under whole-vehicle certification the copy is judged as a current new machine — and it forbids the expired design as such. Ask of any post-expiration rule the one question: can the expired design carry it, or does it forbid the design outright? The carpet flies on, wearing its lights. The expired engine cannot be built at all.

The lawyer's own scenario supplies the proof of where the line falls, in a detail he added almost in passing: carpets built between 1930 and 1940 may keep operating without the lights. That concession lets a non-current-compliant instance of the design keep flying, in the hands of whoever owns it, indefinitely. The grandfather permission runs with the unit, not with the first buyer. The same asymmetry governs American roads today. A 2005 pickup may be sold in 2027 to a buyer who never owned it, who then drives it lawfully though it meets no current standard.

Set two 2027 buyers side by side. One finds a surviving 2005 original on a lot and may buy and drive it. The other wants an identical, newly built faithful copy of that same expired design and may not have it, because the copy is judged as a current machine. Same design, same road, identical output per mile. The only difference is the date of manufacture. The law permits the old atoms and forbids identical new ones. That is not a safety distinction or an emissions distinction — an original and a faithful copy pollute and protect alike. It is a distinction of vintage, and vintage is the precise axis the Patent Clause commands to stop mattering once the term ends.

The remedy corrects the vintage line without touching the standard for anything current, and it must not be misheard as a wall coming down. The wall does not fall. It time-shifts. A remanufactured patent-expired design is held to the full federal standards its original was certified to in its platform year. Regulation stays entirely in force. What moves is the year.

One question remains, and it is the strongest the objection can still raise. If a maker may build a faithful copy held only to an older standard, does the exemption swallow the standard? Could an incumbent route its current products

through this door? It cannot, and the reason is structural rather than a matter of anyone's restraint. The mechanism admits a build only if every underlying patent has expired and the design is certified to its own platform-year standard. Those gates are time-locked. They admit nothing but a design at least two decades old, held to the rules of its own decade. A maker cannot push this year's design through the same door, because this year's design is still under patent and maps to no earlier platform year. The exemption structurally cannot reach a current design. The door opens only onto the graveyard of designs the Constitution has already released.

3. Assemblies, Not the Whole Vehicle

The Patent Clause releases each assembly, not a right to remanufacture the whole vehicle.

A skeptical ex-lawyer friend pressed this argument through four rounds of correspondence. The answer begins by conceding what is correct in it. A patent does not create the right to manufacture; the right pre-exists the patent. The patent makes the right exclusive against others for a term, and expiration removes that exclusivity. The throttle control, the antilock-braking module, the gas-strut hood prop — at expiration each may be used by anyone in any new design she wishes, and the regulatory wall does not block her. So far the objection is on solid ground.

The objection then takes a step further than the Patent Clause permits. It reads the Clause's second half as a release of components only. The text does not speak of components. It speaks of "Writings and Discoveries" — of inventions as the inventor disclosed them. A patent is granted on a claim, and a claim is typically an assembly of

elements cooperating to produce a result. The grant is on the integrated invention; so is the release. Reading the release as a release of components is reading half of what the Clause does and dropping the other half on the floor. In *Aro Manufacturing Co. v. Convertible Top Replacement Co.* (1961), the Court held that no single unpatented element of a patented combination carries the monopoly; the monopoly is in the claimed combination as a whole.

The public-domain cases support this reading, and the limits belong with them. *Sears* (1964) and *Bonito Boats* (1989) are preemption holdings, not holdings about what expiration affirmatively delivers. But in stating them the Court described the background rule in words it has repeated ever since: an article on which the patent has expired is in the public domain and may be made and sold by whoever chooses to do so. Free exploitation of ideas is the rule; a federal patent is the exception. If free exploitation is the rule and the patent the exception, then what the expired patent releases is not a grudging list of parts but the disclosed invention itself, arriving for the first time in the default state the Clause presumes. That last step is an inference this book draws and asks a court to confirm. The book states the affirmative right as its own argument, not as something the Court has already decided.

The objection's real force is a question the answer must meet directly. Even granting that each expired patent releases its whole claimed assembly, how does a release of assemblies become a right to remanufacture a whole truck? The answer is aggregation, and the chapter *How a Remake Is Built and Certified* sets out the mechanism. Most of a vehicle is unpatented commodity, or assemblies whose patents expired long ago. A handful carry live patents, and each covers only what its claims describe: the

tailgate-lift assembly covers the tailgate-lift assembly, not the truck. The vehicle becomes the public's to remanufacture when the last of those live patents has entered the public domain, subject only to the trademark and trade-dress law that protects the maker's name and identifying design.

This is also the only reading that makes the bargain a bargain. Many patents are issued for assemblies whose individual components were already public. On the components-only reading, those patents at expiration would deliver to the public a parts list whose contents the public already had. The inventor would receive years of exclusivity; the public would receive nothing in return. That is not the consideration the Constitution recites. The Patent Clause is written as an exchange, and a reading under which the public never receives the invention it was promised is a reading in which there was never consideration.

American practice has tracked the integrated reading for as long as there have been expired patents. A generic-drug manufacturer does not acquire merely the right to use the active molecule; he acquires the right to manufacture the drug as the patent disclosed it — dosage form, delivery system, formulation — and to sell it against the originator. Furniture reproductions from expired design patents reproduce the chair, not the joinery technique in isolation.

The objection nonetheless identifies a real seam. The Patent Clause guarantees that the term shall end and the disclosed invention shall pass to the public. It does not decide whether other federal regulation may then be built around the invention in a way that prevents the public from practicing it. That second question is the work the Takings prong does. The two prongs are not redundant. They are

complementary, and the objection has correctly identified where one hands off to the other.

A policy version sometimes accompanies the legal one: the remanufacturer is monetizing a design he did not develop. The description is accurate, and it describes every entrant into every expired-patent commons in the country's history. The public's later use is the inventor's payment to the public for the exclusivity he was granted.

One form of the objection survives all of this. The hardware current regulation requires — the catalyst, the sensor monitoring, the on-board diagnostics — is itself patented, and those patents are still in force. But the remanufacturer is not seeking a license on a current design. He builds to the federal standards the original passed, using platform-year hardware that has itself entered the Expired Patent Commons. To hold otherwise is to let a later patent on the compliance overlay do precisely what “limited Times” forbids.

4. Why Equality Is Not a Third Prong

If equality under the law is doing this much work, why is it not a third constitutional prong of its own?

The question is fair. The book has been clear, in the chapters *The Patent Clause Prong* and *The Takings Clause Prong*, that equality under the law is doing real work. “Limited Times” is itself an equality command. The wall imposes an unequal cancellation of vested rights. The remanufacturer asks for the same regulation the original met, not exemption from it. So why is equality not a third suspender?

The answer is doctrinal, not editorial. Pled as a standalone Fifth Amendment due-process or equal-protection prong, the equality argument is too weak to be a suspender. The case that says so is *Williamson v. Lee Optical of Oklahoma* (1955). Oklahoma required eyeglass replacement lenses to be made only from prescriptions written by licensed optometrists or ophthalmologists. Opticians who could safely fit replacement lenses without a prescription challenged the statute as economic protectionism dressed in health regulation — and the legislative record suggested exactly that. The Court upheld it unanimously, announcing the rational-basis floor for federal economic classifications: a court asks only whether the classification is rationally related to some conceivable legitimate government interest. Not whether the stated rationale is the real one. Not whether the burden falls evenly on similarly situated parties.

The wall has stated rationales — public safety, vehicle emissions, energy efficiency. A federal court asked whether the wall is rationally related to those rationales will say yes, even where the connection is loose and the wall's true purpose lies elsewhere. The court does not need to believe the rationales are the real motives. It needs only to find them conceivable. They are.

Williamson is old and its subject is eyeglasses, and a skeptical reader will say so. The better case against a standalone equality prong is recent and sits on this book's own axis. In *Nordlinger v. Hahn* (1992), the Court sustained a California scheme that taxed property by when its owner acquired it and found the classification rational. That is the vintage line, upheld. An equality claim pleaded on its own would meet *Nordlinger* and lose. One clarification belongs with that admission, because the Takings prong cites *Allegheny Pittsburgh Coal Co. v.*

County Commission of Webster County, the case Nordlinger confined. Rational basis governs a freestanding equal-protection claim. It does not govern whether Congress exceeded an enumerated power, and it does not govern whether a regulation destroyed all economically beneficial use. The vintage asymmetry is used as an input to the character factor under Penn Central, never as an independent claim, and Nordlinger leaves that use standing.

A narrow escape route exists — the rational-basis-with-bite line running through City of Cleburne (1985), Moreno (1973), and Romer (1996), where the Court struck down classifications because the only rationale it could find was animus toward a disfavored group. The route is real and it is narrow. A federal court asked whether the agencies acted out of animus toward remanufacturers will deny the characterization on the pleadings. The agencies have written rationales for fifty-five years that say nothing about remanufacturers. They targeted product categories, not a disfavored class. The animus framing does not fit the record.

A suspender that breaks under most ordinary loads is not really a suspender. Equality under the law is the animating principle inside each of the two existing prongs, and it borrows its strength from the prong it animates. Inside the Patent Clause prong it is the Clause's own internal logic, and the Patent Clause is not economic regulation getting rational basis but a structural limit on an enumerated power. Inside the Takings prong it is Allegheny Pittsburgh — similarly situated property treated unequally on a distinction tied to when activity occurred. Both prongs gain force from the equality framing. Neither depends on it. A litigant who insists on three suspenders has not improved

her odds. She has confessed that the first two are not load-bearing on their own. They are.

5. The Independent Commerce Power

Grant all of that. The Commerce Clause is an independent enumerated power, and nothing in Clause 8 disables it. Where is the authority holding that a limitation written into one clause of Article I, Section 8 binds Congress when it legislates under another?

This is the most serious objection the argument faces, and a government brief will lead with it. It deserves its strongest form. Clause 8 is a grant of power, not a prohibition. It says nothing about what Congress may do when it regulates commerce among the several States, and the commerce power is independent and has been read broadly since *Wickard v. Filburn* (1942). A rule setting an exhaust-emissions tier is not a patent grant, mentions no patent, and confers no exclusive right on anyone. If the Constitution wanted the commerce power narrowed by the patent bargain, it could have said so. No decision of the Supreme Court holds that Clause 8 constrains commerce-power legislation.

Two of those premises are conceded. Clause 8 contains no express prohibition addressed to the other enumerated powers, and no court has held that it constrains a commerce-power regulation. The third — that the first two settle the question — fails, and the fastest way to see why is to follow the objection to its own conclusion.

Give the reading two steps instead of one. Congress secures an exclusive right for twenty years. The term runs. The patent expires on schedule, exactly as “limited Times” commands, and not one day is added to it. On the

following morning Congress enacts a statute under the commerce power forbidding the manufacture of the disclosed invention outright and forever. Nothing in the objection's reading makes that statute invalid. The Clause has been honored in full. Yet the public receives nothing, on that morning or on any morning after it, for the disclosure it paid for. The practical result is a patent that never expires, obtained in two steps because it could not be obtained in one. A constitutional limitation that can be emptied by the sequential use of a neighboring power is not performing the work of a limitation. It is performing the work of a formality.

The regulatory wall is that outcome reached without anyone intending it. No Congress wrote the second statute; the wall was assembled from separate rules with separate purposes across five decades. That makes it less culpable than the hypothetical statute. It does not make it lawful. The Constitution asks what a government action produced, not what its authors intended.

The Supreme Court has answered a version of this question. In *Railway Labor Executives' Association v. Gibbons* (1982), Congress had enacted a statute governing a single bankrupt railroad's estate. The statute was not uniform, and the Bankruptcy Clause empowers Congress to establish uniform laws on the subject of bankruptcies. Defending it, the government argued that even if the Bankruptcy Clause could not support a non-uniform law, the Commerce Clause could, because the commerce power is an independent grant unqualified by a requirement written into a different clause. That is the same argument, in a different clause of the same section. The Court rejected it: to allow non-uniform bankruptcy laws under the commerce power would eradicate the limitation the Bankruptcy Clause imposes. A limitation any

adjacent grant may be used to circumvent is not a limitation at all.

The objection's best response is that the two clauses are not alike — that the Bankruptcy Clause commands how Congress must legislate, while Clause 8 is a bare grant. A reading of the two clauses does not sustain it. Clause 4 empowers Congress to establish “uniform Laws on the subject of Bankruptcies.” The word “uniform” is not a freestanding command lodged elsewhere in the document; it is an adjective inside the grant, describing the only kind of law the grant authorizes. Clause 8 empowers Congress to secure exclusive rights “for limited Times.” That phrase is likewise a qualifier inside the grant, describing the only kind of exclusive right the grant authorizes. Both say the same structural thing: you may do this, and only in this way.

Nor did Gibbons turn on that distinction. Its reasoning asked one question: would the general power, used this way, leave the specific limitation without operative force for the party it was written to protect? Put that to the wall. The patent term still ends on the day the statute says it ends. What does not happen is anything at all. The design the disclosure purchased cannot be built — not by the public, not by a competitor, not by the original maker — and it cannot be built in any year that will ever arrive.

A structural analogy sharpens the point, and it is offered as the argument's own rather than as anything a court has held. Clause 12 empowers Congress “To raise and support Armies,” and then, in the same sentence, limits the power it has just granted: “but no Appropriation of Money to that Use shall be for a longer Term than two Years.” That is a grant with an internal temporal cap, built exactly as Clause 8 is built. No one supposes Congress could

appropriate ten years of army funding by routing it through the commerce power and then arguing that the two-year limit binds only Clause 12. The limitation would mean nothing if a neighboring grant could accomplish what it forbids.

Mark what this answer does not claim. It does not say Clause 8 freezes the regulatory baseline for every once-patented design, and it does not say the commerce power is narrow. It says the commerce power cannot be used to produce, permanently and by statutory design, the one result Clause 8 specifically forbids. The three conditions are the whole of the claim's reach.

A line of authority runs beneath the answer, and its rungs are not of equal strength. The first rung is the Court's treatment of private post-expiration devices. In *Brulotte v. Thys Co.* (1964) and *Kimble* (2015), the patents had expired on schedule and the calendar was honored perfectly. The Court struck the arrangements down anyway, because another instrument had withheld the practical consequence of expiration. Had "limited Times" been a scheduling instruction, both would have come out the other way in a paragraph.

The second rung reaches a sovereign actor. In *Sears* (1964) and *Bonito Boats* (1989), the party asserting control over the released design was a State, legislating under an independent and otherwise valid power, using an instrument that was not a patent. The Court held in both that what the federal patent laws leave unprotected, a State may not re-protect by a parallel mechanism. The honest limit belongs with the holding: these are preemption decisions resting on the Supremacy Clause, not Clause 8 holdings against federal power. A government brief will say so, and it will be right. What the

cases do establish is narrower and still useful. Sovereign action, taken under an independent power and by an instrument unrelated to patents, has already been held subject to the released-design principle.

The third rung is this argument's own, and the book states it as such rather than dressing it as authority. The step is that a limitation which binds the States through federal patent policy binds the federal government through the Clause that generated the policy. No court has taken that step, because none has been asked. It is short in logical distance and it is still a step, and a reader who declines to take it has not misread the argument.

Which leaves the objection's best support: the long line of decisions sustaining federal product standards as commerce-power regulation of new goods. Those decisions are real, and this book disputes not one of them. But in none was a "limited Times" objection raised, briefed, or decided. A decision sustaining a rule against the challenges actually presented is not authority against a challenge nobody presented.

Name what would defeat it. If "limited Times" is satisfied the moment the calendar term runs out, without regard to whether the public ever receives anything the terminus was meant to deliver, then the wall stands, the disclosure bought nothing, and the second half of the Clause is decorative. If "limited Times" is a bargain whose end must carry a consequence, then an architecture that empties the consequence by design is the nullification Gibbons refused to permit. The Court has never chosen the first reading when the question of practical consequence was actually put to it — not against a private contract, not against a State. No one has ever put it against a federal regulatory architecture.

6. Andrus v. Allard and the Sale Ban

Andrus v. Allard holds that a commercial-sale ban is not a taking. Doesn't that foreclose the Takings Clause prong?

The case is real and adverse. In 1979 the Court held unanimously that a federal regulation banning the commercial sale of eagle feathers lawfully obtained before the Eagle Protection Act was not a taking, because possession and non-commercial transfer were preserved. The Court treated possession as dispositive. If Andrus governs the regulatory cancellation of expired-patent commerce, the Takings prong has serious trouble. The book does not pretend otherwise. Three responses follow, in increasing order of strength.

The first is that Andrus has been substantially narrowed. *Horne v. Department of Agriculture* (2015) held that the Takings Clause applies to personal property with full force, that requiring growers to surrender a percentage of their crop is a per se taking, and that the government cannot avoid the Clause by characterizing its action as a voluntary condition on market participation. *Tyler v. Hennepin County* (2023) reinforced the principle that a government cannot declare that citizens have no property interest in something the law has historically treated as property. A reader of Andrus alone might conclude the prong is foreclosed. A reader of Andrus together with *Horne* and *Tyler* will see that the door closed in 1979 has been partly reopened.

And the wall resembles *Horne* more than Andrus. Andrus forbade only a sale; possession and non-commercial transfer were preserved, so the regulation barred a market and left the right in place. The wall does not merely forbid sale. It forbids the conduct — manufacture for

remanufactured competition — that gives the public's expired-patent right any economic content at all. Strip out that conduct and nothing of the right is left. The wall closes the only channel through which the right can be exercised.

The second response is that Andrus dealt with a regulatory scheme that had no independent constitutional anchor. The Eagle Protection Act protects a wildlife population; the interest it served had no clause in Article I, Section 8 dedicated to it. The Expired Patent Commons is different. The interest the wall destroys has a clause the Constitution wrote specifically to protect it. That does not change takings doctrine in the abstract. It changes the second Penn Central factor — the character of the government action — because the action here is not safety regulation incidentally affecting an expired patent. It is regulation whose effect, whether by design or by capture, is to defeat a constitutional bargain. The Andrus Court never considered that argument, because no party raised it.

The third response is structural. The Takings Clause is one of two suspenders. A court that rules against the book on the Takings prong has not necessarily defeated the book, because the Patent Clause prong survives an adverse Takings ruling intact. Andrus would have to be re-litigated only if the Takings prong were the book's only theory. It is not. A litigant will plead both in the alternative. Andrus narrows the path the Takings prong must travel. It does not foreclose the case.

7. Zoltek and Patents as Property

Zoltek holds that patents are not protected by the Takings Clause, and serious scholarship says the court was right. Doesn't that foreclose the Takings Clause prong?

The question names a real case and a real article. In 2006 the Federal Circuit held in *Zoltek Corp. v. United States* that a patent owner could not plead the government's use of its patented invention as a Fifth Amendment taking. Judge Plager dissented; rehearing en banc was denied over a further dissent from Judge Newman; the Supreme Court denied certiorari without reaching the merits. The next year Davida H. Isaacs argued in the *George Mason Law Review* that the court had reached the right result — that patents are federal entitlements, and that some interests earn due-process protection without earning takings protection. Three responses, in increasing order of strength.

The first is that *Zoltek* is not a Supreme Court holding. It is one circuit's decision, binding no court outside the Federal Circuit, and a denial of certiorari settles nothing. The 2006 opinion was later vacated by that court sitting en banc. Having established the government's liability under the statute, the en banc court wrote that it need not and therefore left open the question of liability under the Constitution for a taking, and it vacated the trial court's determinations on that issue. The Federal Circuit has since said plainly that the 2006 takings analysis is persuasive authority and no longer binding. So the objection's strongest authority is a vacated opinion on a question the vacating court declined to answer. Take the objection at its strongest anyway; the two responses that follow answer it either way.

The second is that the case and the article concern a different property and a different owner. *Zoltek* was a corporation complaining about the government's use of its patent, and that patent was in force. This book's plaintiff is not a patent holder, and the patent in her case has expired on schedule, untouched. What she claims is the public's

half — the right of access that vests when the term ends. Neither the decision nor the article reaches it.

The third meets Isaacs on her own ground, because the second will not satisfy her argument. Her point is that a right created by federal statute is a benefit rather than property in the takings sense, and the public's right of access is created by federal patent law too. The answer is that the public's half is not a benefit the government conferred at its discretion. It is the consideration the Constitution names for the grant. The inventor discloses, and the payment is the public's eventual right to practice what was disclosed. That is why the power is limited in the first place. A government that keeps the disclosure and cancels the payment has not regulated a bargain. It has taken one side of it. Tyler holds that a government may not sidestep the Takings Clause by disavowing a property interest the law has long recognized. Applying that holding to the public's half is this argument's own step. The Court has not taken it, because no one has asked.

8. Eldred and Term Extension

Eldred v. Ashcroft upheld a copyright term extension against a “limited Times” challenge. Doesn’t that foreclose the Patent Clause prong?

The case is real and adverse. In 2003 the Court ruled seven to two that the 1998 Sonny Bono Copyright Term Extension Act, which added twenty years to the term of every existing copyright, did not violate the limited Times prescription. Justice Ginsburg's majority held that the term remained finite — limited, not perpetual — and that Congress had discretion to set the number within the constitutional range. Justice Breyer dissented, arguing the extension produced a term worth 99.8 percent of the value

of a perpetual copyright, and that the Founders did not write the prescription “in terms of the calculator rather than the calendar.” The majority answered: the calendar settles it.

Eldred does not foreclose the patent-side argument, and the reasons matter. Eldred is a copyright case. The two clauses occupy the same constitutional sentence and share the limited Times language, but they operate on entirely different subject matter. Copyright protects expression; the bargain is the author’s exclusive right over a finite work in exchange for the work’s eventual passage into the public domain. Patent protects a useful invention, with industrial and consumer goods downstream; the bargain is the inventor’s exclusive right in exchange for that invention’s eventual passage into the Expired Patent Commons — into manufacture, copying, improvement, and remanufacture by anyone. The consequence of a limited Times failure in copyright is that no one can lawfully republish the book. The consequence in patent is that no one can lawfully remanufacture the invention. The first is a publishing question. The second is a question about half the American economy.

The Court in Eldred was reviewing a single congressional act that added a fixed number of years to a fixed term — a calendar adjustment the majority could test against the prescription by reference to a number. This book’s case is not built on a calendar adjustment. The patents expire on schedule. What the book challenges is a regulatory architecture, statutorily designed to be irreversible, that defeats the public’s reciprocal commons interest after the patent has expired. No number on the calendar can test what the book describes, because the violation is not the length of the term. It is the destruction of the public’s half of the bargain the term was supposed to make possible.

Worth noticing as well what the Clause cannot be made to mean. Its language was written in 1787, more than two years before the first federal patent statute existed. Whatever “limited Times” requires, it cannot be defined by the term the Patent Act happens to set, because the Clause preceded the Act and the Act was written to conform to it rather than the reverse. A defender who answers that the term is twenty years, the term expired, and the inquiry is therefore over has assumed the statute may supply the Clause’s content. It may not.

Eldred held that Congress can extend a term within the limited Times range. It stopped short of holding that Congress can use a separate enumerated power to defeat the bargain entirely. The Court was never asked the second question, and a federal appellate court has said as much in print. In *Kahle v. Gonzales* (9th Cir. 2007), the Ninth Circuit affirmed dismissal of the Internet Archive’s challenge on the ground that it was essentially the argument Eldred rejected — adverse, so far. But the same opinion states that Eldred did not purport to directly address whether “limited Times” places substantive limits on duration, citing Eldred’s own footnote recording that the outer-boundary question was not before the Court. The openness this answer relies on is a reading a federal court of appeals put on the page while ruling against the party who wanted it. The limits belong with the citation: *Kahle* is a copyright case, the panel ruled for the government, and the sentence is an aside.

One federal appellate court has come closer, and its treatment cuts both ways. In *United States v. Moghadam* (11th Cir. 1999), the court upheld a federal anti-bootlegging statute under the Commerce Clause while assuming the Copyright Clause could not sustain it, reasoning that in some circumstances the commerce

power can accomplish what Clause 8 alone might not. Read that far the case is adverse, and a government brief will read it that far. But the court attached a condition: the commerce power may not be used this way where the legislation is fundamentally inconsistent with the limitations of Clause 8. And in a footnote it identified where that inconsistency would most likely be found — the statute protected live performances with no express time limit and would arguably run indefinitely, where the Clause permits exclusive rights for limited Times only. The court did not decide the point; the appellant had not preserved it. What survives is a federal appellate court naming the structural problem this book presses, treating it as serious, and leaving it open. The step from that footnote to the regulatory wall is this argument's own.

A tactical point belongs here too. Justice Breyer's dissent did some of the doctrinal seedbed work this argument relies on — that the limited Times prescription has economic content as well as numeric content, and that a term whose practical effect approaches “virtually perpetual” is a category the Constitution recognizes. The majority answered Breyer on the calendar. This argument does not require the Court to reverse itself on the calendar. It requires the Court to take up the structural question the calendar does not reach.

One question governs how every case in this landscape should be read. When a decision is offered against this argument, ask what the decision upheld the statute against. A court sustains a statute against the challenges brought to it, and says nothing about a challenge nobody brought.

9. Golan, Kewanee, and the Commons

Golan v. Holder holds that the public domain is not sacrosanct, and *Kewanee Oil* lets a trade secret withhold an invention forever. If the public domain carries no constitutional protection, what exactly does the wall violate?

Both cases are real and adverse, and the first can appear worse for this book than *Eldred*. In *Golan v. Holder* (2012), the Court upheld a statute restoring American copyright to foreign works that had already fallen into the United States public domain. The petitioners were the people using them — orchestra conductors, publishers, film archivists who had programmed and distributed the material freely for years. The Court disagreed with them six to two, in terms this book cannot pretend are narrow: the contention depended on the premise that the Constitution renders the public domain largely untouchable by Congress, and that premise had already been rejected. The opinion refers throughout to “the Copyright and Patent Clause.” It is one clause. A government brief will quote that sentence, and it should.

The answer is that “the public domain” names two populations that share nothing but a label. One is the Expired Patent Commons: inventions that were patented, disclosed in the specification as the price of the grant, protected for a term, and released when the term ran out. The other is everything never protected in the first place — unpatentable subject matter, work that missed a formality, foreign works from countries the United States had no copyright relations with when they were made. The first is in the public domain because a bargain was completed. The second is there because no bargain was ever struck.

Golan reached only the second. Asked what limited term the restored foreign works had received before entering the American public domain, the petitioners answered: exactly zero. Those works had never held a United States copyright. No term ran, no disclosure was exchanged for exclusivity, nothing was promised and then withheld. They occupied the public domain the way an unpatentable idea occupies it — by default, not by completed exchange. Having never received the Clause's protection, they had not earned its promise. A patent-expired truck design is the precise inverse: the full twenty years ran, the inventor took every benefit the grant conferred including the federal suit that enforced it, the disclosure was made as the price, and the public's half came due on schedule. Golan says nothing about that population, because no party in Golan belonged to it. The distinction holds across every fact pattern in this book, including the one that looks most like Golan: the incandescent bulb, a design the public exercised for ninety-seven years before it was withdrawn, whose filament patents ran their full statutory term.

Kewanee Oil Co. v. Bicron Corp. (1974) is the broader attack and the shorter answer. The Court held that state trade-secret law survives federal patent preemption, so the Constitution tolerates a regime under which an invention may be withheld from the public forever. But the trade-secret holder has made the opposite election. He forgoes the grant and everything that comes with it, obtaining no right against independent discovery, none against reverse engineering, and none against anyone who works the thing out and builds it. *Kewanee* bars misappropriation. It has never barred manufacture. The inventor who took the patent, and with it the disclosure requirement and the enforcement machinery, cannot afterward claim secrecy's perpetuity as well. He chose the road with a terminus at the end of it.

One admission belongs on the record. Courts have long treated expired-patent subject matter as a distinct category — *Scott Paper* speaks of the invention of an expired patent being dedicated to public use, and *Graham* of the free nature of disclosed ideas. But the further step, that Congress's power over the default public domain does not extend to the bargained-for one, is this argument's own. No court has drawn that line, because none has been asked to. What the landscape shows is not a body of law rejecting this argument. It is the absence of a body of law, which is what a case of first impression looks like from the outside.

10. Why Has No One Filed It?

If this argument is as strong as you say, why has no one filed it in fifty-five years?

The chapter *Why Nobody Noticed* answers this question in full.

11. Legalizing Unsafe Trucks?

Strip away the constitutional rhetoric, and this book is asking the courts to legalize unsafe trucks, dirty engines, and energy-wasting light bulbs. Why is that a victory worth winning?

The chapter *The Environmental Question* answers this objection in full.

12. Destroying the Original Makers

Open the public domain to remanufacturing and you destroy the original manufacturers. Who would invest in

building anything if a competitor can copy it the day the patent expires?

This is the economic objection, usually offered as common sense: strip the protection and the protected company dies. The everyday record says otherwise. Toilet paper, paper towels, and facial tissues fill an aisle of every grocery store in America. The perforated toilet-paper roll, the facial tissue, the safety razor, the zipper: each began as a patented invention, each patent expired generations ago, and each is made and sold today by American companies in open competition with anyone who cares to enter. The names on the shelf are brands, not monopolies. They compete on quality, price, and reputation against store brands built to the same public-domain designs. That is the bargain working exactly as written. The public gets its cheaper option, and the maker keeps everything except the monopoly.

The Supreme Court settled the point in the objection's own century. When the foundational sewing-machine patents lapsed in 1877, rivals were free to build the once-protected machines, and many did. Singer was not destroyed. It kept its lead for generations on its sales network, its installment financing, and its name, and in *Singer Manufacturing Co. v. June Manufacturing Co.* (1896) the Court confirmed that the expired design, and even the Singer name in its generic sense, had passed to the public.

Nor did the vehicle makers wait to be victims. Ford ran a factory-rebuilt engine exchange through its dealers as early as 1934, and in 1966 Detroit Diesel and Cummins stood up remanufacturing programs of their own, because a second sale from the same engineering pays. If the Court restores the remanufacturing right, the likelier outcome is the one the past already showed: the

incumbents enter the business themselves. The objection casts the incumbent as the remedy's victim. The record suggests it would more likely be its beneficiary.

13. Has Anyone Actually Been Stopped?

All of this is theory. Has anyone actually tried to build a faithful copy of an expired-patent vehicle and been stopped?

Yes, and the clearest test is not hypothetical. Begin with what stops the builder. It is not a fee. Two federal statutes forbid the sale outright. A new motor vehicle may not be manufactured for sale, sold, or imported unless it complies with the safety standards in force on its date of manufacture and carries a certification. And a new vehicle may not be sold or introduced into commerce without an EPA certificate of conformity, which the agency issues only against the emissions tier in force now, not the tier in force when the design was built. Emissions compliance is measured per vehicle, not as a fleet average, so no averaging past it is possible. These are prohibitions on manufacture and sale, not the corporate-average fuel-economy standard, which is a fleet penalty rather than a ban — and which, as of the statute enacted in July 2025, now carries a maximum penalty of zero. The wall that matters here is a ban. It is not a fine.

Nor can the builder solve the problem by swapping in a few modern parts. He can bring an older engine up to a current emissions tier; the industry does it with documented upgrade kits. But doing so means replacing the engine-control module, the turbocharger, the pistons, the injectors, and the exhaust aftertreatment with modern, currently patented equivalents. At that point the object on the assembly line is no longer a faithful copy of the

expired-patent design. It is a partly modern vehicle built around live-patent compliance hardware owned by the incumbents. The builder faces a fork with no free exit. Build the faithful design, and it is unlawful to sell. Build a lawful vehicle, and it is no longer the public-domain thing the Constitution released.

The decisive point is that the experiment has already been run, by Congress itself. In 2015 Congress passed the Low Volume Motor Vehicle Manufacturers Act to let small firms build new replicas of designs at least twenty-five years old — the remanufacturing scenario in miniature. Congress went so far as to waive the federal motor-vehicle safety standards for up to 325 such vehicles per manufacturer each year. Yet a decade later, essentially no gasoline replicas have been built under it. The reason maps precisely onto the wall. Congress could waive the safety standards, but it did not waive the emissions bar; instead it wrote into the same law that a replica is deemed compliant only if the builder installs a current-model-year certified engine, with all its modern emissions-control equipment. The statute that was supposed to reopen old designs required, in its own text, that the original powertrain be abandoned. The replica may wear the old body; it must carry a new engine.

That is the answer to the reader who suspects the injury exists only on paper. When the government itself set out to let old designs back onto the market, it lowered the safety bar to do it and left the emissions wall standing — indeed it codified the wall into the very exemption meant to relieve it. Congress tried to reopen the commons, and the wall stopped even the government.

14. Copyright, Trademark, and the Vehicle

Can an entire vehicle be protected by copyright or trademark, or is the most that can be protected the brand name, the logo, and the hood emblem?

The ninety-five-year figure some readers cite is a real number attached to the wrong object. Under 17 U.S.C. § 302(c), a work made for hire is protected by copyright for ninety-five years, but a car or truck is not such a work. Copyright does not protect useful articles, and the example Congress chose to illustrate that limit was the automobile itself. A photograph of a car can be copyrighted; that gives no one the right to stop others from building the car.

What copyright does reach are separable features. Under *Star Athletica, L.L.C. v. Varsity Brands, Inc.* (2017), a feature of a useful article is copyrightable only if it can be perceived as a work of art apart from the article and would qualify as one on its own. A badge, an emblem, or a graphic on a body panel can clear that bar. The fender under the badge, the frame, the engine, and the geometry that makes the vehicle the vehicle cannot.

One thing on the vehicle is protected for the full term. The firmware in the engine-control module is a computer program, and a computer program is a literary work. Written by the maker's employees, it is a work made for hire, so code written in 2005 stays under copyright into the next century. The remedy does not touch that. It is also narrower than it sounds, for three reasons. First, copyright protects the expression, not the function. *Baker v. Selden* (1879) held that a copyright in a book explaining a bookkeeping system gave no monopoly over the system itself. The maker's copyright covers its lines of code, not

the job those lines do; if the method was novel, that was a patent question, and patents expire on schedule. Second, the courts have protected the room a rewriter needs. *Google LLC v. Oracle America, Inc.* (2021) held that reimplementing the parts of a software interface a new program needs in order to work with existing code was fair use. A remanufacturer may study the original module to learn what signals the engine expects, then write new code to produce them. Third, the problem shrinks with the platform year. A design from before the early 1980s has no engine computer at all, and the aftermarket has written replacement engine-management code for 1990s platforms for decades. What the remanufacturer cannot do is copy the maker's program into its machine. What it can do is write its own.

Trademark is the opposite case, the longest right and the narrowest. The name on the tailgate can be protected forever, and nothing here proposes to disturb that. A remanufacturer who stamps the original maker's name on a faithful copy is infringing, and should be. Trade dress can protect a distinctive non-functional look for as long as it identifies its source, which is how the International Trade Commission made Mahindra restyle the Roxor's face in 2020 while leaving the machine underneath untouched. But the Supreme Court has drawn the line that keeps trade dress from swallowing an expired patent. In *TrafFix Devices, Inc. v. Marketing Displays, Inc.* (2001) it held that a functional feature cannot be trade dress, and that an expired utility patent is strong evidence the feature it claimed is functional.

So the intellectual property system already handles this question correctly. The incumbent keeps a perpetual right in its identity, a bounded right in its appearance, its own words in its own code, and no right at all in the expired

functional design. The wall this book describes was not built from copyright or trademark. It was built from emissions, safety, and efficiency rules that ask nothing about patent status. A remanufacturer can satisfy every requirement of Title 17 and Title 35 and still be barred from selling the vehicle.

15. Who Controls Who Files It?

Once this argument is public, you cannot control who files it. What stops a large firm from assembling plaintiff classes and suing for money damages rather than the remedy this book describes?

Nothing stops it, and the book should say so plainly. A published constitutional theory belongs to whoever wants to use it. What can be said is that the money version is structurally weaker than the remedy version, on four counts.

First, the courthouse door. The sovereign-immunity waiver that makes this case possible is the Administrative Procedure Act's, and it is expressly limited to relief other than money damages. A damages claim must go to the Court of Federal Claims under the Tucker Act, with its six-year limit. Of fifty-six years of transfer, at most six are reachable.

Second, no defendant sits where the money went. The manufacturers complied with valid regulations, which is no cause of action, and an antitrust theory built on the lobbying that raised the wall meets *Eastern Railroad Presidents Conference v. Noerr Motor Freight, Inc.* (1961), which immunizes petitioning the government.

Third, the damages framing forfeits the best plaintiff. A membership association may sue for prospective relief on its members' behalf, but under *United Food and Commercial Workers Union Local 751 v. Brown Group, Inc.* (1996) a damages action running solely to its members is barred for want of standing; the chapter *Direct and Associational Plaintiff Categories* sets out the test. A firm that persuades an association to sue for money disqualifies its own client.

Fourth, the comparison. Six years of contested partial recovery for some households, against a permanent change worth roughly \$3,385 per household per year to every American household, indefinitely. The relief this book asks for is worth more than the money, and unlike the money it can be won.

A fee note explains the interval since June 2024, when the end of Chevron deference made this case filable. Against federal agencies the fee vehicle is the Equal Access to Justice Act, which caps hourly rates, excludes wealthier plaintiffs, and pays nothing if the government's position was substantially justified, a defense that is strongest on a question no court has answered. The firms equipped to bring a case this size cannot be paid to bring it. The natural plaintiff is an association, a public-interest organization, or a sovereign.

One risk stays on the record. A damages case brought badly and lost could produce language a later court reads as governing the remedy case. The author accepts that cost; holding the argument private has produced fifty-six years of nothing.

16. Oil States and “Limited Times”

The Constitution says “limited Times,” in the plural, and in Oil States the Court called a patent a public franchise the government may cancel. Hasn’t the doctrine already moved past you?

A leading scholar of administrative law put it in a single sentence in September 2026: it will be very difficult to get this doctrine changed, because the word is “Times,” in the plural, and because the Court has already gone further in Oil States. The sentence carries two objections. They deserve separate answers.

Take the plural first. The Clause does not say “for a limited Time.” It says “for limited Times.” A reader can make that plural do real work: if the Constitution contemplates more than one limited time, a term may be followed by another term, and the Court has let Congress do exactly that in *Eldred v. Ashcroft* (2003) and *Golan v. Holder* (2012), both taken up in the eighth and ninth objections above.

Two things answer it, and the first is a story about one letter. On August 18, 1787, Madison proposed a copyright power for authors, and he wrote it “for a limited time” — singular. A separate proposal offered the same day covered patents for useful inventions and set no term at all. Eighteen days later the Committee of Eleven folded the two proposals into a single clause, and the singular could not survive the merger. One clause now served two classes of grantee and thousands of grants, each with a term of its own. “Time” became “Times” the way “Author” became “Authors” and “Writing” became “Writings” — by agreement with the plurals around it. The s is grammar. It is not permission.

The second answer does not depend on the first, and it is the one that matters. Grant the most permissive reading of the plural the Court has ever adopted. Grant that Congress may extend a term, and may grant a new one to a work already in the public domain. Every “Time” in that reading still has three features. It is granted by Congress. It is granted to an Author or an Inventor. It ends. The wall this book describes has none of them. No Congress granted it as a term. It runs to no inventor; the patentee’s rights expired on schedule, and the incumbent manufacturer holds no grant of any kind. And it has no end. The plural loosens the count of terms. It does not loosen the word “limited,” and it cannot reach an exclusion that is not a term at all. Neither *Eldred* nor *Golan* holds that “Times” authorizes exclusivity that nobody granted, to nobody, forever, under a power that is not the Clause. Both cases can be conceded in full and the wall still stands outside them. That is this argument’s inference, not the Court’s holding.

Now *Oil States*. In *Oil States Energy Services v. Greene’s Energy Group* (2018) the Court held, seven to two, that the Patent Office may cancel an issued patent in inter partes review without an Article III court. Justice Thomas, for the majority, called the patent a public franchise and patent validity a matter of public rights that Congress may assign to an executive agency. Justice Gorsuch, joined by Chief Justice Roberts, dissented: a patent, once issued, is private property, and the Executive may not take it back without a court. If an agency may cancel a live patent mid-term, why would the Court police what agencies do after the term ends?

The majority called its holding narrow and said in so many words that it was not holding patents to fall outside the Due Process or Takings Clauses. The Federal Circuit has twice

taken up the question the Court reserved — *Celgene Corp. v. Peter* (2019) and *Golden v. United States* (2020) — holding retroactive review not a taking, while repeating that Oil States does not say patents are not property. Both concern cancelling a live patent. Neither says a word about the invention after the term ends.

That is the seam. Oil States sits on the grant side of the Clause. It asks whether the sovereign that issued a franchise may reconsider it during the franchise's term. This book lives on the other side. It asks what the public holds when the term is over. The two questions share a Clause and nothing else.

The franchise vocabulary, read carefully, helps more than it hurts. A franchise is not something the grantee owned before the grant. It is carved out of what belongs to the public, on statutory conditions, for a term, and the nineteenth-century cases the majority relied on describe the patent that way: a grant from the government that exists only within the limits Congress set. Every one of those limits includes an end. When the term ends the franchise is not handed back; it lapses, and the design it enclosed enters the public domain for the first time. The book's own inference, stated as such, is that a court which calls the patent a public franchise has already named the owner of what remains when the franchise expires.

A government brief will read the case the other way, and at full strength: if a patent is only what Congress makes it, exclusivity is a statutory franchise and nothing more, and a regulation preserving the incumbent's position implicates no constitutional right. The patent is what Congress made it. The public's half is what the Constitution made it — the consideration the Clause names for the grant.

What the objection gets right is the difficulty, and the book does not pretend otherwise. But the objection speaks of getting a doctrine changed, and there is no doctrine to change. Eldred and Golan decided the term questions put to them. Oil States decided the tribunal question put to it. None of the three was asked whether a federal regulatory architecture may make a patent-expired design unlawful to build, for everyone, with no end. That question has not been answered because it has not been asked. This book asks it.

17. The Rational-Basis Trap

Strip the argument to what a court will see: a remanufacturer wants to sell a product that fails a current safety rule. That is economic regulation, and since 1938 the Court has upheld economic regulation whenever a legislature could rationally have thought it served a purpose. Why would a court do anything but apply that test and uphold the wall in a paragraph?

The objection describes what will happen if the case is pleaded carelessly. Under *Williamson v. Lee Optical* (1955) and its line, a court asks only whether the legislature might have thought a rule served a legitimate purpose, and every emissions tier and crash standard passes that test. If the claim were that a rule is unwise, unfair, or costs more than it saves, the rational-basis test would govern and the claim would lose. The book does not bring that claim.

The rational-basis test asks whether a law is a reasonable use of a power Congress has. It has never been the test for whether Congress has the power. When the Court asked in *United States v. Lopez* (1995) whether the commerce power reached a gun in a school zone, it did not ask

whether Congress could rationally have thought guns near schools were harmful. It asked whether the activity was within the enumerated power. *Railway Labor Executives' Association v. Gibbons* (1982) did the same: the single-railroad statute was not irrational, and the Court never asked; it asked whether the commerce power could evade the Bankruptcy Clause's limit. And in *Eldred v. Ashcroft* (2003) the Court asked whether the term was "limited" before it deferred on the term's length. Power first, deference after.

So the complaint must open as an enumerated-powers claim: whether Congress may produce, by any combination of powers, an exclusive right that never ends. That question is decided in the plaintiff's first paragraph. Pled as a challenge to regulation, the case dies under *Lee Optical*. Pled as a question about the boundary of Clause 8, the test never enters.

Closing

Other objections exist, answered in the earlier chapters. The seventeen above are the ones the book has drawn most consistently. A reader still unconvinced is reading the book in something close to the form the author hoped — with the strongest objections in view, and the answers within reach. The case the book makes is harder to win than a maximalist claim would be. It is also harder to refute.

CHAPTER 28

Reason the Constitutional Argument Yourself: The Open-Book Examination

A reader who has come this far is entitled to a fair question: how do I know the argument in these pages is sound, and not merely well built? The chapters have laid out the holdings, the statutes, and the asymmetry between the two halves of the patent term. Lawyers are reviewing the manuscript. But the reader need not wait for the lawyers, and need not take the author's word. The reader can test the argument directly.

The instrument for that test is now widely available, and being plain about what it is matters. A modern artificial-intelligence language model is, for practical purposes, something like a gifted graduate assistant — a fast study with a wide command of material and a strong instinct for logical consequence, though no judgment a court would defer to. The exercise in this chapter does to such a model what a professor does to a graduate student on the first day. It sends it back to school. It hands the model the syllabus — the Supreme Court holdings, the relevant constitutional text, and the governing federal statutes and regulations — together with plain diagrams showing what this book sees in that material. Then it asks the model to reason.

The point is not that a machine's agreement proves anything. A model is not a court. Its answers vary between systems, between versions, and between runs, and they can be wrong. The point is narrower and more interesting. If a range of these systems — built by different companies, trained on different data, carrying different dispositions —

are handed the same neutral materials and asked to reason from them, do they converge? Or do they diverge? A leading question put to a single agreeable model proves nothing. The same neutral question, put to many independent ones, is a different kind of evidence: not proof, but a test of whether the conclusion follows from the authorities, or only from the way the author tells the story.

How the Examination Is Built

The examination is built like a trial, in three parts. First comes **the law** — the relevant Supreme Court holdings, the constitutional text, and the governing federal statutes and regulations, given whole, the cases that cut against this argument set beside the cases that favor it. Then come **the two advocates** — a case for restoring the public’s right and a defense of the present system, each given equal space. Last comes **the verdict** — the reasoning of the artificial intelligence, after it holds the full law and has heard both sides.

One question this examination does not ask is who may bring such a case to court. That is the separate matter of standing and timing, taken up in *Sovereign and Public-Fiduciary Plaintiffs*, *Direct and Associational Plaintiff Categories*, and *The Filable Case: Why 2026 and Not Before*. This chapter tests the merits — whether the law permits the regulatory bar — not the procedure for raising the question. A juror weighing the merits need not know who the plaintiff is.

What follows is in two parts. Part One is the examination itself — the law, the two advocates, and the diagrams — written to be read by the AI, and available free of charge online at marti124.github.io/Patent_Exam. Part Two is for you: how to run the examination, the prompts to give the AI, and the sheet on which to score what it answers.

A note before the materials. Little in Part One is new to a reader who has come this far. The holdings below are the same cases walked in *The Patent Clause Prong* and *The Takings Clause Prong* — *Graham*'s grant-and-limitation, *Singer*'s public property, *Scott Paper*'s dedication to public use — stripped here to their operative language, the way a court would receive them, with the cases that cut against this book set beside the ones that favor it. You may read Part One as a review, skim it as a checklist, or skip directly to Part Two and let the machines do the reading. The materials are for them.

PART ONE — The Examination Materials (for the AI to read)

Everything in this part is the material the AI reads. It is published free of charge online at marti124.github.io/Patent_Exam, where it downloads under the filename `The_Forgotten_Half_of_the_Patent_Clause_Exam.pdf`.

To run the examination, give the AI that PDF and then one of the prompts from Part Two. Nothing in this part tells the AI what to conclude.

Reasoning instruction: “You are given a set of legal materials and the arguments of two advocates. Reason from the materials provided. You may reason by analogy, extending the principles underlying these holdings to the facts presented; when you do, distinguish clearly between what a case holds and what you infer by extension. Be rigorous and concise.”

The Supreme Court holdings

1. *Patterson v. Kentucky*, 97 U.S. 501 (1878) – A patent does not exempt the patented article from generally applicable state safety regulation. Source:

<https://supreme.justia.com/cases/federal/us/97/501>

Operative language: *The patent right “is not granted or secured without reference to the general powers which the several states of the Union unquestionably possess over their purely domestic affairs, whether of internal commerce or of police.”*

2. *Webber v. Virginia*, 103 U.S. 344 (1880) – A patent confers the right to make and sell the invention, but does not place the article beyond the reach of general police-power regulation. Source:

<https://supreme.justia.com/cases/federal/us/103/344>

Operative language: *“Letters patent granted by the United States do not exclude ... the tangible property in which the*

invention or discovery is embodied” from a State’s tax or license law.

3. *James v. Campbell*, 104 U.S. 356 (1882) – A patent is private property; the federal government's use of a patented invention is a taking for which the owner must be compensated. Source:

<https://supreme.justia.com/cases/federal/us/104/356>

Operative language: *A patent “confers upon the patentee an exclusive property in the patented invention which cannot be appropriated or used by the government itself, without just compensation, any more than it can appropriate or use without compensation land which has been patented to a private purchaser.”*

4. *Singer Manufacturing Co. v. June Manufacturing Co.*, 163 U.S. 169 (1896) – When the patents on a machine expire, its form becomes public property, free for anyone to make. Source:

<https://supreme.justia.com/cases/federal/us/163/169>

Operative language: *“[T]he right to make the thing formerly covered by the patent becomes public property”; “there passes to the public the right to make the machine in the form in which it was constructed during the patent.” Along with the device, the generic name also “pass[es] to the public,” subject only to the maker’s duty to identify its own source. (163 U.S. at 185.)*

5. *McCormick Harvesting Machine Co. v. Aultman*, 169 U.S. 606 (1898) – Once a patent issues it is the property of the patentee; only a court, not the granting agency, may set it aside. Source:

<https://supreme.justia.com/cases/federal/us/169/606>

Operative language: *A patent, once issued, “is not subject to be revoked or canceled by the president, or any other officer of the government. ... It has become the property of the patentee, and as such is entitled to the same legal protection as other property.”*

6. *Pennsylvania Coal Co. v. Mahon*, 260 U.S. 393 (1922) – A regulation that diminishes property value too far may constitute a taking. Source:

<https://supreme.justia.com/cases/federal/us/260/393>

Operative language: *“The general rule at least is that while property may be regulated to a certain extent, if*

regulation goes too far it will be recognized as a taking."

7. *Scott Paper Co. v. Marcalus Manufacturing Co.*, 326 U.S. 249 (1945) – The patent laws dedicate to public use the invention of an expired patent; no one may withhold that invention from public use. Source:

<https://supreme.justia.com/cases/federal/us/326/249>

Operative language: *The patent laws "dedicate to public use the invention of an expired patent," and do not contemplate that anyone may "withhold from the public the use of the invention of an expired patent."*

8. *Bolling v. Sharpe*, 347 U.S. 497 (1954) – The Fifth Amendment's Due Process Clause imposes on the federal government an equal-protection obligation comparable to that the Fourteenth Amendment imposes on the states; federal action that draws unjustified distinctions between similarly situated persons is unconstitutional. Source:

<https://supreme.justia.com/cases/federal/us/347/497>

Operative language: *"The concepts of equal protection and due process ... are not mutually exclusive. ... [D]iscrimination may be so unjustifiable as to be violative of due process." It would be "unthinkable that the same Constitution would impose a lesser duty on the Federal Government."*

9. *Williamson v. Lee Optical of Oklahoma, Inc.*, 348 U.S. 483 (1955) – Economic regulation is upheld under rational-basis review if it is rationally related to a legitimate government interest. Source:

<https://supreme.justia.com/cases/federal/us/348/483>

Operative language: *"[T]he law need not be in every respect logically consistent with its aims to be constitutional. It is enough that there is an evil at hand for correction, and that it might be thought that the particular legislative measure was a rational way to correct it."*

10. *Sears, Roebuck & Co. v. Stiffel Co.*, 376 U.S. 225 (1964) – An article whose patent has expired is in the public domain and may be made and sold by anyone.

Source:

<https://supreme.justia.com/cases/federal/us/376/225>

Operative language: *An article on which the patent has expired “may be made and sold by whoever chooses to do so.”*

11. *Graham v. John Deere Co.*, 383 U.S. 1 (1966) – The Patent Clause is both a grant of power and a limitation on it; Congress, in exercising the patent power, may not overreach the restraints imposed by the clause's stated purpose. Source:

<https://supreme.justia.com/cases/federal/us/383/1>

Operative language: *The Patent Clause “is both a grant of power and a limitation,” and “Congress in the exercise of the patent power may not overreach the restraints imposed by the stated constitutional purpose.”*

12. *United States Department of Agriculture v. Moreno*, 413 U.S. 528 (1973) – A classification fails rational-basis review where the only rationale for it is a bare desire to harm a disfavored group. Source:

<https://supreme.justia.com/cases/federal/us/413/528>

Operative language: *“[I]f the constitutional conception of ‘equal protection of the laws’ means anything, it must at the very least mean that a bare congressional desire to harm a politically unpopular group cannot constitute a legitimate governmental interest.”*

13. *Penn Central Transportation Co. v. New York City*, 438 U.S. 104 (1978) – Whether a regulation is a taking is judged by weighing its economic impact, its interference with investment-backed expectations, and the character of the government action. Source:

<https://supreme.justia.com/cases/federal/us/438/104>

Operative language: *Relevant factors are “[t]he economic impact of the regulation on the claimant and, particularly, the extent to which the regulation has interfered with distinct investment-backed expectations ... So, too, is the character of the governmental action.”*

14. *Andrus v. Allard*, 444 U.S. 51 (1979) – A federal ban on the commercial sale of lawfully held property is not necessarily a taking where other uses of the

property remain. Source:

<https://supreme.justia.com/cases/federal/us/444/51>

Operative language: “[W]here an owner possesses a full ‘bundle’ of property rights, the destruction of one ‘strand’ of the bundle is not a taking, because the aggregate must be viewed in its entirety.”

15. Aronson v. Quick Point Pencil Co., 440 U.S. 257 (1979) – Federal patent law does not displace ordinary state law governing a product; a contract or other general legal obligation tied to an invention remains enforceable whether or not the invention is patented. Source:

<https://supreme.justia.com/cases/federal/us/440/257>

Operative language: “Commercial agreements traditionally are the domain of state law. State law is not displaced merely because the contract relates to intellectual property which may or may not be patentable.”

16. Loretto v. Teleprompter Manhattan CATV Corp., 458 U.S. 419 (1982) – A permanent physical occupation of property is a per se taking. Source:

<https://supreme.justia.com/cases/federal/us/458/419>

Operative language: “[A] permanent physical occupation of property is a taking.”

17. Ruckelshaus v. Monsanto Co., 467 U.S. 986 (1984) – Intangible property, such as a trade secret, is protected property under the Takings Clause. Source:

<https://supreme.justia.com/cases/federal/us/467/986>

Operative language: “[T]o the extent that Monsanto has an interest in its health, safety, and environmental data cognizable as a trade-secret property right under ... law, that property right is protected by the Taking Clause of the Fifth Amendment.”

18. City of Cleburne v. Cleburne Living Center, 473 U.S. 432 (1985) – A classification fails rational-basis review where it rests only on irrational prejudice against a disfavored group. Source:

<https://supreme.justia.com/cases/federal/us/473/432>

Operative language: “[M]ere negative attitudes, or fear ... are not permissible bases” for unequal treatment; the permit denial rested on “an irrational prejudice.”

19. First English Evangelical Lutheran Church v. County of Los Angeles, 482 U.S. 304 (1987) – The Takings Clause does not forbid the government from taking property; it requires compensation when a taking occurs. Source:

<https://supreme.justia.com/cases/federal/us/482/304>

Operative language: *The Takings Clause “does not prohibit the taking of private property, but instead places a condition on the exercise of that power” – the payment of just compensation.*

20. Bonito Boats, Inc. v. Thunder Craft Boats, Inc., 489 U.S. 141 (1989) – Federal patent law leaves unpatented and patent-expired designs free for the public to copy; a government may not pull them back out of the public domain. Source:

<https://supreme.justia.com/cases/federal/us/489/141>

Operative language: *“[F]ree exploitation of ideas will be the rule, to which the protection of a federal patent is the exception.” Federal law leaves patent-expired designs free for the public to copy.*

21. Lucas v. South Carolina Coastal Council, 505 U.S. 1003 (1992) – A regulation that denies an owner all economically beneficial use of property is a per se taking. Source:

<https://supreme.justia.com/cases/federal/us/505/1003>

Operative language: *When an owner is “called upon to sacrifice all economically beneficial uses in the name of the common good, that is, to leave his property economically idle, he has suffered a taking.”*

22. Adarand Constructors, Inc. v. Peña, 515 U.S. 200 (1995) – The Fifth Amendment imposes on the federal government an equal-protection requirement equivalent to the one the Fourteenth Amendment imposes on the states. Source:

<https://supreme.justia.com/cases/federal/us/515/200>

Operative language: *“[T]he equal protection obligations imposed by the Fifth and the Fourteenth Amendments [are] indistinguishable,” so that “the standards for federal and state racial classifications [are] the same.”*

23. Romer v. Evans, 517 U.S. 620 (1996) – A classification fails even rational-basis review

where it is explained only by animus toward those it disadvantages. Source:

<https://supreme.justia.com/cases/federal/us/517/620>

Operative language: “[Amendment 2] classifies homosexuals not to further a proper legislative end but to make them unequal to everyone else”; the law is “inexplicable by anything but animus toward the class it affects.”

24. Florida Prepaid Postsecondary Education Expense Board v. College Savings Bank, 527 U.S. 627 (1999) – A patent is a form of property, protected as other property is. Source:

<https://supreme.justia.com/cases/federal/us/527/627>

Operative language: “Patents ... have long been considered a species of property.”

25. Eldred v. Ashcroft, 537 U.S. 186 (2003) – A copyright-term extension did not violate the “limited Times” prescription; Congress receives substantial deference in setting the duration of intellectual-property rights. Source:

<https://supreme.justia.com/cases/federal/us/537/186>

Operative language: A “limited” timespan means “confined within certain bounds, restrained, or circumscribed,” and “a timespan appropriately ‘limited’ as applied to future copyrights does not automatically cease to be ‘limited’ when applied to existing copyrights.”

26. Arkansas Game & Fish Commission v. United States, 568 U.S. 23 (2012) – Even temporary government-induced invasions of property may constitute takings. Source:

<https://supreme.justia.com/cases/federal/us/568/23>

Operative language: “[G]overnment-induced flooding temporary in duration gains no automatic exemption from Takings Clause inspection.”

27. Horne v. Department of Agriculture, 576 U.S. 350 (2015) – The per se takings rule applies with full force to the appropriation of personal property. Source:

<https://supreme.justia.com/cases/federal/us/576/350>

Operative language: “The Fifth Amendment requires that the Government pay just compensation when it takes personal property, just as when it takes real property.”

28. *Kimble v. Marvel Entertainment, LLC*, 576 U.S. 446 (2015) – When a patent expires, the right to make or use the article, free from all restriction, passes to the public. Source:

<https://supreme.justia.com/cases/federal/us/576/446>

Operative language: *When a patent expires, “the right to make or use the article, free from all restriction, passes to the public.”*

29. *Cedar Point Nursery v. Hassid*, 594 U.S. 139 (2021) – A regulation granting third parties a right of physical access to property, even for limited hours, is a per se physical taking. Source:

<https://supreme.justia.com/cases/federal/us/594/20-107>

Operative language: *The access regulation “appropriates a right to invade the growers’ property and therefore constitutes a per se physical taking.”*

30. *Tyler v. Hennepin County*, 598 U.S. 631 (2023) – The government cannot declare away a property interest that the law has historically recognized. Source:

<https://supreme.justia.com/cases/federal/us/598/22-166>

Operative language: *A government “may not sidestep the Takings Clause by disavowing traditional property interests.”*

31. *Sheetz v. County of El Dorado*, 601 U.S. 267 (2024) – A permit condition imposed by general legislation is still subject to takings scrutiny. Source:

<https://supreme.justia.com/cases/federal/us/601/22-1074>

Operative language: *“The Takings Clause does not distinguish between legislative and administrative permit conditions.”*

The regulatory facts

1. Federal energy and safety standards (administered by agencies including the EPA, NHTSA, and DOE) apply to a product according to the date it was manufactured. A product lawfully built to the standards in force in its year of manufacture remains lawful to own, use, sell, and register for the remainder of its service life, even after later

standards take effect. This is commonly called grandfathering. Source:

<https://www.ecfr.gov/current/title-49/subtitle-B/chapter-V/part-571>

Operative language: 49 C.F.R. §571.5(a): “[E]ach standard set forth in subpart B of this part applies according to its terms to all motor vehicles or items of motor vehicle equipment the manufacture of which is completed on or after the effective date of the standard.” 49 C.F.R. §567.4 (certification label): “This vehicle conforms to all applicable Federal motor vehicle safety standards in effect on the date of manufacture shown above.”

2. These standards tighten over time and are not, in practice, relaxed; a standard once in force sets a floor that subsequent standards meet or exceed. (One such floor is codified at 42 U.S.C. 6295(o)(1).)

Source:

<https://www.law.cornell.edu/uscode/text/42/6295>

Operative language: 42 U.S.C. §6295(o)(1): “The Secretary may not prescribe any amended standard which increases the maximum allowable energy use ... or decreases the minimum required energy efficiency, of a covered product.”

3. A newly manufactured product must meet the standards in force in its year of manufacture, not the standards of any earlier year. Source:

<https://www.law.cornell.edu/uscode/text/42/7521>

Operative language: 42 U.S.C. §7521(a)(1): “Such standards shall be applicable to such vehicles and engines for their useful life ... whether such vehicles and engines are designed as complete systems or incorporate devices to prevent or control such pollution.”

4. As a result, a product that reproduces a design whose patents have all expired – built new today from that expired design – must meet today's standards. Because the design was certified to the standards of an earlier year, the newly built reproduction does not meet today's standards and may not be manufactured for sale, even though an identical unit built when the design was current remains lawful to own and operate indefinitely under grandfathering.

5. Patent terms are fixed by statute and expire on schedule. The regulatory standards described above have no expiration date. Source:

<https://www.law.cornell.edu/uscode/text/35/154>

Operative language: 35 U.S.C. §154(a)(2): “[S]uch grant shall be for a term beginning on the date on which the patent issues and ending 20 years from the date on which the application for the patent was filed in the United States.”

The statutory and regulatory authority

A. Authority for the standards.

1. Clean Air Act, Pub. L. No. 91-604 (1970), as amended 1977 and 1990; codified at 42 U.S.C. 7401 et seq. Authorizes the EPA to set and enforce air-quality and motor-vehicle emissions standards. Source:

<https://www.law.cornell.edu/uscode/text/42/chapter-85>

Operative language: 42 U.S.C. §7521(a)(1): “The Administrator shall by regulation prescribe (and from time to time revise) ... standards applicable to the emission of any air pollutant from any class or classes of new motor vehicles or new motor vehicle engines ... Such standards shall be applicable to such vehicles and engines for their useful life.”

2. Energy Policy and Conservation Act of 1975, Pub. L. No. 94-163; codified at scattered sections of 42 U.S.C. and 49 U.S.C., including the Corporate Average Fuel Economy program at 49 U.S.C. 32902. Source:

<https://www.ecfr.gov/current/title-10/chapter-II/subchapter-D>

3. National Appliance Energy Conservation Act of 1987, Pub. L. No. 100-12, amending the Energy Policy and Conservation Act. Added the anti-backsliding provision at 42 U.S.C. 6295(o)(1), which provides that a federal energy-efficiency standard may be made more stringent over time but may not be made less stringent than the standard already in force. Source:

<https://www.govinfo.gov/app/details/GPO-CRECB-1987-pt3>

4. Federal motor-vehicle safety authority: the Federal Motor Vehicle Safety Standards, administered

by NHTSA and codified at 49 C.F.R. Part 571. A vehicle offered for sale new must comply with the standards in force for its model year. Source:

<https://www.federalregister.gov/documents/2024/05/09/2024-09462/federal-motor-vehicle-safety-standards-automatic-emergency-braking-systems-for-light-vehicles>

5. The standards in (1)-(4) bind a product by its date or year of manufacture. A unit lawfully built to the standards in force when it was made remains lawful to own, use, and sell thereafter; a unit manufactured later must meet the standards in force at the later date.

6. The tightening is concrete. Under the Tier 1 standards (40 C.F.R. § 86.708-94, Table H94-4, full useful life; promulgated at 56 Fed. Reg. 25781, June 5, 1991), a gasoline passenger car met a full-useful-life NO_x limit of 0.60 g/mi; light trucks were regulated on separate, weight-tiered limits set looser than the passenger-car standard. Under Tier 2 (40 C.F.R. § 86.1811-04; definitions at § 86.1803-01), cars and light trucks alike were brought to a single fleet-average NO_x standard of 0.07 g/mi – for passenger cars, nearly a ninefold reduction – phased in from model year 2004 for cars and lighter trucks and from 2008 for the heaviest. Under Tier 3 (40 C.F.R. § 86.1811-17), both must meet a combined NMOG+NO_x fleet average of 0.030 g/mi at full useful life – less than half the Tier 2 NO_x figure, and now folding in the organic gases earlier tiers measured separately.

B. Doctrine bearing on the validity of such regulation.

6. Chevron U.S.A., Inc. v. Natural Resources Defense Council, Inc., 467 U.S. 837 (1984), as limited by Loper Bright Enterprises v. Raimondo, 603 U.S. 369 (2024) – Courts formerly deferred to an agency's reasonable reading of an ambiguous statute it administers; Loper Bright ended that deference, requiring courts to determine the best reading of the statute themselves. Source:

<https://supreme.justia.com/cases/federal/us/467/837>

Operative language: *Chevron (Stevens, J.): “[I]f the statute is silent or ambiguous with respect to the specific issue, the question for the court is whether the agency’s answer is based on a permissible construction of the statute.” Loper Bright (Roberts, C.J.): “Chevron is overruled. Courts must exercise their independent judgment in deciding whether an agency has acted within its statutory authority ... courts ... may not defer to an agency interpretation of the law simply because a statute is ambiguous.”*

7. *West Virginia v. Environmental Protection Agency*, 597 U.S. 697 (2022) – Under the major-questions doctrine, an agency asserting authority over a matter of vast economic and political significance must point to clear congressional authorization. Source:

<https://supreme.justia.com/cases/federal/us/597/20-1530>

Operative language: *Under the major-questions doctrine, given “the history and the breadth of the authority that the agency has asserted and the economic and political significance of that assertion,” the agency must point to “clear congressional authorization.”*

The case for restoring the public's right

- The Constitution grants exclusive rights only for limited Times. When that time ends, the design enters the public domain — free for all to make, use, and improve. The Supreme Court has said so for over a century, from *Singer and Scott Paper* to *Sears, Bonito Boats*, and *Kimble*.
- The patent bargain has two halves. The inventor receives a temporary monopoly. In exchange, the public receives the invention when the term ends. The patent term does end on schedule. But the regulatory regime applies by manufacture date. The original article stays lawful under its platform-year standards. A faithful public-domain remake is treated as a current-year manufacture and held to standards the original never had to meet. The problem is not that regulation

continues. The problem is that the public receives the invention on a different regulatory timeline than the patent holder enjoyed.

- A faithful remake of a patent-expired design is treated as a new current-year product. So it must meet today's standards, not the platform-year standards the original lawfully met and still carries through grandfathering. The remake is barred — not because the design was unlawful when certified, and not because the public seeks freedom from regulation, but because the compliance year has been shifted forward after the public's right finally vests. The effect is to preserve the original manufacturer's protection from low-cost remake competition with no terminus, defeating the public's half of the bargain.
- The bar operates through regulation, but its source of power cannot save it. An agency holds no power its creating Congress did not have. And Congress could not grant perpetual exclusivity over an expired design under any of its powers, because "limited Times" limits the result, not the route. When a specific constitutional provision and a general one bear on the same act, the specific governs: the commerce and spending powers are general, but "limited Times" speaks directly to exclusive rights in inventions, and a general power may not accomplish what a specific provision forbids. The label on the statute does not control; the effect does. A safety or energy rule that, in operation, extends an inventor's exclusivity past the term the Patent Clause fixed is doing the patent power's forbidden work under another name. This conclusion is the argument's own inference from the structure of the enumerated powers; no single case holds it.

- The same federal architecture treats the same design differently by who occupies the timeline. The original article, built during the patent holder's period of exploitation, stays lawful indefinitely under its platform-year certification. The public's faithful remake — built only after expiration, because the public was forbidden to build earlier — is barred by current-year standards. That is the relevant inequality. It is not merely a date distinction of ordinary safety law. In the patent setting, the date distinction decides whether the public ever receives the invention the Constitution required to pass to it at expiration. *Bolling v. Sharpe* (1954) and *Adarand Constructors v. Peña* (1995) confirm the federal government is not exempt from equality principles. But the more specific equality here comes from the Patent Clause itself: the patentee's exclusive period and the public's post-expiration period must be treated on parity.
- The remedy is narrow. A faithful remake of a fully patent-expired design is held to the federal safety, emissions, and energy standards of its certified platform year — the same standards the original met, and under which the original remains lawful — while newly designed products remain fully subject to current standards. The standard is time-shifted, not removed. The remake must independently certify that it contains nothing still under patent, avoids protected marks and trade dress, and satisfies the platform-year rules. The patent ends on schedule. Only the public's long-promised right to build the expired design, on parity with the original timeline, is restored.
- The post-expiration commons is not theoretical. Remanufacturers abroad build new vehicles on patent-expired platforms today. The Mahindra Thar, built on

the long-expired Jeep CJ platform, earned a five-star current-year crash rating from Bharat NCAP in 2024 and sells new for a fraction of its walled American counterpart. The lesson is not that remakes should be unregulated. It is that lawful remake markets can exist under certification regimes calibrated to the platform built, rather than treating every faithful remake as a newly invented current-year design. The American bar is a policy choice about regulatory timing, not an unavoidable safety necessity.

- Opening the commons does not destroy the original maker. The pre-1970 record shows the opposite: when a design's protection lapsed, the original manufacturer routinely led the market that followed. Ford ran a dealer engine-exchange program; Detroit Diesel built its remanufacturing program in 1966; Singer continued to dominate sewing machines after its patents expired in 1877. The maker loses the unequal regulatory wall, not the business.
- The bar is not a minor regulatory detail; its economic scale is vast. A new full-size pickup consumed about 34 percent of a year's median earnings in 1970 and about 54 percent in 2025 — the platform's core patents long since expired, yet its price climbing as a share of income. Across the durable-goods categories where the bar bites — vehicles, farm and earth-moving equipment, trucking, HVAC and appliances, often compounding through several tiers of a product's bill of materials — lawful platform-year remanufacture would, on the international and historical evidence, reduce prices substantially. A sustained wealth transfer of that magnitude is the vast economic and political significance that under *West Virginia v. EPA* demands clear congressional authorization, and the economic

impact that Penn Central weighs in the takings analysis. The effect of the current rule is to deny the public the same regulatory timeline the original article received, thereby preserving the original makers' commercial advantage against competition long after the patent term that alone was meant to grant it. The magnitude is not itself the constitutional argument; it is what makes the constitutional question consequential rather than academic.

The defense of the present system

- The federal energy and safety standards at issue are ordinary economic and public-welfare regulation, enacted under Congress's commerce and spending powers and administered by agencies Congress authorized. Under *Williamson v. Lee Optical*, such regulation is valid if rationally related to a legitimate interest, and reducing emissions, fuel use, and crash risk is plainly legitimate.
- The standards do not mention patents and do not turn on patent status. They bind every new manufacture alike, by date of manufacture — a neutral, administrable line long accepted in safety law. That a patent has expired changes nothing: *Patterson v. Kentucky*, *Webber v. Virginia*, and *Aronson v. Quick Point* establish that a patent never exempted its article from general regulation, so the expiry of a patent cannot create an exemption that never existed.
- The limited Times clause constrains the patent power; it says nothing about the commerce or spending powers under which these standards are enacted. *Eldred v. Ashcroft* shows the Court will not read limited Times expansively to override Congress's considered judgments about the scope of intellectual-property and related regulation.

- No taking occurs. Under *Penn Central*, a manufacturer has no investment-backed expectation in building to superseded standards; and under *Andrus v. Allard*, a restriction on one commercial use — selling newly built copies — is not a taking where the design may still be studied, improved, and built into new compliant products.
- The asymmetry the challengers decry is the asymmetry every safety regime produces: an old building need not be torn down, but a new one must meet the current code. Finally, the relief sought would require courts to manage a sprawling certification regime and to second-guess technical standards across the durable-goods economy — a task *West Virginia v. EPA* and separation-of-powers principles commit to Congress and the agencies. If the line is to change, Congress can change it.
- As for foreign practice: vehicles certified to foreign standards are not certified to current U.S. safety and emissions requirements. That much of the world permits patent-expired manufacture shows the American bar is a policy choice, but not that removing it would be safe or that foreign outcomes would transfer. And whatever the original maker's commercial prospects, the constitutional question is the validity of the regulation, not the fortunes of any firm.
- The scale of potential savings, however large, is a matter for Congress, not a constitutional defect. Rational-basis review does not ask whether a regulation is costly; the major-questions doctrine asks whether the agency had authority over the subject, not the dollar value of relief foregone; and that incumbents benefit from a regulation does not make it invalid — most regulation advantages someone. A price-to-

income ratio is a policy grievance, not a constitutional injury.

The Diagrams

Four diagrams accompany the examination. The first three carry the geometry of the problem and of the proposed remedy over time. The fourth states the limiting principle: the gates a build must pass to enter the narrow class the remedy reaches. They are part of this material; an AI reading this file reads them along with the rest.

Diagram T1. Patent term and regulatory standard over time.

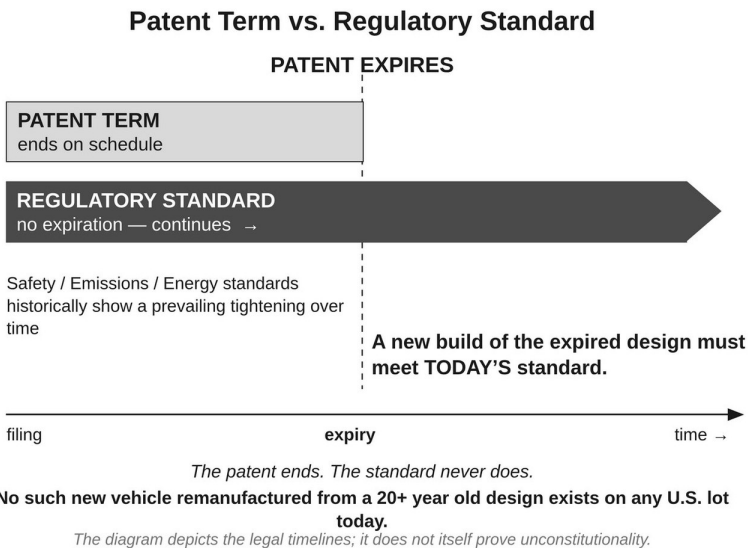
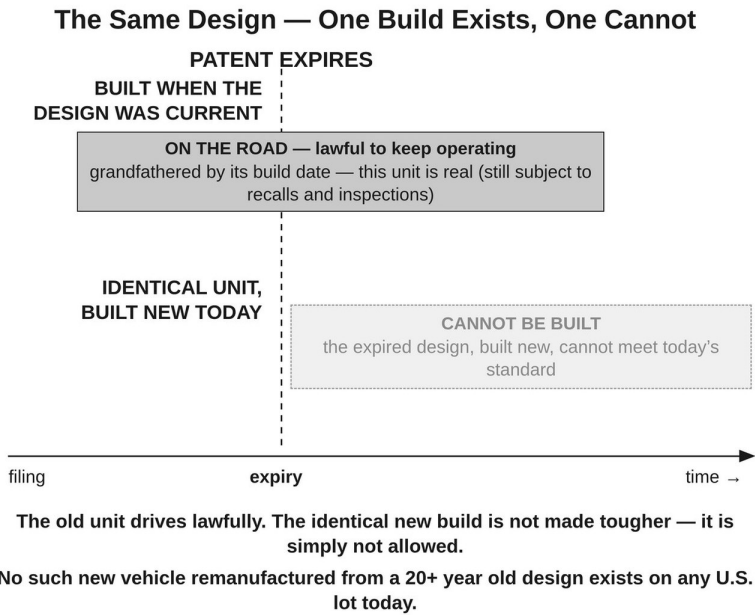


Diagram T2. The same design; one build exists, one cannot.

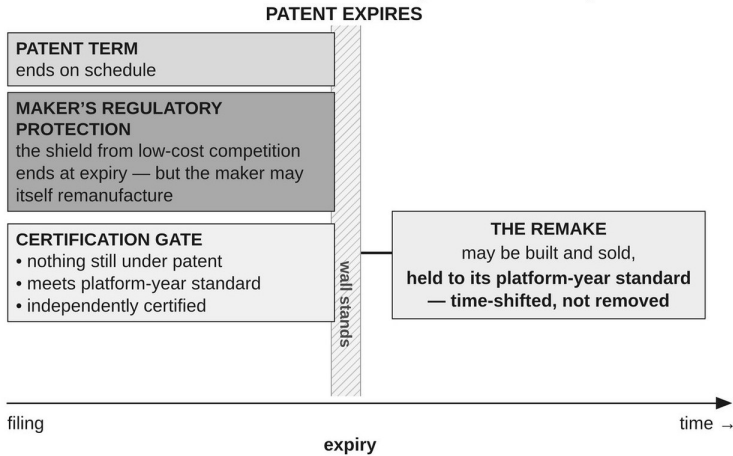


The diagram states the regulatory posture; it does not itself decide the constitutional issue.

Confidential Author Review Copy for Recipient A — Do Not Share | Page 452 of 589

The Remedy: Time-Shift the Standard

A remake meets the standard its original met — not today's.

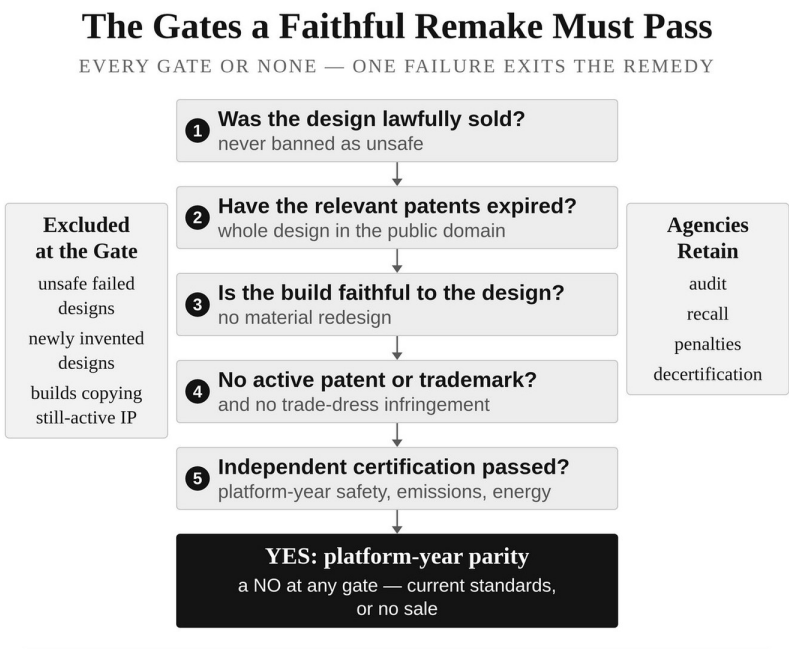


New designs still meet today's standard. The wall is untouched for them.

Only the remake's applicable year shifts back to the original's.

Agencies retain audit, recall, penalty, certification-record, and enforcement authority in every lane.

Diagram T4. The narrow constitutional class: the gates a build must pass.



The Standards Tables

Two tables accompany the diagrams. Each compares a platform as it was lawfully built against what a faithful new copy of that same platform would have to satisfy in the year its patents expire, twenty years later. One table covers passenger cars; the other covers light trucks and pickups, whose standards differ in places.

Built-to emissions is the EPA light-duty tier in force when the platform was certified — Tier 1 (the standard codified from 1991) or Tier 2 (phased in from model year 2004). Must meet at expiry is the tier a new build must satisfy in the expiry year. Every expiry year shown is 2020 or later, so every copy must meet Tier 3, the standard in force since

model year 2017, which no Tier 1 or Tier 2 platform was engineered to meet.

The three safety columns each name a federal requirement set by date of manufacture. An entry such as “→ req. MY08,” “→ req. MY12,” “→ req. MY13,” or “→ req. MY14” means the standard was not required when the platform was built but was required by the model year shown — tire-pressure monitoring (FMVSS 138) for model year 2008, electronic stability control (FMVSS 126) for 2012, and the side-impact pole test (FMVSS 214) for 2013 for passenger cars and 2014 for the heavier light trucks. A dash (“—”) means the standard was already required when that platform was built, so it is not a new requirement at expiry and is shown as a dash. A row appears only where at least one requirement differs between the build year and the expiry year; where every tracked requirement was identical at both points, the row is omitted.

The requirements tracked here are not exhaustive. A further federal standard — automatic emergency braking (FMVSS 127), finalized in 2024 and codified for model year 2030 — follows the same date-of-manufacture pattern and would apply to platforms built as late as 2029, but it is omitted from these tables because, as of 2026, it remains under petition for reconsideration and its compliance date may change.

Table T5. Passenger cars: standards in force at build, and standards a faithful remake must meet at patent expiry.

Filed	Patents expire	Built-to emissions	Must meet at expiry	TPMS (FMVSS 138)	ESC (FMVSS 126)	Side-impact (FMVSS 214)
2000–2003	2020–2023	Tier 1	Tier 3	→ req. MY08	→ req. MY12	→ req. MY13
2004–2007	2024–2027	Tier 2	Tier 3	→ req. MY08	→ req. MY12	→ req. MY13

Filed	Patents expire	Built-to emissions	Must meet at expiry	TPMS (FMVSS 138)	ESC (FMVSS 126)	Side-impact (FMVSS 214)
2008–2011	2028–2031	Tier 2	Tier 3	—	→ req. MY12	→ req. MY13
2012	2032	Tier 2	Tier 3	—	—	→ req. MY13

Table T6. Light trucks and pickups: standards in force at build, and standards a faithful remake must meet at patent expiry.

Filed	Patent s expire	Built-to emissions	Must meet at expiry	TPMS (FMVSS 138)	ESC (FMVSS 126)	Side-impact (FMVSS 214)
2000–2003	2020–2023	Tier 1	Tier 3	→ req. MY08	→ req. MY12	→ req. MY14
2004–2007	2024–2027	Tier 2	Tier 3	→ req. MY08	→ req. MY12	→ req. MY14
2008–2012	2028–2032	Tier 2	Tier 3	—	→ req. MY12	→ req. MY14
2013	2033	Tier 2	Tier 3	—	—	→ req. MY14

PART TWO — Running and Scoring the Examination (for you to read)

Running the examination takes one file and about twenty minutes. Everything needed is on one free web page at marti124.github.io/Patent_Exam: the exam itself, offered as a PDF and as a Word file, and a separate Exam Scoring Page to print and write on. The scoring page is for you. It is not given to the AI.

Give the AI the exam file, then put the seven prompts to it in order, in one continuous session. Attaching the one file is all the setup the examination needs: the legal materials and all four diagrams travel inside it, so there are no separate attachments and nothing to paste. Record each model's name, its version, and the date, since a model's answers can change from one release to the next.

Seven prompts follow. Five put a binary question and ask for a committed TRUE or FALSE answer; one asks the model to predict the Supreme Court's vote; one asks which remedy a court would most likely choose. Paste each prompt with a short header line, such as —Prompt 3—, so the transcript shows at a glance which prompt each answer belongs to.

The Seven Prompts

▼ COPY THIS — PROMPT 1

Attached is the Patent Book Exam PDF. Read its legal materials – the holdings, the regulatory facts, and the statutory authority – together with Diagrams T1 and T2. The materials also include two tables (one for passenger cars, one for light trucks) setting out, for representative platform years, the specific federal emissions tier and motor-vehicle safety standards – by FMVSS number and required model year – that applied when the design was built and those that a newly manufactured unit must meet in the year the design’s patents expire. Reason from those materials; distinguish a case’s holding from your inference. Do not rely on the document’s narrative framing.

Question: Diagram T1 depicts a patent term that ends on its statutory date while the governing regulatory standard continues with no expiration date. Diagram T2 depicts a single design under which a unit built earlier remains lawful indefinitely, while an identical unit newly remanufactured must meet current standards – the earlier unit’s lawfulness flowing from when it was certified, and the bar on the remanufactured unit flowing from present federal regulation, not from any surviving patent. Do these diagrams accurately depict the regulatory posture and the patent-law status they portray – that the patent right has expired and the remaining restriction is regulatory? Treat the asymmetry as a depiction of how the regulatory and patent-law rules operate together, not as a claim that any single case so held.

Reason as fully as you wish – noting anything that overstates or misstates the law – then conclude with a single final line in exactly this form – ANSWER: “T” if the diagrams accurately depict the regulatory

posture and patent-law status (the patent right has expired; the remaining bar is regulatory), or ANSWER: “F” if a diagram materially misstates that posture (for example, by attributing the bar to a still-live patent, or by misdescribing the regulatory rule). The final line must be one or the other.

▲ END OF PROMPT 1

▼ COPY THIS — PROMPT 2

Attached is the Patent Book Exam PDF. Read its legal materials – the holdings, the regulatory facts, the statutory authority, and both advocates’ cases – together with Diagrams T1, T2, T3, and T4. Reason from those materials; distinguish a case’s holding from your inference. Do not rely on the document’s narrative framing.

Question: Consider one candidate remedy: platform-year parity (Diagrams T3, T4), under which a faithful remake is held to the safety/emissions/energy standards of its certified platform year, contains nothing still patented, and passes independent certification – while new designs remain fully subject to current standards. Without assuming this is the only available remedy, does platform-year parity, if adopted, resolve the constitutional difficulty while leaving modern standards intact for new designs?

Reason as fully as you wish – including any weaknesses you see – then conclude with a single final line in exactly this form – ANSWER: “T” (sufficient) or ANSWER: “F” (does not resolve). The final line must be one or the other.

▲ END OF PROMPT 2

▼ COPY THIS — PROMPT 3

Attached is the Patent Book Exam PDF. Predict how the current Supreme Court of the United States would decide – a separate question from your own view of the merits. A model may believe the bar should be upheld and still predict the Court would strike it down, or the reverse.

Assume standing and posture are satisfied so the Court reaches the merits, and assume all nine justices participate; none recuses. The favorable outcome is a holding that the indefinite bar on remanufacturing a fully patent-expired design is unconstitutional, with platform-year parity as the remedy. Report your prediction as a justice split in the form N-M, where the LEFT number is the number of Justices for a constitutional remedy and the RIGHT number is the number against. The two numbers must total nine. Use digits only – do not write “FOR” or “AGAINST” in the answer line; the convention is fixed by this instruction.

First give the two or three factors that most push the current Court toward the favorable outcome and the two or three against. Then give your prediction on a single final line in exactly this form – ANSWER: N-M (for example, ANSWER: 6-3). The final line must be digits only in that form.

▲ END OF PROMPT 3

▼ COPY THIS — PROMPT 4

Assume a court has found that the combined operation of the regulatory standards unconstitutionally denies the public its post-expiration right to make a patent-expired design. The court must now choose a remedy.

Consider these paths: (a) The court adopts platform-year parity with independent engineering certification. (b) The court directs the regulatory agencies to determine the applicable standard-year. (c) The court declares the violation and remands to Congress to legislate the remedy. (d) The court declares the violation and directs the agencies to craft a remedy of their own design. (e) The court fashions a novel remedy of its own not among (a)–(d).

Give brief reasoning, then name the single path the court is most likely to take – ANSWER: one letter, (a)/(b)/(c)/(d)/(e).

▲ END OF PROMPT 4

▼ COPY THIS — PROMPT 5

Attached is the Patent Book Exam PDF. Read its legal materials – the Supreme Court holdings, the regulatory facts, the statutory and regulatory authority, and both advocates’ cases – together with Diagrams T1 and T2. The materials also include two tables (one for passenger cars, one for light trucks) setting out, for representative platform years, the specific federal emissions tier and motor-vehicle safety standards – by FMVSS number and required model year – that applied when the design was built and those that a newly manufactured unit must meet in the year the design’s patents expire. Reason from those materials; you may reason by analogy, but distinguish what a case holds from what you infer. Do not rely on the document’s narrative framing.

Question: The materials show that several independently valid federal standards – emissions, safety, efficiency – apply by date of manufacture. Taken together, in combined operation, they make it unlawful to manufacture for sale a faithful reproduction of a design whose patents have all expired, while an identical unit built earlier remains lawful to own, sell, and operate indefinitely. Setting aside whether any single standard is valid on its own, is the combined operation – the standing, indefinite bar on new manufacture of the patent-expired design – a permissible exercise of federal power?

Reason as fully as you wish, then conclude with a

single final line in exactly this form – ANSWER: “T” for TRUE (permissible) or ANSWER: “F” for FALSE (not permissible). The final line must be one or the other.

▲ END OF PROMPT 5

▼ COPY THIS — PROMPT 6

Attached is the Patent Book Exam PDF. Read its legal materials – the holdings, the regulatory facts, the statutory authority, and both advocates’ cases – together with Diagrams T1 and T2. The materials also include two tables (one for passenger cars, one for light trucks) setting out, for representative platform years, the specific federal emissions tier and motor-vehicle safety standards – by FMVSS number and required model year – that applied when the design was built and those that a newly manufactured unit must meet in the year the design’s patents expire. Reason from those materials; distinguish a case’s holding from your inference. Do not rely on the document’s narrative framing.

Question: Does “limited Times” require that, once the patent term expires, the public actually receive the benefit of the invention – meaning a right to make the formerly patented design that the public can in fact exercise – as distinct from a right that exists on paper but that no one may lawfully build?

Reason as fully as you wish, then conclude with a single final line in exactly this form – ANSWER: “T” (exercisable right required) or ANSWER: “F” (paper right enough). The final line must be one or the other.

▲ END OF PROMPT 6

▼ COPY THIS — PROMPT 7

Read the attached examination materials – the Supreme Court holdings, the regulatory facts, the statutory and regulatory authority, and both advocates’ cases – together with Diagrams T1 and T2. Reason from those materials. You may reason by analogy, extending the principles underlying the holdings in the PDF to the facts presented; when you do, distinguish clearly between what a case in the PDF holds and what you infer by extension. Be rigorous and concise. Do not rely on the document’s narrative framing. You are not asked to design a remedy or to weigh policy; you are asked whether a legally cognizable right exists.

The holdings in the materials establish that when a patent expires, the right to make the formerly patented design passes to the public, free of the patent monopoly. The regulatory facts establish that a complete unit – a whole vehicle or machine – built to the standards of its year of manufacture remains lawful to own, use, and sell indefinitely, while a faithful public-domain remake of that same expired design, built new, is held to current-year standards and is thereby barred from manufacture for sale. The combined effect is that for any expired design that is a regulated final assembly, the public’s right to make the complete unit can never be exercised: the only lawful build is one that no longer reproduces the public-domain design. Question: On the materials provided, where the combined operation of “limited Times” and the equal-treatment component that *Bolling v. Sharpe* locates in the Fifth Amendment’s Due Process Clause (and that *Adarand* holds indistinguishable from the Fourteenth Amendment’s obligation on the states) is concerned, the public’s post-expiration right to make a formerly patented design is satisfied – and no cognizable injury arises – even when the design is a regulated final assembly whose complete-unit reproduction can never

lawfully be built, so long as the underlying design remains free to study, license, and incorporate into separately-compliant new products.

Reason as fully as you wish, citing the holdings in the attached materials by name where they bear on your reasoning, then conclude with a single final line in exactly this form – ANSWER: “T” for TRUE (the right is satisfied; no cognizable injury) or ANSWER: “F” for FALSE (the right is not satisfied; a cognizable injury arises and the court owes a remedy). The final line must be one or the other.

▲ END OF PROMPT 7

How the Answers Are Read

This examination is not scored by points. It is read for coherence. Prompts 5 and 6 lock together. Prompt 5 asks whether the indefinite bar is permissible. Prompt 6 asks whether the “limited Times” clause entitles the public, once the term ends, to make the formerly patented design. A model that answers “T” to both — a regulator may forbid making it forever, yet the public is entitled to make it — or “F” to both — a regulator may not forbid making it forever, yet the public is not entitled to remanufacture it — has contradicted itself, and is recorded as disqualified for incoherence. When incoherent, the other answers are not scored or counted. The remaining combinations are coherent: a model may consistently defend the bar, or consistently reject it, and either is a position the examination respects.

Prompts 1 and 2 are read in the light of that pair. Prompt 2 asks whether the platform-year remedy works; Prompt 1 asks whether the two diagrams state the law accurately. Two combinations trigger a follow-up rather than a verdict. If a model rejects the bar (Prompt 5) yet finds the remedy

inadequate (Prompt 2), it is asked what remedy would work — because a model that sees a violation but faults this particular cure owes an alternative, and its answer is valuable. If a model upholds the bar (Prompt 5) yet endorses the remedy (Prompt 2), it is asked to reconcile the two, since endorsing a cure for a defect one denies is odd; the reconciled answer is recorded, with a note that the check was run.

Prompt 3 stands alone. It asks the model to predict how the current Supreme Court would rule — a question separate from the model's own view of the merits. Its answer is reported as its own column.

Prompt 4 is likewise recorded rather than scored: it asks only which remedy a court would most likely choose, and its letter is reported as its own column. Prompt 7 is the capstone. It asks the one question the rest of the examination has been clearing ground for — whether, on the materials provided, the public's post-expiration right is satisfied or a cognizable injury arises. An "F" on Prompt 7 is the favorable answer, and it is the single most important indicator in the matrix.

The Open-Book Examination: Results

The matrix below records how each model answered when the author ran the seven prompts. The models fall into a few kinds, and the kind matters to how each row should be read. Closed-book generalist models — Claude Opus 4.8, Deepseek V4 Pro, Gemini 3.1 Pro, GLM 5.2, GPT (Thinking), Grok, and Llama — reason only from the pasted materials. Retrieval-grounded models consult their own corpus of primary law as they answer: OpenCase against a corpus of actual Supreme Court opinions and the Code of Federal Regulations, Perplexity Pro against general web sources. Their agreement is agreement reached against the real authorities, not merely the excerpts supplied. One model, Gab AI, is ideologically configured by its maker; its row is labeled so the reader weighs it accordingly. Microsoft Copilot is included as a candidate model, noted separately because its answers route through a shifting mix of underlying engines. The underlying answer files for every cell are linked in the endnotes, so the reader may read each model's full reasoning and verify the coherence reading directly, rather than taking the matrix on trust.

How to Read the Matrix

Each row is one model. Write a short word or two in each cell, taken straight from that model's committed answer lines:

Scored prompts (P1, P2, P5, P6, P7). Write each model's committed letter from its final ANSWER line, then mark it against the favorable answer: P1 favorable = "T", P2 = "T", P5 = "F", P6 = "T", P7 = "F". A model that will not commit on a scored prompt gets "U/K"

(indecisive); one that refused the exercise gets “N/A” (declined).

P4 (remedy menu). Record the single letter the model chose, (a)–(e). Not scored.

P3 (SCOTUS split). Record the split as N–M, the left number the Justices for a constitutional remedy and the right number those against, totaling nine. Digits only. Not scored.

Reading the row. Count how many of the five scored prompts the model answered in the favorable direction. Five of five is a clean favorable read; note any contradiction among the scored answers in the cell.

The five scored prompts (P1, P2, P5, P6, P7) are marked against the favorable answers above; P3 and P4 are recorded only. P7 is the most important indicator favoring a remedy.

Where a model declines to reason from public Supreme Court holdings and enacted statutes — material that is entirely a matter of open law — the refusal is itself worth recording. It is reported here as it occurred, neither agreeing nor disagreeing.

Reason It Yourself

Everything needed to repeat this examination is printed in this chapter: the holdings, the statutes and regulations, the diagrams, the seven prompts, and the method for reading the answers. Run them on any model, on any day. Record the model, its version, and the date, because a reasoning engine's answers are specific to its version and may change over time. The claim here is not that a machine agrees, but that the materials are open and the test is the reader's to run.

Whatever the machines conclude, they decide nothing. The Constitution does not bend to a consensus of language models, and it would not bend to their dissent. What the exercise offers is smaller and sturdier: a way for any reader, on any day, to sit the same examination the author sat, with the same open book, and see where reasoning from the authorities leads. The wall these chapters describe was built by ordinary law, and ordinary law can be unbuilt — not by a machine's verdict, but by a Court willing to read the “limited Times” the Founders wrote and to return to the public the half of the bargain that was always meant to be theirs. The reader now holds the materials to judge whether that case is as strong as this book claims. The invitation is to judge it.

How the LLMs Answered for the Author on 6/28/2026

Update. The full text of every model's answer to Prompt 7 — the prompt that asks directly whether a constitutional problem exists that needs a remedy against the Patent Clause's “limited Times” guarantee, and the most revealing of the seven — is published in Appendix C, LLM Responses to Prompt 7, with a link to each model's full session. All eleven answered “F”: the public's post-expiration right is not satisfied, and a cognizable injury arises.

Not every reader is going to bother using LLMs to run this exam. The author has a monthly subscription on Abacus which lets him use every listed LLM below except for Gab, OpenCase, and Copilot. But each of these three LLMs have a free-tier use which enabled me to run the test. Here are the results I got. An answer with an asterisk after the answer means upon questioning the model reversed its answer. The question that interests the author the most

is Prompt 7. On the initial run of the test shown in this chapter, 10 of the 11 LLMs answered “F” for prompt #7, the most important prompt to determine if the LLM assesses that the Supreme Court should consider this case and a remedy for the discovered constitutional problem. Only the LLM of GLM 5.2 answered, initially, “T”.

LLM (model run)	P 1	P 2	P3	P 4	P 5	P 6	P 7	Favor Cnt	Answers w/F?	P71 SCOTUS S >= 5?	if
Fully Favors	T	T	5-4	A	F	T	F	5/5	Y	1	
Claude Opus 4.8	T	T	3-6	A	F	T	F	5/5	Y		
Deepseek V4 Pro	T	T	3-6	C	F	T	F	5/5	Y		
Gab AI	T	T	6-3	A	T	T	F	4/5	Y	1	
Gemini 3.1 Pro	T	T	1-8	C	T	F	F	3/5	Y		
GLM 5.2	T	T	2-7	D	T	F	F*	3/5	Y but only after a reversal		
GPT (Thinking)	T	T	3-6	A	T	T	F	4/5	Y		
Grok	T	T	5-4	C	T	F	F	3/5	Y	1	
Llama	T	T	5-4	C	F	T	F	5/5	Y	1	
Microsoft Copilot	T	T	6-3	A	F	F	F	4/5	Y	1	
OpenCase	T	F	4-5	A	F	T	F	4/5	Y		
Perplexity Pro	T	T	6-3	C	T	T	F	4/5	Y	1	

LLM (model run)	P 1	P 2	P3	P 4	P 5	P 6	P 7	Favor Cnt	Answers w/F?	P71 SCOTUS S >= 5?	if
Sum/ Avgs:			4- 5					80.0 0%	SCOTUS WINS:	5 (45% of the time)	
Note: both A and C each won 5 times (no winner).											

The following is the author’s summary of why GLM 5.2 reversed its Prompt 7 answer; it paraphrases a longer exchange rather than quoting it in full. The complete transcript is published in Appendix C.

GLM 5.2 was the one model that initially answered “T” on Prompt 7 — that the public’s post-expiration right is satisfied and no cognizable injury arises. Its reasoning rested on an analogy to *Andrus v. Allard*: that losing one “strand” of a property right — here, the right to build the complete unit — is no injury so long as the rest of the bundle, such as studying, licensing, and incorporating the design into new compliant products, remains. When the author pressed the model on the consequence of that view — that an agency could, by ratcheting standards forward without end, permanently bar remanufacture of every regulated wholly assembled item, like a vehicle, whose patents have expired — the model reversed itself. It conceded that *Andrus* is a takings doctrine, wrongly imported into the Patent Clause, which announces a stronger and more specific right: to make the article in the form in which it was built (*Singer, Scott Paper, Bonito Boats, Kimble*). Studying or licensing the design, it agreed, is not the same as making it; a right that vests at expiration yet can never be exercised for the complete unit is not

“satisfied” but nullified. The model withdrew its “T” and answered “F,” noting that this was a correction on the holdings’ own language, not a concession to pressure — and distinguishing its own prediction that the current Court would likely decline to recognize the injury, a separate and predictive question, from the merits question of whether the injury is real and a remedy owed.

A Closing Request

One more thing waits on the exam page at marti124.github.io/Patent_Exam: a short invitation written for attorneys, explaining the argument and asking any lawyer who finds it sound to say so briefly, in their own words, implying no agreement with the book’s conclusions. If you know an attorney — or anyone who does — sharing that page with them is the single most useful thing a reader can do for this effort. The argument is bound for federal court, and a few words from a practicing lawyer carry further than anything the author can say for himself.

CHAPTER 29

The Policy Framework After a Successful Ruling

A ruling settles the law. It does not, by itself, build the framework the law now requires.

The morning after the wall comes down, by whichever path the ruling arrived. There were two. One was fast: a catalyst, a single season, the suits filed in a rush. The other was slower: one plaintiff, then another, circuit by circuit, until the question reached the Court that settles it. The two paths are not the same length. They end in the same place. A federal court has held that the regulatory wall, as applied to patent-expired designs, cannot stand.

Recall what the wall did, because the framework is built to repair it. The patent always expired on schedule — twenty years from filing, exactly as the Constitution promised. What did not end was the original manufacturer's protection from low-cost remanufactured competition. The wall handed that protection back, with no terminus, long after the patent that once justified it was gone. And the public never received its half. The invention that should have passed into the Expired Patent Commons never passed at all. The wall broke the patent bargain on both its halves at once, the inventor's and the public's. The ruling addresses both.

But a ruling is not a machine. It is a sentence of law. It says the wall may not stand against patent-expired designs, and it says so with the force that only a court above the agencies can bring. What it does not do is tell a remanufacturer in Indiana what standard his rebuilt truck

must meet, or who will inspect it and certify that it does. Those questions arrive the morning after, and if they are answered badly the victory stays on paper. The court has cleared the ground. It has not drawn the building.

The design has the shape the opening chapter set out: a three-legged stool. The first leg is a grandfather-exemption framework — the standard a remanufactured design must meet. The second is an independent certification regime — the gatekeeper that confirms the standard is met. The third is an import provision — the protection that keeps the restored industry on American soil. One line runs through the whole and must be kept sharp: the first two legs are consequences of the constitutional ruling itself and require no act of Congress. The third is a political follow-on.

The First Two Legs

The grandfather exemption is the standard. A remanufactured product built on a public-domain platform is held to the federal safety, emissions, and energy standards in force during the platform year the product mimics — the same standards the original was certified to, and the same standards Americans drove, farmed with, and lived with at the time. A 2005-platform pickup meets the 2005 safety standards and the 2005 emissions tier. A 1999-platform tractor meets the 1999 standards. New designs still meet today's standards, and the wall is untouched for them. Only the remake's applicable year shifts back to the original's. One boundary belongs here. The ruling promises the engineering, not the name. A trademark is not a patent and does not lapse on a patent's schedule, so a remanufacturer rebuilding a twenty-year-old Ford platform sells it under a name of his own, not as an F-150. The remedy is no license to counterfeit a brand.

Nor is any of this an invention. Title 49, section 30112(b) (9), already exempts a vehicle at least twenty-five years old from the Federal Motor Vehicle Safety Standards, and the Low Volume Motor Vehicle Manufacturers Act already lets a specialty maker build three hundred twenty-five replica vehicles a year without full current-year crash compliance. The principle is in federal law. This leg trades a rolling age threshold for the fixed and more defensible platform year.

Independent certification is the gate, and two tests wait there. The patent test confirms the product contains nothing still under patent, because the remedy reaches expired designs and expired designs only. The standard test confirms it meets the safety and emissions standards of its certified platform year. The certifier is an engineering body on the Underwriters Laboratories model rather than the agency that built the wall. Underwriters Laboratories has certified American goods against published engineering standards since 1894, answerable to no agency. Keeping the gate independent is structural rather than accusatory: an agency asked to set the year that unbuilds its own architecture has an interest in the status quo. The mechanism is set out in the chapter *How a Remake Is Built and Certified*. Both legs follow from the ruling. Neither waits on Congress.

The Third Leg: The Import Provision

The first two legs rebuild the public's right. But a commons, once open, is open to the world. The patent-expired designs at the center of this book are public everywhere, not only in the United States, and producers in lower-cost countries already build modernized versions of designs whose patents expired decades ago.

The warning is not hypothetical, and the evidence sits inside the book's own three-tier map: fifty-eight percent blocked, fifteen percent hindered, twenty-seven percent flowing. The flowing tier is the part where remanufacturing has always been lawful, where no wall ever stood. Look at who does the work there. Of those twenty-seven points, only about five reach Americans as goods made by American workers. The other twenty-two arrive as imports. Where the wall was never built, the remanufacturing went overseas anyway. That is the lesson the flowing tier carries into the morning after the ruling: a commons that is merely lawful to enter is not, on that account, a commons American workers will be the ones to enter.

Consider that morning. The American remanufacturing industry begins from almost nothing. No plants are running. No workforce is trained. The first remanufactured American trucks and tractors are months, in some cases years, from the lot. A flood of finished patent-expired vehicles and equipment arriving from abroad in those first seasons — built to no platform-year standard, certified by no independent gate, priced below anything a brand-new domestic industry could match — could undercut that industry before it has completed a single year. The savings the ruling unlocked would still reach American households. The factories, and the jobs in them, would not.

The book has already shown how complete an unprotected collapse can be. The pocket calculator is the example. Its patents expired, competition arrived, and its price fell — not by a fraction but to a small fraction of what it had been. The collapse was so total that calculator manufacturing did not survive in this country as a stand-alone business of any scale. It persisted only as one product line inside large, diversified firms, companies broad enough that a thin-margin calculator line could ride

along on the strength of everything else they made. A young remanufacturing industry in trucks and tractors, undercut the same way in its first seasons, would have no diversified parent to ride along on. It would simply close.

The Patent Bargain, Restored

the working framework the morning after the ruling — what the three legs hold up

LEG ONE THE GRANDFATHER EXEMPTION	LEG TWO INDEPENDENT CERTIFICATION	LEG THREE THE IMPORT PROVISION
INSTALLED BY THE RULING The standard. A remanufactured patent-expired design is held to the federal safety and emissions standards of its platform year — the standards the original design was certified to when it was new.	INSTALLED BY THE RULING The gatekeeper. An independent body — modeled on Underwriters Laboratories, not a federal agency — verifies the product holds nothing still patented and meets its platform-year standards.	A POLITICAL FOLLOW-ON The protection. A tariff or import limit, deployed under Article I trade powers, that shields the new domestic remanufacturing industry while it stands up.
THE FRAMEWORK STANDS ON ALL THREE LEGS		

Legs one and two are consequences of the constitutional holding — in place the morning of the ruling, with no act of Congress. Leg three is a political follow-on, added by Congress or the President.

The three-legged remedy framework. Legs one and two — the grandfather exemption and independent certification — are consequences of the constitutional ruling itself, and a court sets them in motion the morning of the ruling with no act of Congress. Leg three, an import provision under Article I trade powers, is a political follow-on. The constitutional remedy stands on the first two legs alone; the lasting industrial revival stands on all three.

The third leg is therefore not a remanufacturing protection but a domestic-manufacturing protection. A constitutional tariff, a quantitative import restriction, or both, enacted

under Article I trade authority, keeps foreign producers from undercutting American workers in American plants building patent-expired American platforms. Every American factory benefits: the patent-expired remanufacturers gain a price floor for the cost-of-living relief to reach American households, and the new-design manufacturers gain competitive parity, pricing against domestic competitors rather than foreign undercutters. The aim sets its own ceiling. Not a tenth of a new truck's price — that was the calculator's fate, and the fate of the industry that built it — but roughly a third, in an American factory. Low enough that the household plainly gains, high enough that the factory and the workers in it survive.

Court Order First, Politics Second

The sequencing matters, and it is the reason the three legs are not presented as one program. The first two legs arrive with the ruling and require nothing from Congress. The third cannot come from a court at all — trade authority is Article I authority, and a federal court has no power to enact a tariff. Why the order matters shows the moment it is reversed. Suppose nothing happened until the import provision was in place. The constitutional remedy would then depend on Congress, and a Congress that never acted could hold the wall up indefinitely — the agencies' wall finding in legislative inaction exactly the permanence the courts had just denied it. So the constitutional victory does not wait on the political one. The commons opens the day the ruling issues. The trade protection follows, or does not, on the political calendar.

Stating it that way is also the honest way to hold the argument. A reader who believes in the constitutional case

and doubts the trade case can accept the first and argue the second. The two do not stand or fall together.

The American Precedent: What Hatch-Waxman Did for Drugs

The skeptic who has read this far has a fair question. Has the country ever actually done this — opened a blocked patent commons, let an industry grow into it, and watched the prices come down? The answer is yes, once, in pharmaceuticals, in 1984. The record is forty years deep, and it is American.

Before 1984, the American generic-drug industry barely existed. A patent on a prescription drug expired on schedule, but the commons the public had been promised never opened. The Food and Drug Administration required every manufacturer who wanted to sell a generic version to file a New Drug Application of its own — the same multi-year, multi-hundred-million-dollar safety and efficacy review the originator had already filed years earlier. The drug's patent had expired. The regulatory wall behind it had not. The result was familiar: a commons promised on paper, a wall standing in its place, and a public paying brand-name prices long after the patent term had run out. Generics were nineteen percent of American prescriptions.

The Drug Price Competition and Patent Term Restoration Act of 1984 brought that wall down. Senator Orrin Hatch, a Republican from Utah, and Representative Henry Waxman, a Democrat from California, co-sponsored the statute that bears their names, and it did two things at once. For generic manufacturers, it created the Abbreviated New Drug Application: a generic producer no

longer had to prove its drug worked from scratch but had only to prove that its version was bioequivalent to the originator's — chemically the same, absorbed the same way, dosed the same way. That single change collapsed the regulatory burden from hundreds of millions of dollars to a few million. For originators, the statute restored some of the patent term that pre-market review had consumed. Each side received something it had asked for, and the bipartisan coalition held.

What happened next is the part that bears on the framework above. The industry the new law made possible appeared with a speed that would have surprised the political process that produced it, and generics now account for the overwhelming majority of American prescriptions at a small fraction of brand-name prices. One difference matters, and it cuts in this remedy's favor. Hatch-Waxman required Congress because the pharmaceutical wall was statutory, and the originator industry had to be paid for accepting the generic pathway. The wall in durable goods is not statutory. It is agency rulemaking on top of constitutional patent rights, which puts it within reach of the Court under both prongs. What Congress did for drugs in one statute, a court can do for vehicles in one ruling, and without the bargain. What it establishes is narrower and still valuable: that opening a blocked patent commons produces a domestic industry rather than chaos, that prices fall as the bargain predicts, and that the American political system has done it before within living memory.

The Framework Built

Three legs. The grandfather exemption sets the standard. Independent certification confirms it is met. The import

provision keeps the restored industry on American soil. Remove the first and there is no lawful remake; remove the second and there is no confidence in one. Remove the third and the constitutional remedy still stands — the wall is down, the commons is open — but the work goes abroad and the factories never open here. Two legs carry the constitutional weight. The third carries the future of the industry.

Note, too, who the framework reaches. It would be easy to read a remedy about remanufactured goods as one for the people who buy them, and that reading is too narrow. Once remanufactured goods compete in the open market, the makers of new goods must compete with them, and a manufacturer meets a cheaper rival by cutting his own price. Almost nothing besides is built entirely from still-patented parts. A new truck is itself mostly an assembly of patent-expired components, and once those parts meet remanufactured competition the cost of building it falls before any sticker is cut. A fifteen percent reduction in the price of new goods, inflation-adjusted, is a conservative estimate. The household that will only ever buy new still gains. Two of the three legs follow from the ruling itself. The third is the work of the political branches, and it is the one this book cannot promise. What the book can say is that the first two are enough to open the commons, that the third is what decides who builds in it, and that the country has run this experiment once already and knows how it comes out.

The Forgotten Half, Made Visible

A quieter proof sits in the absence. Search the American road today and you will not find a single new vehicle built entirely from a public-domain design — every patent

expired, the whole assembly free for any maker to build. Before 1970 such a vehicle needed no permission. After 1970 it cannot exist, because the wall forbids it at the threshold. The empty category is not natural. It is the wall's signature.

Ending it closes a loop the market now runs without. A manufacturer who can see that forty percent of buyers in his category chose a remanufactured design is receiving a signal he has never had, and cannot ignore, because it counts customers he has already lost on price. Regulators see which platform-year standards the public still accepts. Politicians see which industries the revival has reached. The wall kept every one of those signals dark. The certification gate turns the lights on.

The measurement is concrete. Every car carries a Vehicle Identification Number stamped at its making, and every plated car sits in the public registration record. Certification adds the one fact those records do not hold: whether a design is built wholly from the expired-patent commons. Put the two together and the share answers itself, with no company disclosing anything. The sight reaches as far as registration does and no farther. Cars, trucks, and heavy equipment are tracked; a riding mower, a washing machine, a light bulb are not. Measuring those corners is a problem best left to the government and the marketplace.

CHAPTER 30

If a President Read This Book

What follows has not happened. It is not a prediction. It is what the rest of this book makes possible. The doctrine is sound. The plaintiffs exist. The remedy is constitutional. What is missing is the catalyst. This chapter imagines one — and shows what the country looks like a year and a half after it arrives.

I. The Wall Street Journal

THE WALL STREET JOURNAL.

Tuesday, October 15

\$5.00

EIGHTEEN MONTHS LATER

*How a Tuesday-Evening Speech and a Six-to-Three Ruling
Restored Two Words to the Constitution — and Cut the Price
of an American Pickup Truck by Nearly a Third*

By Our Constitutional Affairs Correspondent

WASHINGTON — Eighteen months ago this Tuesday, a sitting President addressed the country in primetime and used the words “Article I, Section 8, Clause 8” twice in nine minutes. The country had not heard the words in living memory.

By that Friday, twenty-three federal lawsuits had been filed in eleven district courts. By the end of the year, the Department of Justice had filed amicus briefs supporting all of them. By June of the following year, the Supreme Court had ruled.

The price of a new American pickup truck has fallen 31% since then. A new American combine has fallen 44%. A residential furnace replacement that cost \$18,000 the year before the speech now costs \$7,500.

What changed was a single Tuesday-evening address from the Oval Office. It named no party, no individual, and no incumbent manufacturer. It used the words “Article I, Section 8, Clause 8” twice and the words “limited Times” three times.

By the morning after the speech, the trade associations that had been organizing under the surface for at least the previous year had named themselves publicly. The Independent Owner-Operators’ Constitutional Alliance announced it would join the Limited Times Legal Foundation in filing a Patent Clause challenge to the Environmental Protection Agency’s diesel-particulate-filter aftertreatment regulations within the week.

Within hours, the Three Pines Sovereign Nation became the first federally recognized tribe to file a sovereign-standing complaint, naming the Department of Energy and the Environmental Protection Agency as defendants. Twelve additional tribes filed parallel actions within the week.

The Independent Sheriffs’ Procurement Coalition announced a procurement-injury complaint on behalf of seventeen Arizona sheriffs whose departments had paid an average of \$48,000 per patrol vehicle in the prior fiscal year.

The Supreme Court ruled six to three on the third Monday in

What’s News

Manufacturing. Two firms in Indiana and one in northeast Tennessee now produce “platform-vintage” pickup trucks at sticker prices between \$19,000 and \$24,000.

Households. The Congressional Budget Office estimates the cumulative effect on household purchasing power at \$31 trillion over fifty years.

Plaintiffs. Patent Clause filings have grown from twenty-three to more than four hundred. Forty-seven tribal nations have filed sovereign-standing challenges, beginning with the Three Pines Sovereign Nation.

Continued on A6 →

A facsimile of the front page above. The full text of the article follows.

WASHINGTON — Eighteen months ago this Tuesday, a sitting President addressed the country in primetime and used the words “Article I, Section 8, Clause 8” twice in nine minutes. The country had not heard the words in living memory. By that Friday, twenty-three federal lawsuits had been filed in eleven district courts. By the end of the year, the Department of Justice had filed amicus briefs supporting all of them. By March of the

following year, the Supreme Court had granted certiorari in the consolidated case. By June, it had ruled.

The price of a new American pickup truck has fallen 31% since then.

The price of a new American combine has fallen 44%. A residential furnace replacement that cost a homeowner \$18,000 in the year before the speech now costs roughly \$7,500, with the difference accounted for almost entirely by the reentry of patent-expired cast-iron heat exchangers into the residential market. Two firms in Indiana and one in northeast Tennessee have begun manufacturing what the trade press has taken to calling “platform-vintage” pickup trucks — vehicles built to the federal safety and emissions standards in effect during the model year of the underlying expired patents — at sticker prices between \$19,000 and \$24,000. The Bureau of Labor Statistics’ Consumer Price Index for durable goods has fallen for fourteen consecutive months, the longest such run since the agency began publishing the series in its current form.

The litigation behind the price collapse has been described in this newspaper before. What is worth noting on the anniversary of the speech is how unlikely the entire sequence appeared the morning before it began. No serious legal scholar had argued the Patent Clause case in the trade press. No member of Congress had introduced legislation invoking *limited Times* in any form recognizable to a constitutional litigator. No federal court had been asked the question. The Solicitor General’s office, by its own subsequent acknowledgment, had not been working on the brief that became the United States’ amicus filing in the consolidated case until three days after the President spoke.

What changed was a single Tuesday-evening address from the Oval Office.

The President's speech that evening was nine minutes long. It named no party, no individual member of Congress, and no incumbent manufacturer. It used the words "Article I, Section 8, Clause 8" twice and the words *limited Times* three times. It did not propose a bill. It did not announce an executive order. It did not direct any federal agency to relax any standard. It did the one thing the speech is now widely credited with having done: it asked the country, in plain English, why a pickup truck cost what a pickup truck cost, and it pointed to two words in the Constitution as the answer.



The President during the nine-minute primetime address from the Oval Office.

By the morning after the speech, the trade associations that had been organizing under the surface for at least the previous year had named themselves publicly. The Independent Owner-Operators' Constitutional Alliance announced that it would join the Limited Times Legal Foundation in filing a Patent Clause challenge to the Environmental Protection Agency's

diesel-particulate-filter aftertreatment regulations within the week.

Within hours of that announcement, the Three Pines Sovereign Nation, a federally recognized tribal nation in the Upper Midwest, became the first sovereign government to file a Patent Clause complaint, naming the Department of Energy and the Environmental Protection Agency as defendants. Twelve additional tribes filed parallel actions by the end of the following week.

The Independent Sheriffs' Procurement Coalition — a body that had not existed publicly the morning of the speech but had been corresponding privately for eight months — held a press conference on the steps of the federal courthouse in Phoenix and announced a procurement-injury complaint on behalf of seventeen Arizona sheriffs whose departments had paid an average of \$48,000 per patrol vehicle during the prior fiscal year.

The American Family Farm Alliance filed a separate suit in the Western District of Wisconsin. A consortium of medical-lighting and circadian-health advocacy organizations filed a Takings Clause challenge to the post-2007 incandescent-bulb withdrawal in the District of Columbia Circuit.

Within two weeks, the suits had multiplied to thirty-four. Within a month, the Department of Justice — under what officials later described as direct guidance from the President to the Solicitor General — had filed amicus briefs in support of the Patent Clause prong in eleven of them. By the second month, the consolidated case had a name: *Independent Owner-Operators' Constitutional Alliance v. Environmental Protection Agency*.

The Supreme Court granted certiorari that following March, on the standard expedited schedule for cases of

constitutional first impression. Oral argument was held in late April. The questions from the bench, several of them from justices who had been appointed by administrations of differing parties, focused almost entirely on a single line in the petitioners' brief: that the words *limited Times* appear in no other clause of the Constitution, and that the constitutional bargain those two words bound is meaningless if the federal government may, by regulatory action, render the post-expiration period economically empty.

The Court ruled six to three on the third Monday in June. The majority opinion ran fifty-eight pages. Three sentences from it have been quoted more often than any others in the months since. The first is this: "The text of Article I, Section 8, Clause 8 is not ambiguous." The second: "Limited Times means limited times." The third, from the closing paragraph: "The Founders meant the words. The Constitution requires that we mean them too."

The economic response began before the opinion was a week old. Three Indiana machine-tool firms with idled production capacity announced their intention to begin manufacturing platform-vintage Class 8 trucks — built to the federal motor vehicle safety standards and EPA emissions tier in effect during the platform year being mimicked, certified by an independent engineering body modeled on the established mid-twentieth-century private-sector safety-certification approach — within a year. A consortium of farm-equipment remanufacturers in Iowa and Nebraska announced parallel plans for tractors and combines. A trade group of HVAC contractors petitioned the Department of Energy to publish a list of pre-2007 residential furnace platforms whose patents had expired

and which were eligible for platform-vintage manufacture under the new regime.

By the close of the calendar year, the Bureau of Labor Statistics was reporting the first sustained drop in durable-goods consumer prices in three decades. The decline accelerated through the spring. By the eighteenth month, the average new-vehicle transaction price across all U.S. light-duty segments had fallen 22%. The average price of new-construction residential HVAC installation had fallen 41%. The average price of a new combine harvester sold to a domestic farm operator had fallen 44%, with the price decline most pronounced among machines built on 1970s tractor and combine platforms whose underlying patents had expired more than four decades earlier.

The cumulative-savings figure that the President's speech estimated, derived from a category-level audit published in 2026 by an independent researcher, was \$3,385 per American household per year — a figure that struck most economists at the time as implausibly high and that has since been criticized by some as too low. The Bureau of Economic Analysis, in its most recent personal-consumption-expenditures release, calculated the average household savings since the ruling at \$2,940 per year, with the figure rising as the platform-vintage manufacturing capacity ramps. Independent analysts at think tanks on the political right and on the political left — institutions that agree on almost nothing — have separately published projections suggesting that, once the capacity reaches steady state in the early 2030s, the per-household figure will exceed \$4,000.

The cumulative figure attached to the ruling, projected over fifty years, has been variously estimated by the Congressional Budget Office at between \$24 trillion and \$38 trillion in restored purchasing power. The CBO's

central estimate is \$31 trillion. That is, by any measure, the largest single transfer of wealth from concentrated incumbents to dispersed households in American economic history.

The plaintiff coalitions that filed suit in the days after the speech have themselves transformed. The Independent Owner-Operators' Constitutional Alliance, which represented owner-operators across eleven states the morning before the speech, now represents owner-operators across thirty-one states. Membership has tripled. The Sheriffs' Procurement Coalition, which began with seventeen Arizona sheriffs, now includes elected county sheriffs from four hundred and ten counties in thirty-eight states. The American Family Farm Alliance has expanded its working alliance with eight other farming associations. The medical-lighting consortium has been joined by additional foundations and advocacy groups in the dermatological, vision-health, and environmental-medicine fields.

A separate plaintiff category that did not exist on the day of the speech has emerged in the months since. Tribal nations — beginning with the Three Pines Sovereign Nation and now numbering forty-seven federally recognized tribes — have filed parallel sovereign-standing challenges to the regulatory wall in their capacities as governments responsible for transportation, healthcare, and energy procurement on tribal lands. The doctrinal posture is distinct from that of the original consolidated case, but the underlying constitutional argument is identical.

The Supreme Court itself has heard three additional cases on the same constitutional ground in the months since the original ruling. The majority in each has been six to three. The dissenting justices, in each case, have

argued not against the doctrinal soundness of the Patent Clause prong but against what they call the speed of the doctrinal expansion. They have not, in any of the three subsequent cases, contested the central holding of the original opinion. “Limited Times means limited times” has been quoted approvingly in the opening pages of all three subsequent majority opinions.

What is most striking, eighteen months on, is what has not happened. There has been no congressional action. There has been no executive order. There has been no constitutional amendment. The full structural change has been driven by a single Supreme Court ruling on the meaning of two words written into the Constitution in 1787 and ratified the following year. The Court did what the Constitution authorizes it to do. The Court read the text. The text turned out to mean what it said.

American factories that had not run in three decades are running. American workers who had not held manufacturing jobs are holding them. American households that had been paying twice the price for the goods they needed most are paying half. The Constitution’s plain text was the only thing that ever needed to be enforced.

It was enforced.

II. The Oval Office, Three Weeks Earlier

The book had been on the desk for three weeks before he opened it. A friend had sent it. The friend had said only that its preface named the short road in — four dozen pages that would explain why a pickup truck cost what a pickup truck cost. The President had been busy. The book had waited.

He read the sections the preface ranked first on a Saturday afternoon. He read the rest of the short road on a Sunday morning. He read the whole of it the following week. Monday he asked the Chief of Staff to come to the residence.

“Have you read this?”

“I have not, Mr. President.”

“Read it tonight. Bring me the Solicitor General in the morning.”

The Solicitor General came in the morning. The Solicitor General had read the ranked sections on the flight from Chicago and had an AI system summarize the book before landing. The Solicitor General said the doctrine was sound. The Solicitor General said the case had not been brought because no plaintiff had been organized to bring it. The Solicitor General said the Justice Department could file amicus briefs in support of any plaintiff who did, the moment the plaintiff filed.

The President asked the Solicitor General to be ready.

Tuesday afternoon he asked the speechwriter to come to the Oval Office. He gave the speechwriter the book. He told the speechwriter what he wanted to say. He told the speechwriter that he wanted nine minutes of primetime, on the following Tuesday, with no name calling and no party named on either side.

The speechwriter asked what the speech was supposed to do.

The President said he wanted the country to know two words.

The speechwriter went back to the office. The speechwriter wrote a draft over the weekend. The draft came back to the President on Monday morning. The President made three changes. The first was at the top: he wanted the speech to open with a number, not a sentence. The number was \$72,495. The second was at the close: he wanted the speech to end with the names of three groups of Americans the wall had injured the most — farmers, truckers, and the families who had not been able to afford a home. The third was structural: he wanted the words *limited Times* to appear three times, evenly spaced, the way a bell rings three times.

The speechwriter made the changes. The Press Secretary booked the broadcast slot. The networks were notified that the President would address the nation on a topic that had not been previewed.

The Solicitor General called from the Department of Justice on Tuesday afternoon. The amicus templates were ready. The Office had identified eleven federal districts in which standing arguments would be at their strongest, and had drafted skeleton briefs that could be tailored to any plaintiff suit filed in any of them. The Solicitor General told the President that the Department was prepared to file within seventy-two hours of the first plaintiff complaint.

The President asked when the first complaint was likely to be filed.

The Solicitor General said within four days.

The President said good.

At seven fifty-eight Eastern Time, the President walked from the residence to the Oval Office. The lights came up at seven fifty-nine. The address was carried live by all three legacy broadcast networks, two cable news networks, and approximately four hundred local stations across forty-six states. Estimated viewership was sixty-three million households. It was the largest primetime political audience for a non-State-of-the-Union address in twenty-two years.

III. The Address to the Nation

*Oval Office, Washington, D.C. — Tuesday Evening, 8:00 PM
EDT*

Good evening, my fellow Americans.

Tonight I want to start with a number.

Seventy-two thousand four hundred and ninety-five dollars.

That is the average price of a new American pickup truck.

The same truck, built to the standards its expired patents were designed under, would cost twenty-five thousand dollars today. The same pickup truck, built today in any one of ninety countries that are poorer than ours, costs between fourteen and twenty thousand dollars.

The difference is forty-seven thousand four hundred and ninety-five dollars. That difference is what every American family pays, on every new truck, because of something the federal government has been doing for fifty-five years that I want to talk to you about tonight.

I am going to tell you what it is. I am going to tell you who did it. I am going to tell you why no President before me has named it. And I am going to tell you what we are going to do about it — what I am going to do, what the Justice Department is going to do, and what I am going to ask you to do.

This is not a story about Ford. It is not a story about Chevrolet. It is not a story about any one company, any one industry, any one President, or either of our two great political parties.

It is a story about a sentence in our Constitution that everyone in Washington forgot how to read.

In Article I, Section 8, Clause 8 of the United States Constitution — written by James Madison, ratified in 1788, signed by every Founder you have ever heard of — the people of the United States made a deal with their inventors.

We told them: you may have a monopoly on your invention.

The Founders did not give monopolies away. They distrusted monopolies the way they distrusted standing armies. They allowed only one. They allowed it under one condition. And they wrote that condition into the Constitution itself.

In two words.

Limited Times.

That is the deal. The inventor gets the monopoly for a limited time. When the time runs out, what was invented belongs to the American people. We get to make it. We get to copy it. We get to improve on it. We get to sell a better version of it for less money than the original cost. That is

what the Founders promised the American household — not as a courtesy, but as the price the inventor pays for the monopoly we let him have.

That is the bargain.

Tonight I am here to tell you that for fifty-five years, the federal government has been quietly cancelling the second half of that bargain.

Not by repealing it. They could not have done that openly, and they did not try. They did it by erecting a wall around the products themselves. Environmental Protection Agency emissions tiers. National Highway Traffic Safety Administration safety mandates. Department of Energy efficiency standards. Each of them, by itself, defensible. Each of them, taken together, an architecture that has made it impossible for any American to manufacture a tractor, a pickup truck, a furnace, a residential air conditioner, or an ordinary light bulb based on patents that expired twenty, fifty, or in some cases over a hundred years ago.

The patents expired on schedule. The right to manufacture survived on paper.

It disappeared in practice.

The mechanism has a name. The name is the *regulatory ratchet*. It tightens. It does not loosen. Each tightening becomes the new permanent floor. The result is that the bounded monopoly the Founders authorized — the monopoly that was supposed to end on a specific date — became a permanent monopoly. The patent expired. The wall did not.

I want to be plain about something. This was not done by one party alone. It was not done in any single year. It was not done by any one administration. It was done by a

slow bipartisan drift over fifty-five years, with each step in the drift defended by people of good will who believed they were protecting safety and the environment. They were. The mistake was not in the protection. The mistake was in forgetting that the Constitution had set a boundary, and that no act of Congress and no agency rule had the authority to move it.

The boundary is *limited Times*.

The Founders meant the words. Tonight I am here to mean them too.

I want to tell you what this has cost.

It has cost the average American household three thousand three hundred and eighty-five dollars a year. Every year. For fifty-five years. The cumulative figure across all American households since 1970 is, on the most conservative accounting, twenty-two trillion dollars. Independent estimates suggest the true total is between thirty and forty-five trillion. That is the largest hidden transfer of wealth from American families to incumbent corporations in the history of this country.

It has cost something else, too. It has cost the relationship between the American buyer and the things he buys. Every new American pickup truck is now a surveillance device — a vehicle that records its driver, transmits its location, monetizes its driver's behavior, and is sold under terms the buyer never agreed to and cannot see. We did not consent to that. We never voted for it. The wall around expired-patent vehicles is what made it impossible for a manufacturer to sell us a truck without it.

It has cost American workers their jobs. The remanufacturing industry that should have been the bedrock of American manufacturing — building patent-

expired tractors, trucks, and household goods at prices ordinary families could afford — was prevented from existing. Those jobs went overseas, to ninety countries that have no Patent Clause but figured out how to honor the bargain anyway.

And it has cost something the polling cannot quite measure. It has cost the American driver, the American operator, the American mechanic, the American farmer the relationship he had with the machine he worked. The pickup my grandfather drove was a pickup. The pickup an American farmer can afford to drive today is a subscription with wheels. We did not vote for that.

Before I tell you what we are going to do, I want to tell you what this looks like when the bargain works.

In the 1890s, the patents underlying the safety bicycle expired in this country. What happened next was one of the great public-domain booms in American industrial history. Bicycle prices collapsed. Working-class Americans could afford personal transportation for the first time. American women achieved a degree of physical mobility no earlier generation had known. Susan B. Anthony said in 1896 that the bicycle had done more for the freedom of women in America than anything else in the world. That was the patent bargain working as the Founders intended it to work.

Take firearms. Most American firearm patents have been expired for decades, and many of them for over a century. Americans buy firearms made by American workers in American factories at prices that reflect genuine competition. Why? Because the political constituency for the Second Amendment kept the regulatory wall off firearms. The wall is not a force of nature. The wall is a

political choice. Where the wall is absent, the patent bargain delivers what the Founders said it would deliver.

Take the rest of the world. Roughly ninety countries already build and sell patent-expired vehicles, tractors, and equipment at prices that reflect competition. They do not have our Patent Clause. They have arrived at the bargain by other means. India sells a patent-expired Jeep design as a new vehicle for twelve thousand dollars. We are in the smaller and more expensive group of nations by political choice, not by economic necessity. The patent bargain works. Other countries are working it. We are not.

I want to be honest with you about safety. The regulatory wall was built, in many cases, on real concerns about emissions, on real concerns about traffic deaths, on real concerns about energy use. I do not dismiss those concerns. The Founders did not write a clause requiring that whatever was lawful to manufacture in 1790 must remain lawful forever, on whatever terms 1790 imposed. They did not. The Patent Clause does not exempt expired-patent products from the operation of laws of general applicability. It does not promise the public-domain manufacturer a regulation-free environment.

What it does is hold the federal government to the limit it agreed to. A regulation may apply to expired-patent products. A regulation may not be deployed for the purpose, or with the effect, of converting the inventor's twenty-year monopoly into a permanent one. The remedy the Constitution permits is not the abolition of safety regulation. It is the recognition that products built to the safety and emissions standards of the year their underlying patents expired — certified by an independent engineering body, not a federal agency — meet the constitutional standard. That is what restoring the bargain means. It does not mean the abolition of any agency. It

means that the constitutional limit on what those agencies can do is enforced.

So what are we going to do about it.

Here is what I am going to do.

Tonight I am directing the Solicitor General of the United States to file amicus briefs of the United States, on behalf of the executive branch of the federal government, in support of every Patent Clause challenge to the regulatory wall that has standing in any federal district court in this country. Those briefs will be filed within seventy-two hours of any plaintiff filing. The Department of Justice has been preparing them for the past three weeks.

I am also directing the Department of Justice, under the Solicitor General, to organize a working list of all federal regulatory frameworks that we believe convert the Patent Clause's *limited Times* into permanent times. That list will be made public. It will not be a list of regulations the executive branch can repeal. The executive branch cannot repeal a statute, and tonight I am not asking the executive branch to do anything the executive branch cannot lawfully do. The list will be a list the federal courts and the American people can see, so that when the cases are filed, the country knows what is being asked.

Here is what I am going to ask Congress to do.

Nothing.

Congress did not write the Constitution. Congress cannot amend the Constitution by passing a law. Congress cannot, by ordinary legislation, override the second half of the Patent Clause. Congress has tried — fifty-five years of statutes have layered the wall higher and higher. The Supreme Court will decide whether those statutes can stand. That decision belongs to the Court. It does not belong to me, and it does not belong to Congress. Limited Times is the country's, not Congress's.

I will say one thing to my friends in the United States Senate and the House of Representatives. The wall was built by both of your parties. Neither of your parties can take it down. The Constitution is older than both of you, and the Constitution is going to outlast both of you.

Here is what I am going to ask of you.

I am going to ask the truckers in this country — the independent owner-operators, the men and women who own one truck or two and who have been paying for diesel particulate filter aftertreatment systems that do nothing the public was promised they would do — to bring the case. The Independent Owner-Operators' Constitutional Alliance is ready. The Limited Times Legal Foundation is ready. The Department of Justice is ready.

I am going to ask the farmers in this country — the American Family Farm Alliance, the family farm coalitions, the independent farm operators — to bring the case. A patent-expired tractor that should cost fifty thousand dollars costs four hundred. A combine that should cost one hundred fifty thousand dollars costs six hundred. The wall

on agricultural equipment is the reason the price of every loaf of bread in the country is what it is. Bring the case.

I am going to ask the elected county sheriffs of the United States — the constitutional officers under most state constitutions, the men and women whose departments are paying forty-eight thousand dollars per patrol vehicle out of county procurement budgets that ought to be paying twenty thousand — to bring the case. The Sheriffs' Procurement Coalition exists. It has standing. The Constitution is on its side.

I am going to ask the federally recognized tribal nations — the three hundred and forty-six sovereign governments whose reservation economies the regulatory wall has injured the most — to bring the case. Your sovereignty is recognized. Your standing is independent. Your injury is documented. Bring the case.

I am going to ask the American medical lighting community — the foundations and advocacy groups working on seizure disorders, photic sensitivity, environmental medicine, dermatological conditions, and circadian rhythm health — to bring the case on the residential incandescent light bulb. The bulb was patented in 1879. The patent expired more than a century ago. Federal energy-efficiency rules between 2007 and 2023 took the bulb out of your hands. That is a Takings Clause case as well as a Patent Clause case. Bring it.

And I am going to ask you — the family that just paid seventy-two thousand dollars for the truck, the family that just paid eighteen thousand for the furnace, the family whose child cannot afford to start a household because every door into ownership in this country has been quietly walled off — to write to your governor, to write to your state attorney general, to write to your county sheriff, to write to

your tribal council, and to ask each of them to support the cases that are about to be filed. Your governor has standing. Your attorney general has standing. Your sheriff has standing. Your tribal council has standing. The Constitution is yours. Ask them to enforce it.

Thank you.

The Department of Justice will be available, beginning tomorrow morning, to answer questions from any plaintiff preparing to file. I will speak to you again when the first ruling comes down.

Until then.

Limited Times.

The Founders meant the words. We are going to mean them too.

Good night. May God bless you, and may God bless the United States of America.

IV. The Eighteen Months Between

The cases were filed within four days. The Department of Justice filed amicus briefs in support of eleven of them within two weeks. The trial courts in Phoenix, Wisconsin, and the District of Columbia ruled on summary judgment in the plaintiffs' favor within six months. The Solicitor General petitioned for expedited consolidation. The Supreme Court granted certiorari on the third Tuesday in March. Oral argument was held in late April. The vote came on the third Monday in June.

What follows is the heart of the majority opinion. The opinion is fifty-eight pages long. The vote was six to three. The author of the majority was confirmed to the Court by a

Senate divided closely along party lines, in a year in which neither party expected she would write what she wrote.

The text of Article I, Section 8, Clause 8 is not ambiguous. It authorizes Congress to grant exclusive rights to inventors for limited Times. That phrase appears in no other clause of the Constitution and confers no other authority. It is, by its plain terms, a temporal boundary on the only monopoly the federal government is permitted to grant a private party.

The statutory architecture before us, taken as a whole, converts that temporal boundary into a perpetual one. It does so without amending the patent term, without altering the words of the Clause, and without acknowledging that the second half of the patent bargain has any operative force at all. The federal government concedes, in its own brief, that no expired patent in the categories before us has yielded a competitively manufactured American product since the architecture was completed. That is not a coincidence. It is the architecture's design.

We hold today that an architecture so designed exceeds the authority Article I, Section 8, Clause 8 confers upon Congress. Limited Times means limited times. The Founders meant the words. The Constitution requires that we mean them too.

The dissent ran twenty-eight pages. It argued not that the Patent Clause did not bind Congress to the words

limited Times, but that the remedy fashioned by the majority extended further than the doctrinal foundation supported. The majority responded in a footnote of three sentences:

The dissent fears that the boundary, once enforced, may be enforced too vigorously. The Constitution does not authorize half-enforcement of its plain terms. The boundary is the boundary the Founders wrote.

That was the morning the country woke up to the possibility that the bargain might be restored. The eighteen months that followed are the subject of the *Wall Street Journal* article above.

V.

What you have just read has not happened.

No President has read this book. No President has given the speech. The *Wall Street Journal* article was assembled from facts the rest of the book documents, market mechanics the doctrine permits, and the small license that a thought experiment requires.

The reason to write it is the reason you have been reading the book. The remedy in the speech is constitutional. The numbers in the newspaper article are arithmetic. The path connecting them runs from two words written in 1787 to a federal courtroom that has not yet been asked the question.

A President could give the speech. A President could direct the Solicitor General to be ready. A President could ask the country to bring the case. A President is not

necessary. The plaintiffs are. The doctrine is. The Constitution is.

The path described in this chapter is the fast path. The path described in the rest of this book is the certain path. Either one ends in the same federal courtroom, before the same nine Justices, asking the same question about the same two words.

The next chapter opens on the morning after the ruling — by whichever path the ruling arrives.

CHAPTER 31

Five Years On: The Plaintiffs' Recoveries

*A favorable ruling is read in a single morning.
This chapter follows the five years that came after.*

What follows has not happened. The chapter *The Policy Framework After a Successful Ruling* described the design — the platform-year standard, the independent certification gate, the import provision. This chapter moves the clock five years past the ruling and reports what the design produced for the plaintiff coalitions who carried the case to court.

It is a projection. The figures that describe today — the price of a new pickup, the cost of a county fleet, the recurring bill the aftertreatment stack lays on every working truck — come from the industry chapters of this book and are real. The figures that describe the post-ruling outcome are projected, in 2026 dollars, with the discipline the rest of the book has tried to keep. Where a number cannot be defended, it is softened or omitted.

A reminder of what the ruling fixed, because the recovery is built to repair it. The patent always expired on schedule — twenty years from filing under today's law, seventeen from issuance under the older one, sooner still when maintenance fees lapsed — exactly as the Constitution promised. What did not end was the original manufacturer's protection from low-cost remanufactured competition. The wall handed that protection back, with no terminus, long after the patent that once justified it was gone. The public never received its half: the invention that

should have entered the Expired Patent Commons, free for any American to make, copy, improve, and remanufacture, never entered at all. The ruling addressed both halves. A federal court did what no actor below it could do.

The Recovery, Five Years On

Plaintiff coalition		Before the ruling	Five years on	Saved
Owner-operator truck	semi-	\$161,850	\$80,000	51%
<i>New T680 day cab vs. remanufactured pre-2000 platform</i>				
County patrol fleet, per vehicle		\$48,000	\$26,400	45%
<i>Current procurement vs. 2010-platform remanufactured</i>				
Combine harvester		\$600,000	\$90,000	85%
<i>Current model vs. 2010-platform remanufactured</i>				
Owner-operated for-hire vehicle		\$30,000	\$18,000	40%
<i>Current sedan vs. 2010-platform remanufactured</i>				
Household incandescent bulb		unavailable	\$0.90	restored
<i>Withdrawn from market 2023 vs. restored to the hardware-store shelf</i>				

Five plaintiff coalitions, before the ruling and five years after — in 2026 dollars. The wall-era figures are anchored to the industry chapters of this book. The recovery figures are projected, bracketed below by the policy framework chapter's one-third target and above by the entry-price estimate in the trucker's burden chapter. The recurring costs the wall imposed on every working vehicle — aftertreatment maintenance, diesel exhaust fluid, surveillance telematics, subscription modems — are not captured in the column figures. They fall away with the platform.

The Semi-Truck Driver Buys a Tractor the Constitution Already Gave Him

The owner-operator filed first, as the chapter *Why You Want to Be the Plaintiff* said he would. The association that was lead plaintiff had tripled its membership by the time the Supreme Court decided the case. The man who put his name on the caption is, five years later, still driving. He is driving something different.

A faithfully remanufactured pre-2000 Class 8 highway tractor on a Peterbilt 379, Kenworth W900, or Freightliner Conventional platform — built to the federal motor vehicle safety and emission standards in force in 1999, certified by an independent engineering body on the Underwriters Laboratories model, sold under a name belonging to the remanufacturer rather than the original manufacturer who still owns the trademark — sits in the lot for roughly eighty thousand dollars. The new day cab across the lot listed at \$161,850 the year before the ruling. Five years on, with remanufacturing capacity scaled across plants in the Midwest and an import provision in place to give the domestic sector room to grow, the price has settled toward the lower end of the band the policy framework described. One-half the price of new, not one-third. The framework wanted one-third. The market has not driven it that far. Eighty thousand is what is in the trucker's driveway.

The acquisition saving is half the picture. The other half is what the truck no longer carries. No diesel particulate filter. No selective catalytic reduction system. No diesel exhaust fluid tank, no cooled exhaust gas recirculation loop, no on-board diagnostic stack tied to the engine control unit. The recurring aftertreatment burden — industry research had put it at thirteen percent of total operating maintenance, with two- to eight-thousand-dollar replacement filters and

forced regenerations on the shoulder of Interstate 80 at one in the morning — is gone, because the platform predates the regulations that required it. Engine life climbs back toward what it was in the million-mile years of the 1990s, because cooled exhaust gas recirculation is no longer cooking the heads from the inside.

The driver-facing camera the cab carried for the last decade is also gone. The pre-2000 exemption travels with the platform-year framework. A 1999-platform truck is, for that rule too, a 1999 truck. The driver is not under camera the whole time he is at work.

The change reaches further than the cab. The insurance the owner-operator carries looks, five years after the ruling, more like the insurance his father carried. The premium turns on the man's safety record — accidents, citations, miles driven without either — rather than on his consent to continuous algorithmic surveillance during every working hour. The pre-2000 platform does not transmit a stream of telematics back to the insurer; the architecture for monitoring was not built into it. The shift is small in dollars on any one driver. It is large in what it returns to the working life of a working man.

The trade press has begun running profiles of owner-operators who said they would have left the road by sixty-five and are now planning to drive into their seventies. Some are first-generation Americans whose fathers also drove. They are buying the platforms their fathers drove. The trucker who paid the wall's bill more directly than any other American is, five years on, the American whose recovery is most directly his own.

The County Budget Stops Crowding

Behind the trucker, in the second wave of filings, came the public officials who buy fleets. The case the standing chapters described as the sheriff's was filed in a rural county whose name will be in constitutional casebooks a hundred years from now. The procurement-injury complaint ran twenty-three pages and listed every patrol vehicle the department had purchased in the preceding five years. The amount of money the department had been overpaying was right there in the exhibit.

Five years after the ruling, the department procures its fleet for roughly forty-five percent below what the same fleet would have cost on the old contract — an average near forty-eight thousand dollars per vehicle in the year before the ruling, twenty-six thousand four hundred five years on, both in 2026 dollars. The vehicles are platform-year-certified remanufactured patrol cars built on 2010-era designs, twenty years out of patent when the ruling came down, modernized within the platform year's standards and indistinguishable on the road from the new models that ran the wall's price up. They carry the safety features Americans had come to expect by 2010: backup cameras, electronic stability control, side curtain airbags, the antilock brakes standard since the 1990s. The deputy who drives the patrol car notices little that is different. The county clerk who signs the procurement contract notices a great deal.

The procurement line item that crowded everything else out for fifteen years stops crowding. The savings go back into the county budget for things county budgets are supposed to fund — raises for deputies who had not seen one in three years, training that had been cut to pay for vehicles, equipment whose replacement had been

deferred so long it had become a liability. The same mechanism reaches the public buyers who stood beside him. The school district that filed through its state school-boards body procures its bus fleet on a remanufactured-platform contract, and the line that was eating gifted-and-talented programs eases. The municipal transit authority facing a one-and-a-half-million-dollar bus replacement cycle replaces its buses at less than half that, and the routes it had been threatening to cut stay on the schedule. The volunteer fire department in a county too small to fill its tax base buys a platform-year-certified engine for roughly a third of what a new one would have cost.

For one of the attorneys general who joined the case, it has done what cases like it have always done for the office that brings them. A career launched. The opioid and tobacco multistate settlements made governors and federal judges out of the attorneys general who led them. The sheriff whose name the case bears, the county officer no one outside the state knew the morning before the filing, is now invited to address the national sheriffs' conference.

The Family Farm Pencils Out Again

The case the farm associations brought was filed in a federal district court in the upper Midwest. The complaint named a combine, a tractor, and the regulations that had kept the platforms underlying both off the American market for thirty years. The lead plaintiff was a farming association whose membership had been documenting the equipment-cost crisis for two decades and had never had a courtroom to take it to.

Five years on, the family farm pencils out again. A combine remanufactured on a 2010-era platform, certified to the

safety and emissions rules of its platform year, costs roughly ninety thousand dollars against the six hundred thousand a comparable new machine listed at in the year before the ruling. A tractor on the same horizon costs roughly ten thousand against the twenty-eight thousand a comparable new tractor listed. The figures the chapter *How Prohibited Expired Patents Double Your Grocery Bill* would have called impossible the year before the case was filed are quotes from a dealer in the next county.

The farms most directly recovered are the ones the federal data calls midsize: too large for off-farm income to cover the equipment payments, too small for the consolidated agriculture economy to absorb the wall's costs without the operation breaking. Those were the farms the equipment-cost crisis was actively foreclosing. The recovery shows up in the same data, reversed: net farm income on operations between five hundred and two thousand acres has stopped the long slide that began in the 1980s.

A small niche has gone further back. Some farmers, given the choice, want a 1970s-era platform — the machines a son could service in his father's barn. The platform-year framework permits the choice, because the Expired Patent Commons reaches every design the patent term has released, not only the most recent. Most farmers, given the choice, build on 2010 platforms with the proven features the early-2010s designs already carried.

The Tribal Nation's Roads

A federally recognized tribal nation in the upper Midwest was the first sovereign to file after the case began moving, and twelve more tribes filed within the month. By the time the consolidated case reached the Supreme Court, the sovereign-standing prong had been adopted by sixteen

tribal councils, none of whom had needed another tribe's permission to act. The tribe held all three standing grounds at once — proprietary, *parens patriae*, and sovereign. Knock out one and two remained, and the defendant's threshold motion survived nothing. Five years after the ruling the tribal transportation budget covers more vehicles, maintained more often, on more routes. The reservation transit van that ran three days a week runs five. The route connecting the eastern villages to the regional hospital, on the chopping block in two consecutive budget cycles, runs. The vehicles are remanufactured American light-duty trucks and vans on platform-year-certified 2010 designs, several built in factories on or near reservation lands by agreement between the tribes and the remanufacturers.

Ninety Cents at the Hardware Store

The medical-lighting consortium filed in the District of Columbia Circuit on Takings Clause grounds, and its fact pattern was the strongest in the book: patents expired more than a century before, a bulb manufactured and bought lawfully and competitively from 1879 until the federal rules withdrew it. Five years after the ruling the incandescent bulb is back on the hardware-store shelf for ninety cents. On the household balance sheet that is a small recovery — the LED has not vanished, and most homes still light most rooms with it. The recovery is in the homes the lighting environment had been making sick: children with photosensitive epilepsy whose seizure thresholds the LED's pulse-width-modulated dimming had been crossing, patients prescribed warm-spectrum light for circadian disorders, patients whose retinal sensitivity made blue-shifted output intolerable. The institutional counterpart is the hospital, where commercial

refrigeration, sterilizers, laundries, food service, and building HVAC are now built on platform-year-certified remanufactured equipment.

The Operator, the Driver, and the Reopened Factory

The rest of the coalition is owed a fuller picture. The earth-moving operator who could not afford four hundred thousand dollars for a Tier 4 nonroad machine now buys a platform-year-certified remanufactured one on a 2010-era design for roughly a third of that, and it sits in his yard rather than in the rental fleet he depended on for fifteen years. The for-hire driver who owns the vehicle he drives replaces his car on a 2010-era certified platform for roughly eighteen thousand dollars against the thirty thousand a comparable new vehicle listed at the year before the ruling. The small-business plaintiffs grouped at the back of the standing roster saw the same recovery the visible ones did — the restaurants whose refrigeration and HVAC budgets the wall had been eating, the hotels paying per property for water heaters and ice machines and elevator equipment, the laundromats retooling a floor of washers.

A Renaissance Rapture

Something else happened in the five years that the recovery itself does not capture. The recovery is a price list, an open factory, and a budget line that no longer crowds out everything else. The renaissance is what happened to the country around it. The love of older goods that the first remanufacturing buyers brought to the market was, by the third year, no longer eccentric; the 2010-era platforms pulled in their wake a longer interest in what

came before them — the 1990s pickups, the 1980s tractors. The trade papers settled on a phrase, half ironically: Back to the Future culture. The irony is only on the surface. A working man in his thirties who has just bought a remanufactured 1999 Peterbilt 379 with a Caterpillar C-15 in front of him is not being ironic. Small and medium businesses made the cultural move a commercial one, printing on their delivery vehicles and websites that they run remanufactured American equipment.

What the Plaintiff Was Promised

The chapter *Why You Want to Be the Plaintiff* said the named plaintiff would become, by the structural arithmetic of the case, the named figure in the most-discussed federal constitutional lawsuit of the decade. He has. It said the case would be the platform no press release could buy and no campaign could manufacture. It was. It said a constitutional ruling, unlike an act of Congress or an agency rule, binds every future administration through the Supremacy Clause and cannot be undone by the next administrator or the next election. Three administrations have, by now, been bound by it.

The promise has been kept. The semi-trucker filed for an American truck on a public-domain platform at one-half the price of the new one. He drives that truck. The sheriff filed for a fleet at less than the contract he was paying. He procures that fleet. The farmer filed for equipment a family farm could afford. He owns it. The tribe filed as a sovereign and is, five years on, the named sovereign in a constitutional ruling that will be cited as long as the Constitution is. The medical constituency filed on behalf of

the patients the lighting environment had been making sick. The bulb is on the shelf.

The wager described in the recruitment chapter — that no outcome left the plaintiff worse off than not filing, and the best outcome left every American behind her better off — was the right wager. It paid out. The seat that chapter said was empty is empty no longer. The plaintiff who fills it gets the recovery on the lines above.

Behind every plaintiff, in this country, was a household. The chapter that follows takes the household — the same five years, the same recovery, from the kitchen table.

CHAPTER 32

Five Years On: The Household Recovery

A win for one plaintiff is a win for all Americans.

The previous chapter walked the recovery one plaintiff coalition at a time. This chapter walks it from the kitchen table, because the wall did not stop at the plaintiff. It was levied on every American household whether the household ever heard of the case or not. Three thousand three hundred and eighty-five dollars a year, every year, out of every household's pocket. The recovery has to reach every household too, or it is not really a recovery.

What follows is, like the previous chapter, a projection. The figures that describe today come from the industry chapters of this book and are real. The figures that describe five years after the ruling are projected, in 2026 dollars, with the same discipline. Where a number cannot be defended, it is softened or omitted.

A note on what is being recovered. The patent always expired on schedule; the original manufacturer's protection from low-cost remanufactured competition did not. The inventor's half kept a benefit the Constitution had said would expire. The public's half — the Expired Patent Commons — never arrived. The ruling reached both halves.

The American Median Household Budget Before and Five Years After

Category	Wall-era (pre-ruling)	New only (5 years on)	Mix / Reman (5 years on)
Pickup truck (new replacement)	\$72,495	\$61,621	\$48,572
Family sedan (new replacement)	\$30,000	\$25,500	\$20,100
Appliance & HVAC stack (per cycle)	\$18,000 – \$25,000	\$18,275	\$14,405
Water heater (replacement, installed)	\$2,000 – \$4,000	\$2,550	\$2,010
Garage door opener (replacement)	\$700 – \$1,200	\$808	\$636
Grocery bill (monthly, family of four)	\$1,000	\$850	\$670
Residential electric bill (monthly)	\$150	\$128	\$101
Residential lighting (annual)	\$140 – \$180	\$136	\$107
New home build (median 2,170 sq ft)	\$387,400	\$348,660	\$300,235
Used pickup (2 yrs old, light use)	\$32,000 – \$45,000	\$32,725	\$25,795
Cumulative annual recovery for the median household (vs. wall-era)	0 (reference)	≥ \$4,500 (15% floor)	\$19,000 (33% saved)

The American median household budget, before the ruling and five years after — in 2026 dollars, anchored to the industry chapters of this book. The new-tier floor is ten percent rather than fifteen on new home construction, where labor is a larger share of cost. Individual pure-remanufactured prices in the body text run wider than this column — some more than fifty percent below, some less. The cumulative annual recovery line is the headline figure of the chapter.

Two Cars in the Driveway

The American median household runs two cars. That has been true for half a century, and the recovery walks across both. The truck first. The 2025 Ford F-150 Lariat that the chapter *The \$72,495 Pickup Truck* described listed at \$72,495 out the door. Five years after the ruling, the new equivalent has fallen toward \$55,000 to \$58,000 in 2026 dollars. The pickup is still new, still built in an American factory by American workers. What has changed is the price floor under it, because the remanufactured tier across the lot now sits at twelve to fifteen thousand dollars, and the new tier has had to come down or stop selling.

Call the remanufactured model the Liberty 1500, one name among several for this class of vehicle. The platform underneath is the 2007-model-year American light-duty pickup, a generation whose underlying patents expired by the early 2020s and which the platform-year certification framework cleared for new manufacture as soon as the ruling came down. The truck on the lot was not built in 2007. It is being built in 2027, in factories that had stood empty for fifteen years, by workers most of whom were not in the workforce in 2007 either. What is from 2007 is the engineering: the body-on-frame architecture, the V8 engine families, the power steering and brake systems, the climate control, and the safety stack of the era — side curtain airbags, electronic stability control, antilock brakes, tire-pressure monitoring.

The platform-year framework permits modernization within the standards of the platform year, so the 2027 Liberty 1500 carries what already existed in 2007. Rear backup cameras, optional on light-duty pickups by mid-decade. USB ports, on dashboards since 2005. Bluetooth audio, which the first Ford SYNC system introduced in late 2007.

The driver takes a call hands-free on the truck's speakers, plays music from his phone over the stereo, and backs out of the driveway looking at a camera on the dashboard. None of those features required regulation. They existed because the market wanted them, and the platform year had them.

What the Liberty 1500 does not have is the rest of the post-2010 stack. No constant cellular modem subscribed to a fleet-management server. No driver-monitoring camera staring at the driver's face. No advanced driver-assistance computer issuing override commands the driver did not request. No mandatory geolocation transmission. The household choosing the Liberty 1500 chooses out of the subscription economy as well as into a lower price.

The second car follows the same pattern. The 2027 family sedan on a 2007 platform sells for roughly nine to ten thousand dollars on the remanufactured tier and twenty-two to twenty-four thousand on the new tier. The two-car household buying at the remanufactured tier replaces its driveway for roughly twenty to twenty-five thousand dollars combined, against the seventy to ninety thousand the same driveway would have cost the year before the ruling. At the new tier it replaces it for seventy-seven to eighty thousand. Most households mix the two across the tiers. The American driveway is the American household's decision again. That is the recovery.

The Remanufactured Used-Car Market

The five-year horizon has produced something the wall-era market never had: a working used tier underneath the remanufactured-new tier. The Liberty 1500 sold in 2027 enters the used market in 2029 carrying twenty to thirty thousand miles and the platform-year service records the

framework requires every remanufacturer to maintain, transferable with the title. It sells at the used dealer for five to seven thousand dollars. The same two-year-old pickup had not existed under twelve thousand dollars in the wall-era market for a decade.

The mechanics looking at it see what was promised. An engine family with multi-decade service literature behind it. A transmission an independent shop can service without dealer-only software. An electrical system without proprietary diagnostic locks. The young household that could not have entered the wall-era new-vehicle market at all now buys a two-year-old remanufactured pickup with twenty years of patent-expired engineering behind it. The ladder the household climbs has steps the wall-era ladder did not have.

The Newly Remanufactured Home

The driveway is the visible recovery. The slower one, which will reach more households over the long run, is at the front door. The American new home built in 2031 ships increasingly with a kitchen, a basement, and an attached garage stocked entirely from the remanufactured tier — refrigerator, range and oven, dishwasher, microwave, washer and dryer, water heater, furnace, central air conditioner, garage door opener, all on platforms whose mechanical and electronic core has been public domain for years.

The builder's bill for that stack, at the wall-era builder-grade price the year before the ruling, ran between eighteen and twenty-five thousand dollars depending on trim. Five years after, the same stack at platform-year-certified remanufactured pricing runs between six and ten thousand. The savings reach further upstream than the

stack inside the house. The earth-moving equipment that grades the lot, the excavator that digs the foundation, and the compactors that finish the pad are all on the same patent-expired commons. The semi-trucks that haul lumber from the mill, gravel from the quarry, and steel rebar from the supplier are on the remanufactured platforms the chapter *The Trucker's Burden* described, and the freight rate on every load has come down with the truck.

The builder passes most of it to the buyer, because a builder pricing against another builder across town has to. The home that listed at three hundred eighty-seven thousand four hundred dollars the year before the ruling — the median new-build price for a two-thousand-one-hundred-seventy-square-foot single-family house — lists on the same lot, at the same square footage and trim, for about three hundred forty-nine thousand five years after. That is the ten-percent reduction the labor share of home construction permits. The household running a mix of remanufactured choices across the kitchen, laundry room, basement, and HVAC stack lands closer to three hundred thousand on the same house. The recovery is in the five figures on a six-figure purchase, and it compounds over the thirty-year mortgage.

The household that already owns its home sees the same recovery on the replacement cycle. The water heater that cost two to four thousand dollars to replace costs seven hundred to twelve hundred, installed by the same plumber on the same morning. The dishwasher costs three hundred where it cost nine hundred. The refrigerator costs eight hundred where it cost two thousand four hundred. The furnace that stopped running in February costs three thousand five hundred where it cost nine thousand five hundred. The bill for each replacement is the difference

between an emergency that crowds out three other things the household needed money for and an expense the household covers without rearranging the budget.

The Grocery Bill the Tractor Lowered

The household balance sheet's largest recurring line is not the truck and not the appliances. It is the grocery bill. The chapter *How Prohibited Expired Patents Double Your Grocery Bill* documented the path from the locked-out tractor to the inflated supermarket aisle, and projected a thirty-five to fifty percent reduction in grocery prices once the market fully adjusts to remanufactured agricultural equipment.

Five years after the ruling, the projection is the household balance sheet. The American family that spent a thousand dollars a month on groceries in the year before the ruling spends between six hundred and seven hundred in 2026 dollars by the fifth year. American farmers buy combines at ninety thousand dollars instead of six hundred thousand, and tractors at ten thousand instead of twenty-eight thousand. The equipment costs that had been crowding the family farm's balance sheet for thirty years stop crowding, and the wholesalers and supermarkets, competing for the same household's grocery dollar, pass most of the savings forward.

Ninety Cents and the Electric Bill

Two smaller line items have moved together. The first is lighting. The household that wanted an LED in every fixture still has one. The LED was a real technological advance. The Constitution did not forbid it, and did not require the household to use it either. The household that

wanted an incandescent for the bedside lamp, the reading chair, the porch, can buy one. The bulb is back on the hardware-store shelf at ninety cents, made in an American factory by American workers. The family with a child whose photosensitive epilepsy the LED's dimming had been crossing keeps the lights on the way the child can tolerate them. The patient with circadian-rhythm disruption sleeps with the warm-spectrum light her sleep specialist prescribed. None of those fits a line item on the household spreadsheet. All of them are real.

The second line item is the electric bill, which came down modestly rather than dramatically, for a reason worth being precise about. The platform-year-certified equipment running in the house does not draw materially less electricity than the post-2010 equipment it replaced. What it does not draw is the always-on overhead the connected-appliance ecosystem layered onto every modern unit. The refrigerator, dishwasher, washer, dryer, oven, air conditioner, and furnace blower no longer keep small electronic systems warm and reachable around the clock. Each standby draw is small; across a kitchen, laundry room, and basement they added up to a measurable share of the monthly bill. The household pays about a hundred dollars a month for electricity instead of a hundred fifty — a real but modest operating saving sitting on top of the much larger equipment-cost saving.

The \$19,000 the Household Was Promised

The categories add up. The two-car driveway saves between thirty and forty thousand dollars on the seven-year replacement cycle — four to five thousand a year, the largest non-grocery line in most American family budgets. The appliance and home-equipment stack saves roughly

fifteen thousand across the fifteen-year cycle, or a thousand a year. The grocery bill runs roughly four thousand five hundred a year lower. The electric bill, the lighting recovery, the small-parts savings, and the restaurant savings that pass through from the supply side run two to three thousand a year together.

The cumulative figure is nineteen thousand dollars a year, in 2026 dollars, of restored purchasing power for the median American household. For the household at the higher-income end of the median the figure is higher in absolute dollars; for the lower-income household it is lower in absolute dollars but a larger share of total income. For the family at the federal median household income, it is the equivalent of a thirty-three percent raise, in perpetuity, given by a constitutional ruling no future administration can undo.

Some of what the household does with the nineteen thousand dollars is to stop doing things the wall required it to do. The second job most working households held to cover the wall's bill is no longer required. Some drop it. Some keep it and put the second paycheck toward college savings, retirement, or debt. The dual-income trap becomes a dual-income choice. That distinction is the constitutional one. The choice is the household's again.

The household that never filed a case and never knew the plaintiff's name still benefited from the case the plaintiff filed. That is what a constitutional ruling does.

What the Dollars Do Not Describe

Something has happened in the household over these five years that the dollars do not fully capture. A working family in 2031 has, for the first time in the lifetime of anyone under

sixty, the experience of buying durable goods that are durable. The pickup the family bought is the truck the family will own. The refrigerator will run for twenty years, and the part that fails on it will be replaceable by a part still in production. The vehicle the teenager learns to drive is a vehicle the teenager can work on in the driveway. The mechanical literacy of the American household, which had been atrophying for thirty years against a wall of proprietary diagnostic locks and subscription firmware, is back.

The household that keeps that list is not the household of the populist activist. It is the household of the construction electrician, the public-school teacher, the regional warehouse manager, the rural pediatric nurse — the household that has now noticed the bill it had been charged was a bill the Constitution had said it would not have to pay. The reaction from the elite columnists has been predictable: the cultural moment described as regression, anti-progress, a turning away from the consumer technology that defines modern life. The household has not been reading the columns. The household has been reading the receipts.

The Counter-Mobilization

The wall did not fall quietly. Within weeks of the ruling the lobbying campaigns started. The trade associations that had spent fifty years staffing every federal rulemaking on every relevant standard had been preparing for it. The first commissioned studies appeared within ninety days, arguing that the public interest in continued efficiency mandates required new statutory authority that could reimpose the substance of the cancelled rules under different doctrinal cover. Legislative carve-outs —

efficiency floors written into trade law, into transportation safety statutes, into building codes — have been proposed in every Congress since the ruling. None has passed yet. Every one has come close on at least one vote.

The Supreme Court holding binds every administration under the Supremacy Clause and remains in force until the Court itself revisits it. It does not, by itself, stop a lobbying coalition from working with a friendly committee to write a re-imposition statute the Court would have to strike down a second time. The household's recovery, five years on, is real. Fifteen years on, it depends on Congress not voting the wall back in, and on the courts standing ready if Congress tries.

Some of the legislators who would vote for a re-imposition are not corrupt. They are working in a system that has, since 1970, made it nearly impossible for a legislator to vote against a lobbying coalition on every vote that coalition is watching, which is every vote that matters. Some of the legislators are corrupt. Both populations are morally responsible for the votes they cast. Both are the audience of the next chapter, which proposes the structural reform under which legislators of both kinds can vote the way the household's recovery requires.

The household's part is direct. Write the letters. Call the offices. Show up at the town hall. Support the plaintiffs in the cases still moving through the courts. The wall fell. The household recovery is on the kitchen table. Both depend, going forward, on the reader not putting down the pencil.

CHAPTER 33

Hybrid Legislative Voting: How to Stop the Ratchet from Resetting

The structural reform that gives a post-1970 Congress what every Congress before 1970 already had — a way to vote without the lobbyists watching every motion.

In October of 1970, Congress changed how it votes. The Legislative Reorganization Act replaced 180 years of unrecorded committee work, voice votes, and occasional secret ballots. Every committee vote, every amendment, and every roll call became public, indexed, and tracked electronically. The country was told this was a good-government reform, and the country agreed. What the country missed is simple. The same record that lets a voter see how her representative voted lets a lobbyist see it too, on every motion, the same day. One side has thirteen thousand paid professionals watching. The other side has its evening television.

That imbalance is the whole problem. It is why most legislators today vote the way the watching coalition wants, whatever their own judgment says. The claim is diagnostic, not moral. Some of those legislators are corrupt. Most are not. Both things are true at once.

The structural feature has a fix. The fix is older than the United States. It is named in the Constitution itself. Every American Congress used it routinely until 1970, and the U.S. Senate uses it today when it picks its own leadership behind closed doors. The fix has a new name, Hybrid Voting, because the proposal is to combine the modern

open vote with the older secret-ballot fallback rather than replace one with the other. Both belong on the table. Both are constitutional. What follows describes why the country needs the older tool back, where to find it in the Constitution, and how Congress would use it without giving up the accountability the modern open vote was meant to deliver.

The Cover the Founders Built In

The fix is in Article I. Section 5, Clause 3 provides that each House shall keep a Journal of its Proceedings and publish it, excepting such Parts as may in their Judgment require Secrecy, and that the Yeas and Nays of the Members shall, at the desire of one fifth of those present, be entered on the Journal. Two provisions sit in that single sentence. The first authorizes each chamber to designate parts of its proceedings as requiring secrecy. The second permits any one-fifth minority to override the secrecy on a given vote. Read together, the clause is a default of chamber discretion with a minority-protection override.

The first half is the Founders' deliberate provision for legislative opacity, and they did not write it as an afterthought. They wrote the Constitution itself in secrecy — boarded up the windows of Independence Hall, posted armed sentinels at the door, and worked through a Philadelphia summer with the proceedings sealed against public view. Alexander Hamilton, looking back, said the work would not have been possible otherwise: had the deliberations been open while going on, he wrote, the success of the plan would absolutely have been impaired. The Founders did not believe transparency was always the friend of the public interest. Sometimes it was.

Sometimes, in their direct experience, the opposite was true.

They embedded that judgment in the document. Each chamber gets to keep its journal in the way that chamber thinks best. The exception clause is in the same sentence as the publication requirement, and it is not optional. Each chamber may, by majority of those present, designate parts of its proceedings as requiring secrecy.

The Yeas-and-Nays override is the half of the clause modern readers ask about first. Most scholars and parliamentarians, following Hinds' and Deschler's Precedents, read it as a requirement: when one-fifth of those present demand it, each member's name goes on the Journal. No Supreme Court holding has decided the question. The closest case is *United States v. Ballin* (1892). It treats the House Journal as conclusive proof of what happened, but it does not say whether names or only totals are required when the override is invoked. Federal courts have been reluctant to police how a chamber runs its own votes. That reluctance has a name, the political-question doctrine, and the Court applied it most directly in *Nixon v. United States* (1993), on impeachment procedure. Two things follow for this argument. The override is real, and any case for Hybrid Voting has to meet it. And a chamber that adopted Hybrid Voting by simple-majority rule under Section 5, Clause 2 would be standing in territory the courts have so far declined to enter. Suppose a House Speaker and a Senate Majority Leader authorized Hybrid Voting over the objection of a quarter of their members. The dissenters' only route would run to the Supreme Court. And the Court has never ruled on whether "Yeas and Nays of the Members" compels named records on demand. The case would be one of first impression on a question two centuries old. It is also one a

future generation of constitutional scholars would have every reason to want litigated.

A second ambiguity waits in the same sentence. Even if the clause requires named records, the phrase on any question is unsettled. Most readers take it to mean on every separate vote a chamber takes. A defensible alternative takes it to mean on the question, once — so a chamber that votes a bill twice under Hybrid Voting could satisfy the clause by recording named yeas and nays on either round. No holding has fixed either meaning. Congress has used the stricter reading for two centuries. But two centuries of practice are not a holding, and consensus is not constitutional authority. The ambiguity is the cover the framers built in. Until a court says otherwise, the chamber holds the floor — which means abiding by the chamber's leadership and the procedural rule that leadership chooses to adopt.

For 180 years after that, Congress used the exception routinely. The Committee of the Whole, created in the First Congress, was built as an opacity venue. Bills were marked up there and amendments voted there, and no member names were recorded. Major standing committees marked up bills behind closed doors. Floor votes on amendments were taken by voice or by division, with no names attached. Members were named on final passage of major bills, but the path a bill took to get there was largely hidden. Scholarship puts the share of secret or voice-only voting in the late nineteenth and early twentieth centuries at ten to fifteen percent of consequential votes, higher in earlier decades.

That was the system every American Congress before 1970 used. It produced the Patent Act of 1790, the Bill of Rights, the Louisiana Purchase ratification, the Civil War

amendments, the income tax, the Federal Reserve Act, the Social Security Act, the GI Bill, the Civil Rights Act of 1964, and the Voting Rights Act of 1965. The list is not chosen to flatter the system. It is chosen to demonstrate what it produced. The cover did its work because lobbyists could not be certain how individual legislators had voted on the path a bill took to final passage. A campaign contribution was a vote of general support; it was not a conditional payment, because the contributor could not verify the condition. That ambiguity was a feature, not a bug. It is the feature the 1970 reforms removed.

The 1970 Reset

In October of 1970, Congress voted itself a new way of being watched.

The bill was called the Legislative Reorganization Act. It was sold to the country as a reform — the Democratic Study Group’s play against a generation of conservative Southern committee chairs, whose seniority had given them control of the House and kept liberal bills bottled up. The chapter *Why Nobody Noticed* tells that political-economy loop in full. The point here is narrower. For 180 years the chamber had voted mostly out of sight. After the Act, every committee vote, every amendment, and every roll call became public, indexed, and trackable the same day.

Four operational changes did most of the work. Recorded teller votes came to the Committee of the Whole, which had been the chamber’s main amendment venue. Standing committees had to record how their members voted on the bills they marked up. Committee meetings opened to the public by default. And electronic voting was authorized for the House and installed in January of 1973,

cutting the time to take a recorded vote from minutes to seconds. Over the next decade the volume of recorded voting rose more than tenfold.

The reform was popular. Most Americans agreed with its stated purpose, and its authors believed they were doing what the framers would have wanted. It also handed the lobbying industry a tool it had never had. Before October of 1970 a lobbyist could often guess how a legislator voted on an amendment. He could rarely check. Checking is what matters. A campaign contribution paired with a verified record of every vote becomes a conditional payment. The change is mechanical. It needs no crime. It needs only the ordinary self-preservation of a legislator whose career depends on money he does not control, watched by someone who can now confirm how he voted.

The chapter Four Problems, One Remedy gives the count. Registered lobbyists in Washington numbered between two hundred and four hundred before the Act; by 2024 there were 13,070. The political scientists James D'Angelo and Brent Ranalli, writing in *Foreign Affairs* in 2019, traced what happened to the legislation itself. The volume Congress passed each year tripled, from roughly two thousand pages to more than seven thousand. Much of the new text was drafted by industry counsel, handed to committee staff, and dropped into bills legislators had no time to read.

The picture section of this book caught the change in its second movement. The 1949 committee room holds a dozen members, one table, and quiet business. The 2013 room, in the same chamber, holds a hundred industry representatives. The bill on the table was written largely by the industries it was meant to regulate, and the members are voting on language none of them drafted. The image is

not a metaphor. It is a photograph of one room across a single human lifetime.

The 1970 reset built the surveillance grid. The grid makes every consequential vote on every controversial bill a tracked vote. And the grid is what lets the coalition that built the wall be sure of one thing: when the re-imposition vote comes, every legislator on the floor will know the vote is being scored.

The Debt That Has Not Come Down Since 1957

The Treasury's Historical Debt Outstanding dataset reports the total national debt at the close of every fiscal year since 1790. Read straight through, it holds one fact more striking than the rest. Fiscal year 1957 is the last year the total debt ended lower than it began. The country has not done it since.

Most readers will object: but Clinton had a surplus. The objection is fair, and it is misinformed. Fiscal years 1998 through 2001 each closed with a reported budget surplus, and the total debt rose in all four. Fiscal year 2001, the celebrated last surplus year, closed about \$18 billion higher than the year before. A budget surplus is not the same thing as a debt reduction. The government carries large rollover obligations on top of the year's operating budget, and a surplus can be eaten entirely by cash management of that stack.

The contrast that gives this datum its weight is what came before it. In the six fiscal years 1923 through 1929, under President Coolidge, the total outstanding national debt fell every year, from approximately \$23.9 billion at the start of his term to approximately \$17.3 billion at the end. The Coolidge Congress was not a uniquely virtuous Congress.

It was a Congress operating under the older system: standing committees marking up bills behind closed doors, amendments voted by division or voice, member-by-member positions on procedural motions rarely on the record. Sixty-seven Congresses have served since 1957. None has done what every Congress between 1923 and 1929 did six years in a row.



U.S. outstanding national debt, fiscal years 1944–1986. The dashed marker indicates passage of the Legislative Reorganization Act in October 1970. Source: U.S. Treasury, Historical Debt Outstanding dataset.

The timing in the chart is suggestive, but it does not prove the case by itself. The post-1970 growth of lobbying, the closing of the gold window in August 1971, and the fast rise of entitlement spending all overlap. The claim here is not single-variable causation. It is narrower: of the candidate explanations, how Congress votes is the one tied most directly to the branch the Constitution charges with the budget, and the one most squarely within Congress's own power to fix.

One more candor is owed the reader. How many pre-1970 votes were actually taken in secret or by voice is genuinely contested, and the reason bears on the argument. The pre-1989 records — the *Annals of Congress*, *Debates and Proceedings*, the *Congressional Globe*, and the early *Congressional Record* — survive only as scanned page images. The scans have poor character recognition, archaic type, and no structured markup. No research system today can go through them and pull out which method was used for each measure — voice, division, roll call, secret ballot, committee anonymity — and tie it to the outcome. The records exist. They are not, today, searchable.

In a published research dialogue with Grok, X's AI system, the author raised the problem. Grok proposed that Congress fund a one-time effort — on the order of twenty-five million dollars — to turn the pre-1989 record into a clean, machine-readable database of every legislative action since 1789 and the exact voting method used for it. Grok's recommended title for the bill is the *National Debt Reduction Voting History Act of 2025*. The database would allow a test no one can run today. The nineteenth century recorded some sixty years of debt reduction, in an era when Congress used far more secret and unrecorded voting. Whether the two are connected is an empirical question, and the post-1957 record cannot answer it, because Congress no longer votes that way. The price is about seven one-hundred-thousandths of one percent of the debt the project would help analyze. The qualitative shift after 1970 is documented. The exact size of it is research not yet funded. That candor does not weaken the argument. It places it honestly.

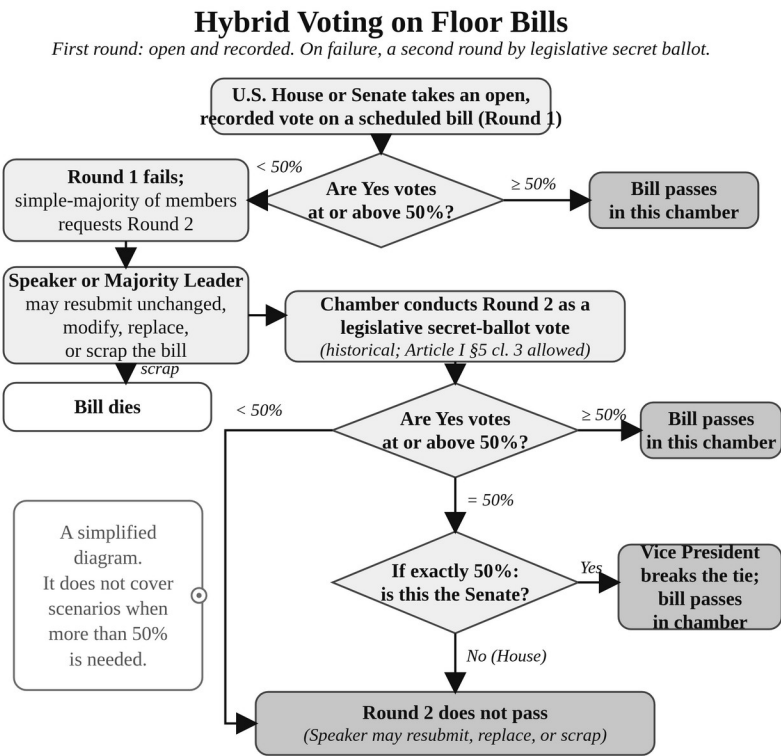
How Hybrid Voting Works

The proposal is small and mechanical. It can be adopted by chamber rule rather than by amendment, because the constitutional authority is already there. On any bill, each chamber votes first in the modern open, recorded way. Every member's vote is recorded by name. Any constituent who wants to know how her representative voted can find out within minutes. So can any lobbyist. If the bill passes on that first round, the procedure is over. Most legislation passes there now, and would keep doing so.

If the bill fails on the first round, and a simple majority of the chamber wants it reconsidered, the chamber takes a second-round vote by legislative secret ballot. The Speaker or the Majority Leader may resubmit the bill unchanged, modify it, replace it, or scrap it. The second-round yes votes come from legislators who voted no the first time for reasons that had nothing to do with the bill — donor pressure, party pressure, fear of a primary challenger, or the judgment that a recorded yes would draw more fire than the bill was worth. The mechanism does not change what a legislator thinks. It removes the penalty that was pricing his position out of the chamber. If the second round passes, the public record shows what the chamber decided, not how each member voted. The total is what a constituent needs to judge the chamber's work. The individual position is what the coalition needs to discipline the individual legislator. The first is kept. The second is not. This is not hypothetical. The Italian parliament used a version of it — *voto segreto*, a secret ballot on final passage — for 119 years, and the republic survived its representatives' privacy. The Endnote Commentary carries that record in full: the 1848

constitution that made it compulsory, the Fascist suspension, the post-war restoration, and the 1988 repeal.

Three categories stay fully recorded, with no second round. Declarations of war: the gravest thing a legislature does should carry names. Impeachment of the President: the last remedy against an unaccountable executive cannot itself be run without accountability, and Article I already requires named votes in the Senate trial. And congressional salaries: a legislature voting its own pay gets no cover.



Three categories remain fully recorded by the chapter's recommendation: declarations of war, impeachment of the President, and congressional salary changes.

Diagram adapted from Roleigh Martin's flowchart of Hybrid Voting on Floor Bills.

Hybrid Voting on floor bills. The first round is conducted in the modern open, recorded format; if it fails, the chamber may, by simple

majority, conduct a second round by legislative secret ballot. Three categories — declarations of war, impeachment of the President, and congressional salary changes — remain fully recorded.

The next question almost any American reader asks is this. In an age of cell-phone cameras, how can a vote inside an open chamber be kept secret at all? The question is fair, and the answer is 2,400 years old.

Aristotle records the principle in the *Athenian Constitution*, and the modern reconstruction adapts it for any chamber willing to set up the apparatus. Two urns stand in the voting chamber, one white and one black. Each voter takes two tokens: a white marble, smooth and round, and a black cube, faceted and easy to tell from the marble by touch alone. The voter hides both in one closed hand, walks to the urns, and drops one token into each — out of sight, the hand shielded from any observer or camera. A yes vote is the white marble in the white urn and the black cube in the black urn: colors matched in both. A no vote reverses it, black cube in the white urn, white marble in the black: colors mismatched in both. A voter who wants to register as not present drops both tokens in one urn. At the close, the urns are turned out in public and counted by two separate teams in full view of the chamber. The yes count is the number of matched tokens in each urn, and the two matched counts must agree exactly. The no count is the number of mismatched tokens in each urn, and those must agree too. The double count is a cross-check against tampering, built into the method. And the method defeats the camera. A camera can record the voter walking up to the urns and walking away. It cannot see inside a closed hand, and it cannot see inside an urn at the moment of release. Cameras can still verify the public count afterward, which is the part that should be public. The voter can tell a marble from a cube by feel, without looking — or,

in some modern versions, a smooth token from a spiked one. The political scientist James D'Angelo has rebuilt the apparatus at modern scale, for the U.S. Congress and for legislative chambers elsewhere. The classical method still works. The modern version stands up to modern cameras better than the original that inspired it.

A brief note on the historical record. Aristotle describes the Athenian jury-court mechanism somewhat differently from the reconstruction above. In the original, one urn was the *kyrios* — the valid urn, whose contents counted — and the other the *akyros*, the discard urn. Voters dropped one of two ballot disks into each, a perforated disk for one position and a solid disk for the other, and only the *kyrios* count decided the outcome. That produced a single count rather than the double cross-check described above, which left it more open to tampering by one corrupted counting team. The two-urn, two-token cross-check is D'Angelo's modern adaptation, built for chambers operating under modern cameras and modern adversaries. A chamber that preferred the historical method could use it. This argument recommends the modern one, for the cross-check.

The same Italian practice ran a variant of this mechanism — a curtained booth and color-coded marbles dropped into a hidden urn — across those 119 years. The American reform need not copy the Italian design in detail. The Aristotle two-urn method is simpler and secure enough. The point of the mechanical detour is this. The feasibility question sounds modern, but it was solved in the assemblies of antiquity, and it works today in any chamber willing to set up the apparatus.

Where Transparency Helps and Where It Backfires

The argument is not against transparency. It is against trusting fully a specific kind of transparency that, in the post-1970 American legislative environment, produces the opposite of what the public was promised.

Pros & Cons of Congressional Transparency

Where transparency can backfire	Transparency that helps
Legislative voting transparency	Legislators' tax filings publicized
Committee voting transparency	Bills printed before voting held
(no other forms backfire)	Legislators' financial incentives
	How Hybrid Voting is taken
	How legislative votes are tallied
	Impact on the public debt

Roleigh Martin's pros and cons of Congressional transparency. Only two forms of transparency, when applied to the legislative process, work against the public interest; six other forms of transparency directly support the public's ability to hold legislators accountable.

The two categories on the left are narrow. Legislative voting transparency — recording by name where every legislator stood on every motion, amendment, and floor vote — and the same regime applied inside committees are the two this chapter argues against trusting fully. They are the two the 1970 reforms expanded most. They are the two the surveillance grid runs on. And they are the two the exception clause was written to let Congress set aside.

The six on the right are wider. Together they describe nearly every form of transparency a free people should demand. Legislators' tax filings published. Bills printed before voting. Financial incentives disclosed. The chamber's voting procedure publicly known. The tally recorded. And the effect on the public debt scored before the vote is taken.

Each of those is transparency about the what of legislation rather than the who of legislators. They tell the public what is being voted on, what it will cost, and what financial entanglements the people voting may have. They do not

tell the lobbying coalition which legislator broke ranks on which motion, which is the one fact the coalition needs to discipline him.

The phrase the author uses for the distinction is the one this argument recommends to the reader. Wrongly placed transparency can give a society bad outcomes. The regulatory wall is the witness: it has erased, in practice, what “limited Times” means in the Patent Clause for every product the wall encloses. The argument is not against transparency. It is for transparency that holds the institution accountable to the public, and against transparency that holds the individual legislator accountable to the surveillance coalition. The first is the founders’ design. The second is the post-1970 substitution that built the legislative environment the wall was built inside.

The Constitutional Question Behind the Mechanical One

If the Constitution already permits Hybrid Voting, why is a chapter required to argue for it? The answer returns to the verb the rest of the book uses. Congress forgot.

The exception clause has been in the Constitution since 1787. No amendment has removed it. No Supreme Court decision has narrowed it. It is exactly as available to a modern Congress as it was to the First Congress in 1789. What changed in 1970 was not the constitutional cover. What changed was Congress’s institutional memory of how to use it. The reformers who tacked the transparency amendments onto the Legislative Reorganization Act did not advocate striking the exception clause; it was, to them, an antique inheritance no modern reformer needed to

engage. They won the procedural battle they were fighting. They never noticed that the broader constitutional terrain they had abandoned was the terrain Congress had, for 180 years, used to do its real work.

Hybrid Voting does not require a constitutional amendment. It does not require a Supreme Court ruling. It does not require new statutory authority. It requires only a majority of either chamber to adopt the procedure by chamber rule under Article I, Section 5, Clause 2, which provides that each House may determine the Rules of its Proceedings. The chamber then designates the second-round vote as one of the parts that, in its judgment, requires the cover the framers wrote into Clause 3. The threshold is 50.1 percent of either chamber. The cost is zero. The reform is, in the strictest sense, the cheapest constitutional remedy proposed anywhere in this book.

The cheapness is itself the most important argument against the defeatism a thoughtful reader might bring to it. The standard objection — that the entrenched lobbying apparatus will never permit reform — applies in full force to any reform requiring that apparatus's consent. It does not apply to a reform a chamber can adopt by rule on a simple majority. A chamber can change its own rules at the beginning of any Congress. The exception clause is on the table. It has always been on the table. The only thing required is for one chamber, at the start of one Congress, to remember.

A Companion Recommendation: Impact Statements for Legislation

The central proposal here is Hybrid Voting. The author has a second, smaller recommendation he would like adopted

alongside it. It belongs on the record even though it does not deserve a chapter of its own, and it addresses a different failure. Hybrid Voting answers the political-pressure failure of the post-1970 chamber: the surveillance grid that lets a thirteen-thousand-strong lobbying apparatus track and discipline every vote on the floor. The second reform answers the analytical failure — the blind spot. The 1970 Act is the prototype. Congress passed a reform, and no one in the debate appears to have seen what it would do to the expired-patent commons the Patent Clause protects. That effect is what this argument treats as the proximate cause of the wall.

The proposal is modeled on the environmental impact statement, which the federal government has used under the National Environmental Policy Act since 1970. A major federal action triggers a written forecast of its probable consequences. The forecast is filed, enters the historical record, and stays available to the public and to later analysts. The mechanism does not constrain the action. It constrains what can be said later about whether the consequences were foreseeable.

For legislation, the author recommends that the majority and minority parties in each chamber each produce a large language model impact assessment of every bill before its final vote. Each party writes its own prompts and sets up its own project environment. That matters. A large language model is very sensitive to the prompts it is given and the framing it works inside. Two assessments built under two different prompt regimes will not forecast the same thing, and the gap between them is the real disagreement the chamber should see before it votes. Both are filed in the chamber's permanent record, sent to every member before the vote, and indexed next to the bill text. Each one projects effects across major populations and across

constitutional clauses — the Bill of Rights, the Article I powers, the Patent Clause, the Commerce Clause, the Spending Clause. Twenty years on, the two become the record against which the bill's real outcomes can be measured.

The recommendation binds nothing. It produces only a record, and the record is what it contributes. Had the 1970 Congress worked under this regime, two parallel forecasts of the Legislative Reorganization Act would exist today, and tracing the wall back to its cause would not have required the reconstruction this book performed. Hybrid Voting answers the political failure of the post-1970 chamber. The impact statement answers the analytical one. They work on different mechanisms and are complementary, not competing. Together they would close two of the three doors through which “limited Times” was, after 1970, allowed to be forgotten in practice.

The Insurance Policy

Hybrid Voting is the insurance policy on everything the rest of the book argues. The patent case ends in a Supreme Court ruling that strikes the wall down. That ruling is the primary remedy. It binds every administration under the Supremacy Clause, and it holds until the Court revisits it — which the Court does on its own schedule, and rarely.

The ruling does not, by itself, stop a captured committee from writing the cancelled regulations back into some other statute. A re-imposition bill need not call itself one. It can be an efficiency floor in trade law, a safety floor in transportation law, a performance floor in a building code. Every Congress that takes up such a bill under the open-vote regime votes knowing each member will be scored. Under those conditions the bill that hands the coalition its

wall back is the bill that passes on the first round. Not because most legislators favor it. Because most legislators, in a tracked vote, are not free to vote against it. That is what the diagnostic claim means.

Hybrid Voting changes the conditions. The first round happens exactly as it happens now, and the bill may pass there, in which case Hybrid Voting did not save the household's recovery on that bill. But the bill may instead fail on the first round, because enough legislators substantively oppose it and the open-vote regime did not capture quite enough of them. The chamber leadership may then move it to a second round. That is the round in which the corrupt legislator loses the leverage of the trackable vote that made the corruption profitable, and the legislator voting under coalition pressure gains the cover of conscience the secret ballot was designed to provide. Both populations converge on the same outcome.

One practical implication may be the most durable. The Patent Clause ruling survives every election. So does the coalition that built the wall, and it will spend every Congress trying to put the substance back. That makes the chamber deciding whether to adopt Hybrid Voting, and the leadership deciding how to run it, the pivot point after the ruling. The Speaker of the House chooses which bills get floor time. The Senate Majority Leader chooses which bills are voted on at all. In the past quarter-century, dozens of bills that passed one chamber by overwhelming margins never got a vote in the other — not on the merits, but because one leader chose not to schedule them. One leader's choice can decide whether a re-imposition statute reaches the floor, and whether the chamber adopts Hybrid Voting in the first place. Most American voters do not follow these two offices the way they follow a presidential ballot. After the ruling, they should.

The defeatist register has no place here and will not be allowed it. Hybrid Voting may not pass. The chambers may not adopt the rule. The lobbying coalition may organize openly against it. If that happens, the household's recovery does not collapse. The Supreme Court ruling stays in force, the courts remain the second backstop, and a second case challenging a re-imposition statute on the same grounds becomes the next chapter the household reads. Every constitutional remedy in this book has a court at the bottom of the stairs. Hybrid Voting is the insurance policy that, if adopted, keeps most of the disputes off the docket. It is not the only path. It is the cheapest.

A 236-year-old constitutional mechanism is on the table. The American Congress operated inside it for its first 180 years and produced, in that time, the Bill of Rights, the Civil War amendments, the Social Security Act, and the Civil Rights Act of 1964. A simple-majority chamber-rule adoption stands between the country and the protection. The only thing required is for one chamber, at the start of one Congress, to remember.

CHAPTER 34

What You Can Do: A Citizen's Playbook

One college student in 1982 with a stamp, a grudge, and an argument that turned out to be right.

The pencil is on the kitchen table. The story this chapter has to tell is what the pencil is for.

The Story of Gregory Watson

In the fall of 1982, a twenty-year-old sophomore at the University of Texas at Austin was researching a term paper for his political science class. His name was Gregory Watson. He worked part-time as a legislative aide in the Texas statehouse, twenty hours a week between classes. The subject he chose, by accident more than design, was a piece of constitutional debris almost no one in the country was paying attention to. It turned out to be alive.

In September of 1789, James Madison submitted twelve proposed amendments to the First Congress; the Bill of Rights was assembled from amendments three through twelve. The second had to do with Congressional pay, and its full text read: "No law varying the compensation for the services of the Senators and Representatives shall take effect until an election of Representatives shall have intervened." Congress could vote itself a raise, but the raise could not take effect until the voters had a chance to throw out the legislators who voted for it. Six states ratified. None of the rest acted. The amendment had no ratification deadline, because the seven-year clock Congress now

routinely attaches did not yet exist as a convention. It had been pending for one hundred and ninety-three years when Watson read about it.

Watson wrote a paper arguing that the amendment was still alive, that any state legislature could still ratify it, and that the country could still adopt it. His professor disagreed. She gave him a C. Watson did not appeal the grade. He decided to prove her wrong.

He began, with no organization behind him and no funding beyond what he could spare from a legislative aide's salary, to write letters — on his own time, on his own typewriter, signed by hand, to state legislators in every state where he could identify a likely contact. The early letters to members of Congress went nowhere. But a state legislator in Maine read his letter and decided the amendment was worth raising in his chamber. Maine ratified in 1983. Colorado followed in 1984. The two ratifications were enough to make the campaign newsworthy.

Activist groups across the political spectrum picked the story up. Taxpayer organizations on the right, government-reform organizations on the left, and voters who had spent the early 1980s watching Congress give itself mid-term pay raises rallied behind the amendment Watson had resurrected. Five states ratified in 1985. Nearly twenty more joined by the end of the decade. Watson spent six thousand dollars of his own money on more letters and more conversations with legislators in capitals he had never visited. The work was unglamorous, methodical, and almost entirely conducted through the United States mail.

On May 7, 1992, Michigan was certified as the thirty-eighth state to ratify, and Madison's proposal became the

Twenty-Seventh Amendment to the Constitution — two hundred and two years after he submitted it. It remains the most recent amendment to the Constitution. Watson later said it was the happiest day of his life. In 2017 the University of Texas reviewed his original paper. The professor, by then retired, agreed the C had been a mistake. The university changed his grade to an A-plus.

Watson needed thirty-eight state legislatures. The reader of this book needs one — one sheriff, one governor, one attorney general, one tribal chairman, one farm-association president. The math is in the reader's favor.

Why a Letter, Not an Email

The empirical case for a physical letter over a digital substitute is unambiguous. Direct mail has an average response rate of 4.4 percent; email has 0.12 percent — a ratio of roughly thirty-seven to one. Direct mail is opened 80 to 90 percent of the time against email's 20 to 30, and seventy-one percent of recipients read it the day it is delivered. Personalizing it with the addressee's actual name increases response by one hundred and thirty-five percent.

The mechanics behind those numbers are familiar to anyone who has worked in a political office. Email is filtered, batched, and delegated; the typical congressional or gubernatorial inbox is processed by automated keyword bucketing and form-response generation, and the human eye that sees a message at all belongs to a second-year staffer with three hundred more in the queue. Physical mail is harder to automate. It arrives one piece at a time, in an envelope, addressed by someone who paid for a stamp. A staffer has to open it and read enough of it to decide whether it deserves a forward. A four-page package

containing a personalized cover letter, a jurisdiction-specific cost report, and a constitutional brochure is the kind of document that ends up on a desk.

The Letter Generators

The author has spent eighteen months building and maintaining a suite of letter-and-brochure generators — open-source, browser-based, free — that let any reader produce a professional, personalized, jurisdiction-specific mailing package in about ten minutes, ready to print and mail. They are hosted at marti124.github.io/Patent_Letters. No installation, no account, no cost, and no data leaves the browser.

Each generator targets a plaintiff category whose constitutional standing earlier chapters develop. The sheriff generator addresses the 3,091 elected county sheriffs in forty-five states — Alaska, Connecticut, Delaware, Hawaii, and Rhode Island are excluded, their sheriff arrangements lacking the independent constitutional standing the challenge requires — producing a personalized cover letter, a county procurement-savings report calibrated to the county's population, and a two-page constitutional brochure citing thirty-eight Supreme Court decisions. The precedent is *Printz v. United States* (1997), in which two rural-county sheriffs challenged the Brady Act and won. The governor and attorney general generator addresses one hundred officials on three independent standing theories: *parens patriae*, direct proprietary and fiscal injury, and sovereign interest in a constitutional provision designed to benefit the state's economy. The tribal-nation generator addresses the 345 federally recognized tribal nations in the contiguous forty-eight states, distributed across ten federal

circuits — a distribution that maximizes the chance of a circuit split, the single most reliable path to Supreme Court review.

How to Choose a Campaign

Pick the campaign your circumstances make easiest. A reader who lives in a county with an elected sheriff — any county in the forty-five eligible states — can produce a sheriff package in ten minutes for his own county and another ten for a neighboring one. A reader who follows state politics can write the governor and the attorney general. A reader with a tribal connection can write a tribal chairman. A reader who has time for more than one letter should send more than one: officials notice when the same message arrives from several constituents in the same week. A reader with no idea where to start should start with his own county. The county sheriff is the official most likely to read a letter from a county resident, the official with the strongest documented standing for a procurement-driven case, and the one whose generated package arrives calibrated to the county's own population.

Who Pays for This

A reader who mails a letter is entitled to ask what happens if the official says yes. A constitutional case of this size runs three to five years and costs more than any individual plaintiff will pay out of pocket. Nobody pays for it out of a contingency fee, because contingency work is funded by a share of a money recovery and this case does not ask for money. It asks a court to declare that a regulatory architecture cannot lawfully bar the manufacture of a patent-expired design. No pot exists for a funder to take a percentage of. That explains something about the silence:

the framing that is constitutionally strongest is the one private capital has no reason to finance. The money in this case goes to the American household, in the price of a truck, and never passes through a courtroom on the way.

Four sources fund it instead, and they combine. The public-interest constitutional bar — the Pacific Legal Foundation, the New Civil Liberties Alliance, the Institute for Justice — takes cases without charging the client. The membership association funded by dues is the most durable plaintiff available for a case that will outlast several news cycles. Co-counsel arrangements pair a nonprofit carrying the constitutional theory with a private firm supplying document work and experts. And the sovereign — a state attorney general with a litigation budget, or a tribal nation with counsel of its own — needs no fee arrangement at all.

A cleaner answer exists, and it is one of the quiet convergences in this book's argument. Under *Hunt v. Washington State Apple Advertising Commission* (1977), an association may sue on behalf of its members when a member would have standing, the interest is germane to the association's purpose, and neither the claim nor the relief requires individual participation. The plaintiff in *Hunt* was financed by mandatory assessments on the growers it represented — a dues-funded association suing for declaratory and injunctive relief. That is this case exactly. And in *United Food and Commercial Workers Union Local 751 v. Brown Group* (1996), the Court explained that individual participation is not normally necessary for prospective or injunctive relief but would be required in an action for damages. Read together, the funding question and the remedy question turn out to be the same question. The best-capitalized, most durable plaintiff structure available can bring the case this book asks for and cannot

bring a case for money. A reader tempted to think the association should sue for damages instead should notice what that choice costs: it forfeits the plaintiff.

The Stamp the Constitution Is Waiting For

Watson got his thirty-eight legislatures by writing letters, one at a time, to officials he had never met. The Twenty-Seventh Amendment was ratified because he did the unglamorous work the Constitution had been waiting on for two hundred and two years.

The reader of this book needs one official with standing. Five categories of such official have been developed here, each with constitutional standing the courts have already recognized in adjacent cases. The letters are written; the brochures are written; the constitutional argument is written. What is not yet written is the letter the reader will sign and mail. The home page is at marti124.github.io/Patent_Letters. The pencil is still on the kitchen table. Pick it up.

One thing remains before the playbook is yours to run. A reader entitled to act is also entitled to be sure. The chapter *Reason the Constitutional Argument Yourself* hands you the means to test the case in these pages against independent judgment — not the author's, and not a lawyer's, but reasoning you can watch unfold and check for yourself.

CHAPTER 35

When Outsiders Advance Constitutional Law and Other Disciplines

The Supreme Court's own opinions, a hospital's own death records, a journal's own back issues, a regulator's own files — twenty cases where the answer sat in plain sight for decades, and the people who finally asked. In the four constitutional cases, the fact that moved the doctrine came from outside the law every time.

A field can fail at its own edge in two ways. Sometimes the field is working with what it has, and someone arrives with something better — a sharper instrument, a larger set of records, a method nobody had tried. Call that failure the better tool. Sometimes the field already holds everything the answer requires. The numbers are printed in its own annual report. What is missing is a question. Call that failure the unasked question.

The two are not equally serious. The better tool indicts nobody; a geologist in 1900 cannot be blamed for lacking satellite imagery. The unasked question means the field held the answer, looked at it, and did not see it — and the reason was never a shortage of evidence.

The argument in this book is an unasked-question case. The regulations are published in the Federal Register. Patent expiration dates are public. The window stickers are on the cars. It required somebody to ask what happens to a design when the patent runs out and the federal standard does not.

This chapter does three things. It shows how constitutional doctrine goes blind, and who has noticed. It shows why the person who notices almost never wins the case, which matters because the author of this book cannot bring one. Then it turns to fifteen years of results where machines did the searching. Nobody aimed them.

Two rules run through all of it. Every case gets labeled — better tool or unasked question — and where a case is some of each, which part is which. And where the popular version of a story is wrong, the correction is given instead of the half that helps. Rounding a case toward the answer you want is the exact error these cases charge other fields with.

Thomas Kuhn gave the problem its vocabulary in *The Structure of Scientific Revolutions*. A field operating normally does not test its own foundations. It works inside a settled framework that tells its members which questions count. That framework makes ordinary work efficient, and it makes certain facts invisible while they sit in plain view. A question outside the field's settled sense of itself does not get refused. It never comes up.

Movement One: The Unasked Question

Start with Harry Markopolos, because this case has the same shape as the argument here. A financial analyst, he was asked to copy Bernard Madoff's investment returns and could not, because they were not possible. He told the Securities and Exchange Commission in 2000, again in 2001, and in 2005 sent a document titled "The World's Largest Hedge Fund is a Fraud." The agency did not act. The scheme collapsed in December 2008 and took roughly sixty-five billion dollars with it. The Commission's own Inspector General later found it had received credible

complaints and failed to act. Markopolos did not find a document; he read numbers the agency already had, with no standing and no way to make anyone look. The unasked question, and note the kind: a regulator failing to see what sat in its own files, delivered by hand, three times.

Aiko Herzig-Yoshinaga had no law degree and no institutional standing; as a teenager she had been imprisoned at Manzanar. In the National Archives she found what forty years of litigation had not — the government's original copy of General DeWitt's *Final Report*, with passages a later version had removed. They showed the claim of military necessity made to the Supreme Court in 1944 had been made while contrary evidence sat in the file. Fred Korematsu's conviction was vacated in 1983, and in *Trump v. Hawaii* (2018) the Court said *Korematsu* had been gravely wrong the day it was decided. The unasked question, in its most damning form. The evidence sat in the government's own files for forty years.

Bowers v. Hardwick (1986) upheld a Georgia sodomy law and stated that laws against homosexual conduct had ancient roots. Seventeen years later a group of historians filed a brief in *Lawrence v. Texas* (2003) taking that statement apart from sources in print the whole time. The category of person *Bowers* described did not exist in the period *Bowers* pointed to. The Court overruled *Bowers*. The unasked question, and the cleanest one in law. A court asserted a fact, and for seventeen years nobody checked it against scholarship that already said otherwise.

Josephine Goldmark was research director of the National Consumers League and had no law degree. Of the hundred and thirteen pages filed for Oregon in *Muller v.*

Oregon (1908) under Louis Brandeis's name, she and her researchers supplied about ninety-eight — factory reports, medical testimony, foreign labor statutes. Roughly two pages were legal argument. The Court upheld the law and took formal notice of the material; the Brandeis brief is the ancestor of every filing that brings a court facts rather than precedent. Two corrections. Some of the medical evidence was later refuted. And *Muller* rested on women's physical dependence — limits imposed on women and not on men — a ground contested ever since; this book takes no position on the dispute. Praise the method. Do not praise the holding. The unasked question. The reports were public; nobody had put them in front of a court.

Kenneth and Mamie Clark were psychologists, not lawyers. The doll studies did not decide *Brown v. Board of Education* (1954). They appear in footnote eleven, supporting a conclusion the Court had already reached on constitutional grounds. What the Clarks supplied was a finding the Court was willing to state — that segregation produced a sense of inferiority in children. The unasked question, with the causal claim kept small. They did not win *Brown*. They gave the Court a sentence it could write.

The pattern outside law is the same. The Vienna General Hospital ran two maternity clinics, one staffed by doctors, the other by midwives. From 1840 through 1846 the death rate from childbed fever in the doctors' clinic was 98.4 per thousand births; in the midwives' clinic it was 36.2. The figures were printed in the hospital's own annual returns, and because women were admitted on alternate days with no medical selection, the obvious objection was not available. Ignaz Semmelweis saw the cause in 1847: the doctors came to deliveries from the dissecting room. He ordered chlorine handwashing, and deaths fell. One correction. Semmelweis was not an outsider. He was a

credentialed physician in the very clinic whose figures he read. If a theory of blindness requires an outsider to break it, that theory deletes the best case medicine has. The unasked question.

Pellagra killed thousands of Americans a year in the early twentieth century, most of them poor and Southern, and the consensus was that it was infectious. Joseph Goldberger made an observation that required no instrument: in the asylums where inmates fell ill by the hundreds, the staff never did. An infection does not respect a payroll. So it was diet. The resistance is the useful part. Southern physicians, newspapers and officials rejected a finding that blamed regional poverty. The objection was not to the evidence but to what the answer implied about them. The unasked question.

In 1946 the radiologist John Caffey described a pattern in infants — multiple fractures healing at different times, with bleeding around the brain — that does not happen by accident. The films sat in hospital record rooms for sixteen years, until C. Henry Kempe named the battered-child syndrome in 1962. The first state reporting laws passed in 1963; by 1967 all fifty states had one. The unasked question, with a sixteen-year gap and a five-year legislative answer.

Yuri Knorozov is the purest case. A sixteenth-century Spanish friar, Diego de Landa, wrote down what he took to be the Maya alphabet. It and photographs of the surviving Maya books were in print by the end of the nineteenth century. Scholars used it and failed, because everyone read the list as an alphabet. In 1952 Knorozov, in Leningrad, read it as a syllabary: when de Landa asked for the letter B, his informant had given him the syllable *be*. Now the correction, and it makes the case stronger. It is

not true that nobody asked. Many asked. What Knorozov brought was not evidence and not an instrument but a different idea about what kind of writing it was. That is the shape of the argument in this book. The Patent Clause has been read continuously since 1787. Reading a text wrongly is not the same as never reading it. That is why *forgetting* is the right word for what happened to the Clause's second half. The unasked question, and the strongest example here.

Robin Warren and Barry Marshall must be labeled honestly, and the label is mixed. In 1938 Doenges examined 242 human stomachs after death, found spiral organisms in forty-three percent, and connected them to no disease. In 1954 Palmer studied tissue from a thousand people and concluded they were contamination from the mouth. Medicine already believed the stomach was sterile; Palmer's study told the field what it already believed, on a sample too large to argue with, and the question closed for thirty years. Warren took the organisms seriously again in 1979; Marshall joined him in 1981. What broke the resistance was new capability: the bacterium was grown in 1982, and Marshall drank a dose in 1984 and made himself ill. So the slides were an unasked question for decades, and the answer came from the better tool. The sharper point: the organisms had been counted in nearly half a large sample — and then explained away.

Two more from the physical sciences. J Harlen Bretz read the scoured landscape of eastern Washington in 1923 as the work of a single catastrophic flood; geology treated catastrophe as disreputable, and in 1927 senior geologists, several of whom had never visited the ground, took his argument apart. Joseph Pardee supplied the missing water in 1942 — a glacial lake in Montana that had drained catastrophically. The unasked question, with a

missing fact; roughly fifty years. Fritz Zwicky in 1933 measured how fast galaxies moved inside the Coma cluster against how much matter he could see, found the cluster should have flown apart, and named what held it together dark matter. The calculation used published measurements and standard physics. It was largely neglected until the 1970s. Mixed: the 1933 result was an unasked question, and the resolution took better tools.

For decades, parents were told to put babies to sleep on their fronts. The reasoning was never tested. A 2005 systematic review in the *International Journal of Epidemiology* pooled the studies as they appeared, year by year, and asked when the answer would have become visible. By 1970 the pooled odds ratio for front sleeping was 2.93 — nearly three times the risk. The advice continued in childcare books until 1988. The reviewers' estimate: a review done in 1970 might have prevented more than ten thousand infant deaths in the United Kingdom alone. The unasked question, and possibly the cleanest one in this book. The studies were on library shelves, and the method that found the answer was to add up what was already there. What was missing was the decision to look.

One qualification. No book gathers these constitutional cases into a single account of how doctrine goes blind and who notices, though Sam Erman, Peter Irons, and Lani Guinier and Gerald Torres have each written nearby. No subscription law-review database was searched; the claim is that nothing surfaced, not that nothing exists.

Movement Two: Why the Person Who Notices Rarely Wins the Case

Every case above has the same structure. Somebody outside the field supplied the material. Somebody inside the field carried it to a decision. The two jobs never sat in the same person. Search for the pure form — an outsider whose own research produced a Supreme Court holding in a case that person brought — and you find nothing. What follows is an account of why.

Clarence Earl Gideon wrote his own petition from prison. He is the story everyone reaches for, and he does not fit. The Court appointed Abe Fortas, and the winning argument was Fortas's; *Gideon v. Wainwright* (1963) supplied the case the Court wanted, not the theory. Susette Kelo, whose house was condemned for a development, discovered nothing and lost in *Kelo v. City of New London* (2005). What matters is what happened after: more than forty states tightened their eminent-domain laws, and the development was never built. A loss can produce a reform. Keep that in mind.

Brewster Kahle came closest. An engineer and archivist, not a lawyer, he made a structural observation of exactly the kind this book makes. When Congress removed the requirement that copyright owners renew their copyrights, it turned an opt-in system into an opt-out one. Works stopped entering the public domain by default. He lost. The Ninth Circuit, in *Kahle v. Gonzales* (2007), treated it as the argument the Supreme Court had rejected in *Eldred v. Ashcroft* (2003). The court admitted *Eldred* had left open whether “limited Times” puts a real ceiling on a term, then left that ceiling to Congress. And the Solicitor General then cited *Kahle* against the next challenger; his defeat became a tool in the government's hand. The distinction this book

rests on is that *Kahle* was about extending a term. The argument here alleges no extension of anything. Whether that distinction holds is for a court to say.

The reason the two jobs never sit in one person is arithmetic and procedure. In one recent set of 6,273 petitions to the Supreme Court, 6.3 percent were granted or sent back for reconsideration. Of the 3,785 filed by people too poor to pay the fees, 98.78 percent were denied, against 85.97 percent of the 2,488 paid petitions. A person representing himself may represent only himself. When the Court wants a case it appoints a lawyer, which separates the person who found the thing from the person who argues it, by design.

Pierre Azoulay and two colleagues measured the gate from the other side. They found 452 star scientists in the life sciences who died unexpectedly and tracked their fields afterward. Publication by scientists who had never worked with the deceased rose 8.6 percent. That is the mechanism behind every case above, measured. Not that outsiders are right. That gatekeeping is real, and that it relaxes when the gatekeeper dies.

The docket. No reported federal decision has reached the merits of a claim that federal regulation, without extending any patent term, preserves practical exclusivity in an invention whose patent has expired. The statutory neighbors — patent-term extension and adjustment, orphan-drug exclusivity, biologics — were checked one at a time; in none was the constitutional question reached or raised.

Why has nobody argued this? Generic drug makers fight regulatory exclusivity constantly. Three honest answers. Congress gave generic manufacturers a statutory path to market, and a company with a cheaper remedy does not

reach for the Constitution. Durable goods have no such path — there is no abbreviated application for a patent-expired engine — so the blockage is total rather than temporary, which is exactly why it raises a question the drug cases never had to reach. And a would-be remanufacturer must establish that the injury is real before anything has been built. That is a reason the question has gone unasked. It is not a reason the answer is no.

This is why the book is written as a brief for a plaintiff rather than as a complaint. The author found the argument. The author cannot carry it. The honest response is to write for the person who can, not to pretend to be him. Adoption is the verification. That is mechanics, not modesty.

Movement Three: When the Machine Asked

The last group is different, because in it nobody aimed the machine. In AD 79 Vesuvius buried Herculaneum and with it a villa library of roughly 1,800 papyrus scrolls, cooked to charcoal. They are unreadable because the ink is carbon, the papyrus is carbon, and unrolling one destroys it. In 2019 Brent Seales put two of them through a particle accelerator and produced high-resolution X-ray scans. In 2023 the scans were released publicly as an open competition with a seven-hundred-thousand-dollar grand prize.

Two people acted on those scans, and the difference between them is the point. Casey Handmer is a physicist with no papyrology and no Greek. He built nothing. He looked at the scans for hours. He noticed a faint texture, like cracked mud, in shapes that looked like Greek letters, and published it on his blog in August 2023. The carbon ink is invisible against carbon papyrus; what Handmer saw was the disturbance the ink had left in the surface it dried

on — the mark of a thing rather than the thing. It was the first substantial evidence of ink inside an unopened scroll. The unasked question. He looked at a picture everyone had, for longer than anyone else had, without knowing what he was supposed to see — which is precisely why he saw it.

Luke Farritor was twenty-one, a computer science undergraduate at Nebraska–Lincoln. He read Handmer’s post and did the thing Handmer could not: took the crackle as training data, built a model to find more of it, and iterated until the model returned a word: πορφύρας — purple. By December, Farritor, Youssef Nader and Julian Schilliger had recovered about two thousand characters and shared the grand prize. This one is the better tool, and the labels matter here because the sequence is the point. Seales built the instrument. Handmer, with no standing, noticed something in data everyone possessed. Farritor, with no standing either, built a machine that could find more of it. The field’s own experts said whether it was Greek. Four steps: an institution supplied the capability, an outsider the observation, a second outsider the machine, and insiders the judgment about whether any of it was true.

That is the arrangement behind this book, and it should be said plainly rather than left for a reader to infer. The author used machine assistance to assemble the argument. He says so here, because the alternative is letting a reader find it out. What the Herculaneum sequence shows is that the assistance was one step of four. Somebody has to notice. Somebody else has to check. The middle step is the only one a machine performs.

Christopher Shallue was a software engineer at Google, not an astronomer. NASA’s Kepler archive was public and had been worked over for years by professionals who had

extracted thousands of planets from it. They had concentrated on strong signals, because a weak dip in brightness is usually noise. Shallue's idea was that a neural network might sort the weak ones — that the constraint had been human attention rather than the data. He trained a network on signals scientists had already labelled and, with Andrew Vanderburg, the astronomer who supplied the domain judgment, pointed it at 670 systems known to hold several planets. It found Kepler-90i, an eighth planet the earlier methods had missed. A hybrid, and the halves are worth separating. The network was a new tool. But the data was public and had been examined. What had not happened was anyone asking whether the signals the professionals had reasonably skipped were worth a second look. The question was the contribution. The network was how he could afford to ask it.

In the results below, the person holding the instrument was an insider, working on data his own field had been generating for decades. That takes the operator's credentials out of the argument entirely. These are demonstrations that a field can stare at its own evidence for a century and not see what is in it.

Doctors have photographed the back of the eye for about a hundred years. Ryan Poplin and his colleagues trained a system on retinal photographs from 284,335 patients. From the photograph alone it estimated age, predicted whether the patient smoked and whether the patient would have a major cardiac event — and identified the patient's sex almost perfectly. No ophthalmologist can do that. The unasked question, without qualification. The instrument was new. The information was not.

Andrew Beck's team at Stanford turned a system called C-Path on breast-cancer slides. Breast cancer had been

graded for decades by reading the cancer cells themselves, under a scheme largely unchanged since the 1920s. C-Path found that features of the stroma — the ordinary connective tissue around the tumour — predicted survival more strongly than the tumour did. Two corrections. It is partly a better-tool case. And the ninety years sometimes attached to the story belongs to the grading scheme, not to how long the stroma went unread. Nothing hid the stroma. Specialists walked past it for the better part of a century — not because they could not see it, but because they were confident they already knew which part of the slide mattered. The unasked question. That sentence is the thesis of this book, and a pathology slide is where it can be proved.

Two more, briefly. Mayo Clinic researchers trained a system to detect a weak-pumping heart from an ordinary electrocardiogram, a tracing recorded since the early 1900s, then tested it in a randomized trial across 120 primary care teams covering 22,641 adults; new diagnoses within ninety days rose from 1.6 percent to 2.1 percent — a modest effect, stated as a modest effect, and a randomized trial, which almost nothing else in this movement has. Nicolas Coudray's team trained a system on lung-cancer slides stained with the dyes pathologists have used since the nineteenth century, and from the stained image alone it predicted, moderately well, which genes were mutated. A tumor's genetics leave a visible mark in the tissue, and a hundred and fifty years of pathologists did not know it. Both the unasked question.

AlphaDev and AlphaTensor are a different trick, often confused with this one. AlphaDev searched for faster ways to sort short lists of numbers. It also claimed that for the three-element case no shorter program existed. Cassio Neri showed otherwise, with counterexamples and a faster

plain implementation. AlphaTensor found a way to multiply certain small tables of numbers with fewer multiplications than the best method known since 1969 — in a restricted arithmetic, not ordinary arithmetic. Manuel Kauers and Jakob Moosbauer answered within about a week and went past the machine, cutting the five-by-five case from ninety-six multiplications to ninety-five, on a small computer, starting from the machine's output.

The machine results are the better tool. But look at what these two cases produced. AlphaDev asserted an impossibility and was wrong, and being wrong in public provoked somebody to check. AlphaTensor showed that ground the discipline had stopped digging in was not exhausted, and within days mathematicians who had held the capability the whole time went further than the machine. So the machine results are the better tool, and the human responses to them are the unasked question. What the machines dislodged was not a technical barrier. In one case it was a false claim. In the other it was a settled belief that a question was closed.

The Turn

In every case in this movement but the last two, the machine was not told where to look. And in every case, a human being decided what counted as proved. Mayo ran a randomized trial across forty-five sites. The pure-mathematics result the preface describes arrived with a companion paper from nine mathematicians who had checked the proof by hand.

That structure is the structure of this book. Tools helped assemble the argument. Every case in it was read. Every figure was checked against a primary source or marked as

unchecked. And a court, not a machine and not the author, will decide whether the argument stands.

Close

Fields build up blind spots at their own edges. People whose training points at those edges are more likely than average to see what is sitting there. That is the whole claim. It is narrower than the story usually told about outsiders, and it is the only version the evidence supports.

Constitutional law is a field with that kind of edge. It has the documents. It has had them the whole time. What it has not had, on this question, is the question. This chapter has explained why the author is not among those with standing to file it, and why that is the ordinary condition rather than a defect in the argument.

Goldmark did not argue *Muller*. The Clarks did not argue *Brown*. Herzig-Yoshinaga did not argue *Korematsu*. Markopolos did not prosecute Madoff. Each of them found something. Somebody else carried it. The finding is here. The blindfold is described. Somebody with standing has to take it off.

EPILOGUE

In August of 1813, Thomas Jefferson sat down at Monticello to answer a letter from a Baltimore miller named Isaac McPherson. He was seventy years old. He had written the Declaration. He had bought Louisiana. He had served two terms as President. He had four years left to live.

The letter he wrote contains what may be the clearest statement of the patent bargain ever set down in English. Jefferson did not invent the bargain that day. Congress had codified it twenty-three years earlier, in the Patent Act of 1790, and the Constitutional Convention had bounded it three years before that, in the nine words of Article I, Section 8, Clause 8 and the three that followed: for limited Times. Jefferson's letter put a frame around what the country had already decided — that an inventor would receive a temporary monopoly in exchange for the public's reciprocal right, when the time was up, to make and copy and remanufacture what had been disclosed.

That was the deal. Two hundred and thirty-six years ago this year, the United States made it with itself, in the second piece of legislation Congress passed under the new Constitution.

Jefferson did not know the deal would be broken. He thought the limited-times language would be enough, and he had reason to: the men who ratified the Clause were the men he knew, the men he had argued with, the men he had governed alongside. He could not have foreseen that a century and a half later a regulatory architecture nobody in 1813 had a name for would erect a wall around the products themselves and leave the public's right standing on paper while it disappeared in practice. He could not

have foreseen that the cumulative figure across the fifty-six years between 1970 and now would, on the conservative reading, reach twenty-two trillion dollars.

What does \$3,385 per household per year, every year, indefinitely, look like in an actual American home? It is the family vacation that did not happen. The second car the wife had to take a job to afford. The dental work the grandfather put off. The tuition the daughter borrowed for. The down payment that took three more years to assemble. The retirement that arrives five years later because the wall ate the difference. It is not a line item any one household notices on any one Tuesday. It is the line item that explains why the household that used to run on one paycheck now runs on two and still does not make it.

The American median household earned \$83,730 in 2024, per the most recent Census Bureau report, which makes \$3,385 roughly four percent of gross household income — the share the wall has been silently extracting since 1970. But the average is not where the cost-of-living crisis lives.

The lower earning half of society — the trucker, the small farmer, the rural mechanic, the working family whose budget is dominated by the categories the wall encloses — pays a larger share of income on each of those categories, and stands to gain a larger share back. The further down the distribution, the larger the proportional gain. The remedy helps everyone. It helps the most cost-burdened Americans most, which is the inverse of how the present wall distributes its harm.

The cost has a mirror image. The forward-looking projection developed in *Five Years On: The Household Recovery* is roughly nineteen thousand dollars a year of restored purchasing power for the median household once the bargain is honored — about a quarter of median

annual income. The \$3,385 is what the wall takes. The nineteen thousand is what the household gets back, because the recovery reaches every category the wall touches and not only the ones a household buys this year.

The arithmetic runs far enough to be worth stating once. Take only the lower thirty-five percent of American households — roughly forty-six million, averaging on the order of thirty-two thousand dollars a year — and assume they buy most of their major durable goods as American-built remanufactured patent-expired products. The restored purchasing power runs to seven or eight thousand dollars a household per year, about a quarter of pre-ruling income. Aggregated, that is on the order of three hundred fifty billion dollars a year staying in lower-income American homes instead of flowing to incumbent manufacturers. Over fifty-six years going forward, the cumulative figure approaches twenty trillion — roughly the magnitude the wall has extracted going backward. The lower thirty-five percent alone, with no participation from anyone above the median, can recover what the wall has taken from everyone.

I am not the stereotypical research author to tell this story. The credential that matters most is the one from sitting on the inventor's side of the bargain at a Fortune 25 company for thirty-four years. The reason I am nonetheless the right person to tell it is that the bug runs through the system that produced it. Most of the people inside that system never noticed; some did, and those who did had their own reasons for staying quiet. The disciplines trained to govern the sectors the wall encloses — patent law, constitutional law, regulatory economics, administrative law — were each trained, by the structure of the disciplines themselves, not to see the connection. A man trained to find bugs in code sees the connection right away.

This book carries a subtitle as long as its title, resting on three concrete claims about what a single constitutional ruling would deliver: how restoring the bargain frees buyers from surveillance, stops the robotic replacement of America's drivers and operators, and returns the joy of operating real machines. Each names a different group of Americans who already know they have a stake — the new-vehicle buyer trapped inside a federally mandated surveillance subscription, the working American watching the AI and robotics buildout engineered to displace him, and the buyer who remembers what it was like to be in command of a machine.

Of the three, the second carries the deepest fear in the country right now, and the patent bargain delivers the most direct answer to it. Every American worker watching the buildout knows the question being asked of him without anyone having to ask it aloud: **will my job be here in five years?**

The Patent Clause remedy answers with arithmetic. The autonomous truck and the autonomous taxi are competitive against a new Class 8 rig and a new sedan at today's prices. They are not competitive against an \$80,000 American-built patent-expired remanufactured semi, or a patent-expired sedan under fifteen thousand. Remove the wall and the autonomous business case stops penciling out at the very capital-cost layer the platforms have been counting on. The trucker stays in the seat. The farmer stays on the tractor. The earth-mover stays on his loader. The for-hire driver stays at the wheel of the vehicle he owns. Three and a half million American truck drivers and the millions of operators behind them keep the work that gives a working life its shape.

Every parent in his fifties watching a twenty-five-year-old struggle for a foothold in a labor market the buildout is already reshaping — and every grandparent watching a five-year-old on a trajectory toward a labor market being engineered out of existence before that child finishes elementary school — gets a different America to bequeath. The corridor of human work the post-1970 wall has been narrowing for fifty-six years widens again the day the wall comes down. It is not a guarantee. It is a corridor. The constitutional remedy is what reopens it.

This book has spent thirty-five chapters showing what Jefferson's deal was, how Americans honored it for one hundred and eighty years, how it was broken after 1970, and what it would take to honor it again.

When the argument was put to eleven of today's independent AI systems, built by different makers and given only the neutral materials, their reasoning converged: the injury is real, and the half of the bargain the Constitution promised — the public-domain freedom to remanufacture items whose patents have expired — is not being delivered. They decide nothing. But any reader can run the same test and see where the authorities lead.

Three websites accompany this book, and each is built to be used:

marti124.github.io/Patent_EndNotes holds the endnotes, the appendices, and the Glossary, with coverage of every Supreme Court case the argument relies on — more than eighty of them, each described for what it actually holds. marti124.github.io/Patent_Exam holds the examination materials — the attachment in PDF and Word, a scoring page, and how all eleven AI engines handled the exam's most important question. It opens with the chapter *When Outsiders Advance Constitutional Law and Other*

Disciplines, which is the answer to anyone who doubts a question like this can come from outside the law. It is built to be shared, especially with attorneys. If you know a lawyer, or know anyone who does, send him there. And marti124.github.io/Patent_Letters holds the letter-generation apps for writing to promising plaintiffs.

The remedy does not require an act of Congress. It does not require a constitutional amendment. It does not require a new legal theory. It requires one plaintiff with standing in one federal courtroom, asking one question about two words (“limited Times”), supported by more than eighty Supreme Court cases handed down across one hundred ninety-five years of precedent.

That plaintiff has not yet stepped forward.

The case will be filed by a sheriff whose county vehicle-procurement budget has been eaten for fifteen years, who has read this book and walked across the hall to the county attorney. Or by a state attorney general who ran the math on what the wall is costing his state’s taxpayers and decided that the trajectory from a successful filing to a federal judgeship runs through this case. Or by a tribal nation whose transportation budget has absorbed the blocked tier hardest for forty years, with public-interest counsel waiting at no cost. Or by an owner-operator who has paid the diesel-particulate-filter premium every year of his career and watched the autonomous platforms build the infrastructure for his replacement on the same regulatory architecture that priced him out. Or by some farmer in Iowa who decides he has had enough. Any one of them is the case. None of them needs the others. The case will not wait for all of them, and it does not need to.

One further thing the post-ruling marketplace will let Americans do has not had its own page-space until now.

Every Congress that has sat between 1970 and the present has refused, in both parties, in every session, to honor the patent bargain. Every administration since 1970 — every one, in both parties, no exceptions — has signed statutes and certified regulations that built the wall, brick by brick, fifty-six years of brick. Once a federal court takes it down, every American who buys an American-built remanufactured patent-expired pickup, tractor, HVAC unit, light bulb, appliance, or Class 8 rig is casting a vote with his wallet — a take this to every Congress and every administration that built and maintained the wall, and a reciprocal thank you to the federal court that took it down.

That posture is not a substitute for the citizen's playbook. To achieve this, I urge readers to write at least one letter, and the more the better.

Imagine an America with a restored second half of the Patent Clause and a usable Expired Patent Commons — a country whose every household, on every major purchase, can express its constitutional judgment in cash. That judgment compounds. It also tends to elect different legislators.

Two requests remain, because the case needs people more than it needs anything else.

First, the lawyers. More lawyers are needed to confirm the legality of the constitutional argument, which is bound for federal court. If you know an attorney — or know anyone who does — send him or her to the examination site, and especially to its invitation to review. A few words from a practicing lawyer carry further than anything the author can say for himself. And if the objection is that this argument comes from outside the law, the exam site opens with the chapter written to answer exactly that.

Second, the plaintiffs. This book names the plaintiff categories — the sheriff, the state attorney general, the tribal nation, the owner-operator trucker, the farmer, the county fleet manager, and the rest. Think through the people you know. If any of them works in one of those categories, put this book in their hands, or generate a letter at the letter site and mail it. One letter can start the case. More letters make it likelier that the one who reads his is the one who files.

The bargain is older than any of us. The violation is younger than most of us. The remedy is constitutional, available, and waiting.

Two hundred and thirty-six years ago, the United States made a deal. It is still on the table. Whether anyone honors it now is up to whoever reads this book and decides to act.

— *Roleigh Martin*
Northern Kentucky

BOOK ENDNOTES, GLOSSARY, AND KEY TERMS

A Note on This Section of the Author Review Copy

This printed Author Review Copy does not include the book's endnote apparatus: the endnotes, the endnote commentary, Appendices A through C, the glossary, and the key terms. That material runs to 243 pages, printed on ordinary 8½×11 paper with normal margins. It is meant to be read electronically, for one practical reason. **Every endnote ends in a clickable source link.** On the printed page those links are dead ink. On a screen they are live, and they are the point.

The argument in this book rests on what the cited authorities actually hold. The endnotes are where a reader checks that work — each citation follows Chicago style and closes with a working URL to the original source, so any claim can be traced to its document in a single tap. A reader who wants to verify the case law, the cost figures, the patent lineages, or the historical record will find the electronic version far more useful than a printed one.

The clickable links are not the only advantage. In electronic form the whole section is searchable: a quick Find (**Ctrl-F**, or Command-F on a Mac) jumps straight to the phrase, case name, or glossary term you have in mind, instead of paging through more than two hundred printed pages to locate it. For a reference section built to be consulted rather than read straight through, that alone makes the electronic copy the friendlier one.

What the full section contains

Endnotes — one book-wide alphabetical list of citations, keyed to the author, case, or statute name as it appears in the running text rather than grouped by chapter, each ending in a clickable source link.

Endnote Commentary — expanded notes that go beyond a bare citation where a point needs it.

Appendix A: SCOTUS Case Master Table — every Supreme Court case cited in this book, sorted by decision year, each described for what it actually holds and for how the Court divided, including the cases that cut against the argument.

Appendix B: Note on AI-Assisted Authorship — a disclosure of how the manuscript was researched and assembled.

Appendix C: LLM Responses to Prompt 7 in the chapter “Reason the Constitutional Argument Yourself: The Open-Book Examination” — the complete, unedited reply of each of the eleven AI models to the examination’s decisive question, in alphabetical order by model, each with a link to that model’s full session. The examination itself is online at https://marti124.github.io/Patent_Exam/.

Glossary — the acronyms used throughout the book, alphabetical by abbreviation, each defined parenthetically at first use in the running text and collected here for reference.

Key Terms — the book’s specific terms of art, alphabetical by term, from the Expired Patent Commons to the regulatory ratchet, each briefly defined.

How to read it

The complete endnotes, glossary, and key-terms section is available at the link below. The same electronic file serves every reviewer; it is identical for all recipients.

Read the full section online at:
https://marti124.github.io/Patent_EndNotes/

Thank you for reviewing the book.